

**MONITORING OF MARINE MAMMALS IN
HONG KONG WATERS (2024-25)**
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EXECUTIVE SUMMARY

In the past three decades, a longitudinal study on Chinese White Dolphins and Indo-Pacific finless porpoises has been conducted in Hong Kong. With the funding support from the Agriculture, Fisheries and Conservation Department of the Hong Kong SAR Government, the present monitoring study represents a continuation and extension of this research programme, covering the period of April 2024 to April 2025.

During the one-year monitoring period, 119 line-transect vessel surveys with 3,358.1 km of survey effort were conducted among six survey areas in Hong Kong. A total of 162 groups of 544 Chinese White Dolphins and 64 groups of 174 Indo-Pacific finless porpoises were sighted during the vessel surveys.

Overall pattern of dolphin distribution in 2024-25 indicated that they frequently occurred along the coastal waters to the west and southwest of Lantau Island, extending from Tai O Peninsula toward Kau Ling Chung and Shek Pik. The most important dolphin habitats in 2024 were mostly located along the west coast of WL, but they were nearly absent in the North Lantau region with extremely low level of occurrences.

For finless porpoises, concentration of their sightings in South Lantau waters was found all around the Soko Islands, especially on the eastern side of Siu A Chau in 2024-25. The most heavily utilized porpoise habitats in 2024 were higher at the eastern and southwestern sides of the Soko Islands, as well as between Shek Kwu Chau and Cheung Chau. Temporal changes in porpoise habitat use revealed that there was a sign of recovery in their usage near Shek Kwu Chau with a steady rebound in 2023 and 2024.

In 2024, the combined estimate of dolphin abundance in Hong Kong waters in the four main survey areas of dolphin occurrences (i.e. SWL, WL, NWL and NEL) was 38 (the combined estimates for the last six years, i.e. 2018 to 2023, were 32, 52, 37, 40, 34 and 34 respectively). Large declines in dolphin abundance were detected over the past two decades in both NEL and NWL survey areas, and such number has remained relatively stable in WL waters in the past four years.

Passive acoustic monitoring (PAM) was also conducted under the present study with a focus on deploying F-POD units among four locations within BMP and SCLKCMP as a complementary data collection approach to the vessel-based survey programme that has been increasingly ineffective to monitor dolphin occurrence in

North Lantau waters through abundance estimation. During the study, the four F-POD units collected PAM data from January 2024 to January 2025 for a combined total of 1,413.16 logged days. The results revealed that a total of 717 DPMs (all were classified as dolphin detections) were recorded among the four North Lantau sites. Most detection were from Lung Kwu Chau N, where dolphins were detected on approximately one quarter of the logged days. Both BMP sites had very minimal detection throughout the study period. There were clear differences between different seasons (higher detections during the dry season) and different times of day (more detection at night) at both marine parks. Year-over-year comparisons utilizing PAM data showed an overall decrease in dolphin detections at the four sites over the past eight years from 2017 to 2024, and such decline was still continuing in 2024 even at a very low level of detection. The detected pattern of continuing decline corresponded well (but not perfectly) with the decline in annual estimated abundance of CWD in the North Lantau region during the same period as determined from line-transect vessel survey data.

Using the photo-identification technique, 102 individual dolphins were identified with 372 re-sightings during the 2024-25 monitoring period. More than two-third of all re-sightings were made in WL waters. A total of six new individuals have been added to the photo-ID catalogue.

HKCRP researchers delivered a total of eleven education seminars at local schools regarding the conservation of local dolphins and porpoises. Through this integrated approach of long-term research and publicity programme, the Hong Kong public can gain first-hand information from the researchers.

行政摘要 (中文翻譯)

一項有關本地中華白海豚及印度太平洋江豚的長期研究，在過去三十年以來一直在進行中，而此項為期一年 (由 2024 年 4 月至 2025 年 4 月)、獲香港特別行政區政府漁農自然護理署資助的研究工作，正是這長期監察項目的延伸。

在十二個月研究期間，研究員共進行了 119 次樣條線船上調查，在全港六個調查區航行 3,358.1 公里，並在船上共觀察到 162 群中華白海豚 (總數 544 隻) 及 64 群江豚 (總數 174 隻)。

在 2024-25 年的監察年度，中華白海豚的整體分佈模式顯示牠們經常出沒在大嶼山西面及西南面的近岸水域，由大澳半島延伸至狗嶺涌及石壁一帶。中華白海豚在 2024 年的重要棲身地主要集中在西大嶼山近岸水域，但在北大嶼山水域卻錄得極低的海豚使用量。

另一方面，江豚在 2024-25 年間的目擊記錄集中在索罟群島附近水域，尤其是在小鴉洲東側水域較多出現。在 2024 年錄得最高使用量的江豚棲身地，主要集中在索罟群島的東面及西南面、及長洲與石鼓洲之間的水域。江豚於石鼓洲附近水域的使用量在 2023 及 2024 年間由極低水平顯著回升。

在 2024 年，中華白海豚在大嶼山西南、西、西北及東北四個調查區域的整體數目估計為 38 隻 (過去六年的年度數目分別為 32、52、37、40、34 及 34 隻)。在過去二十年，大嶼山東北及西北調查區域的海豚數量均錄得大幅度減少，同樣地在大嶼山西面調查區的海豚數目在過去四年已轉趨平穩。

在本研究中還進行了「被動水底聲音監察方法」(PAM)，重點是在北大嶼山水域的大小磨刀和沙洲龍鼓洲海岸公園中的四個位置部署 F-POD 裝置，作為一種補充資料收集方法，以監測海豚出現情況。因為基於過去一直沿用的船上調查方法來估計海豚在大嶼山北部水域的數量，成效越來越低，無法通過數量估計來監測北大嶼山水域的海豚出現情況。於 2024 年 1 月至 2025 年 1 月期間，四個 F-POD 水底監聽器共搜集了 1,413.16 天的研究數據。分析數據顯示共錄得 717 個 DPM (註：每一分鐘內存有至少一次海豚卡嗒聲音調即為一個 DPM)，全部均屬於中華白海豚的發聲記錄。這些海豚發聲記錄大部份來自龍鼓洲北的監察點，在這處有四分之一的日數均錄得海豚的出沒記錄；此外，兩個位處大小磨刀海岸公園的監察點的海豚發聲記錄均十分稀少。海豚發聲記錄的模式於不同季節及日夜之間在兩個海岸公園均存有明顯差異 (主要在旱季及夜間出現)。在 2017 至 2024 年間，於四個監察點的海豚發聲記錄數據整體呈現下降趨勢 (在 2024 年在極低水平下仍持續下降)，跟同期透過船上調查估算出來的大嶼山北部水域海豚年度數量所呈現

的趨勢大致吻合。

於 2024-25 監察年度期間，研究員利用照片識別技術辨認出 102 隻個別海豚、共 372 次的目擊紀錄。超過三分之二的目擊紀錄是出現在大嶼山西面水域。2024 年內共有六隻新的個別海豚被加入相片辨認名錄。

在本年度，研究員為本地中小學主持了共十一場講座，內容主要圍繞香港中華白海豚及江豚的最新保育狀況。透過揉合長期研究監察及公眾教育活動，香港市民可從研究員獲得更多有關鯨豚的最新資訊。

1. INTRODUCTION

In the past three decades since 1995, HKCRP has been conducting a longitudinal study on Chinese White Dolphins (*Sousa chinensis*) and Indo-Pacific finless porpoises (*Neophocaena phocaenoides*) in Hong Kong and the Pearl River Delta region. Such multi-disciplinary research study in the past two decades has been primarily funded by AFCD as well as various government departments and NGOs, aiming to provide critical scientific information to the Hong Kong SAR Government for formulation of sound management and conservation strategies for the local populations of dolphins and porpoises. In addition, HKCRP has been extensively involved in numerous environmental consultancy studies to assess potential impacts of marine construction works on cetaceans in Hong Kong waters and the Pearl River Estuary, and to provide suggestions and guidance on mitigation measures to lessen the pressures of the development projects on dolphins and porpoises. Results from these integrated studies have been used to establish several systematic databases, which can be used to estimate population size, to monitor trends in abundance, distribution, habitat use and behaviour over time, and to keep track of levels and changes in mortality rates of local cetaceans (e.g. Hung 2008, 2022, 2023, 2024; Jefferson et al. 2002, 2009, 2012; Wang and Hung 2018, 2022, 2023, 2024).

The present monitoring project represents a continuation and extension of this research programme, with funding support from AFCD of HKSAR Government. The main goal of this one-year monitoring study is to collect systematic monitoring data for in-depth analysis and assessment of spatial and temporal patterns on distribution, abundance and habitat use of Chinese White Dolphins and Indo-Pacific finless porpoises in Hong Kong, to take photographic records of individual dolphins, to analyze the monitoring data for better understanding of various aspects of local dolphins and porpoises, and to conduct passive acoustic monitoring to supplement the vessel surveys in North Lantau waters. The one-year project covers the period of 18 April 2024 to 17 April 2025, and this final report is submitted to AFCD as a summary of this monitoring project, covering the entire 12-month study.

2. OBJECTIVES OF PRESENT STUDY

The main goal of this one-year monitoring study was to collect systematic monitoring data for an in-depth analysis and assessment of distribution, abundance and

habitat use of Chinese White Dolphins (CWD) and Finless Porpoises (FP) in Hong Kong, to take photographic records of individual dolphins, and to analyze the monitoring data for better understanding of various aspects of local dolphins and porpoises. To achieve this, several specific objectives were set for the present study. The first objective was to assess the spatial and temporal patterns of distribution, abundance and habitat use of CWD and FP in Hong Kong in detail. This objective was achieved through data collection on dolphins and porpoises by conducting regular systematic line-transect vessel surveys and helicopter surveys.

The second objective was to identify individual CWD by their natural markings using photo-identification technique. This objective was achieved by obtaining high-quality photographs of CWD for photo-identification analysis. Photographs of re-sighted and newly identified individuals were compiled and added to the current photo-identification catalogue, with associated descriptions for each newly identified individual. Photographic records of FP were also taken during surveys for educational purposes.

The third objective was to analyze the monitoring data for better understanding of the various aspects of local populations of CWD and FP. This objective was achieved by conducting various data analyses, including line-transect analysis, encounter rate analysis, distribution analysis, behavioural analysis and quantitative grid analysis to assess the spatial and temporal patterns of abundance, distribution and habitat use and trends of occurrence of CWD and FP using vessel survey data. The fourth objective was to conduct ranging pattern analysis and residency pattern analysis to study individual core area, ranging pattern, habitat use and movement pattern of CWD based on the data obtained from both the line-transect vessel surveys and the associated photo-identification works.

The fifth objective was to carry out passive acoustic monitoring (PAM) to supplement the line-transect vessel surveys in assessing the spatial and temporal patterns of Chinese White Dolphins and finless porpoises. This objective was achieved through repeated periodic deployments, retrieval and redeployments of F-POD units within two marine parks in North Lantau waters, and utilizing specialized computer software to conduct objective automated detection and identification of echolocation click trains produced by dolphins and porpoises in the PAM data obtained from this study. The final objective was to educate the members of the public on local dolphins and porpoises, by disseminating the study findings from the long-term monitoring research programme. This objective was achieved by providing public

seminars to local primary and secondary school students through the arrangement of AFCD.

3. RESEARCH TASKS

During the study period, seven tasks were completed to satisfy the objectives set for the present marine mammal monitoring study. These seven specific tasks were:

- to collect monitoring data for assessment on spatial and temporal patterns of distribution, abundance and habitat use of local dolphins and porpoises through systematic line-transect vessel surveys;
- to analyze line-transect survey data for assessment on spatial and temporal patterns of distribution, abundance, habitat use and trends of occurrence of dolphins and porpoises in Hong Kong;
- to take photographic records of CWD for photo-identification analysis and update the photo-identification catalogue;
- to analyze photo-identification data of individual CWD to assess their ranging patterns, core area use and movement patterns;
- to conduct passive acoustic monitoring to supplement line-transect survey data in assessing the spatio-temporal patterns of CWD and FP;
- to take photographic records of FP; and
- to assist AFCD in arousing public awareness on local dolphins and porpoises through school seminars.

4. METHODOLOGY

4.1. *Vessel Survey*

The survey team used standard line-transect methods (Buckland et al. 2001) to conduct regular vessel surveys, and followed the same technique of data collection that has been adopted in the past three decades of marine mammal monitoring surveys in Hong Kong developed by HKCRP (Hung 2005, 2023, 2024; Jefferson 2000a, b; Jefferson et al. 2002). The territorial waters of Hong Kong Special Administrative Region are divided into twelve different survey areas, and line-transect surveys were conducted among six survey areas (i.e. Northwest (NWL), Northeast (NEL), West (WL), Southwest (SWL) and Southeast Lantau (SEL), and Deep Bay (DB); Figure 1).

For each vessel survey, a 15-m inboard vessel with an open upper deck (about 4.5 m above water surface) was used to make observations from the flying bridge area. Two experienced observers (a data recorder and a primary observer) made up the on-effort survey team, and the survey vessel transited different transect lines at a constant speed of 13-15 km per hour. The data recorder searched with unaided eyes and filled out the datasheets, while the primary observer searched for dolphins and porpoises continuously using 7 x 50 *Fujinon* or *Steiner* marine binoculars. Both observers searched the sea ahead of the vessel, between 270° and 90° (in relation to the bow, which is defined as 0°). One to two additional experienced observers were available on board to work in shifts (i.e. rotating every 30 minutes) in order to minimize fatigue of the survey team members. All observers were experienced in small cetacean survey techniques and identifying local cetacean species and had participated in rigorous at-sea training program provided by the principal investigator.

During on-effort survey periods, the survey team recorded effort data including time, position (latitude and longitude), weather conditions (Beaufort sea state and visibility), and distance traveled in each series (a continuous period of search effort) with the assistance of a handheld GPS. When dolphins or porpoises were sighted, the survey team would end the survey effort, and immediately record the initial sighting distance and angle of the dolphin/porpoise group from the survey vessel, as well as the sighting time and position. Then the research vessel was diverted from its course to approach the animals for species identification, group size estimation, assessment of group composition, and behavioural observations. The perpendicular distance (PSD) of the dolphin/porpoise group to the transect line was later calculated from the initial sighting distance and angle.

The line-transect data collected during the present study were compatible with the long-term databases maintained by HKCRP in a way that it can be analyzed by established computer programmes (e.g. all recent versions of DISTANCE programme including version 6.0, ArcView[®] GIS programme) for examination of population status including trends in abundance, distribution and habitat use of CWD and FP in Hong Kong waters.

4.2. Photo-identification Work

When a group of CWD were sighted during the line-transect vessel survey, the survey team would end effort and approach the group slowly from the side and behind to take photographs of them. Every attempt was made to photograph each dolphin in the group, and even photograph both sides of the dolphins, since the colouration and

markings on both sides differ. One or two professional digital cameras (e.g. *Canon EOS 7D Mark II* model), each equipped with long telephoto lenses (100-400 mm zoom), were available on board for researchers to take sharp, close-up photographs of dolphins as they surfaced. The images were shot at the highest available resolution and stored on Compact Flash memory cards for downloading onto a computer.

All digital images taken in the field were first examined, and those containing potentially identifiable individuals were sorted out. These photographs would then be examined in greater details, and were carefully compared to all identified dolphins in the Pearl River Estuary CWD photo-identification catalogue compiled and curated by HKCRP. Chinese White Dolphins can be identified by their natural markings, such as nicks, cuts, scars and deformities on their dorsal fin and body, and their unique spotting patterns were also used as secondary identifying features (Jefferson 2000a; Jefferson and Leatherwood 1997). All photographs of each individual were then compiled and arranged in chronological order, with data including the date and location first identified (initial sighting), re-sightings, associated dolphins, distinctive features, and age classes entered into a database. Any individuals not in the current catalogue were given a new identification number and added to the catalogue along with their data and text descriptions including age class, gender, any nickname or unique markings. The updated photo-identification catalogue incorporated all new photographs of individual dolphins taken during the present study.

4.3. *Passive Acoustic Monitoring*

In recent years, the dolphin numbers in North Lantau waters have dramatically dropped to extremely low level (Hung 2023, 2024). As a result, estimating their abundance in this region with reasonable precision becomes increasingly difficult to near impossible. As part of a rethink of monitoring strategy to avoid the ineffectiveness in monitoring dolphin occurrence in North Lantau waters through the traditional line-transect vessel survey method adopted in nearly three decades, passive acoustic monitoring was conducted under the present marine mammal monitoring study with a focus on deploying F-POD units among four locations within the two marine parks in North Lantau waters. Recent PAM studies demonstrated that this monitoring approach is extremely valuable in assessing dolphin occurrences, especially in North Lantau waters where such occurrence has remained at historical low and near-absent level in recent years. Another consideration for such complementary approach to adopt the “hybrid” approach of shipboard and passive acoustic monitoring is related to the fact that most dolphin activities within the two marine parks in North Lantau, namely the Sha Chau and Lung Kwu Chau Marine Park (SCLKCMP) and the Brothers

Marine Park (BMP), occurred primarily outside of daily hours (see Wang and Hung 2018, 2022, 2023, 2024). The distinct diel pattern of dolphin occurrence in North Lantau waters provides strong argument that PAM method is evidently superior to the visual monitoring method in assessing dolphin occurrence and associated temporal trend in this specific region.

For the present study, besides the on-going line-transect vessel surveys in North Lantau waters, the HKCRP team also deployed and installed four F-POD units within SCLKCMP and BMP for a period of 12 months from mid-April 2024 to mid-April 2025. These four sites (“Lung Kwu Chau N” and “Sha Chau SE” in SCLKCMP; “Siu Ho Wan” and “Tai Mo To” in BMP) for the F-POD deployments are shown in Figure 1 and are the same sites (with exactly the same deployment locations) used in previous PAM studies funded by AFCD during the eight consecutive years of 2017-25 (see Wang and Hung 2018, 2022, 2023, 2024).

During the first deployment on April 17th, 2024, the F-PODs with their associated underwater frames were installed by the professional dive team from Oceanway Corporation Limited at each of the four sites within BMP and SCLKCMP. Subsequently, three events of retrieval and data recovery of the units were arranged by the dive team on July 24th, 2024, October 17th, 2024 and January 20th, 2025 (each with approximately three-month interval of data collection). During these events, HKCRP researchers were onboard to replace the batteries and SD cards, and recalibrate the clock for each unit after retrieval and refurbishment of the unit surface. The units were then re-deployed and re-attached to the underwater frame immediately so minimal time was lost for data collection. Spare F-POD units were brought on board for each dive trip, and in the event of any loss of, or damage to, the units during the retrieval process, the spare unit could be deployed immediately to avoid or reduce any data gaps in deployment period at each location. The serial number of all F-POD units as well as the time and date of deployment were recorded during each deployment. One final trip was made on April 6th, 2025 to recover the units and underwater frames from the four deployment sites with the final three months of data being retrieved.

4.4. Data Analyses

4.4.1. Distribution pattern analysis

The line-transect survey data were integrated with a Geographic Information System (GIS) to visualize and interpret different spatial and temporal patterns of dolphin and porpoise distribution using their sighting positions collected from vessel and helicopter surveys. Location data of dolphin and porpoise groups were plotted on

map layers of Hong Kong using a desktop GIS (ArcView[®] 3.1) to examine their distribution patterns in detail. The dataset was also stratified into different subsets to examine distribution patterns of dolphin groups with different categories of group sizes, fishing boat associations, young calves and behavioural activities. Data from the long-term sighting databases were used to compare past distribution patterns of dolphins and porpoises in recent years to data from the present study period.

4.4.2. Encounter rate analysis

Since the line-transect survey effort was uneven among different survey areas and across different years, the encounter rates of CWD and FP (number of on-effort sightings per 100 km of survey effort) were calculated separately for each survey area to correct for the uneven survey effort. As such, encounter rates could be useful indicators of the relative importance of different regions within the study area to the dolphins and porpoises.

4.4.3. Line-transect analysis

Density and abundance of CWD for the four main areas of their occurrences (i.e. NEL, NWL, WL and SWL) in 2024 were estimated by line-transect analysis using systematic line-transect vessel survey data collected during the present study. To examine the long-term trend in dolphin abundance, the annual estimates in 2024 from the four areas were also compared with the ones generated for every year since 2001 in the NWL and NEL survey areas and since 2003 in the WL survey area, as well as the biennial and annual periods since 2002 in the SWL survey area.

For the line-transect analysis, survey effort in each single survey day was used as the sample. Estimates were calculated only from dolphin sightings and effort data that were collected during conditions of Beaufort 0-3 (see Jefferson 2000a) and using standard line-transect methods (Buckland et al. 2001). The estimates were made using the computer program DISTANCE Version 6.0, Release 2 (Thomas et al. 2009). The following formulae were used to estimate density, abundance, and their associated coefficient of variation:

$$\hat{D} = \frac{n \hat{f}(0) \hat{E}(s)}{2 L \hat{g}(0)}$$

$$\hat{N} = \frac{n \hat{f}(0) \hat{E}(s) A}{2 L \hat{g}(0)}$$

$$CV = \sqrt{\frac{\text{var}(n)}{n^2} + \frac{\text{var}[\hat{f}(0)]}{[\hat{f}(0)]^2} + \frac{\text{var}[\hat{E}(s)]}{[\hat{E}(s)]^2} + \frac{\text{var}[\hat{g}(0)]}{[\hat{g}(0)]^2}}$$

where D = density (of individuals), n = number of on-effort sightings, $f(0)$ = trackline probability density at zero distance, $E(s)$ = unbiased estimate of average group size, L = length of transect lines surveyed on effort, $g(0)$ = trackline detection probability, N = abundance, A = size of the survey area, CV = coefficient of variation, and var = variance.

A strategy of selective pooling and stratification was used in order to minimize bias and maximize precision in making the estimates of density and abundance (see Buckland et al. 2001). Distant sightings were truncated to remove outliers and accommodate modeling, and size-bias corrected estimate of group size was calculated by regressing $\log_e$ of group size against distance. Three models (uniform, half-normal and hazard rate) were fitted to the data of perpendicular distances to estimate $f(0)$ and the resulting dolphin density and abundance (Buckland et al. 2001). The best model (and thus its associated values for these parameters) was determined by the lowest Akaike's Information Criterion (AIC) value.

4.4.4. Quantitative grid analysis on habitat use

To conduct quantitative grid analysis of habitat use (Hung 2008), positions of on-effort sightings of CWD and FP were retrieved from their long-term sighting databases, and then plotted onto 1-km² grids among the nine survey areas on GIS. Sighting densities (number of on-effort sightings per km²) and dolphin/porpoise densities (total number of dolphins/porpoises from on-effort sightings per km²) were then calculated for each 1 km by 1 km grid with the aid of GIS. Sighting density grids and dolphin/porpoise density grids were further normalized with the amount of survey effort conducted within each grid. The total amount of survey effort spent in each grid was calculated by examining the survey coverage on each line-transect survey to determine how many times the grid was surveyed during the study period. For example, when the survey boat traversed through a specific grid 50 times, 50 units of survey effort were counted for that grid. With the amount of survey effort calculated for each grid, the sighting density and dolphin/porpoise density of each grid were then normalized (i.e. divided by the unit of survey effort).

The newly-derived unit for sighting density was termed SPSE, representing the number of on-effort sightings per 100 units of survey effort. In addition, the derived unit for actual dolphin/porpoise density was termed DPSE, representing the number of dolphins/porpoises per 100 units of survey effort. Among the 1-km² grids that were partially covered by land, the percentage of sea area was calculated using GIS tools, and their SPSE and DPSE values were adjusted accordingly. The following formulae were used to estimate SPSE and DPSE in each 1-km² grid within the study area:

$$SPSE = ((S / E) \times 100) / SA\%$$

$$DPSE = ((D / E) \times 100) / SA\%$$

where S = total number of on-effort sightings, D = total number of dolphins/
porpoises from on-effort sightings, E = total number of units of survey effort,
SA% = percentage of sea area

Both SPSE and DPSE values are useful for examining relative dolphin/porpoise usage within a one square kilometre area. For the present monitoring study, both SPSE and DPSE values were calculated in each 1-km² grid among all survey areas for the entire one-year period in 2024 for both CWD and FP, and in the past five years of monitoring (i.e. 2020-24) for FP.

4.4.5. Behavioural analysis

When dolphins were sighted during vessel surveys, their behaviours were observed. Different behaviours were categorized (i.e. feeding, milling/resting, traveling, socializing) and recorded. These data were then input into a separate database with sighting information, which was used to determine the distribution of behavioural data using a desktop GIS. Distribution of sightings of dolphins engaged in different activities and behaviours would then be plotted on GIS and carefully examined to identify important areas for different activities, and compared with past distribution patterns of such activities. The behavioural data was also used in the quantitative analysis on habitat use to identify important dolphin habitats for feeding and socializing activities and examine trends in habitat use over the past two decades.

4.4.6. Ranging pattern analysis

For the examination of individual ranging patterns, location data of identified dolphins with 10 or more re-sightings and sighted during the present study period were obtained from the dolphin sighting database and photo-identification catalogue. To deduce home range for individual dolphins using the fixed kernel methods, the program Animal Movement Analyst Extension, created by the Alaska Biological Science Centre, USGS (Hooge and Eichenlaub 1997), was loaded as an extension of ArcView[®] 3.1 along with another extension Spatial Analyst 2.0. Using the fixed kernel method, the program calculated kernel density estimates based on all sighting positions, and provided an active interface to display kernel density plots. The kernel estimator then calculated and displayed the overall ranging area at 95% UD (utilization distribution) level. The core areas of individuals at two different levels (50% and 25% UD) were also examined to investigate their range use in greater detail.

4.4.7. Residency pattern analysis

To examine the monthly and annual occurrence patterns of individual dolphins with 15 or more re-sightings, their residency patterns in Hong Kong were carefully evaluated. “Residents” were defined as individuals that were regularly sighted in Hong Kong for at least eight years in the past 12 years (i.e. 2013-2024), or five years in a row within the same period. Other individuals that were intermittently sighted during the past years were defined as “Visitors”. In addition, monthly matrix of occurrence was also examined to differentiate individuals that occurred year-round (i.e. individuals that occur in every month of the year) or seasonally (i.e. individuals that occur only in certain months of the year). Using both yearly and monthly matrices of occurrence, “year-round residents” were the individual dolphins that were regularly sighted in Hong Kong throughout the year, while “seasonal visitors” were the ones that were sighted sporadically in Hong Kong and only during certain months of the year within the study period.

4.4.8. PAM data analysis

Upon retrieval and refurbishment of each F-POD unit in each recovery event, their lids were opened, and the SD cards with the F-POD data were removed from the units for data download. The data were then opened on the *FPOD.exe* software for further analyses by the data analyst, Mr. Daniel Murphy, who is uniquely authorized to carry out the data analysis on behalf of *Chelonia Limited*. The raw click data would first be converted to “FP1” files, and then processed to identify click trains and their likely sources using the new KERNO-F classifier. Notably, the two resident cetacean species in Hong Kong’s waters can be differentiated with fairly high confidence because the click trains of porpoises are characterized by being narrow band, high frequency (NBHF), containing many cycles per click and are comparatively quieter than dolphins. In contrast, dolphins produce shorter clicks (i.e. less cycles per click) that are broadband across the detection range. Dolphin clicks are also much louder than porpoises in general, so multi-path detections (i.e. when the reflections of clicks off the water surface or other objects are detected) are more likely to occur. Although possible, it appears rare for dolphins to emit click trains that are more characteristic of porpoises.

The classified click trains were recorded into “FP3” files, and their visual validation were performed to assess the overall rate of false positive Detection Positive Minutes (DPM) as identified by the classifier. This validation was based on a representative sampling of all detections and examining the characteristics of clicks, multi-path clusters, and click trains. Additional criteria based on the characteristics of

the ambient noise regime could also be used, particularly in relation to boat sonar and sediment transport noise, which can generate a large number of ultrasonic “clicks”

After visual validation, parameters such as DPMs were then assessed for dolphin and porpoise occurrences at each of the four deployment locations within SCLKCMP and BMP. In particular, the DPM was used by calculating the total amount of minutes where at least one click train is detected within a one-minute period, in order to measure the duration of time at least one dolphin (or porpoise) spent in an area. The DPM is useful to detect diel and seasonal patterns in dolphin (and porpoise) occurrences, and proper statistical testing were then performed to identify the effects of location, season, time of day and known possible impacts on such occurrences based on various parameters such as DPM and related metrics.

Two common metrics related to DPM are DPD % of logged days (the proportion of all logged days with one or more detections) and mean DPM/day. Discrepancies between the rank order of sites with regards to DPD % of logged days and mean DPM/day suggest that at different sites, animals may differ in how long they remained at a site, their group size or their rate of production of detectable sounds (e.g., mean DPM/day may be inflated if animals spent a greater amount of time foraging and thus emitting more clicks, and more often, at some sites). Repeated consistent discrepancies between these two metrics over a longer period may help to shed insight into the behaviour of the animals at each site as well as the importance of each site to the animals.

For the comparison to the vessel-based line-transect survey data, the PAM data from the four North Lantau sites (Siu Ho Wan and Tai Mo To at BMP, and Sha Chau SE and Lung Kwu Chau N at SCLKCMP) covering the time period from January 30th, 2024 (from the previous deployment as part of AFCD PAM study) to January 20th, 2025 were included in the analysis for this marine mammal monitoring project.

5. RESULTS AND DISCUSSIONS

5.1. *Summary of Data Collection*

5.1.1. Survey effort

During the entire one-year monitoring period, a total of 119 line-transect vessel surveys were conducted among six survey areas in Hong Kong waters from April 2024 to March 2025. These included eight surveys in DB, 10 surveys in NEL, 10 surveys in NWL, 34 surveys in WL, 36 surveys in SWL and 21 surveys in SEL. An additional helicopter survey arranged by AFCD and Government Flying Services was also conducted on December 11th, 2024 to cover the southern and eastern waters of Hong Kong. The details of these survey effort data collected are presented in Appendix I.

As in the past decade, more effort was allocated to survey areas outside of North Lantau waters during the 2024-25 monitoring period, since additional surveys have already been conducted in NWL and NEL survey areas concurrently under the Hong Kong Link Road (HKLR) regular line-transect monitoring surveys as part of the EM&A works for the Hong Kong-Zhuhai-Macau Bridge (HZMB) construction. These additional HZMB-related marine mammal monitoring surveys employed the same HKCRP personnel, survey methodology and research vessels to ensure consistency and full compatibility with the AFCD long-term marine mammal monitoring programme. In order to increase the overall sample size for the present monitoring study, such EM&A data were combined with the AFCD monitoring data for various data analyses presented throughout this report, thereby providing valuable supplementary information on dolphin and porpoise occurrences.

From April 2024 to March 2025, a total of 393.0 hours were spent collecting 3,358.1 km of survey effort during the AFCD vessel line-transect surveys among the six survey areas. Notably, 98.9% of all survey effort was conducted under favourable sea conditions (Beaufort 3 or below, with good visibility). The extremely high percentage of survey effort conducted in favourable conditions is crucial to the success of the marine mammal data collection programme in Hong Kong, as only such data can be used in various analyses to examine the encounter rates and habitat use of both dolphins and porpoises, as well as to estimate the density and abundance of dolphins.

During the 12-month monitoring period in 2024-25, a total of 3,241.1 km of survey effort was also conducted in NEL and NWL survey areas under the HKLR EM&A dolphin monitoring surveys. This brings the total survey effort to 3,961.6 km for the combined dataset from AFCD and HZMB-related surveys in North Lantau

waters. Over 98% of the survey effort of HZMB-related EM&A surveys was also conducted under favourable sea conditions, which can be combined with the AFCD monitoring data for various analyses.

Since 1996, the long-term marine mammal monitoring programme coordinated by HKCRP has collected 294,862 km of line-transect survey effort in Hong Kong and Guangdong waters of the Pearl River Estuary under different government-sponsored monitoring projects, consultancy studies and private studies, with 47.1% of the total effort funded by AFCD. The survey effort in 2024 alone comprised 2.3% of the total survey effort collected since 1996.

5.1.2. Marine mammal sightings

Chinese White Dolphins - From the AFCD monitoring surveys alone, 162 groups of 544 dolphins were sighted during the 2024-25 monitoring period (see Appendix II). Combined with the additional sightings (six groups of six dolphins) contributed from the HKLR EM&A surveys, a total of 168 groups of 550 dolphins were sighted altogether during the same 12-month period. Among them, 155 groups of 525 dolphins were sighted during on-effort line-transect vessel surveys.

In 2024-25, the majority of dolphin sightings were made in the WL (110 sightings) and SWL (44) survey areas, but only six and seven sightings (all were single animals) were made in NWL and SEL survey areas respectively. Notably, an extremely rare sighting of a lone dolphin was made in NEL survey area during a HKLR survey in April 2024, while none was sighted in DB survey area.

Besides the extremely rare occurrence of a single dolphin NEL in April 2024, the virtual absence of dolphins in NEL for the entire year of 2024 was similar to the survey results from recent years of marine mammal monitoring work in this area. Notably, the passive acoustic monitoring (PAM) conducted concurrently by HKCRP with the funding support of AFCD revealed that dolphins have not completely abandoned this area (at least not around the Brothers Islands where the PAM units were deployed) in recent years (Wang and Hung 2018, 2022, 2023, 2024) and during the present monitoring period in 2024-25 (see Sections 5.1.4 and 5.7). Even though dolphin detections have been extremely low and still declining around the Brothers Islands in NEL in recent years, the continuing usage by dolphins (especially outside of the daylight hours) of this once-important habitat should not be overlooked, and the complementary approach of passive acoustic monitoring is critical to fill important data gaps in monitoring dolphin occurrences 24 hours a day within BMP as well as for the

NEL survey area.

Finless Porpoises – A total of 64 groups of 174 porpoises were sighted from vessel surveys during the 2024-25 monitoring period (see Appendix III). Among these porpoise sightings, all except two sightings were made during on-effort search, which can be used in the encounter rate analysis and habitat use analysis. The porpoises were much more frequently sighted in SWL (50 groups) than in SEL survey area (14 groups). As in previous monitoring periods, the porpoises were absent from the WL, NWL, NEL and DB survey areas, where dolphins primarily occur throughout the past three decades.

5.1.3. Photo-identification of individual dolphins

During the 2024-25 monitoring period, approximately 22,000 digital photographs of CWD were taken during the AFCD monitoring surveys for the photo-identification of individual dolphins. All photographs taken in the field were compared with existing individuals from the photo-identification catalogue compiled and curated by HKCRP since 1995. All new photographs identified as existing or new individuals during the study period, as well as any updated information on gender and age class of individual dolphins, were incorporated into the photo-identification catalogue. Some photo-identification data was also contributed from the HZMB-related surveys during the same study period.

Up to January 2025, a cumulative total of 1,149 individual dolphins (including 17 that were confirmed to be dead) have been identified by HKCRP researchers in Hong Kong waters and the rest of the Pearl River Estuary. The current catalogue contained 599 individuals being first identified within Hong Kong's territorial waters and another 550 individuals being first identified in Guangdong waters of the Pearl River Estuary. In 2024, six newly identified individuals from Hong Kong waters were added to the catalogue, while one well-known individual (WL61) was found dead on December 11th, 2024.

The catalogue summary revealed that 271 individuals have been seen 15 times or more; 171 individuals have been seen 30 times or more; and 111 individuals have been seen 50 times or more. In contrast, 38.6% of the identified individuals have only been seen once or twice, with most of these being first identified in Guangdong waters (326 out of 443 individuals). Temporal trends in the total number of identified individuals, the total number of re-sightings made, and the number of individuals within several categories of number of re-sightings showed steady advancement in the photo-

identification works during the 2024-25 monitoring period (Figure 2).

Between April 2024 and March 2025, a total of 102 individual dolphins, sighted 372 times altogether, were identified during the AFCD regular vessel surveys (Appendix IV). With the addition of the HZMB-related monitoring survey data collected in the NWL survey area, there was a combined total of 103 individual dolphins being identified 375 times during the same ten-month period. More than two-third (70.1%) of all re-sightings made during the AFCD/HZMB surveys were in the WL survey area. A total of 101 and seven re-sightings were made in SWL and SEL respectively, while three individuals were re-sighted four times in NWL waters. On the contrary, no individual dolphin was sighted or identified in NEL and DB survey areas during the 2024-25 monitoring period.

Among the 103 identified individuals from the AFCD/HZMB combined dataset, the majority of them were re-sighted only a few times, but some of them were repeatedly re-sighted, indicating their strong reliance on Hong Kong's waters as an important part of their home range. For example, 27 individuals were re-sighted five times or more, while nine individuals (CH38, CH108, WL42, WL72, WL79, WL109, WL152, WL220 and WL336) were re-sighted ten times or more in the combined dataset during the 12-month study period. Furthermore, only ten individuals were re-sighted with their calves during the 2024-25 monitoring period (including the well-known female WL61 that was found dead in December 2024, two months after the last re-sighting with her older calf). The mothers that were re-sighted with young calves (either unspotted calf or unspotted juvenile) should be closely monitored, as their survival is critical for the long-term viability of the dolphin population.

5.1.4. Deployment summary for passive acoustic monitoring

Upon examination of the PAM data that were retrieved for the four deployment periods between January 2024 and January 2025, all four F-PODs appeared to have functioned normally with the exception of a stoppage of the F-POD at Lung Kwu Chau N for approximately 10 days from April 7th to April 17th, 2024. There appears to have been a malfunction with the unit's automatic switching to the second set of batteries after the first set had been drained. The unit started operating again when it was retrieved and resupplied with new batteries that allowed full operation until the next retrieval period. The four F-PODs combined for a total of 1,413.16 logged days (i.e., the number of days the F-PODs were on and recording) during the period from the first deployments on January 30th, 2024 to the last retrievals on January 20th, 2025. The number of days logged by individual F-POD units varied from 345.39 (Lung Kwu

Chau N) to 355.94 (Siu Ho Wan) (Table 1). The numbers of logged days at the other sites were 355.90 at Sha Chau SE and 355.93 at Tai Mo To. The total logged days in each area were 711.87 in BMP and 701.29 days in SCLKCMP (Table 1).

During this sampling period, minimal time lost was registered by the F-PODs. The amount of time lost is related to the minute click limit being reached (i.e., “maxed out”) so that no further clicks are detected/recorded by the F-POD until the start of the next minute. The click limit per minute can be exceeded in noisy environments. The most common causes of such time loss for click detections include: loud boat sonar, sediment transport noise (caused by the collisions of small particles of sediments during movement), snapping shrimps and occasionally storms or rougher sea conditions. The percentage of time lost varied from 0.00% (Siu Ho Wan and Lung Kwu Chau N) to 1.22% (Sha Chau SE) while Tai Mo To recorded 0.89% time lost (Table 1).

Although both boat sonar (around 48, 50 and 100 kHz) and sediment transport noise were detected by F-PODs at the four sites, the amount of noise only appeared to max out very few minutes during the deployment period of the F-PODs as illustrated by the very low percentage time lost that was observed (Table 1), even at sites that were generally identified as “noisy” in the past by C-PODs. The absence or very low amount of time lost by the F-PODs during the current deployment was not surprising given the superior ability of the F-POD to filter out boat sonar without compromising cetacean detections (this was also observed consistently in the F-POD data obtained during the 2021-22, 2022-23 and 2023-24 AFCD PAM studies and this same study in 2024-25). Although these are very positive results of the abilities of the F-POD, it is unknown if the noisier conditions in Hong Kong’s waters in the past would have resulted in any time lost by the F-POD. With so little noise, none of the F-PODs had stopped operating due to batteries being fully drained or SD cards reaching capacity (at Lung Kwu Chau N, the F-POD stopped recording data about 10 days prior to retrieval on April 17th but this appeared to have been due to a malfunction in the automatic switching of batteries sets rather than complete drainage of battery power).

Furthermore, the false positive rates of CWD detections were assessed by visual validation and found to be zero at all sites and consequently, there could not be any false positives for the two marine park areas. However, there were notably very few DPMs at Siu Ho Wan (51), Tai Mo To (20) and Sha Chau SE (92) (Table 1). Nevertheless, the lack or very low levels of false positives provides high confidence in the dolphin detections.

A summary of the data obtained from the four retrieved F-PODs along with detection statistics for dolphins are shown in Table 1. It was assumed that all dolphins detected belonged to Chinese White Dolphins because they are the only delphinid species that are residents of Hong Kong's waters while the occurrence of other similar species are rare or non-existent. A total of 717 dolphin DPMs (and zero DPMs with porpoise characteristics) were recorded. Although finless porpoises are present in Hong Kong's waters, they have not been confidently confirmed from the waters of these two marine parks in North Lantau waters and none were detected during this monitoring period. A few DPMs with porpoise characteristics have been recorded from North Lantau sites previously but none of these DPMs could be confirmed to have originated from porpoises as more research is needed into the possibility of dolphins mimicking porpoise clicks.

5.2. Distribution

5.2.1 Distribution of Chinese White Dolphins

Similar to the previous monitoring periods, dolphin frequently occurred along the coastal waters to the west and southwest of Lantau Island in 2024-25, extending from Tai O Peninsula in WL toward Kau Ling Chung and Shek Pik in SWL (Figure 3). They also occurred to the west and around the Soko Islands, and a lone individual dolphin (NL306) occurred seven times within Pui O Wan in SEL waters. On the contrary, they seldom occurred in the southern and eastern portions of South Lantau waters, and mostly absent from the entire North Lantau region, with the exception of two rare sightings made to the west of the airport platform. No dolphins were sighted at all in DB or NEL where are part of the historically known dolphin habitat (Figure 3).

In 2024 alone, from the combined effort of the AFCD and HZMB-related surveys, the level of dolphin occurrence was the highest along the west coast of Lantau, while they also occurred regularly along the northwestern coastline within the South Lantau waters (Figure 4). A closer look at the North Lantau region revealed that dolphin sightings were scattered at the southwestern end of the region as well as the waters in proximity to Lung Kwu Chau. An exceptionally rare sighting of a lone dolphin was also found to the north of Siu Mo To in NEL waters (Figure 5). As consistently recorded in recent years of monitoring surveys in North Lantau waters, the dolphin was almost non-existent in the central and eastern portions of the region, including the peripheral area of the airport's third runway land mass, as well as the footprints of the Hong Kong Boundary Crossing Facilities at the juncture of the NWL and NEL survey areas.

In WL waters, dolphins occurred much more frequently and evenly distributed throughout the area in 2024 (Figure 6). Higher concentrations of dolphin sightings can be found in inshore waters near Tai O Peninsula, Kai Kung Shan, Peaked Hill and around the Fan Lau Peninsula. As in recent years, the dolphins generally occurred more frequently along the inshore waters of WL, but less frequently further offshore along the western territorial border. Notably, dolphins appeared to avoid the northern end of the WL survey area with the overlap of the HKLR alignment.

In the South Lantau region, concentrations of dolphin sightings were found toward the northwestern end near the Fan Lau Peninsula, near Kau Ling Chung and Shui Hau Peninsula, while other sightings were scattered around and to the west of the Soko Islands as well as within Pui O Wan (Figure 6). On the contrary, the dolphins have mostly disappeared in the southern half of the SWL and most of the SEL survey area (except the lone individual that was repeatedly sighted within Pui O Wan).

5.2.2. Distribution of finless porpoises

From April 2024 to March 2025, concentration of porpoise sightings in South Lantau waters was mainly around the Soko Islands, especially on the eastern side of Siu A Chau (Figure 7). On the contrary, the rest of the porpoise sightings were scattered to the south of Shui Hau Peninsula, near Shek Kwu Chau and Cheung Chau. Examination of temporal changes in porpoise distributions in South Lantau Waters in the past four years (2021-24) revealed that such distribution is more restricted at the southern and eastern sides of the Soko Islands in 2021-23 but become more widespread in 2024, especially to the south of Shek Kwu Chau and southwest of Cheung Chau (Figure 8). Notably, porpoise occurrences also appeared to be much higher in summer and autumn months during 2023 and 2024 when compared to the previous years.

5.3. *Habitat Use*

5.3.1. Habitat use patterns of Chinese White Dolphins

Dolphin habitat use patterns were examined using the quantitative grid analysis (Hung 2008), to calculate the SPSE and DPSE values (i.e. sighting densities and dolphin densities, respectively) in all grids among the six survey areas where they occurred regularly in the past. These patterns were also compared to the annual patterns observed in recent years.

In 2024, the grids with higher dolphin densities were mostly located along the west coast of Lantau, especially near Tai O Peninsula, to the west of Kai Kung Shan and around the Fan Lau Peninsula, and extending toward Kau Ling Chung in SWL

waters (Figure 9). On the contrary, the dolphins were nearly absent in the North Lantau region (including Deep Bay), with only four grids recorded extremely low level of dolphin occurrences. Their occurrences in the southern portion of South Lantau waters (especially most of the SEL survey area) were extremely rare.

Temporal changes in dolphin habitat use patterns

A comparison was made among the dolphin habitat use patterns over the past nine years (2016-24) to examine the temporal changes in their densities in the western waters of Hong Kong. In WL, intense habitat use with high densities was consistently recorded across the majority of grids throughout the nine-year period (Figure 10). However, dolphin usage in the northern portion of the WL survey area, which overlapped with the HKLR09 alignment, have been consistently lower, when compared to the earlier years before and during the initial phase of HKLR construction. Dolphin usage should be continuously monitored in their priority habitats in WL waters, especially for the examination of the long-term impact of the bridge alignment on the north-south movement of individual dolphins between the North and West Lantau regions (see also Section 5.8.2).

In SWL waters, besides the generally lower and more restricted occurrence in 2018, 2020 and 2022, the dolphin habitat use pattern have been fairly consistent in the past decade, with only the grids between the Fan Lau Peninsula and Kau Ling Chung being the only area with high densities (Figure 10). Furthermore, in some years dolphin usage around the Soko Islands appeared to be more frequent (e.g. 2016 and 2017, 2019) while in other years, they only utilized this area at a low level.

In the North Lantau region, the dramatic decline in dolphin habitat use pattern has been well documented for more than a decade, with greatly diminished dolphin occurrences throughout the entire region. Such trend continued in 2024, with dolphin occurrence declining to extremely low level in this region (Figure 11). In recent years, dolphin usage has been increasingly confined to the western end of the North Lantau region, and in 2020-22, their habitat utilization was mostly restricted to the SCLKCMP in generally low densities. And in both 2023 and 2024, such occurrence around the marine park was close to non-existent (Figure 11). The continuous near-absence of dolphins in the central and eastern portions of the region since 2015 is of great concern, as there has been no sign of recovery in dolphin habitat use for nearly a decade after the completion of marine works associated with the HZMB construction as well as the massive reclamation works associated with the 3RS project.

Temporal changes in dolphin habitat use patterns among six key habitats

Temporal trends in dolphin usage at six key habitats were examined for the past two decades (2005-24), which included the four existing marine parks around Sha Chau and Lung Kwu Chau, the Brothers Islands, the southwestern corner of Lantau, and the Soko Islands, as well as two other “hot spots” at Tai O and Black Point where dolphins regularly occurred in the past (Figure 12). To examine dolphin usage over these six key habitats that encompass a suite of grids, the number of on-effort sightings and amount of survey effort were pooled together from those grids, to calculate dolphin densities (DPSE) as a whole for each year during the 2005-24 period and track any changes over the years.

There has been a continuous decline in dolphin usage within SCLKCMP (with 17 grids) since 2013 (Figure 13). Such usage fell even further to an extremely low level in the past four years of 2019-22, before dropping to zero dolphin density (i.e. complete absence) for the first time in 2023; such rare occurrence of dolphins within this marine park continued in 2024. This alarming decline raises serious concerns because the SCLKCMP has long been considered important dolphin habitat in Hong Kong (Hung 2008). Similar to the situation in BMP, the traditional visual monitoring survey may no longer be a viable research tool to examine long-term trend in dolphin usage at this once-important habitat when dolphin occurrences dropped to the near-absent level, and passive acoustic monitoring has become extremely valuable and increasingly critical as a complementary approach to detect whether there is any sign of recovery in dolphin usage within this marine park (see PAM results at this marine park in Section 5.7)

Furthermore, after a dramatic decline in dolphin usage since 2011, the BMP (with 15 grids) recorded zero dolphin density in ten consecutive years in 2015-24 (Figure 13). Although dolphin usage was expected to recover after the completion of marine works associated with HZMB construction a decade ago, the establishment of the BMP in December 2016 as well as the more recent completion of massive reclamation works for 3RS project (just a few kilometers to the west of BMP), their occurrence around the Brothers Islands remains non-existent in the past decade. Notably, passive acoustic monitoring revealed that a very low level of dolphin detections was recorded within this marine park in the past several years, and most of these rare detections were made during night-time (Wang and Hung 2023, 2024, and present study as detailed in Section 5.7). With the near-absence of dolphins in this dolphin habitat for quite a long time now, continuous acoustic monitoring of this area has become even more critical to have a chance of detecting any sign of recovery in dolphin usage.

Besides the noticeable spike in dolphin usage in 2014 within the Southwest Lantau Marine Park (SWLMP, with 15 grids), such usage has remained fairly steady and high in the past decade (Figure 13). Notably, this marine park has consistently recorded the highest levels of dolphin usage among all key dolphin habitats in western Hong Kong in the past two decades, and this last remaining stronghold of the top priority dolphin habitats should be closely monitored to examine any sign of temporal changes in their future usage.

After a dramatic decline in dolphin densities was detected in the South Lantau Marine Park (SLMP, with 30 grids) between 2014 and 2018, dolphin usage there rebounded to a slightly higher level in 2019, before another steady decline to the lowest in 2022. However, their usage has steadily rebounded once again in 2023 and 2024 (Figure 13), despite the resumption of the high-speed ferry traffic in the South Lantau Vessel Fairway after the Covid-19 restriction has been lifted. The long-term trend in dolphin usage within this marine park should be continuously monitored to determine whether this recovery in dolphin usage remains steady over time.

Once identified as a critical dolphin habitat in the western waters of Hong Kong, the waters around Tai O Peninsula (with four grids) recorded a steady decline in dolphin densities from the highest in 2009 to the lowest in 2022, although there have been more fluctuations in recent years with slight rebounds recorded in 2019, 2021, 2023 and 2024 (Figure 13). In addition, the dolphin usage at Black Point (with four grids) fluctuated greatly in earlier years with no apparent trend. After a near-complete absence from this area between 2014 and 2018, there was a small spike in dolphin usage in 2019 before falling to a very low level in 2020 and then a complete absence once again in four consecutive years of 2021-24 (Figure 13). As this area is situated at the border of a proposed large-scale reclamation site at Lung Kwu Tan and only a few kilometres away from SCLKCMP, special attention on dolphin habitat use in this area in the near future is warranted.

5.3.2. Habitat use patterns of finless porpoises

Habitat use patterns of finless porpoises for the entire year of 2024 revealed that their densities were higher at the eastern and southwestern sides of the Soko Islands, as well as between Shek Kwu Chau and Cheung Chau (Figure 14). However, it should be cautioned that with the relatively small amount of survey effort per grid in this region during this one-year period, the porpoise density pattern deduced from the entire year of 2024 could be potentially biased.

In order to increase the overall sample size, the survey effort and porpoise data collected from 2020-24 for SWL and SEL survey areas were pooled and analyzed for a longer period in order to obtain sufficient survey data to provide a better representation of porpoise habitat use pattern South Lantau waters in recent years. Since the porpoises in Hong Kong exhibit pronounced seasonal patterns of distribution, with rare occurrence in each survey area during certain periods of the year (Hung 2005, 2008; Jefferson et al. 2002), the five-year dataset for SWL and SEL survey areas was further stratified into winter/spring (December through May) and summer/autumn (June through November) periods to deduce habitat use patterns of porpoises for the respective dry and wet seasons.

During the dry season (i.e. winter and spring months) in 2020-24, the grids with high porpoise densities in South Lantau waters mainly clustered at the offshore waters to the southwest and east of Tai A Chau (Figure 15). However, the porpoises seldom utilized the western and northern portions of the South Lantau region. On the contrary, the porpoise usage during the wet season in South Lantau waters was somewhat different from the dry season (Figure 16). With the exception of a few grids near Shui Hau Peninsula, almost all grids recorded porpoise usage in South Lantau waters were located at the offshore waters with low to moderate during the wet season. It should also be noted that the grids on the southern and eastern sides of the Soko Islands were consistently utilized by the porpoise during both wet and dry seasons.

Temporal changes in porpoise habitat use patterns

To examine the recent temporal changes in porpoise densities in South Lantau waters, comparisons were made on annual patterns of porpoise habitat use across a nine-year period in 2016-24 (Figure 17). In the earlier years of 2016-17, porpoise usage at the offshore waters between Shek Kwu Chau and the Soko Islands was at a consistently high level. However, such usage has progressively diminished in more recent years, with noticeable decline near Shek Kwu Chau (possibly related to the reclamation works of IWMF phase I during 2018-22), but there was a sign of recovery in porpoise usage near Shek Kwu Chau in 2024.

Three key porpoise habitats in South Lantau (i.e., SLMP, Shek Kwu Chau and Pui O Wan; see Figure 12) were examined for the temporal trends in their habitat use across the 17-year period between 2008 and 2024. Similar to the CWD, to examine temporal trends in porpoise usage of these key habitats that encompass a suite of grids, the number of on-effort porpoise sightings and unit of survey effort were pooled together from those grids, to calculate porpoise densities (DPSE) as a whole for each year from

2008-24.

Porpoise usage fluctuated greatly at the SLMP (with 30 grids) in the past years, starting at a lower level in 2008-09 (Figure 18). Since then, there was a steady increase in porpoise usage of this area to the highest level in 2014. However, in the recent past years, porpoise usage within this marine park has become less stable, with notable drops in 2016, 2019, 2021 and 2022, followed by a steady increase to higher levels in 2023 and 2024. Continuous monitoring would be needed to determine any long-term trend in porpoise occurrence within this marine park.

The inshore waters of Pui O Wan (with nine grids) were more intensely used by porpoises in the earlier years in 2008-09, but then such usage became infrequent between 2010 and 2013 (Figure 18). Even though the porpoise usage rebounded to a higher level in the subsequent years, there were noticeable drop and remained at a low level in the past five years of 2020-24.

Furthermore, the surrounding waters of Shek Kwu Chau (with eight grids) were consistently utilized by the porpoises as an important habitat in the first decade of the 15-year period (i.e. 2008-17), especially with a steady increase in porpoise usage starting from 2013 to reach a much higher level in 2016 (Figure 18). However, there was a sharp decline in porpoise usage since 2018 to reach to the lowest level with near-complete absence in 2021 and 2022, then followed by a steady rebound in 2023 and 2024. The dramatic decline in porpoise usage at this once critical porpoise habitat after 2017 has been closely linked to the construction activities near Shek Kwu Chau in association with the reclamation works for IWMF Phase I since mid-2018 and this raises some grave concerns. Future trends in their habitat use near Shek Kwu Chau as well as for the entire southern waters of Hong Kong should be closely monitored after the completion of the reclamation works for IWMF Phase I in 2022, especially in light of the recent recovery in porpoise usage in this area.

In recent years, a marine infrastructure project, namely the Hong Kong Offshore LNG Terminal (HKOLNG) construction, commenced in late 2020 through late 2022 at the offshore waters in South Lantau region adjacent to SLMP. To examine the potential impact of jetty construction and terminal operation for the project, porpoise usage within the project footprint (overlapped with three grids O35, P34 and P35) is examined for the past eight years with consistent annual survey effort among the three grids overlapping the project site. After an increase in porpoise usage to the highest level in 2018, such usage dropped noticeably in 2019 before a marked increase in 2020

and 2021, followed by another drops in 2022 and 2023 and then a complete absence in 2024 (Figure 18). It appeared that the fluctuation in porpoise usage in these offshore waters of South Lantau did not correspond well with the construction schedule of HKOLNG, as the drop in porpoise usage during 2019 was observed well before the commencement of construction, while the rebound in porpoise usage in 2020 and 2021 happened during the bulk of the construction. Nevertheless, with the diminished porpoise usage in the past three years (especially with the absence in 2024), it should be investigated further whether the operation of the HKOLNG jetty may have resulted in negative impact on porpoise usage in this area.

5.4. Group Size, Calf Occurrence and Activities

5.4.1. Group sizes of dolphins and porpoises

During the 2024-25 monitoring period, the group sizes of CWD varied from singletons to 13 animals, with an overall mean of 3.3 ± 2.57 ($n=168$). Among the areas where dolphins regularly occurred in 2024-25, the mean group sizes were the lowest in NWL and SEL (all singletons sighted, but based on a very small sample size of six and seven sightings respectively) and the highest in WL (3.6) and SWL (3.1) (Table 2a). An exceptionally rare sighting of a lone dolphin was also made in NEL during HKLR monitoring survey. Overall, the mean group sizes were slightly lower in autumn (3.02) and winter (3.09) but slightly higher in summer (3.49) and spring (3.53). Similar to past monitoring periods, the majority of dolphin groups sighted in 2024-25 were small, with 50% of the groups composed of 1-2 animals, and 78% of the groups with fewer than five animals. Only six out of the 168 dolphin groups consisted of ten or more animals (Figure 19).

The examination of long-term trend in annual mean dolphin group sizes in the past decade revealed that the mean group sizes in recent years have stabilized with remarkably similar levels (i.e. 3.23-3.28) during the six consecutive years of 2016-21, which was then followed by a modest increase in the following consecutive years in 2022 (3.55) and 2023 (3.55), before falling back to the previous level in 2024 (3.25) (Figure 20). Temporal changes in mean dolphin group sizes should be continuously monitored, as this could be indicative of changes in their foraging strategies in response to increased disturbance from various sources or changes in prey distribution and overall prey resources in the western waters of Hong Kong.

Distribution of dolphins in different group size categories in 2024 is shown in Figure 21. Larger dolphin groups occurred predominantly near Tai O Peninsula and Kai Kung Shan along the WL coastline, around the tip of Fan Lau Peninsula and near

Kau Ling Chung, with the very large groups (10+ dolphins per group) occurring to the west of airport platform, near Kai Kung Shan, Fan Lau, Kau Ling Chung and Siu A Chau. In contrast, distribution of the smaller dolphin groups closely resembled the overall distribution around Lantau waters, even though almost all groups that occurred at the periphery of the overall distribution (e.g. in NWL waters and within Pui O Wan) were mostly singletons (Figure 21).

For the finless porpoises, their group sizes during the 2024-25 monitoring period varied from singletons to 14 animals, with an overall mean of 2.7 ± 2.22 (n=64). The majority of these groups were very small, with 59.4% of them being composed of 1-2 animals, and all except ten groups (or 84.4% of all groups) had fewer than five animals (Figure 22). Within the two survey areas in South Lantau region, their mean porpoise group sizes were very similar (2.72 in SWL and 2.71 in SEL) (Table 2b).

Temporal trend in annual mean porpoise group sizes were examined in the past decade (Figure 23). Over the decade, mean porpoise group sizes fluctuated but after a spike in 2020, it dropped to the lowest level in 2021 and 2022, before rising noticeably to the highest level in 2023 and 2024 (Figure 23). It would be critical to continuously monitor the trend in porpoise group sizes, as that would provide insights on whether there are any changes in the porpoises' foraging strategies in response to anthropogenic impacts such as increased vessel traffic disturbance or changes in prey distribution and resources.

Distribution of porpoises in different group sizes in 2024 showed that the several larger porpoise groups scattered to the southern and eastern side of the Soko Islands, to the south of Shek Kwu Chau as well as in the waters to the south of Pui O Wan (Figure 24). On the contrary, the smaller porpoise groups resembled the overall distribution pattern, and all porpoise groups at the peripheral area of general distribution were generally small (Figure 24).

5.4.2. Occurrence of dolphin calves

Of the 537 dolphins sighted altogether in 2024, a total of five unspotted calves (UC, or newborn calf) and 24 unspotted juveniles (UJ, or older calves) were identified with their respective age classes, with these young calves (i.e. UC and UJ combined) comprising 4.5% of the total. After a large and steady decline between 2013 and 2018, the annual percentage of young calves in Hong Kong waters appeared to rebound slightly in 2019-21 and 2023-24 despite a slump in 2022 (Table 3; Figure 25). In fact, such percentage in 2024 marked the highest level in the past decade; this encouraging

sign should be closely monitored in the future. Notably, when compared to the earlier years, even though the calf occurrence in recent years was at a very low level, there was a notable increase of UCs and UJs observed in 2023-24 and such increase for UJs in 2024 was particularly pronounced (Figure 25).

Distribution pattern of young calves in 2024 revealed that their sightings were mostly concentrated along the stretch of coastal waters between Tai O Peninsula and Fan Lau Peninsula, while these young calves also occurred near Kau Ling Chung, Shum Wat, to the west of the airport platform, and around the Soko Islands (Figure 26). On the contrary, no young calves were found throughout the North Lantau Region and for the most part of the South Lantau region.

5.4.3. Activities of dolphins

In 2024, 29 (or 17.6% of the total) groups of dolphins were observed to be engaged in feeding activities, while there was only one group each engaged in socializing and traveling activities respectively. Annual percentages of socializing activities remained at similarly low levels over the past eight consecutive years, before dropping to the lowest level in 2024 since the start of the monitoring of such activity in 2002 (Figure 27). On the other hand, after a noticeable rebound of the percentage of feeding activities in the three consecutive years in 2021-23, there was a slight dip in 2024 but still remained at a relatively high level in the past decade or so. The trend of these activities should be continuously monitored.

Distribution of dolphins engaged in different activities in 2024 is shown in Figure 28. Besides the single group near Lung Kwu Chau, and a few groups scattered within Pui O Wan, around the Soko Islands and near Shek Pik, the majority of dolphin groups associated with feeding activities were found along the stretch of WL coastlines, with particularly higher concentrations to the north of HKLR alignment, near Kai Kung Shan and Fan Lau Peninsula. On the contrary, the two lone dolphin sightings associated with socializing and traveling activities were both located near Kai Kung Shan.

5.4.4. Dolphin associations with fishing boats

Of the 168 groups of dolphins sighted during the 2024-25 monitoring period, 14 (or 8.3% of all groups) were associated with operating fishing boats. The majority of these groups (11) were associated with purse-seiners, while two groups were associated with gill-netters and another group was associated with a shrimp trawler.

In 2024, the few dolphin groups associated with operating gill-netters and shrimp trawler were scattered to the north of HKLR alignment, near Kai Kung Shan and Peaked Hill (Figure 29). On the contrary, the occurrence of dolphin groups associated with operating purse-seiners were mostly found near Kai Kung Shan and Tai O Peninsula, with other groups also sighted near Shui Hau Peninsula, Kau Ling Chung and Tai O Peninsula.

5.5. Encounter Rate

5.5.1. Encounter rates of Chinese White Dolphins

To calculate the encounter rates of CWD, only survey data collected in Beaufort 0-3 conditions was included in the analysis as in past monitoring periods. From April 2024 to March 2025, the combined encounter rate of dolphins from the four survey areas of NEL, NWL, WL and SWL was 2.6, which was very similar to the figures recorded in the previous three monitoring periods (Figure 30a & Table 4). Notably, the combined dolphin encounter rates recorded in these recent periods were also at the lowest level among all monitoring periods in the past two decades. Among different survey areas, the encounter rates in NWL during the past six monitoring periods were at the historic low, while the 2024-25 encounter rate in the WL survey area rebounded to a slightly higher level after remaining at the lowest level in the recent past monitoring periods (Figure 30b). A small but steady rebound in dolphin encounter rate in SWL was also noted in the last three monitoring periods.

As consistently recorded in past monitoring periods during the past two decades, WL continued to have the highest dolphin encounter rate (14.3) among the four survey areas with their occurrences, and was considerably higher than the ones in SWL (3.4) and NWL (0.2) (Table 4). The encounter rate in NEL was nearly zero in 2024-25, since only a single dolphin was sighted during 1,521.4 km of survey effort.

Temporal trend in annual encounter rate

Temporal trends in annual dolphin encounter rates in the past two decades were examined for the overall combined areas (i.e. NEL, NWL, WL and SWL), as well as in the North and West/Southwest Lantau regions. After a slight rebound in 2021 from the previous low level in 2018-20, another steady decline was observed in the past two years with a drop to the lowest level in 2023 before a slight rebound in 2024 (Figure 31a). Furthermore, the dolphin encounter rate in the entire North Lantau region (NEL and NWL survey areas combined) in 2024 stayed at the same lowest level as in 2022 and 2023, and such rate has remained at an exceptionally low level in six consecutive years from 2019-24 (Figure 31b). The dolphin encounter rate for the West/Southwest

Lantau region have been fluctuating at a lower level since 2016 and the one in 2024 remained at similar level as in previous years (Figure 31b).

5.5.2. Encounter rates of finless porpoises

Porpoise encounter rates were calculated using data collected in Beaufort 0-2 conditions as in past monitoring periods. From April 2024 to March 2025, the combined porpoise encounter rate of SWL and SEL survey areas was 4.2 sightings per 100 km of survey effort (Table 5). When compared across the past 18 monitoring periods since 2007-08, the porpoise encounter rate has remained fairly stable in the earlier years with the exception of a dip in 2015-16. However, there was a sharp drop in porpoise encounter rate between 2019-20 and 2020-21, followed by a small increase in 2021-22 and 2022-23 before a notable increase in 2023-24 and 2024-25 (Figure 32). Among the two survey areas, porpoise occurrence in SWL (5.7) was much higher than in SEL (2.3) during the 2024-25 monitoring period (Figure 33b).

Annual porpoise encounter rate from the combined areas of SWL and SEL indicates that porpoise usage in South Lantau waters have steadily increased from the lowest in 2009 to the highest in 2014, then followed by a steady decline from 2014 to the lowest level in 2021; subsequently there was a steady increase from 2021 to 2024 (Figure 33a). Furthermore, great fluctuation in their annual encounter rates was evident between the two survey areas in South Lantau waters, apparently the one at SWL has steadily increased in the past four consecutive years (Figure 33b).

5.6. *Density and Abundance*

5.6.1. Estimates of dolphin density and abundance in 2024

Densities and abundance of CWD were estimated for the NEL, NWL, WL and SWL survey areas using the line-transect analysis method, following similar approach as in previous years of dolphin monitoring in Hong Kong (see Hung 2023, 2024). The annual estimates deduced from the 2024 vessel survey data can be used to assess the long-term temporal trend in dolphin occurrence in Hong Kong. Only effort and sighting data collected from the four areas during Beaufort 0-3 conditions were used in the analysis and this included 5,820.8 km of survey effort and 142 dolphin groups from the four areas for density and abundance estimation in 2024 (Table 6a).

Among the four survey areas, WL recorded the highest dolphin density (88.88 individuals/100 km²), which was more than five times and 95 times higher than the densities in SWL and NWL, respectively (Table 6a). In recent years, the WL figures have fluctuated with no consistent trend, but they remained at similar level in 2021,

2022 and 2024 despite a slight dip in 2023 (Table 6b). On the contrary, after the dolphin density in SWL fell in four consecutive years of from 2019 to 2023, there was a slight rebound in 2024. In contrast, after dropping to the lowest level in 2022, the density estimate for NWL in 2023 returned to similar levels in 2019-22, but then dropped once again to the lowest level in 2024. Furthermore, as in the previous eight years, dolphin density and abundance could not be estimated for NEL in 2024 since only a single dolphin was sighted in this area for the entire year.

In 2024, the abundance estimates of CWD were zero, one, 12 and 25 dolphins in the NEL, NWL, SWL and WL survey areas, respectively. As a result, the combined estimate for the four areas was 38 dolphins (Table 6b). After a steady decline in combined abundance estimates from 188 dolphins in 2003 to the lowest of 32 dolphins in 2018, a sharp rebound was observed in 2019 (52 dolphins), but followed by noticeable drops in subsequent years to the second lowest level in 2022 and 2023 (34 dolphins), before a slight rebound occurred in 2024 (Figure 34; Table 6b).

5.6.2. Temporal trends in dolphin abundance

Temporal trends of annual dolphin abundance in NWL and NEL (2001-24), SWL (2002-24) as well as WL (2003-24) were further examined, where consistent amount of survey effort (at least 500 km of annual survey effort) has been conducted in these four areas of major dolphin occurrence. In SWL, temporal trend of annual estimates was only examined since 2010 but not for a longer period, as consistent survey effort (at least 500 km of survey effort per year) was not collected annually prior to 2010. Alternatively, biennial estimates were deduced in SWL for 2002-03, 2004-05, 2006-07 and 2008-09 to examine the overall temporal trend in dolphin abundance over a longer period.

Firstly, the temporal trend in SWL showed fluctuations across the years, with a marked decline from the highest in 2002-03 (30 dolphins) to the lowest in 2006-07 (six dolphins) (Figure 35). Since then, the annual abundance estimates have remained at a lower level in subsequent periods, before a noticeable rebound in 2014 and 2015. Thereafter, abundance estimates dropped to much lower levels in the three subsequent years of 2016-18, before another noticeable rebound occurred in 2019, then followed by a steady decline in subsequent years of 2020-23 and a slight rebound in 2024 (Figure 35; Table 6b). Notably, the associated CVs of the annual abundance estimates in SWL have been moderate and within the range of 20-40% (except for the biennial estimates in 2002-03 (45%) as well as the annual estimates in 2010 (67%) and 2012 (54%)), so the estimates should be reliable for most years.

In WL, dolphin abundance steadily decreased from 54 dolphins in 2007 to only 17 dolphins in 2012 (Figure 36; Table 6b). After a rebound in 2013 and 2014 (with 23 and 36 dolphins, respectively), there was another steady decline in the following years of 2015-17 to the lowest level of 16 dolphins in 2017. Thereafter, the annual abundance varied moderately within the range of 19-29 dolphins in the past seven years of 2018-24 and remained relatively stable in the past four years (Figure 36).

In contrast, dolphin abundance in the North Lantau region has shown a dramatic and consistent decline in the past two decades. In NEL, the decline was appalling, dropping from the highest in 2001 (20 dolphins) to only one dolphin in 2014 and then to zero for nine consecutive years (2015-24) (Figure 36). Dolphin abundance in NWL also dropped steadily and steeply from the highest in 2003 (84 dolphins) to an extremely low level in the past six years with only 1-4 dolphins, and such decline was more than 95% since 2003, or 90% since 2012. Notably, even there was a slight rebound in dolphin abundance in NWL between 2022 (with one dolphin) and 2023 (with four dolphins), such abundance fell once again to the lowest level in 2024 (with one dolphin).

Furthermore, it should be cautioned that the associated CVs for the abundance estimates in 2022 (44%), 2023 (54%) and 2024 (47%) were all quite high in NWL due to the small sample size of on-effort dolphin sightings for these years (i.e. 11 in 2022, nine in 2023 and six in 2024). In fact, as the dolphin numbers in North Lantau waters have dropped to extremely low levels (with zero dolphin in NEL since 2015, and only a few dolphins in NWL in recent years), estimating their abundance with reasonable precision becomes increasingly difficult to near impossible. Although increasing survey effort may compensate for decreasing sighting rates, there have already been large amount of survey effort allocated to North Lantau waters with the additional HZMB-related EM&A data, and it is still impossible to estimate dolphin abundance with any reliability when the number drops to extremely low numbers. Therefore, a change in monitoring strategy has been implemented starting from the previous and present monitoring periods in 2023-24 and 2024-25 to adopt a hybrid and complementary approach of conducting both line-transect vessel surveys and passive acoustic monitoring to paint a better picture of dolphin occurrences and associated temporal trend in North Lantau waters. If resources are limited in the future and a decision is needed to be made between the two monitoring methods, there should be a strong consideration to continue the PAM studies instead of vessel-based monitoring for the long-term monitoring of cetaceans in North Lantau waters.

5.7. *Dolphin Occurrences Assessed by Passive Acoustic Monitoring*

5.7.1. Overall occurrence in BMP and SCLKCMP

Based on the 717 CWD DPMs recorded among the four North Lantau sites between January 2024 and January 2025, detailed assessment was conducted to understand the occurrences of Chinese White Dolphins through passive acoustic monitoring as a complementary approach to the vessel-based line-transect monitoring method with the results detailed in Sections 5.2 to 5.6.

Dolphin activity, as measured by the proportion of all logged days with one or more detections (DPD % of logged days) varied greatly: 3.37% of 355.94 logged days (Siu Ho Wan); 3.37% of 355.93 logged days (Tai Mo To); 9.27% of 355.90 logged days (Sha Chau SE); and 25.94% of 345.39 logged days (Lung Kwu Chau N) (Table 1). Comparing across areas, BMP had very few logged days with detections (only 3.37% of 711.87 logged days), whereas SCLKCMP had 17.50% of 701.29 logged days with at least one detection.

The mean dolphin DPM/day metric also varied greatly (in increasing order): 0.06 (Tai Mo To), 0.14 (Siu Ho Wan), 0.26 (Sha Chau SE) and 1.59 (Lung Kwu Chau N) (see Table 1). By area, the mean DPM/day for BMP was only 0.10 ± 0.74 SD (Coefficient of Variation (CV) = 7.4) and for SCLKCMP was 0.91 ± 3.99 SD (CV = 4.4). The general (but usually imperfect) pattern observed in past years (where the rank order of sites based on DPD% of logged days matched the rank order of sites based on mean DPM/day) was also seen during this deployment. This general pattern was also observed across the areas. Only one discrepancy between the rank orders of dolphin DPD% of logged days and mean dolphin DPM/day was observed (that being the rank position of Tai Mo To being the same as Siu Ho Wan with regards to DPD % of logged days while mean dolphin DPM/day was lower than found at Siu Ho Wan). However, given how few DPMs were recorded at Siu Ho Wan and Tai Mo To, any interpretations of these metrics would be premature.

As in previous years of PAM works, it was clear that the Lung Kwu Chau N site continued to have the highest dolphin detection metrics (by far) of these four sites. It was followed next (but distantly) by the other SCLKCMP site at Sha Chau SE. Both BMP sites had very minimal detection. Although there have been very few DPMs since January 2020, there may still be some seasonal regularity in the presence of dolphins at BMP sites, with most occurring during the dry season (see Figure 37). This same pattern of greater detections being found during the dry season was even clearer for the SCLKCMP sites (Figure 38).

5.7.2. Overall diel patterns of occurrence

The F-POD data obtained during this study revealed some clear differences in peak activity periods during different times of the day and at different sites. For Siu Ho Wan and Tai Mo To, there was a clear diurnal pattern with higher CWD activity periods occurring during the overnight hours from about 22:00 to 04:00 (with another smaller peak of activity at about 07:00) and this pattern appears to be maintained regardless of season (Figures 39a-c). However, with the few DPMs overall, the same cautious caveat remains with regards to interpreting these data.

At SCLKCMP, the daily patterns are less pronounced with the main (and broad) peak of activity being observed from about 22:00 to 09:00) and a sharp peak occurring mid-day (about 11:00 to 13:00) (Figure 40a). Because very few DPMs were recorded during the wet season, the data at SCLKCMP sites (Figure 40b) need to be interpreted cautiously. And given that the majority of the detections were made during the dry season, it is not surprising that the dry season pattern (Figure 40c) is a close reflection of the overall pattern (Figure 40a). Because Lung Kwu Chau N had an overwhelmingly higher number of detections (compared with Sha Chau SE), the overall diel patterns for SCLKMP were also driven primarily by the patterns observed at Lung Kwu Chau N.

5.7.3. Statistical analyses on the spatio-temporal effects on occurrence

To identify the effects of location, season, and time of day, proper statistical testing of the F-POD data was also performed. However, given the low levels of detections at Siu Ho Wan, Tai Mo To and Sha Chau SE, the results must be interpreted cautiously. For location, the data were grouped based on area or specific sites within an area. For the seasons, the groupings were based on the wet and dry monsoon seasons (with clear differences in monthly rainfall amounts over many years in the region according to data maintained by the Hong Kong Observatory) rather than the four solar/astronomical seasons (i.e., spring, summer, autumn, winter) because the former appears to be more biologically important to the local CWDs and FPs. However, the two methods of seasonal classification correspond closely with each other in Hong Kong (i.e., the wet season spanned from about April to September, which matches well with the solar spring and summer used for the analyses associated with the vessel-based survey data, and the dry season extended from about October to March, which represent autumn and winter). For the diel periods, the hours of the day were separated into “day” and “night”, and six hours were omitted around the “transitional” periods when sunrise and sunset occurred (three hours each for sunrise and sunset). The omitted hours included the hours immediately preceding and following, as well as

the hour in which sunrise/sunset occurred. The omission of the “transitional” periods helps to reduce the confounding issues of including possible different activity levels during transitional periods and autocorrelations between two periods of time being adjacent to one another.

To examine interannual changes, F-POD data from 2024 (mostly obtained during the current marine mammal monitoring study but supplemented by another PAM study also funded by AFCD) were compared with available F-POD data from past studies: Siu Ho Wan (2020, 2021, 2022 and 2023), Tai Mo To (2021, 2022 and 2023), Lung Kwu Chau N (2020, 2021, 2022 and 2023), Sha Chau SE (2020, 2021, 2022 and 2023). Daily DPM data from the same sites and calendar dates (and times to the nearest hour) were compared year-over-year to determine if and how CWD acoustic activities have changed over time at each site. For comparison, data for each calendar year (January 1st to December 31st) were used.

All subsets of data, which were used in various comparisons to better understand the effects of location, season and diel period on dolphin acoustic activity as well as the comparisons with data from past years where available (for interannual differences), were assumed to deviate from either the normal distribution or homogeneity of variance assumption of parametric analyses. This assumption was based on a long history of testing C-POD and F-POD data for these characteristics and observing an overwhelming number of the subsets of C-POD and F-POD data used for analyses were either significantly different from normality or were heteroscedastic (see Wang and Hung 2018, 2019, 2020, 2021, 2022). As such, non-parametric statistical methods were employed. When comparisons were made across two independent groups, the Mann-Whitney U test was employed. When comparisons were made between or across dependent groups (i.e., for repeated measures where pairs of data are available for the same sample, such as the comparisons of data obtained during the day vs. night for the same day, which was the sample, or comparing data obtained in different years but on the same calendar day), the Wilcoxon matched pairs test (for paired groups) or the Friedman ANOVA test by ranks (for multiple groups) was used.

With the exception of the summary of data from the current study period (January 30th, 2024 to January 20th, 2025) presented in Table 1, all other statistical analyses were performed based on the calendar year (from January 1st to December 31st) rather than based on logistical deployment/retrieval dates that have little bearing on biology. And although the calendar year also likely has minimal importance to biology, the calendar year facilitates comparisons with data from the long-term vessel-based monitoring

program in Hong Kong. Given that the current study approximated the calendar year 2024, data for only January 2024 (obtained from the previous deployment) were added to complete the calendar year while data obtained after December 31st, 2024 during this PAM work were omitted from current statistical analyses. To better visualize long-term changes and patterns, available data from past years were also included in graphs of dolphin detections across time (year-over-year).

Comparison of dolphin occurrence among sites and between different areas

Not surprisingly, statistical analyses (Mann-Whitney U test) showed significant differences in daily CWD DPM between the two marine park areas $U=215683.5$ ($p<<0.0001$). These results strongly support that for further examinations of sites as well as diel and seasonal patterns, data from the different areas should be analyzed separately to prevent sites with large numbers of detections from overwhelming sites with many fewer detections. Mann-Whitney U tests were used to determine if the two sites within each of the BMP and SCLKCMP areas differed from each other with regards to daily dolphin DPMs. Siu Ho Wan had slightly more DPMs than Tai Mo To but the two BMP sites were not statistically different from each other ($U=63687$, $p=0.966$). At SCLKCMP, Lung Kwu Chau N had significantly more CWD detections than Sha Chau SE ($U=51200$, $p<<0.0001$).

These PAM results are consistent with the findings of visual surveys in the past years where dolphin density was consistently higher in SCLKCMP than in BMP (Hung 2022, 2023, 2024). The current F-POD data were also consistent with the general patterns of the C-POD data from past years with regards to relative spatial differences in CWD detections (e.g. Wang and Hung 2022, 2023, 2024).

Comparison of dolphin occurrence across different seasons

There were clear seasonal differences in the daily CWD DPM metrics at both marine park areas with the dry season having more detection (Table 7; Figures 37 and 38). However, the seasonal differences observed at SCLKCMP was significant ($U=42717$, $p<<0.0001$) but not at BMP ($U=55120$; $p=0.061$). For all sites, there were more detections during the dry season than the wet season. However, only for Lung Kwu Chau N was the difference statistically significant ($U=7412.5$; $p<<0.0001$). The seasonal differences for the other three sites were not significant: Siu Ho Wan ($U=13789.5$; $p=0.171$); Tai Mo To ($U=13770$; $p=0.198$); and Sha Chau SE ($U=14044.5$; $p=0.915$) (Table 7). The general seasonal spatial pattern in 2024 was higher detections during the dry season in the waters of North Lantau, which is similar to the pattern observed in the F-POD data for 2021-22, 2022-23 and 2023-24 and C-POD data

for 2020-21 but different from that observed in the 2019-20 C-POD dataset. However, the differences were not as clear as in past years and this may be due the decline in detections to very low (Tai Mo To) or record low (Siu Ho Wan and Sha Chau SE) levels.

Comparison of dolphin occurrence between different times of day

From the statistical analyses of the two marine park areas, the results showed a strong overall diel pattern, with significantly more dolphin DPMs at night than during the day at BMP (Wilcoxon $T=5.00$, $z=3.258$, $p<0.005$) and SCLKCMP (Wilcoxon $T=2527$, $z=3.010$, $p<0.005$) (Table 8 and Figures 39 and 40). The same general pattern of high and low acoustic activities for night time and daytime hours seemed to be maintained regardless of the seasons but this pattern was not significant at BMP during the wet season (Table 8). However, very few DPMs were recorded from BMP overall and during each season as well as from one of the SCLKCMP sites (Sha Chau SE) during the wet season so conclusions should be considered cautiously.

For all four sites, the same overall pattern of higher DPMs at night was observed, but only Siu Ho Wan (Wilcoxon $T=0.00$, $z=2.666$, $p<0.01$) and Sha Chau SE (Wilcoxon $T=45.5$, $z=3.448$, $p<0.001$) were significant. And although both Tai Mo To (Wilcoxon $T=3.00$, $z=1.859$, $p=0.063$) and Lung Kwu Chau N (Wilcoxon $T=1746$, $z=1.835$, $p=0.067$) were not significant, their p-values were close to 0.05 (Table 8). When the diel patterns were analyzed separately for wet and dry seasons, all sites showed more detections at night regardless of season. During the wet season, only Lung Kwu Chau N (Wilcoxon $T=1.00$, $z=1.992$, $p<0.05$) and Sha Chau SE (Wilcoxon $T=17$, $z=1.992$, $p<0.05$) had significantly more DPMs during the night compared with detections during the day time [Note: there were insufficient detections at Siu Ho Wan during the wet season for analyses]. During the dry season, only at Siu Ho Wan (Wilcoxon $T=0.00$, $z=2.521$, $p<0.05$) and Sha Chau SE (Wilcoxon $T=8.00$, $z=2.794$, $p<0.01$) were the detections significantly higher at night than during the day. Because there were considerably more detections during the dry season for all sites (and areas) (Table 7), the diel patterns observed during the dry season were the main drivers of the patterns observed at each site (and area) when season was not considered.

Considering all the results together, it is clear that strong diel patterns exist in CWD DPMs and the diel patterns may be very different seasonally at different sites. In general, there appears to be more detection at night. An interesting reversal of this most common pattern was found at Lung Kwu Chau N last year but not during this study. Similar reversals were observed previously at this site during the wet season in

2017-18 (Wang and Hung 2018) and at several sites in 2018-19 (Wang and Hung 2019), 2019-20 (Wang and Hung 2020), 2020-21 (Wang and Hung 2021) and 2021-22 (Wang and Hung 2022), especially during the wet season and at SWLMP sites. The biological reasons for these diel differences and at different sites should be investigated further when more data become available.

Comparison of dolphin occurrence across different years

For each site, data from the current study (i.e., 2024) were compared with available F-POD data from earlier years obtained from previous studies: Siu Ho Wan (2020, 2021, 2022 and 2023), Tai Mo To (2021, 2022 and 2023), Lung Kwu Chau N (2020, 2021, 2022 and 2023), Sha Chau SE (2020, 2021, 2022 and 2023) (see Figures 41-44). The summary deployment and detection statistics for each year and site are shown in Table 9. Although some data from 2025 were collected during the current project, they were too limited for meaningful interannual comparisons and so were omitted here. Comparisons across sites with widely-varying levels of dolphin detections are less meaningful than within each site, especially when determining if temporal changes have occurred (and without adding potential confounding spatial factors when examining temporal changes). Thus, all sites were compared year-over-year using the calendar year (i.e. January 1st to December 31st). Some sites had data gaps due to equipment loss, malfunction, early stoppage due to battery drainage or memory capacity being exceeded as a result of excessive environmental noise, different deployment/retrieval dates, or other reasons (see Figures 41-44). Dates with missing data in any of the calendar years being compared were omitted from the year-over-year pairwise analyses. Such year-over-year comparisons between even just two or three years can be useful for checking for short-term changes at these sites because massive, construction activities continue at a rapid pace in waters adjacent to the BMP and SCLKCMP.

Although it was not possible to compare detection metrics obtained from C-PODs and F-PODs directly at this time, the general trends in each dataset across multiple years provide an understanding of the trajectories of the detections over time. The trends are as important, if not more so, in understanding the status of marine mammals inhabiting Hong Kong's water because currently, there is no reliable method to determine the absolute number of individuals from acoustic detection data.

At Siu Ho Wan, CWD detections from 2020 to 2024 showed no significant difference across the years using a Friedman ANOVA test ($\chi^2=12.806$, $n=349$, $df=4$, $p=0.123$). Thus, year-over-year pairwise comparisons were not needed because none

would be significant (Table 10). The DPD % of logged days increased slightly from 2020 to 2021 and 2022 (both 2021 and 2022 had the same level) before a larger decline in 2023 from the 2021 and 2022 levels and this decline continued into 2024 (Table 9). The DPM and mean DPM/day metrics also showed this general pattern of increase from 2020 to a peak in 2022 before declining sharply in 2023 and then further again in 2024. Unlike the large and continuing decline in detections recorded by C-PODs from 2017-18 to 2020-21, which was already much reduced compared to data obtained in 2013-14 (see Wang and Hung 2021), the F-POD data did not show any declines until 2023 (and which has continued into 2024) and in fact, there were slight increases in detections each year from 2020 to 2022 (Figure 45). However, the current, ongoing pattern is of a relatively large and continuing decline since 2022 but these data must be interpreted carefully as total DPMs were very low every year since 2020 (Table 9).

For Tai Mo To, CWD detections from 2021 to 2024 also showed no significant difference across the years using a Friedman ANOVA test ($\chi^2=6.667$, $n=126$, $df=3$, $p=0.083$). Thus, year-over-year pairwise comparisons were also not needed because none would be significant (Table 10). The DPD % of logged days increased slightly from 2021 to 2022 before a larger decline in 2023 to the lowest level before increasing slightly in 2024 (Table 9). The DPM and mean DPM/day metrics also showed this general pattern of increase from 2021 to 2022 before declining in 2023 to levels lower than in 2022 (but not as low as in 2021). DPM and mean DPM/day in 2024 were slightly lower than but very similar to the values in 2023. Unlike the large and continuing decline in detections recorded by C-PODs from 2017-18 to 2020-21, the F-POD data did not show a continuing decline and in fact, a slight increase in detections was observed from 2021 to 2022 (Figure 46). However, the current, ongoing pattern is of a relatively large and continuing decline since 2022 but there were far too few DPMs each year (Table 9) for meaningful statistical analyses.

All year-over-year F-POD data for BMP sites must be interpreted carefully because total annual detections were very low so even a single rare event can have a disproportionate influence on annual metrics and thus, the observed trends. Statistically, the lack of significance across the years indicated no noticeable changes in detections and any indications of an increase in 2021 and/or 2022 was not supported by the data obtained in 2023 and 2024, which further reinforces the need for cautious interpretation of early results. The only more reliable conclusion that can be made currently is that detections have declined greatly from 2017-18 to very low levels in 2020 and appears to be unchanged since then. The levels of detections maybe too low for statistical detection of slight declines (or increases) in later years.

At Lung Kwu Chau N, the Friedman ANOVA test showed significant year-over-year changes in dolphin detections at Lung Kwu Chau N from 2020 to 2024 ($\chi^2=215.02$, $n=214$, $df=4$, $p<<0.0001$). Pairwise comparisons showed a slight but non-significant increase from 2020 and 2021 (Wilcoxon $T=8349.5$, $z=0.095$, $p=0.924$) but there were significantly fewer detections in 2022 relative to 2020 (Wilcoxon $T=4855.5$, $z=7.976$, $p<<0.0001$) and to 2021 (Wilcoxon $T=2714$, $z=6.799$, $p<<0.0001$) (Table 10). The lowest values of the detection (for all metrics) were reached in 2024 with significantly fewer detections than in 2020 (Wilcoxon $T=2653.5$, $z=8.296$, $p<<0.0001$), 2021 (Wilcoxon $T=1703$, $z=8.252$, $p<<0.0001$), 2022 (Wilcoxon $T=4420.5$, $z=4.430$, $p<<0.0001$) and fewer, but not significantly so, detections than in 2023 (Wilcoxon $T=5067$, $z=1.624$, $p=0.104$). Previously, the lowest values were reached in 2023, which had significantly fewer detections than in 2020 (Wilcoxon $T=4633.5$, $z=8.620$, $p<<0.0001$), 2021 (Wilcoxon $T=1640.5$, $z=8.296$, $p<<0.0001$) and 2022 (Wilcoxon $T=7001$, $z=2.625$, $p<0.01$). DPMs and mean DPM/day values increased from 2020 to 2021 but then decreased each year in 2022 and 2023 before falling to the lowest levels in 2024; the same pattern was also reflected in DPD % of logged days (Table 9). Although there appeared to be a recovery in 2021, CWD detections (as determined by F-PODs) dropped to the lowest levels each year in 2022, 2023 and 2024. These results further support that the overall large and statistically significant decline in detections observed in the C-POD data from 2017-18 to 2020-21 (see Wang and Hung 2021) is still continuing (Figure 47) although the most recent decline from 2023 to 2024 was not significant. The lack of statistical significance in the decline from 2023 to 2024 maybe partially due to detection levels being too low for statistical analyses to be able to detect small changes in detections. Thus, great caution must be exercised when making conclusions about the cessation of the decline based on a lack of statistical significance because there maybe a lack of statistical power in the analyses due to much reduced detections.

At Sha Chau SE, the Friedman ANOVA test showed significant changes in dolphin detections from 2020 to 2024 ($\chi^2=41.920$, $n=168$, $df=4$, $p<<0.0001$). Year-over-year pairwise comparisons showed no significant changes from 2020 to 2021 (Wilcoxon $T=753$, $z=0.996$, $p=0.319$) but there were significantly more detections in 2022 as compared with 2020 (Wilcoxon $T=351.5$, $z=3.074$, $p<0.005$) and a significant decrease from 2021 to 2022 (Wilcoxon $T=3209$, $z=2.68$, $p<0.01$) (Table 10). The lowest values of the detection metrics were reached in 2024 with significantly few detections than in 2020 (Wilcoxon $T=210.5$, $z=3.872$, $p<0.0005$), 2021 (Wilcoxon $T=1426$, $z=5.772$, $p<<0.0001$), 2022 (Wilcoxon $T=1129.5$, $z=3.572$, $p<0.0005$) and fewer, but not significantly so, detections than in 2023 (Wilcoxon $T=997$, $z=0.887$,

$p=0.375$). Previously, the lowest values of the detection metrics were reached in 2023, which had significantly fewer detections than in 2020 (Wilcoxon $T=89$, $z=4.738$, $p<<0.0001$), 2021 (Wilcoxon $T=1614$, $z=5.371$, $p<<0.0001$) and 2022 (Wilcoxon $T=1595.5$, $z=2.677$, $p<0.01$). DPMs and mean DPM/day values increased from 2020 to 2021 but then decreased each year in 2022, 2023 and to the lowest levels in 2024; the same pattern was also reflected in the DPD % of logged days (Table 9). Although there appeared to be a recovery in 2021, CWD detections (as determined by F-PODs) dropped to the lowest levels each year in 2022, 2023 and 2024. These results further support that the overall large and statistically significant decline in detections observed in the C-POD data from 2017-18 to 2020-21 (see Wang and Hung 2021) is still continuing (Figure 48) although the decline from 2023 to 2024 was not significant. The lack of statistical significance in the decline from 2023 to 2024 is likely partially due to detection levels being too low for statistical analyses to be able to detect small changes in detections. Great caution must be exercised when making conclusions about the cessation of the decline based on a lack of statistical significance because there is likely a lack of statistical power in the analyses due to limited detections.

The significant declines in CWD detections observed in previous C-POD data showed early signs of possible recoveries in detection levels at Siu Ho Wan, Tai Mo To, Lung Kwu Chau N and Sha Chau SE. However, F-POD data from 2023 and now in 2024 have shown those early signs of recovery were premature and the declines appear to be continuing. The large decrease in CWD detections at these sites over the past eight years from 2017 to the lowest levels in 2024 corresponded well (but not perfectly) with the decline in annual estimated abundance of CWD in the North Lantau region during the same period (as determined from line-transect vessel survey data) from 21 dolphins in 2017 to six in 2018, four in 2019, three in 2020, four in 2021, only one in 2022 (the lowest level since 2001), four in 2023 and only one again in 2024. The annual estimated abundance for North Lantau waters being in single digits since 2018 suggest the occurrence of dolphins in these waters has nearly completely stopped. Only seven sightings of CWD were made in 2024 during more than 3,900 km of survey effort (Table 6a). This near disappearance is in stark contrast to the highest annual abundance estimate for North Lantau waters of 102 dolphins in 2003 (only about 20 years ago; Table 6b), when dolphins were seen often in these waters before the onslaught of the multiple large-scale development projects that has occurred.

The lack of perfect correspondence between the two datasets may be related to visual detections being only during the day whereas more acoustic detections were recorded during the night time. Furthermore, the PAM study in North Lantau waters

were confined to four specific locations within the two marine parks, while the survey areas in NWL and NEL encompass a much larger area (see Figure 1). For example, while there was only one dolphin estimated in 2024 for the entire North Lantau region, the majority of sightings made were away from the SCLKCMP and BMP and some were even located at the southwest corner of NWL survey area far away from the closest marine park at SCLKCMP (Figure 4). Nevertheless, the general pattern of a large decrease in detections from 2017 to 2018 and then remaining at extremely low levels since 2018 is well supported by compelling evidence from both the vessel survey and PAM studies.

Comparison of diel pattern of dolphin occurrence across different years

Although there were few CWD detections recorded at Siu Ho Wan, the diel pattern appears to remain the same across the years (including in 2024) with most of the detections occurring between about 21:00 and 07:00 (Figure 49a). There were even fewer CWD detections at Tai Mo To for comparing diel patterns across years but there is some suggestion of a pattern that is similar to that of Siu Ho Wan with most of the detections being during the night time hours and the 2024 data were also consistent with this pattern (Figure 49b).

At both Lung Kwu Chau N and Sha Chau SE, there was no strong consistent diel pattern in CWD detections that spanned across the years. Instead, there seemed to be less defined diel patterns and the patterns seem to differ across the years (Figures 50a and 50b). In some years there were strong peaks of activity during specific times of the day but these were not present in other years; 2024 was no different and did not seem to follow closely any other year's pattern. It is unclear if this lack of clear diel patterning at these two sites was the result of a rapid decline in CWD numbers over the last few years due to increased disturbances in the general area.

5.8. *Range Use, Residency and Movement Patterns of Individual Dolphins*

5.8.1. Individual range use, residency pattern and core area use

Individual Range Use

In order to examine the range use of individual dolphins based on the photo-identification data collected during the present monitoring period, the 95% UD kernel ranges of 107 individuals that occurred in Hong Kong's survey areas in 2024 (as identified through photo-identification works) were deduced using the fixed kernel method, and their ranging patterns are shown in Appendix V. In addition, 92 of these individual dolphins that occurred in 2024 and also had a history of being sighted ≥ 15 times were further examined for their range use and residency patterns (Table 11).

Among these 92 individuals, all of them had occurred in WL in the past, while the majority of them had also occurred in NWL (59.8%) and SWL (81.5%), but to a lesser extent in NEL (10.9%) and DB (12.0%) (Table 11). In contrast, only four individuals had been re-sighted in the SEL survey area as part of their historical range, and none was re-sighted in EL survey area at all. Furthermore, 70 of these 92 individuals (or 76.1%) occupied ranges that spanned the waters of Hong Kong and the Mainland (Table 11), indicating cross-boundary movements by many individual dolphins that occur regularly in Hong Kong's waters. However, some of these individuals occurred just to the west of the territorial boundary without venturing much further into Lingding Bay (see Appendix V).

Residency Pattern

The residency patterns of the 92 individuals were further assessed by examining their annual and monthly occurrences in Hong Kong. Overall, 54 and 25 individuals were identified as year-round and seasonal residents respectively (Table 11), and all these individuals were considered residents in Hong Kong, as they have been sighted consistently in the past 12 years (i.e. 2013-24), or in at least the past five consecutive years. In contrast, nine individuals were considered seasonal visitors that have occurred intermittently in the past 12 years. Furthermore, based on the monthly occurrences of these individuals, more than one third of them only occurred in Hong Kong during certain months of the year, while the rest occurred here year-round (Table 11).

In addition to their residency patterns, attempts were made to classify the 92 individuals into the two social clusters that occurred regularly in Hong Kong (see Dungan et al. 2012), based on their overall range use at 95% UD level as well as core area use at 50% UD and 25% UD levels. Results indicated that seven and 72 individuals belonged to the northern and southern social clusters, respectively (Table 11). In addition, there were also 13 individuals that spanned their range use more or less evenly across North and West Lantau waters with frequent occurrences in both waters, and some of them (e.g. NL123, NL156, NL261, WL11) actually shifted their range use from North Lantau waters to WL and SWL waters in the past decade (see Appendix V).

Core Area Use

The analysis on individual core area use revealed that four major core areas of dolphin activities are located around Lung Kwu Chau, the Brothers Islands, in SWL waters, and along the coast of West Lantau, with the latter further subdivided into Tai O,

Peaked Hill and Fan Lau. Among the 92 individuals, 18 and 15 individuals occupied Lung Kwu Chau as their 50% and 25% UD core areas, respectively, while only two individuals (NL33 and NL123) occupied the Brothers Islands as their 50% core area as well as 25% UD core area in the earlier years before abandoning this habitat as part of their ranges nearly a decade ago (Table 11). Notably, nearly half of these individuals that utilized Lung Kwu Chau and the Brothers Islands as their core areas belonged to the northern social cluster, with the other half spanning their range use across North and West Lantau waters.

In contrast, 83 individuals utilized the waters along the west coast of Lantau as their 50% UD and 25% UD core areas, respectively, with the vast majority of them belonging to the southern social cluster (Table 11). As there has been a surge of individuals expanding or shifting their range use into SWL waters in the past decade, there were also twelve and two individuals that have utilized South Lantau waters as their 50% and 25% UD core areas, respectively.

5.8.2. Individual movement pattern

By combining all photo-identification data collected through the present monitoring study and other studies, movement patterns of individual dolphins within Hong Kong territorial waters in 2024-25 were broadly examined. From April 2024 to March 2025, 111 individuals were re-sighted a total of 492 times, with 82 individuals being re-sighted more than once (i.e. occurred at more than one location).

The examination of individual movement patterns between re-sightings revealed that only 47 individuals moved across different survey areas around Lantau in 2024-25. That included only one individual (NL202) that occurred across NWL and WL survey areas, and 46 individuals that were re-sighted in both SWL and WL survey areas (Table 12). Even though some individuals occurred in all three areas of NWL, WL and SWL in past monitoring periods, such occurrence was not recorded during the present study period. As in recent monitoring periods, no re-sighting of known individual was made in NEL during the 2024-25 monitoring period so there was no movement of individuals into this once-important habitat.

With an extensive amount of photo-identification data being collected from different surveys, there were still a significant portion of individual dolphins sighted repeatedly within just a single survey area and did not range into neighbouring areas. These included 32 individuals that occurred exclusively in the WL survey area, two in the SWL survey area and one in the NWL survey area. The restricted movement

within the small survey area of WL by many individuals raises some concern, as this could be related to potential obstructions to movements across different survey areas as a result of human activities (e.g. vessel traffic) or infrastructure projects (e.g. reclamation platform or bridge structure as physical barrier).

The temporal trend in individual movement patterns across different survey areas was examined among the past 14 monitoring periods (Figure 51; Table 12), in order to provide insights into temporal changes in their intensity of movements as a result of various anthropogenic factors. Besides the dramatic decline in dolphin movements between NEL and NWL survey areas due to the absence of dolphin occurrence in NEL in recent years, there were other notable changes. For example, there was a continuous sharp decline in dolphin movements across the NWL and WL survey areas in recent monitoring periods, and such level in 2024-25 study period reach the lowest ever among all monitoring periods. Furthermore, even though there was a study decline in dolphin movements across SWL and WL from the highest in 2016-17 (81 individuals) to the lowest in 2021-22 (31 individuals), such movement between the two areas was on a rebound in the past three years.

6. SCHOOL SEMINARS AND PUBLIC AWARENESS

During the 2024-25 monitoring period, HKCRP researchers were able to deliver 11 education seminars at local primary and secondary schools (with some held online) on behalf of AFCD to increase public awareness on the conservation of local cetaceans. PowerPoint presentations were prepared for the school talks with up-to-date information on both dolphins and porpoises gained from the present long-term monitoring programme. The talk materials included content such as the threats faced by local cetaceans, and conservation measures that AFCD has implemented to protect them in Hong Kong. Through this integrated approach of the long-term monitoring programme and publicity/education programme, the Hong Kong public can gain first-hand information from our HKCRP researchers, and their support will be vital to the long-term success in conservation of local cetaceans.

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8. LITERATURE CITED

- Buckland, S. T., Anderson, D. R., Burnham, K. P., Laake, J. L., Borchers, D. L., and Thomas, L. 2001. Introduction to distance sampling: estimating abundance of biological populations. Oxford University Press, London.
- Hooge, P. N. and Eichenlaub, B. 1997. Animal movement extension to ArcView (version 1.1). Alaska Biological Science Center, United States Geological Survey, Anchorage.
- Hung, S. K. 2005. Monitoring of finless porpoise (*Neophocaena phocaenoides*) in Hong Kong waters: final report (2003-05). An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 95 pp.
- Hung, S. K. 2008. Habitat use of Indo-Pacific humpback dolphins (*Sousa chinensis*) in Hong Kong. Ph.D. dissertation. University of Hong Kong, Hong Kong, 266 p.
- Hung, S. K. 2022. Monitoring of marine mammals in Hong Kong waters: final report (2021-22). An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 147 pp.
- Hung, S. K. 2023. Monitoring of marine mammals in Hong Kong waters: final

- report (2022-23). An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 121 pp.
- Hung, S. K. 2024. Monitoring of marine mammals in Hong Kong waters: final report (2023-24). An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 147 pp.
- Jefferson, T. A. 2000a. Population biology of the Indo-Pacific hump-backed dolphin in Hong Kong waters. *Wildlife Monographs* 144:1-65.
- Jefferson, T. A. (ed.) 2000b. Conservation biology of the finless porpoise (*Neophocaena phocaenoides*) in Hong Kong waters: final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government.
- Jefferson, T. A. and Leatherwood, S. 1997. Distribution and abundance of Indo-Pacific hump-backed dolphins (*Sousa chinensis* Osbeck, 1765) in Hong Kong waters. *Asian Marine Biology* 14: 93-110.
- Jefferson, T. A., Hung, S. K., Law, L., Torey, M. and Tregenza, N. 2002. Distribution and abundance of finless porpoises in waters of Hong Kong and adjacent areas of China. *Raffles Bulletin of Zoology, Supplement* 10: 43-55.
- Jefferson, T. A., Hung, S. K., Robertson, K. M. and Archer, F. I. 2012. Life history of the Indo-Pacific humpback dolphin (*Sousa chinensis*) in the Pearl River Estuary, southern China. *Marine Mammal Science* 28: 84-104.
- Jefferson, T. A., Hung, S. K. and Würsig, B. 2009. Protecting small cetaceans from coastal development: Impact assessment and mitigation experience in Hong Kong. *Marine Policy* 33: 305-311.
- Thomas, L., Laake, J. L., Rexstad, E. A., Strindberg, S., Marques, F. F. C., Buckland, S. T., Borchers, D. L., Anderson, D. R., Burnham, K. P., Burt, M. L., Hedley, S. L., Pollard, J. H., Bishop, J. R. B. and Marques, T. A. 2009. Distance 6.0 Release 2. Research Unit for Wildlife Population Assessment, University of St. Andrews, UK.
- Wang, J. Y. and Hung, S. K. 2018. Passive acoustic monitoring of Chinese White Dolphins within the Sha Chau and Lung Kwu Chau Marine Park and the Brothers Marine Park: final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 51 pp.
- Wang, J. Y. and Hung, S. K. 2019. Passive acoustic monitoring of marine mammals in Hong Kong waters (2018-19): final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 74 pp.
- Wang, J. Y. and Hung, S. K. 2020. Passive acoustic monitoring of marine mammals in Hong Kong waters (2019-20): final report. An unpublished report submitted

- to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 83 pp.
- Wang, J. Y. and Hung, S. K. 2021. Passive acoustic monitoring of marine mammals in Hong Kong waters (2020-21): final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 91 pp.
- Wang, J. Y. and Hung, S. K. 2022. Passive acoustic monitoring of marine mammals in Hong Kong waters (2021-22): final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 131 pp.
- Wang, J. Y. and Hung, S. K. 2023. Passive acoustic detection services of Chinese White Dolphins and finless porpoises in Hong Kong western waters (2022-23): final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 104 pp.
- Wang, J. Y. and Hung, S. K. 2024. Passive acoustic monitoring of marine mammals in Hong Kong waters (2023-24): final report. An unpublished report submitted to the Agriculture, Fisheries and Conservation Department of Hong Kong SAR Government, 87 pp.

Table 1. Summary data statistics on detection of Chinese White Dolphins as recorded by F-PODs during the current study (from January 30th, 2024 to January 20th, 2025) for Brothers Marine Park (BMP) and Sha Chau & Lung Kwu Chau Marine Park (SCLKCMP) and each site within these areas. (Note: "Logged Days": no. logged days the C-POD was on and recording; "DPD% of Days": detection positive days as a percentage of logged days; DPM: detection positive minutes, minutes where at least one dolphin click train was detected; "% Time Lost": percentage of time lost because the minute click limit has been reached and no subsequent data could be recorded for that minute)

Site / Area	Logged Days	% Time Lost	DPD% of Days	DPM	Mean DPM/Day	% False Positive DPM
BMP	711.87	0.45	3.37	71	0.10	0.00
SCLKCMP	701.29	0.62	17.50	646	0.91	0.00
Siu Ho Wan (BMP)	355.94	0	3.37	51	0.14	0.00
Tai Mo To (BMP)	355.93	0.89	3.37	20	0.06	0.00
Lung Kwu Chau N (SCLKCMP)	345.39	0	25.94	554	1.59	0.00
Sha Chau SE (SCLKCMP)	355.90	1.22	9.27	92	0.26	0.00
Totals:	1413.16	0.53		717	0.50	0.00

Table 2a. Mean group size of Chinese White Dolphins among different survey areas in recent monitoring periods (* denote the mean group size calculated from a sample size of one group)

Monitoring Period	Overall	Deep Bay	NE Lantau	NW Lantau	W Lantau	SW Lantau	SE Lantau
2013-14	3.4	4.0	3.2	3.5	3.4	3.2	N/A
2014-15	4.1	5.1	2.7	3.5	4.4	4.0	1.0
2015-16	3.8	2.0	1.0*	4.1	3.8	3.7	2.5
2016-17	3.3	N/A	1.0*	3.8	3.5	2.4	1.4
2017-18	3.0	3.7	5.0*	3.3	3.0	2.8	1.5
2018-19	3.1	2.3	N/A	2.4	3.6	2.7	1.0
2019-20	3.2	2.0	N/A	2.7	3.2	3.6	1.0
2020-21	3.1	N/A	N/A	2.4	3.3	3.1	1.0*
2021-22	3.4	N/A	N/A	2.9	3.7	2.4	N/A
2022-23	3.6	N/A	N/A	4.4	3.4	4.1	1.0*
2023-24	3.5	N/A	N/A	2.9	3.6	3.5	1.0
2024-25	3.3	N/A	1.0*	1.0	3.6	3.1	1.0

Table 2b. Mean group size of finless porpoises among different survey areas in recent monitoring periods (* denote the mean group size calculated from a sample size of one group)

Monitoring Period	Overall	SW Lantau	SE Lantau	Lamma	Po Toi	Ninepins
2013-14	2.3	2.8	1.9	2.6	N/A	1.3
2014-15	2.7	3.5	2.6	3.1	1.9	2.6
2015-16	3.1	3.1	2.9	4.4	2.5	1.7
2016-17	2.7	2.4	2.7	3.3	3.3	2.2
2017-18	2.5	2.8	2.5	1.9	2.7	1.5
2018-19	2.7	2.1	3.1	2.3	2.0	3.0*
2019-20	2.6	2.7	2.2	2.4	3.4	3.5
2020-21	2.7	2.8	3.2	2.1	1.8	3.5
2021-22	2.0	1.5	2.6	2.1	2.1	2.0
2022-23	2.2	2.5	1.6	1.0*	2.2	N/A
2023-24	3.2	3.2	3.3	N/A	N/A	N/A
2024-25	2.7	2.7	2.7	N/A	N/A	N/A

Table 3. Occurrences of unspotted calves (UC) and unspotted juveniles (UJ) in Hong Kong, including the their annual total number and percentage of the total

Year	No. of UC	UC% of total	No. of UJ	UJ% of total
2002	13	1.0%	74	5.5%
2003	22	1.0%	153	6.9%
2004	18	1.1%	75	4.7%
2005	29	1.4%	123	5.9%
2006	24	1.1%	97	4.4%
2007	11	0.8%	56	4.1%
2008	12	1.0%	58	4.7%
2009	6	0.5%	87	6.9%
2010	4	0.3%	91	7.2%
2011	26	1.2%	80	3.7%
2012	27	1.5%	59	3.2%
2013	21	1.0%	102	4.8%
2014	15	0.7%	64	2.9%
2015	12	0.6%	32	1.6%
2016	1	0.1%	20	1.7%
2017	1	0.1%	20	1.7%
2018	2	0.2%	14	1.3%
2019	3	0.3%	23	2.2%
2020	3	0.4%	14	1.8%
2021	1	0.2%	17	2.6%
2022	1	0.2%	10	1.7%
2023	7	1.1%	11	1.7%
2024	5	0.9%	24	4.5%

Table 4. Encounter rates (no. of on-effort sightings per 100 km²) of Chinese White Dolphins among different survey areas in the past 23 monitoring periods

Monitoring Period	Overall	NE Lantau	NW Lantau	W Lantau	SW Lantau
2002-03	8.6	4.6	10.8	22.6	2.4
2003-04	10.8	5.0	11.3	25.9	2.5
2004-05	8.2	2.9	8.3	21.4	2.6
2005-06	7.8	2.7	8.7	20.2	1.6
2006-07	6.9	2.3	5.7	20.6	1.0
2007-08	9.9	4.7	10.5	26.1	3.7
2008-09	7.2	2.2	7.2	17.9	2.4
2009-10	6.3	1.7	4.9	18.0	2.2
2010-11	6.8	2.6	7.5	13.4	2.4
2011-12	7.7	5.0	8.7	15.3	2.6
2012-13	7.3	1.6	7.8	19.2	3.5
2013-14	7.2	0.7	6.3	19.6	6.8
2014-15	5.5	0.1	3.6	18.4	5.6
2015-16	4.7	0.1	2.2	15.5	5.5
2016-17	4.0	0.0	1.9	14.9	3.2
2017-18	3.4	0.0	2.4	11.8	4.1
2018-19	3.0	0.0	1.7	13.0	2.0
2019-20	3.5	0.0	0.7	13.3	3.8
2020-21	3.3	0.0	1.1	11.6	3.1
2021-22	2.5	0.0	0.5	13.9	2.7
2022-23	2.5	0.0	0.2	13.5	2.7
2023-24	2.6	0.0	0.3	12.1	3.5
2024-25	2.6	0.1	0.2	14.3	3.4

Table 5. Encounter rates (no. of on-effort sightings per 100 km²) of finless porpoises in South Lantau waters in the past 18 monitoring periods

Monitoring Period	Overall	SW Lantau	SE Lantau
2007-08	3.8	2.7	5.1
2008-09	2.2	2.8	1.4
2009-10	3.9	1.9	6.1
2010-11	3.9	2.7	5.4
2011-12	4.1	3.0	5.8
2012-13	7.4	5.9	8.4
2013-14	9.6	7.4	12.5
2014-15	5.6	2.6	8.7
2015-16	3.0	2.3	5.3
2016-17	4.5	2.8	8.1
2017-18	4.9	3.9	6.2
2018-19	4.1	2.9	5.1
2019-20	4.7	3.9	5.6
2020-21	2.6	2.8	2.3
2021-22	2.9	3.2	1.7
2022-23	2.8	3.2	1.8
2023-24	3.3	4.2	2.2
2024-25	4.2	5.7	2.3

Table 6a. Line transects parameters and estimates of density and abundance for Chinese White Dolphins in western waters of Hong Kong in 2024

¹unit for encounter rate: number of on-effort sightings per 100 km of survey effort;

²unit for individual density: number of dolphins per 100 km²)

	NE Lantau	NW Lantau	W Lantau	SW Lantau
Effort	1517.8	2405.5	802.4	1095.1
Number of Sightings	1	6	100	35
Average Group Size	1.00	2.17	3.44	3.26
Encounter Rate ¹	0.07	0.25	12.46	3.20
Individual Density ²	0.15	0.94	88.88	17.67
Abundance	0	1	25	12
95% C.I. (Abundance)	N/A	0-2	18-34	7-19
%CV	N/A	46.6	17.0	24.1

Table 6b. Annual abundance estimates of Chinese White Dolphins from each survey area in western waters of Hong Kong in 2003-24

(figures in red derived from biennial estimates; figures in blue indicate no or only one on-effort sighting made in that area for that year)

Year	Combined	NE Lantau	NW Lantau	W Lantau	SW Lantau
2003	188	18	84	56	30
2004	143	9	62	51	21
2005	128	7	58	42	21
2006	113	9	54	44	6
2007	130	10	60	54	6
2008	108	11	42	43	12
2009	100	5	40	43	12
2010	86	7	35	33	11
2011	88	11	39	28	10
2012	80	4	40	17	19
2013	73	3	36	23	11
2014	87	1	24	36	26
2015	65	0	10	31	24
2016	47	0	11	27	9
2017	47	0	21	16	10
2018	32	0	6	19	7
2019	52	0	4	29	19
2020	37	0	3	19	15
2021	40	0	4	24	12
2022	34	0	1	24	9
2023	34	0	4	21	9
2024	38	0	1	25	12

Table 7. Values of U statistics for comparing differences in daily DPMs for Chinese White Dolphins as detected by F-PODs during wet and dry seasons from January 1st to December 31st, 2024 using the Mann-Whitney U test for BMP and SCLKCMP and the sites within these areas. (Note: red font – significant at the $\alpha = 0.05$ level; blue font – atypical pattern; *=very few dolphin detections).

Areas	U-values	p-values	direction
*BMP	55120	0.061	dry > wet
SCLKCMP	42717	<<0.0001	dry > wet
Sites	U-values	p-values	direction
*Siu Ho Wan	13789.5	0.171	dry > wet
*Tai Mo To	13770	0.198	dry > wet
Lung Kwu Chau N	7412.5	<<0.0001	dry > wet
Sha Chau SE	14044.5	0.915	dry > wet

Table 8. Wilcoxon T statistics for comparing differences in hourly DPM of Chinese White Dolphins as recorded by F-PODs during the day vs night for all data and during the wet and dry seasons from January 1st to December 31st, 2024 for BMP and SCLKCMP and the sites within these areas. (Note: red font – significant at the $\alpha = 0.05$ level; blue font – atypical pattern; * = very few dolphin detections).

Sites or Areas	ALL			WET SEASON			DRY SEASON		
	T-values	p-values	direction	T-values	p-values	direction	T-values	p-values	direction
*BMP	5	<0.005	night > day	0	0.109	night > day	5	<0.005	night > day
SCLKCMP	2527	<0.005	night > day	26	<0.01	night > day	1959	<0.05	night > day

Sites or Areas	ALL			WET SEASON			DRY SEASON		
	T-values	p-values	direction	T-values	p-values	direction	T-values	p-values	direction
*Siu Ho Wan	0	<0.01	night > day	-	-	-	0	<0.05	night > day
*Tai Mo To	3	0.063	night > day	0	0.18	night > day	3	0.22	night > day
Lung Kwu Chau N	1746	0.067	night > day	1	<0.05	night > day	1594.5	0.13	night > day
Sha Chau SE	45.5	<0.001	night > day	17	<0.05	night > day	8	<0.01	night > day

Table 9. Summary deployment data and statistics on annual (standardized to the calendar year, January 1st to December 31st) F-POD detections of Chinese White Dolphins at all sites for 2020 (where available), 2021, 2022, 2023 and 2024. (Note: "Logged Days": number of logged days the F-POD was on and recording; "DPD% of Days": detection positive days as a percentage of logged days; DPM: detection positive minutes, minutes where at least one cetacean click train was detected).

Site	Year	Logged Days	DPD% of Days	DPM	Mean DPM/Day
Siu Ho Wan	2020	348.36	6.02	62	0.18
	2021	364.91	6.3	155	0.42
	2022	364.92	6.3	268	0.73
	2023	364.92	3.01	101	0.28
	2024	364.94	2.74	43	0.12
Tai Mo To	2021	178.35	3.35	6	0.03
	2022	310.75	3.85	38	0.12
	2023	364.92	2.19	18	0.05
	2024	364.92	2.74	15	0.04
Lung Kwu Chau N	2020	348.43	61.32	2661	7.63
	2021	237.95	65.27	5371	22.48
	2022	364.89	41.1	1429	3.92
	2023	364.91	32.33	1328	3.64
	2024	354.38	29.21	773	2.17
Sha Chau SE	2020	167.44	24.4	125	0.74
	2021	364.87	26.3	554	1.52
	2022	364.92	16.99	359	0.98
	2023	364.93	11.51	107	0.29
	2024	364.89	9.04	89	0.24

Table 10. Year-over-year (based on the calendar year of January 1st to December 31st) comparisons of F-POD detections of Chinese White dolphins using the Wilcoxon paired test at all sites for 2020 (where available), 2021, 2022, 2023 and 2024. (Note: red font – significant at the $\alpha=0.05$ level); blue font – higher detections from earlier year(s); * – because the Friedman ANOVA test resulted in no significance, pairwise comparison tests were not needed because none would be significantly different).

Sites	Direction	T-values	p-values
*Siu Ho Wan	2020 < 2021	-	n.s.
	2020 < 2022	-	n.s.
	2020 < 2023	-	n.s.
	2020 > 2024	-	n.s.
	2021 < 2022	-	n.s.
	2021 > 2023	-	n.s.
	2021 > 2024	-	n.s.
	2022 > 2023	-	n.s.
	2022 > 2024	-	n.s.
	2023 > 2024	-	n.s.
*Tai Mo To	2021 < 2022	-	n.s.
	2021 < 2023	-	n.s.
	2021 ~ 2024	-	n.s.
	2022 > 2023	-	n.s.
	2022 > 2024	-	n.s.
	2023 ~ 2024	-	n.s.
Lung Kwu Chau	2020 < 2021	8349.5	0.924
	2020 > 2022	4855.5	<<0.0001
	2020 > 2023	4633.5	<<0.0001
	2020 > 2024	2653.5	<<0.0001
	2021 > 2022	2714	<<0.0001
	2021 > 2023	1640.5	<<0.0001
	2021 > 2024	1703	<<0.0001
	2022 > 2023	7001	<0.1
	2022 > 2024	4420.5	<<0.0001
	2023 > 2024	5067	0.104
Sha Chau SE	2020 < 2021	753	0.319
	2020 < 2022	351.5	<0.005
	2020 > 2023	89	<<0.0001
	2020 > 2024	210.5	<0.0005
	2021 > 2022	3209	<0.01
	2021 > 2023	1614	<<0.0001
	2021 > 2023	1426	<<0.0001
	2022 > 2023	1595.5	<0.01
	2022 > 2024	1129.5	<0.0005
	2023 > 2024	997	0.375

Table 11. Range use (50%/25% UD core areas and sighting coverage) and residency patterns of 92 individuals with 15+ sightings and appeared in 2024

(abbreviations: SR=Seasonal Resident; YR=Year-round Resident; SV=Seasonal Visitor; UD= Utilization Distribution; LKC = Lung Kwu Chau Marine Park; CLK= northeast corner of airport; BR= Brothers Islands; TO= Tai O; PH= Peaked Hill; FL= Fan Lau; SL= South Lantau; WL= West Lantau; DB= Deep Bay; EL= East Lantau; NEL= Notheast Lantau; NWL= Northwest Lantau; SWL= Southwest Lantau; SEL= Southeast Lantau; CH=Chinese waters; * denotes individuals that have their gender determined by biopsy sampling)

ID#	# STG	Gender	Residency	Primary Range	Occurrence in Survey Areas								50% UD Core Area						25% UD Core Area					
					DB	EL	NEL	NWL	WL	SWL	SEL	CH	LKC	BR	TO	PH	FL	SL	LKC	BR	TO	PH	FL	SL
CH12	111	F?	YR	WL					✓	✓		✓					✓	✓				✓	✓	
CH38	162	?	YR	WL					✓	✓		✓					✓	✓				✓	✓	
CH105	37	F	SV	WL				✓	✓			✓			✓						✓			
CH108	172	F	YR	WL				✓	✓	✓		✓					✓	✓					✓	
CH112	33	?	SR	WL				✓	✓	✓		✓			✓		✓	✓				✓	✓	
CH113	76	F	SR	WL				✓	✓	✓		✓			✓		✓					✓		
CH141	70	F	YR	WL					✓	✓		✓					✓	✓				✓	✓	
NL33	167	F*	YR	NL			✓	✓	✓	✓			✓	✓						✓				
NL46	102	F*	SR	NL				✓	✓	✓		✓	✓						✓					
NL104	148	F	YR	NL	✓		✓	✓	✓	✓		✓	✓						✓					
NL123	204	F	YR	NL/WL	✓		✓	✓	✓	✓		✓	✓	✓					✓	✓				
NL156	103	?	YR	NL/WL				✓	✓	✓		✓			✓		✓	✓			✓	✓		
NL182	142	F	YR	NL	✓		✓	✓	✓	✓			✓						✓					
NL202	167	F	YR	NL	✓		✓	✓	✓			✓	✓						✓					
NL224	82	?	SR	NL/WL	✓		✓	✓	✓	✓		✓	✓		✓		✓		✓		✓	✓		
NL233	70	F	YR	NL	✓		✓	✓	✓			✓	✓		✓				✓					
NL236	48	?	SR	NL/WL				✓	✓			✓	✓						✓					
NL255	16	?	SV	NL/WL	✓			✓	✓	✓		✓	✓		✓		✓	✓	✓	✓		✓	✓	
NL256	43	F	YR	NL/WL	✓			✓	✓			✓	✓		✓		✓		✓		✓			
NL261	125	M?	YR	NL/WL	✓		✓	✓	✓	✓		✓	✓						✓					
NL269	84	?	YR	NL/WL				✓	✓	✓		✓					✓	✓				✓		
NL293	50	?	SR	WL				✓	✓			✓			✓						✓			
NL296	83	F?	SR	NL/WL			✓	✓	✓	✓		✓	✓		✓		✓		✓			✓		
NL299	36	?	SR	WL	✓			✓	✓	✓		✓			✓		✓				✓	✓		
NL306	71	?	YR	WL				✓	✓	✓	✓						✓	✓				✓	✓	✓
NL311	42	?	YR	WL				✓	✓	✓		✓			✓		✓	✓			✓	✓	✓	✓
NL317	21	?	SR	NL/WL				✓	✓			✓	✓		✓		✓		✓		✓	✓		
NL321	48	?	YR	NL	✓			✓	✓	✓		✓	✓						✓					
NL332	17	?	SV	WL				✓	✓	✓								✓					✓	
SL40	146	F	YR	WL					✓	✓		✓						✓					✓	
SL44	105	?	YR	WL					✓	✓		✓			✓		✓	✓				✓	✓	
SL58	27	?	YR	WL					✓	✓					✓		✓				✓	✓		
SL59	44	?	YR	WL				✓	✓	✓		✓			✓		✓	✓			✓	✓	✓	
SL60	102	?	YR	WL					✓	✓		✓						✓					✓	
SL66	17	?	SV	WL					✓	✓							✓	✓					✓	
SL68	22	?	YR	WL					✓	✓							✓	✓				✓		
WL11	76	F*	YR	NL/WL			✓	✓	✓			✓	✓		✓				✓					
WL21	89	F	SR	WL				✓	✓	✓		✓			✓		✓	✓			✓	✓		
WL42	195	?	YR	WL				✓	✓	✓		✓					✓	✓				✓	✓	
WL58	23	?	SV	WL					✓			✓			✓		✓	✓			✓	✓	✓	
WL61	165	?	YR	WL				✓	✓	✓		✓					✓	✓					✓	
WL66	34	F	SR	WL					✓			✓			✓		✓				✓			
WL72	170	F	YR	WL				✓	✓	✓		✓					✓	✓				✓	✓	
WL79	175	?	YR	WL				✓	✓			✓			✓		✓				✓			
WL91	163	?	YR	WL					✓	✓	✓	✓					✓	✓				✓	✓	
WL92	65	?	YR	WL					✓	✓		✓					✓	✓					✓	
WL94	104	F	YR	WL					✓	✓		✓					✓	✓					✓	
WL109	168	F	YR	WL				✓	✓	✓		✓			✓		✓	✓				✓	✓	
WL114	116	F?	YR	WL				✓	✓	✓		✓					✓	✓				✓	✓	
WL118	95	F	YR	WL					✓	✓		✓					✓	✓				✓		
WL123	205	F?	YR	WL				✓	✓	✓		✓					✓	✓					✓	
WL128	68	?	SR	WL					✓	✓		✓					✓	✓				✓	✓	
WL129	45	F	SR	WL					✓	✓		✓					✓	✓				✓	✓	

Table 11. (cont'd)

ID#	# STG	Gender	Residency	Primary Range	Occurrence in Survey Areas								50% UD Core Area						25% UD Core Area					
					DB	EL	NEL	NWL	WL	SWL	SEL	CH	LKC	BR	TO	PH	FL	SL	LKC	BR	TO	PH	FL	SL
WL130	150	?	YR	WL				✓	✓	✓				✓							✓			
WL131	203	F?	YR	WL				✓	✓	✓		✓		✓	✓				✓	✓				
WL142	106	?	YR	WL					✓	✓				✓							✓			
WL145	75	F	YR	WL				✓	✓	✓		✓					✓							
WL152	201	F?	YR	WL				✓	✓	✓				✓	✓						✓			
WL159	36	F	SV	WL				✓	✓			✓		✓	✓			✓	✓					
WL166	37	?	SR	WL					✓	✓				✓	✓				✓					
WL167	22	F	SR	NL/WL				✓	✓			✓		✓	✓			✓						
WL169	26	F	SR	WL					✓	✓				✓	✓				✓					
WL171	48	F	SR	WL					✓	✓				✓	✓				✓	✓				
WL179	71	F	YR	NL/WL				✓	✓	✓		✓	✓	✓	✓				✓	✓	✓			
WL200	30	F	SR	WL				✓	✓	✓		✓		✓	✓	✓	✓		✓	✓	✓			
WL206	16	?	SR	WL				✓	✓	✓		✓		✓	✓				✓					
WL208	72	F	YR	WL				✓	✓	✓				✓	✓	✓			✓	✓				
WL210	44	F?	YR	WL					✓	✓		✓		✓	✓	✓			✓	✓				
WL213	39	F	SR	WL				✓	✓					✓	✓				✓					
WL220	143	?	YR	WL					✓	✓				✓	✓				✓	✓				
WL221	101	?	YR	WL				✓	✓	✓		✓		✓	✓	✓	✓			✓				
WL229	46	?	YR	WL					✓	✓		✓		✓	✓	✓			✓	✓				
WL236	22	?	SR	WL					✓	✓		✓		✓	✓				✓					
WL243	64	?	YR	WL				✓	✓	✓		✓					✓				✓			
WL245	19	?	SV	WL				✓	✓			✓	✓	✓	✓			✓	✓					
WL249	19	?	SV	WL					✓			✓		✓	✓			✓						
WL250	72	F	YR	WL					✓	✓		✓			✓	✓				✓				
WL254	48	F	YR	WL					✓	✓				✓	✓				✓					
WL256	31	?	SR	WL					✓	✓				✓	✓				✓					
WL272	19	?	YR	WL					✓	✓				✓	✓	✓			✓					
WL273	60	F?	YR	WL				✓	✓	✓		✓			✓	✓				✓				
WL283	27	?	SR	WL				✓	✓			✓			✓			✓						
WL294	35	?	YR	WL				✓	✓	✓		✓		✓	✓	✓	✓		✓	✓				
WL299	18	?	YR	WL					✓	✓				✓	✓	✓	✓		✓					
WL301	22	?	SR	WL				✓	✓	✓				✓	✓			✓						
WL302	17	?	SV	WL					✓	✓				✓	✓	✓	✓		✓	✓				
WL304	27	?	SR	WL				✓	✓	✓				✓					✓					
WL305	29	?	YR	WL					✓	✓				✓	✓				✓	✓				
WL317	30	?	N.D.	WL					✓	✓				✓					✓					
WL330	15	?	N.D.	WL					✓	✓				✓	✓	✓	✓		✓	✓				
WL331	34	?	N.D.	WL					✓	✓				✓	✓	✓	✓			✓				
WL336	15	?	N.D.	WL					✓	✓					✓	✓	✓			✓				

Table 12. Number of individual dolphins involved in movements across different survey areas around Lantau in the past 15 mointoring periods

Monitoring Period	Total No. of Ind.	NEL-NWL	NWL-WL	WL-SWL	NEL-NWL-WL	NWL-WL-SWL	NEL-NWL-WL-SWL
2010-11	169	29	23	14	9	1	0
2011-12	217	50	66	40	16	8	1
2012-13	200	39	50	34	18	3	1
2013-14	199	19	52	52	12	9	2
2014-15	227	6	62	72	5	14	0
2015-16	210	1	35	87	1	9	0
2016-17	208	0	50	81	0	20	0
2017-18	185	5	48	65	2	17	1
2018-19	172	0	37	52	0	9	0
2019-20	168	0	19	69	0	8	0
2020-21	129	0	25	61	0	13	0
2021-22	111	0	12	31	0	4	0
2022-23	117	0	18	41	0	9	0
2023-24	101	0	11	41	0	5	0
2024-25	111	0	1	46	0	0	0

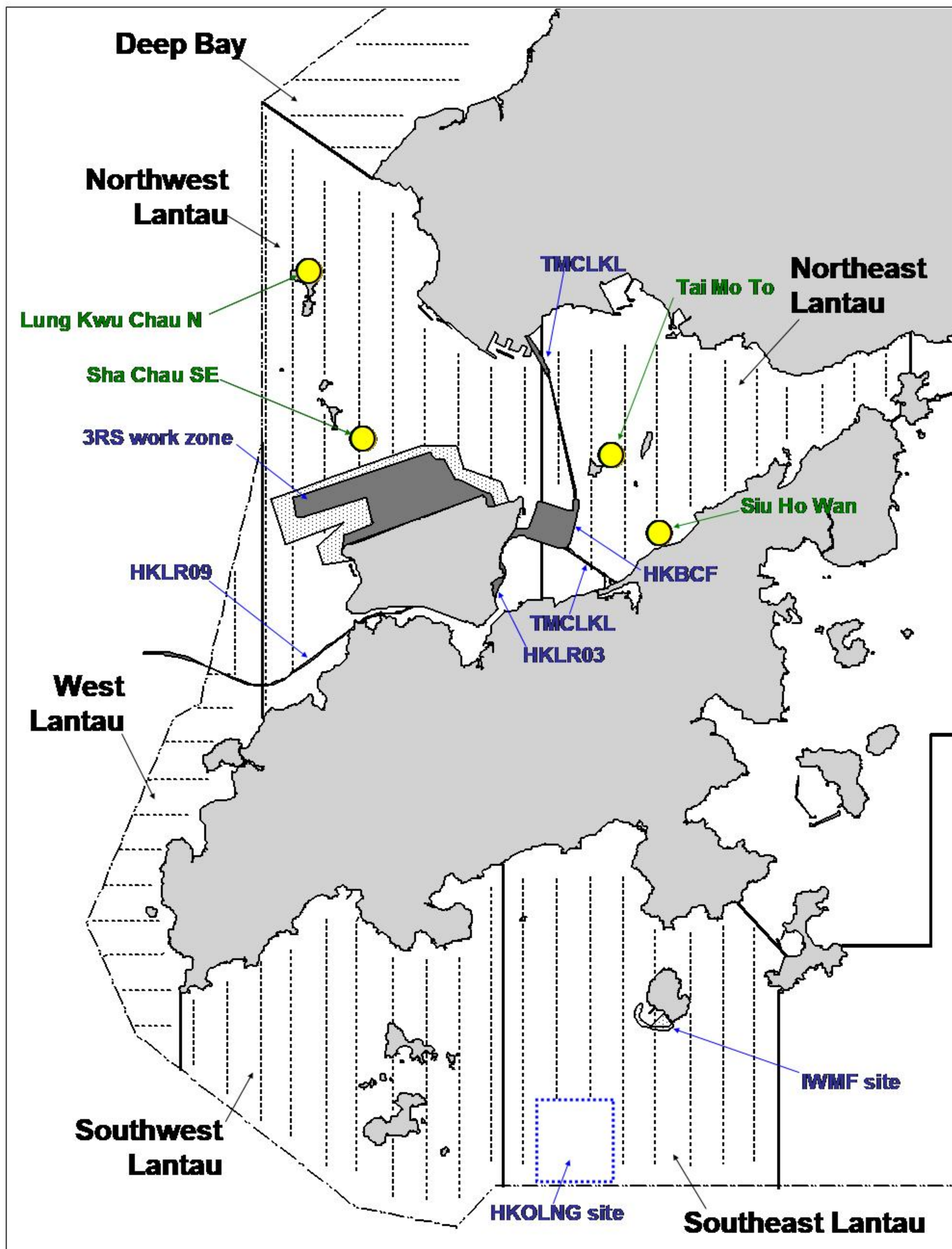
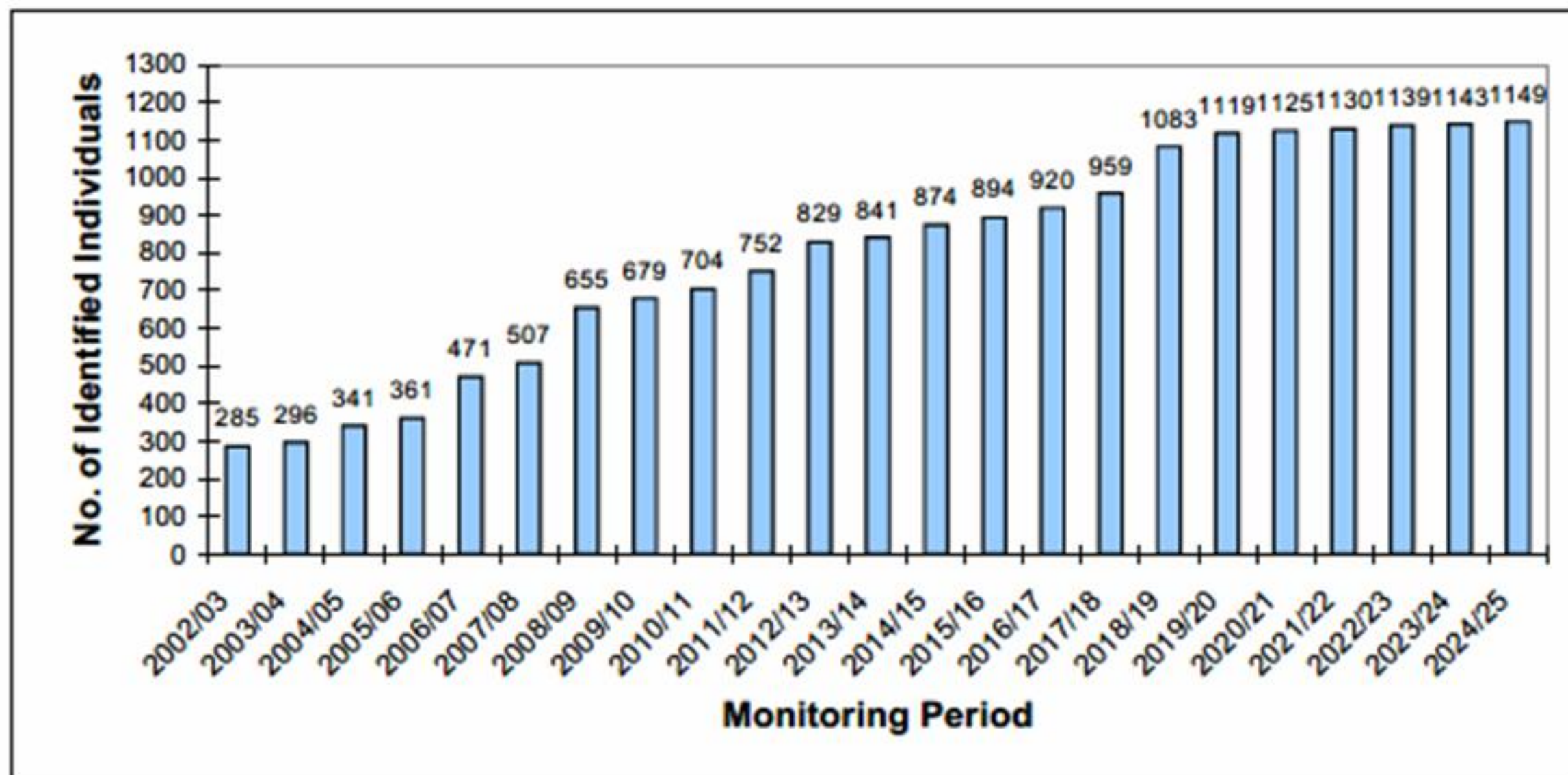
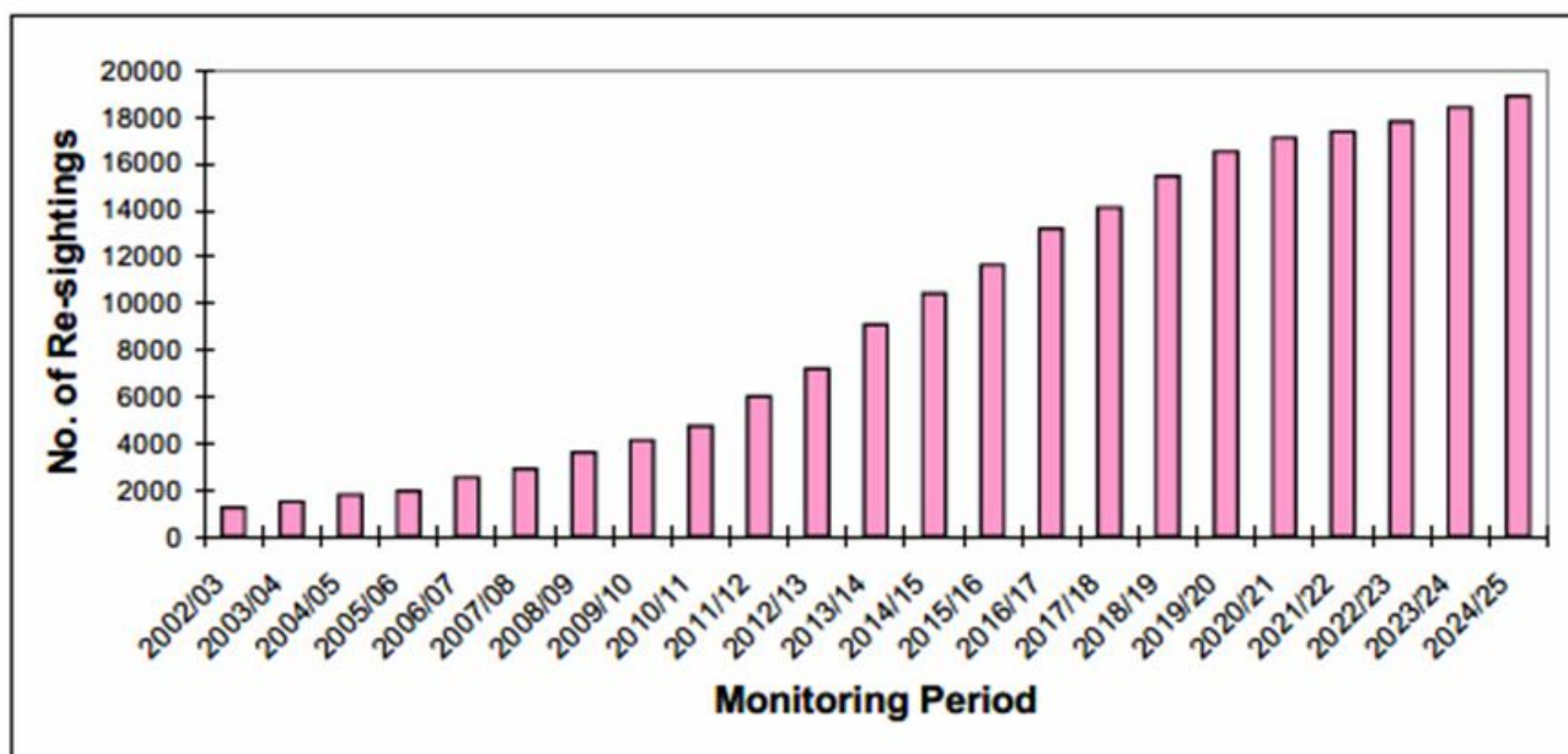


Figure 1. Six Line-Transect Survey Areas and Four Deployment Locations (yellow dots) within two Marine Parks in North Lantau Waters for the 2024-25 AFCD Monitoring Study

(a)



(b)



(c)

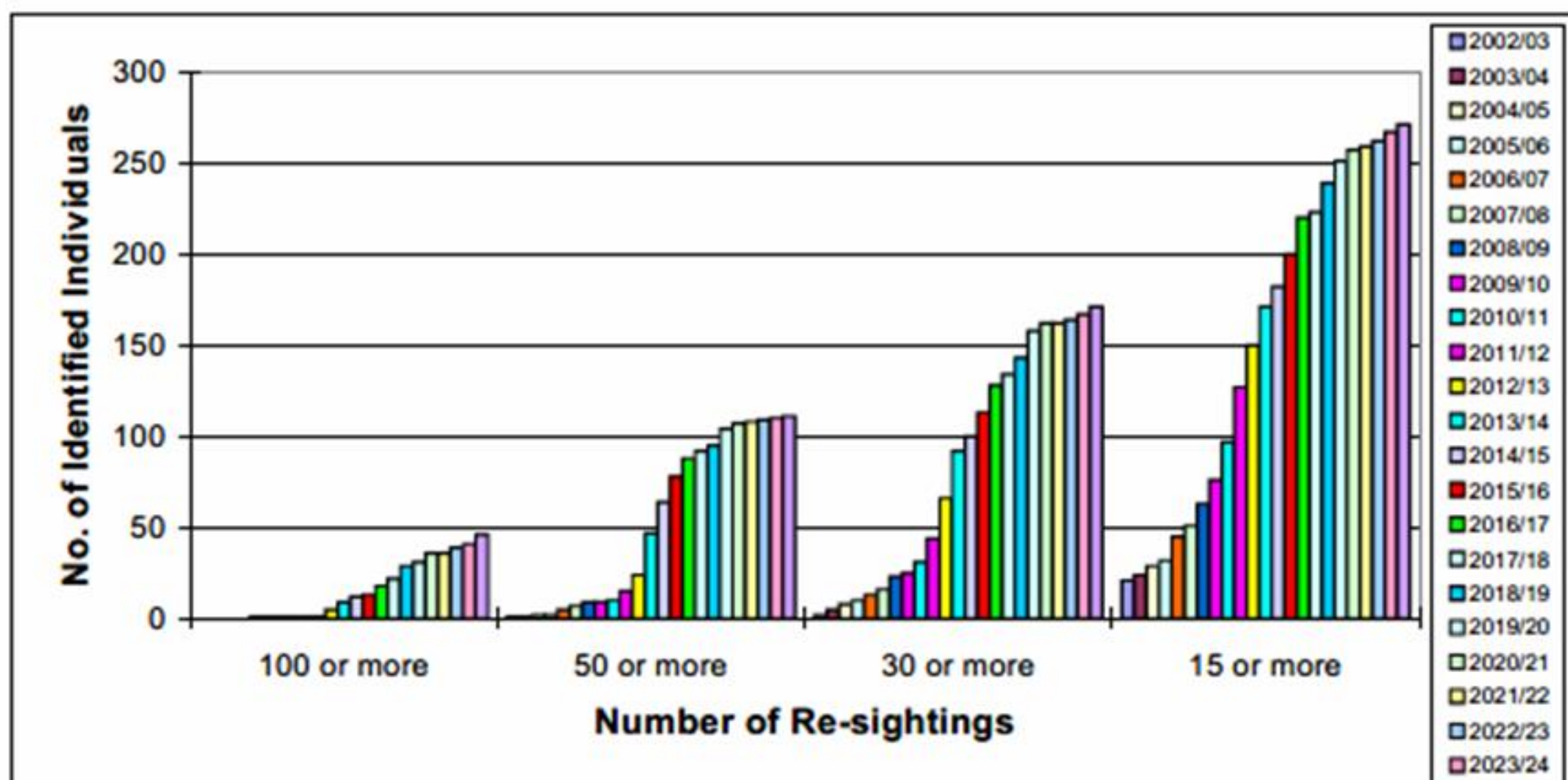


Figure 2. Temporal trends of (a) cumulative number of identified individuals; (b) total number of re-sightings made; and (c) number of identified individuals within several categories of number of re-sightings in the past 23 monitoring periods (2002-2025)

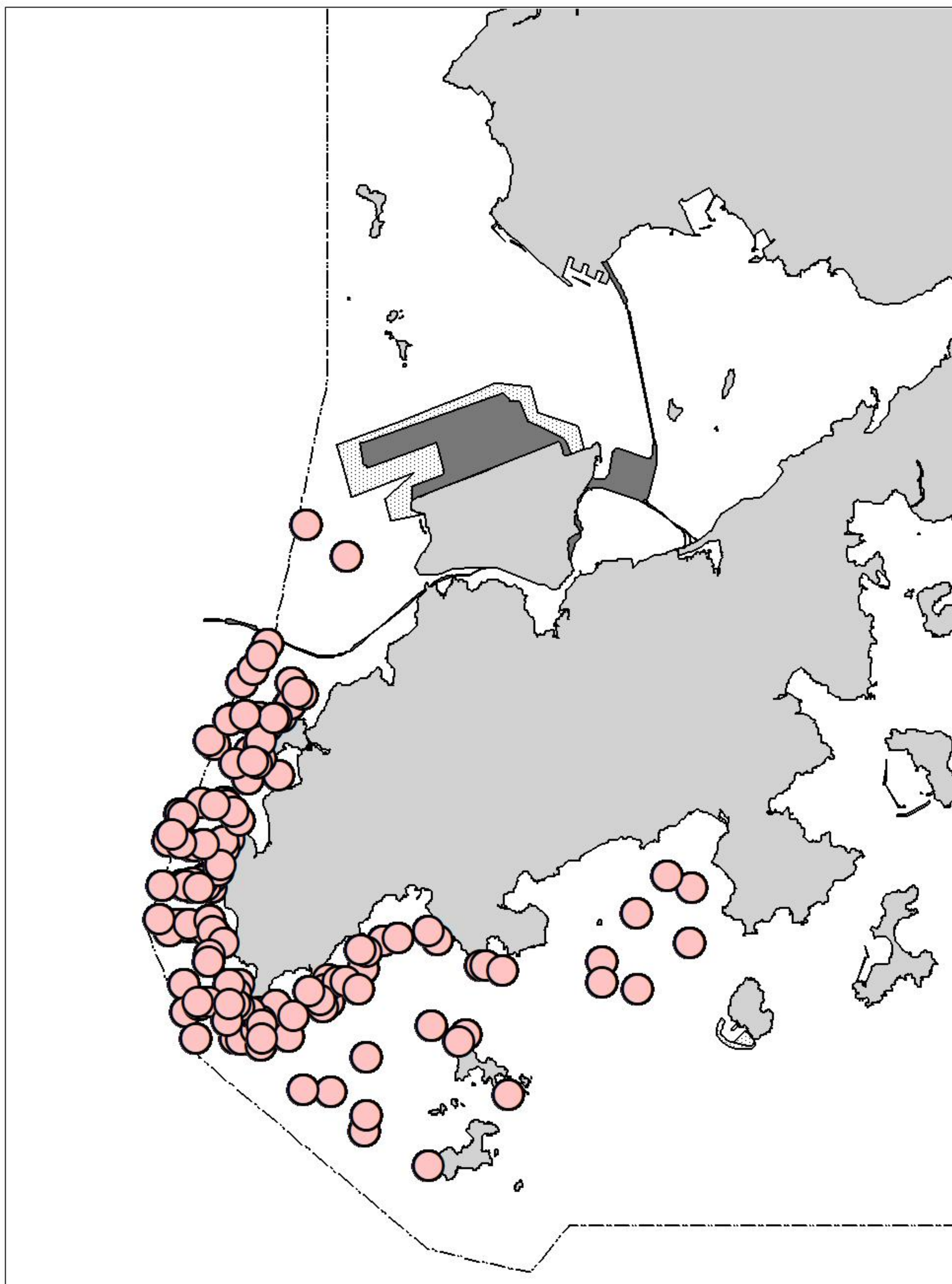
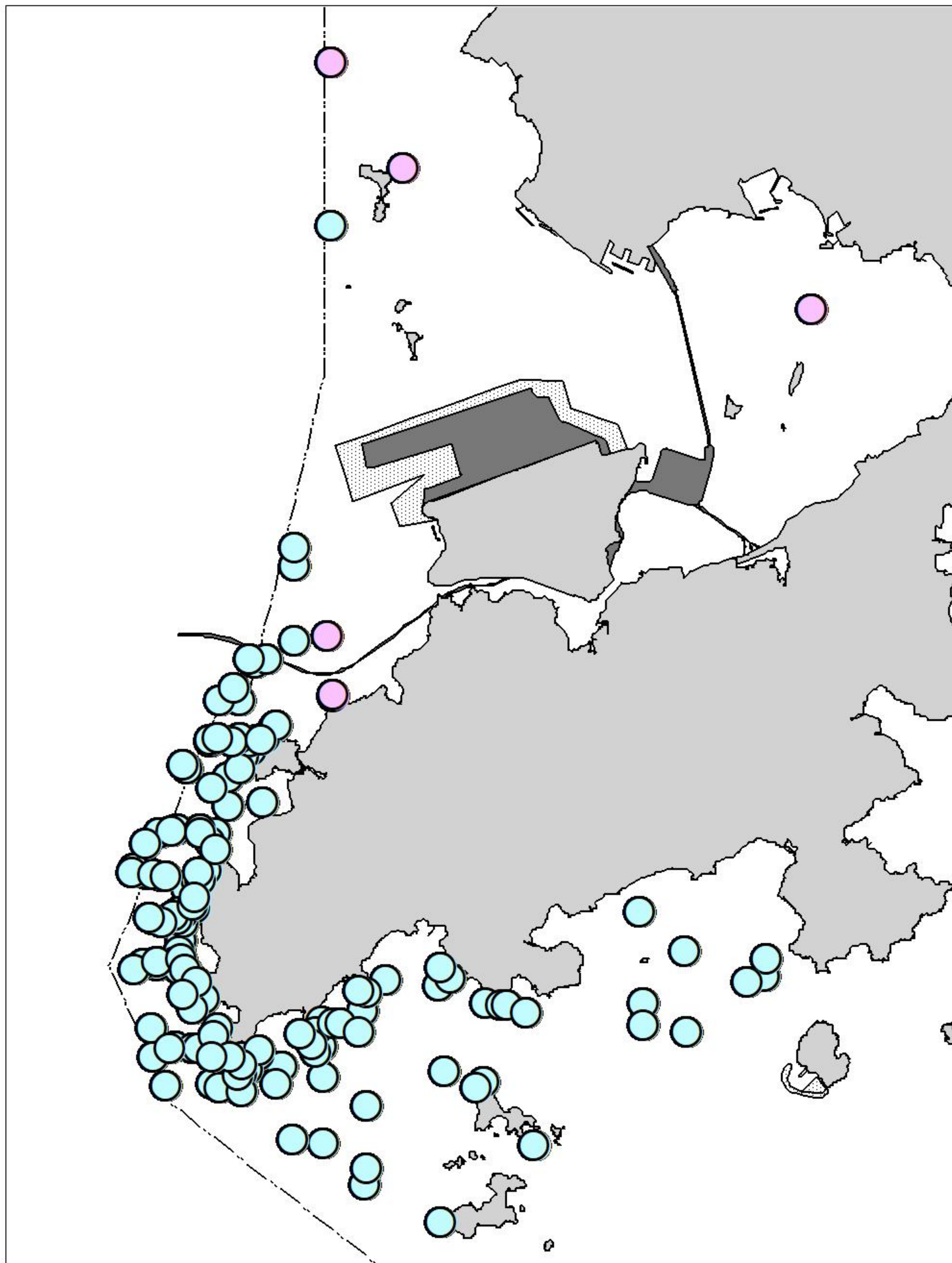


Figure 3. Distribution of CWD sightings in Hong Kong waters during AFCD monitoring surveys (April 2024 – March 2025)



**Figure 4. Distribution of all CWD sightings in Hong Kong waters in 2024
(blue dots: AFCD survey sightings; purple dots: HZMB survey sightings)**

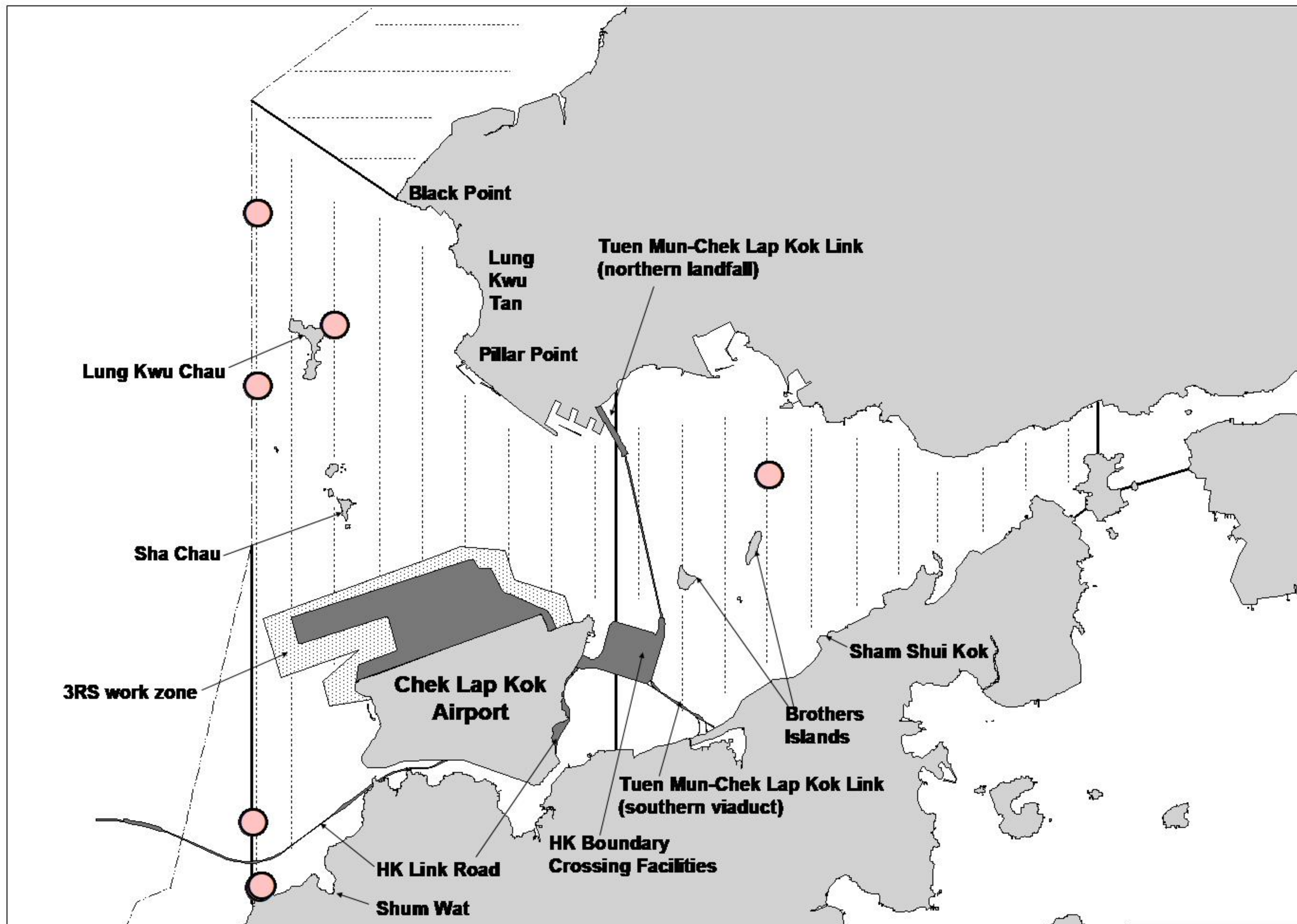


Figure 5. Distribution of Chinese White Dolphin sightings in North Lantau (2024)

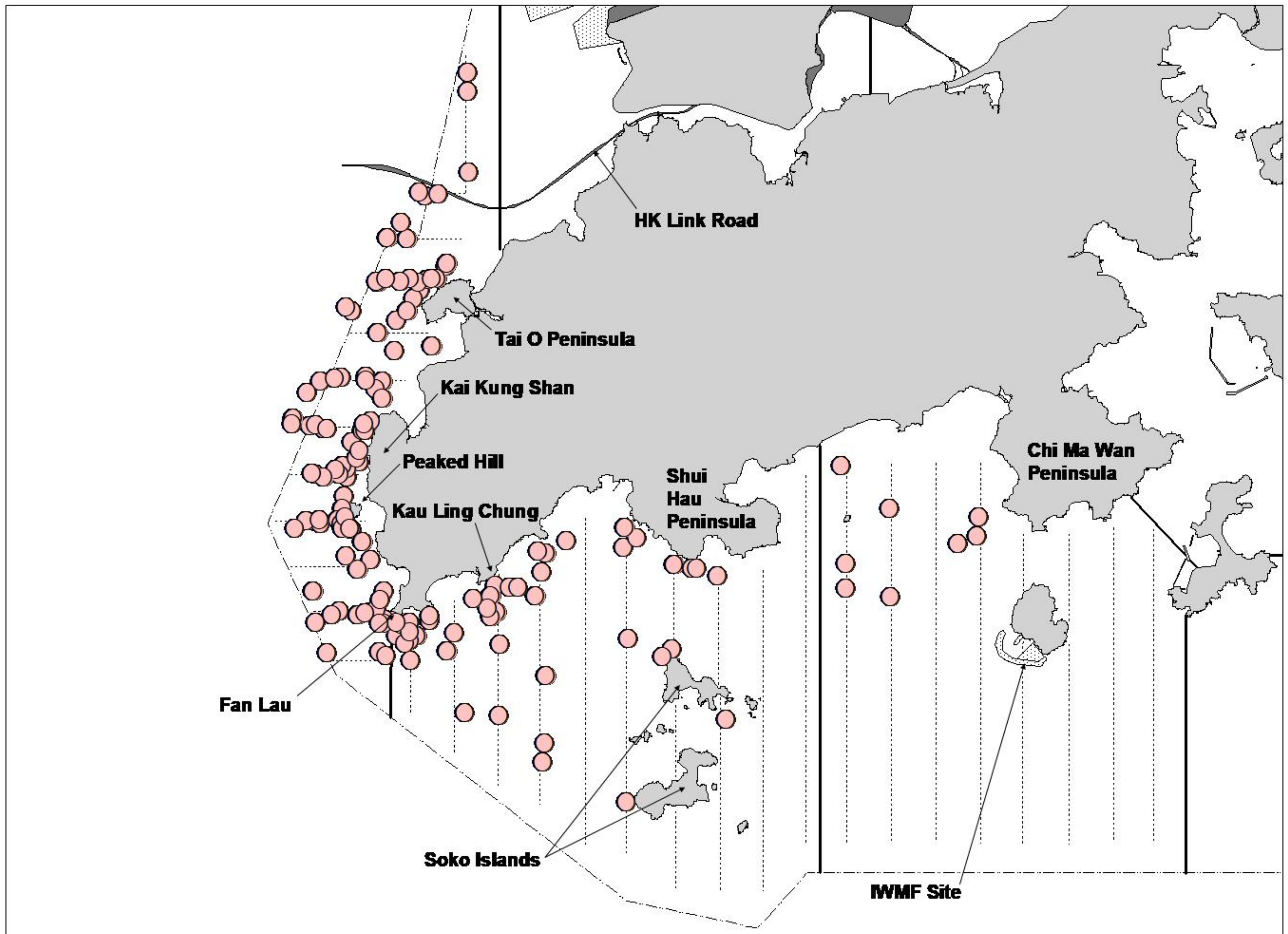


Figure 6. Distribution of Chinese White Dolphin sightings in West and South Lantau waters (2024)

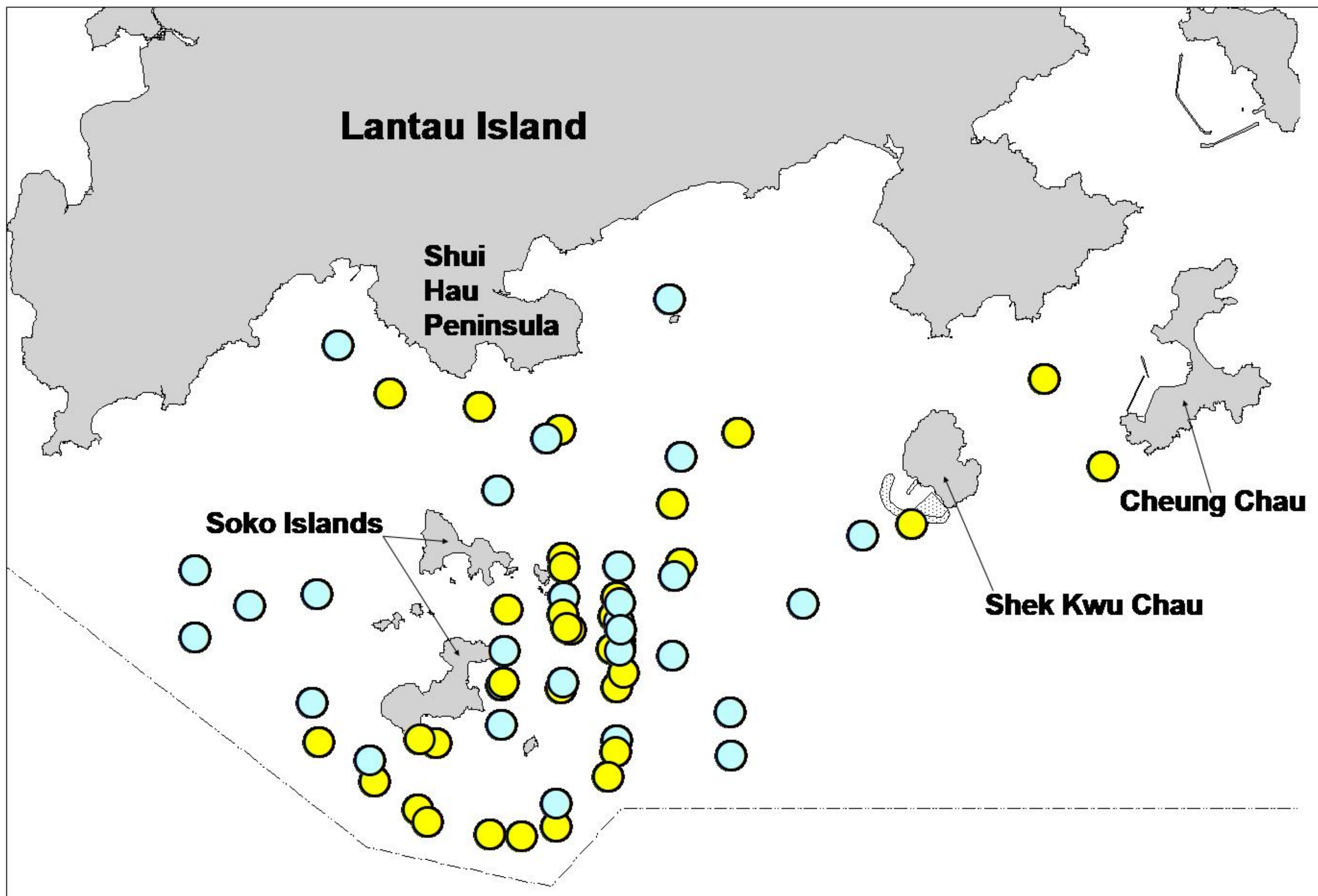


Figure 7. Distribution of finless porpoise sightings made during AFCD surveys (April 2024 – March 2025)
(blue dots: sightings made during winter/spring months; yellow dots: sightings made during summer/autumn months)

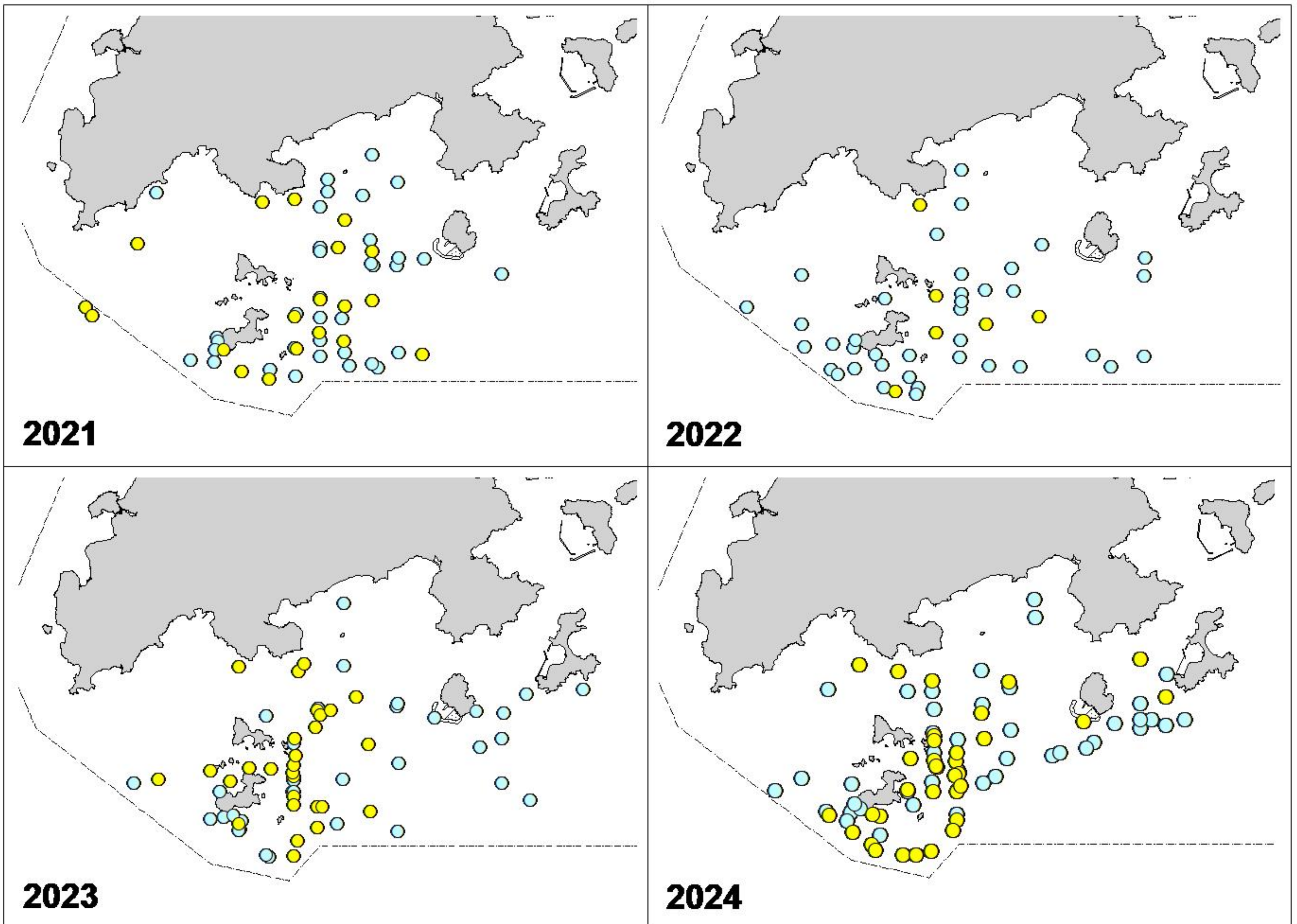


Figure 8. Comparison of annual porpoise distribution patterns in South Lantau waters from 2021-24 (blue dots: sightings made during winter/spring months; yellow dots: sightings made during summer/autumn months)

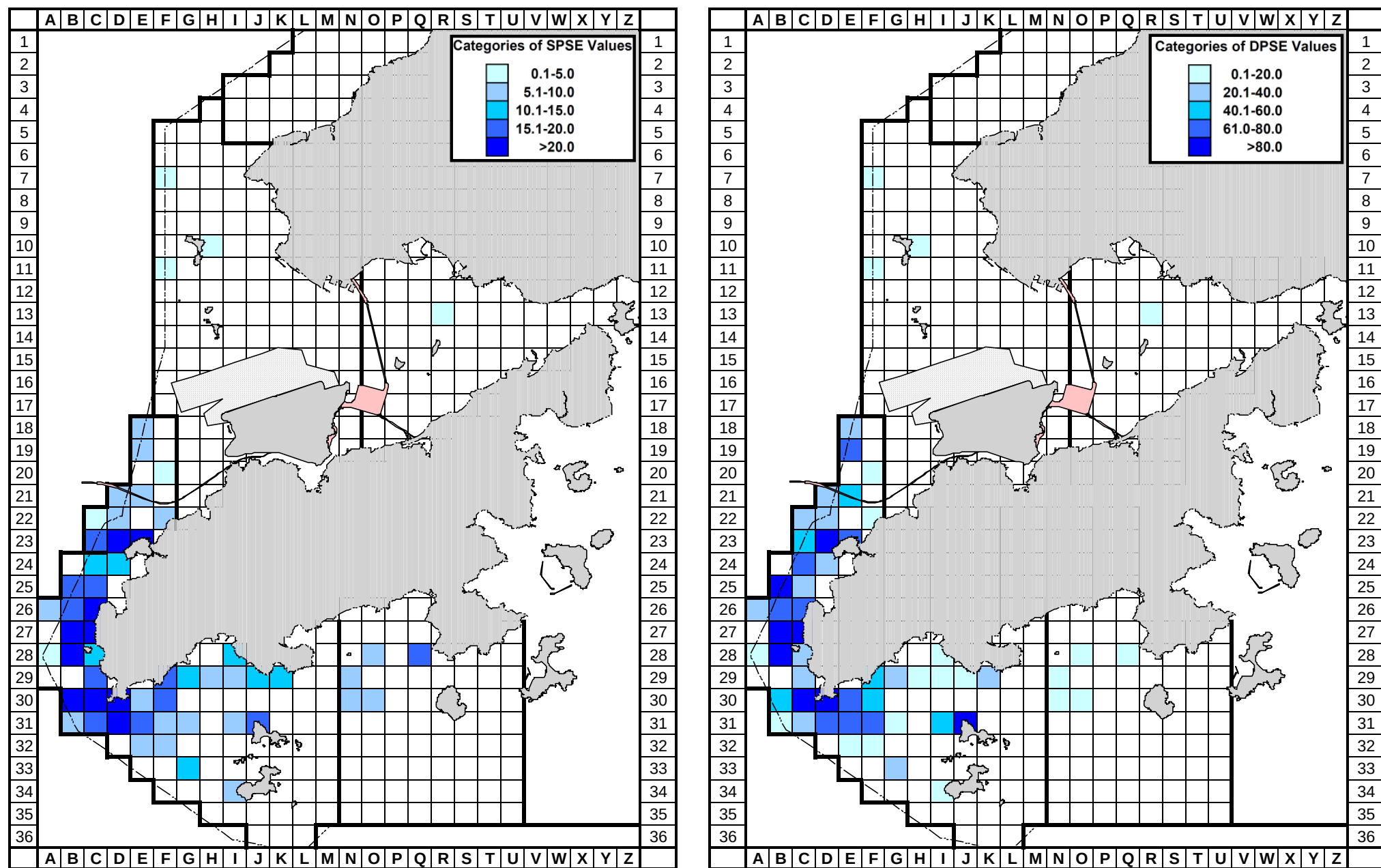


Figure 9. (left) Sighting density of Chinese white dolphins with corrected survey effort per km² in waters around Lantau Island (number within grids represent "SPSE" no. of on-effort dolphin sightings per 100 units of survey effort) (using data from January - December 2024)

(right) Density of Chinese white dolphins with corrected survey effort per km² in waters around Lantau Island (number within grids represent "DPSE" = no. of dolphins per 100 units of survey effort) (using data from January - December 2024)

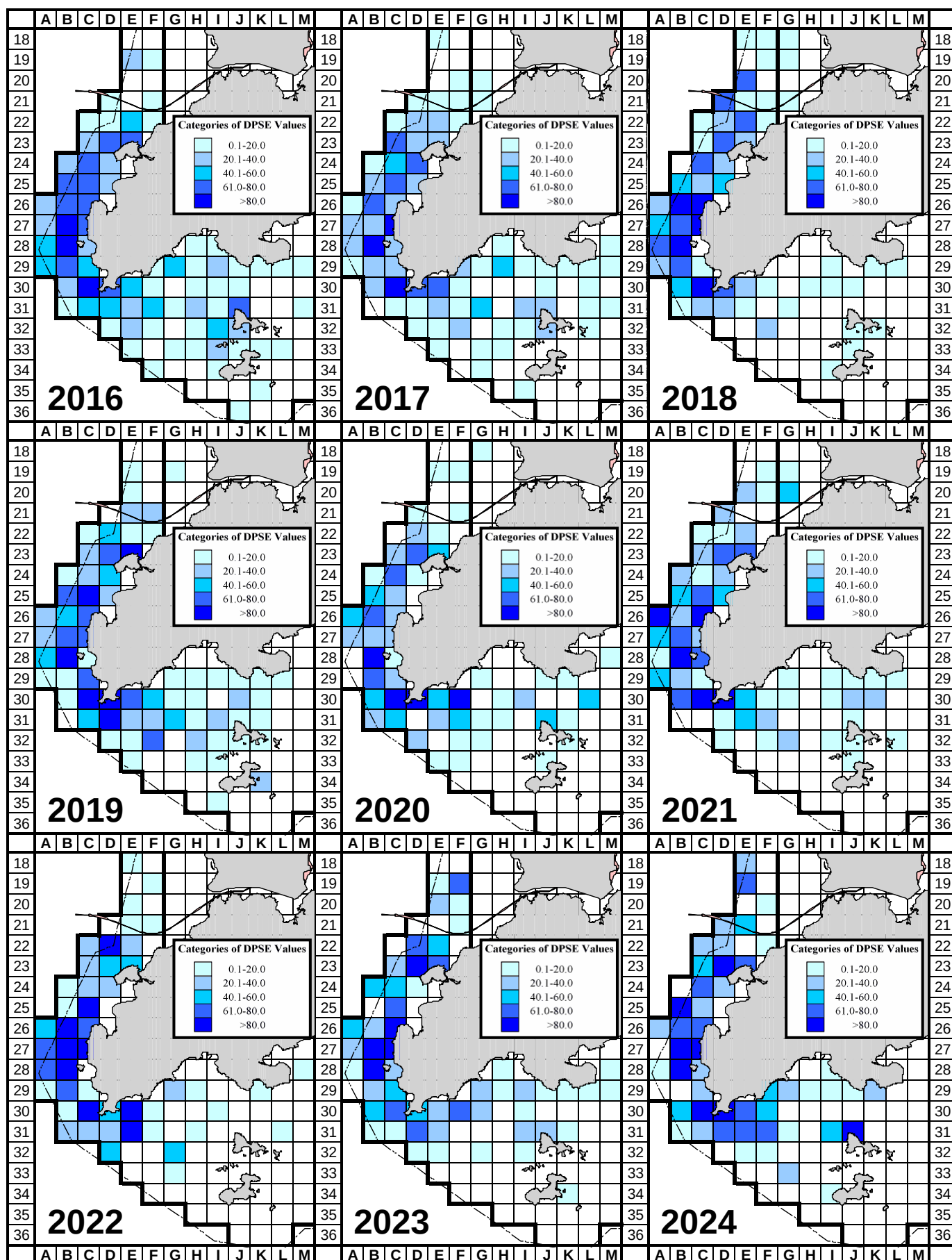


Figure 10. Comparison of Chinese White Dolphin densities with corrected survey effort per km² in West and Southwest Lantau Waters in 2016-24 (number within grids represent "DPSE" = no. of dolphins per 100 units of survey effort)

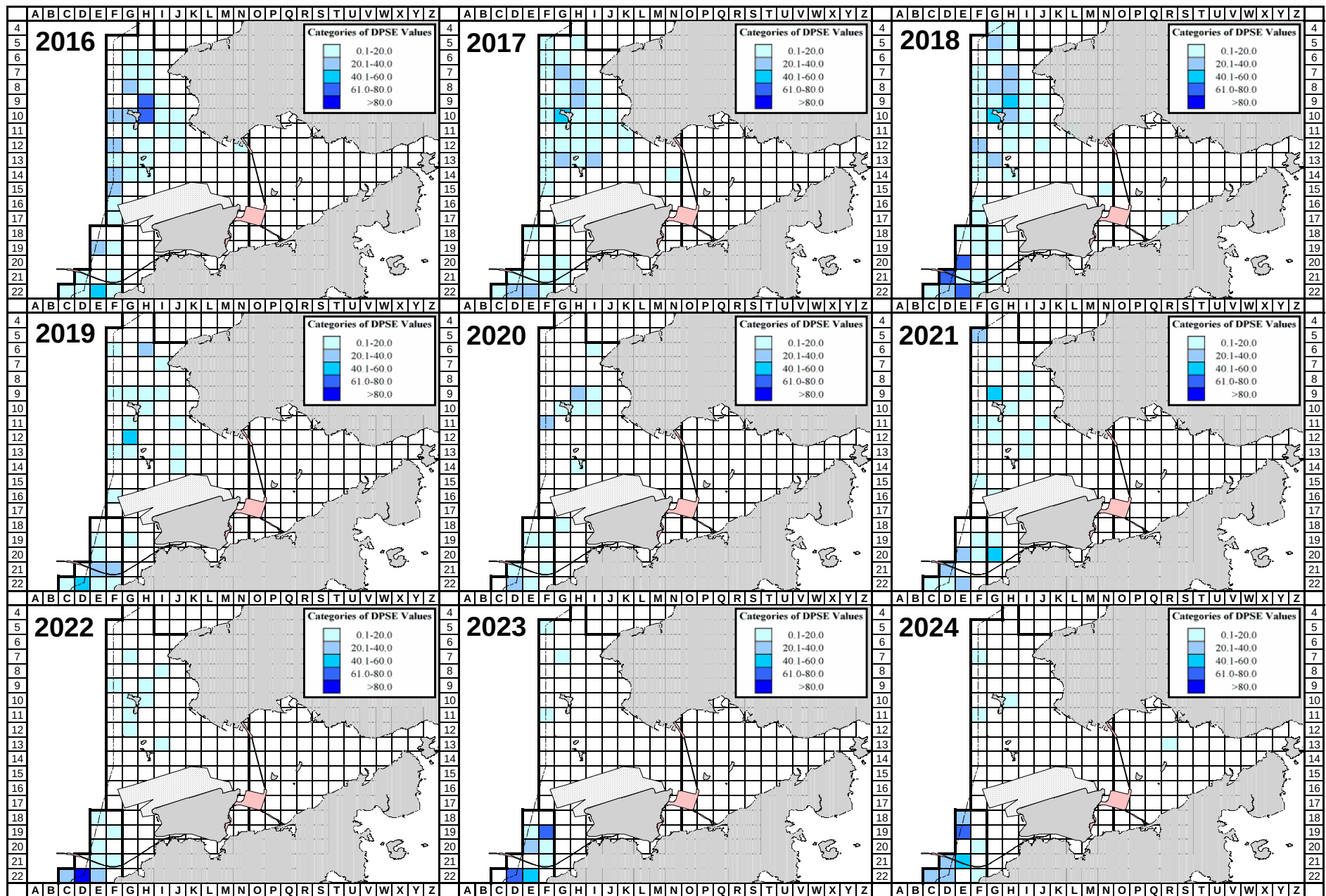


Figure 11. Comparison of dolphin densities with corrected survey effort per km² in North Lantau waters in 2016-24 (number within grids represent "DPSE" = no. of dolphins per 100 units of survey effort)

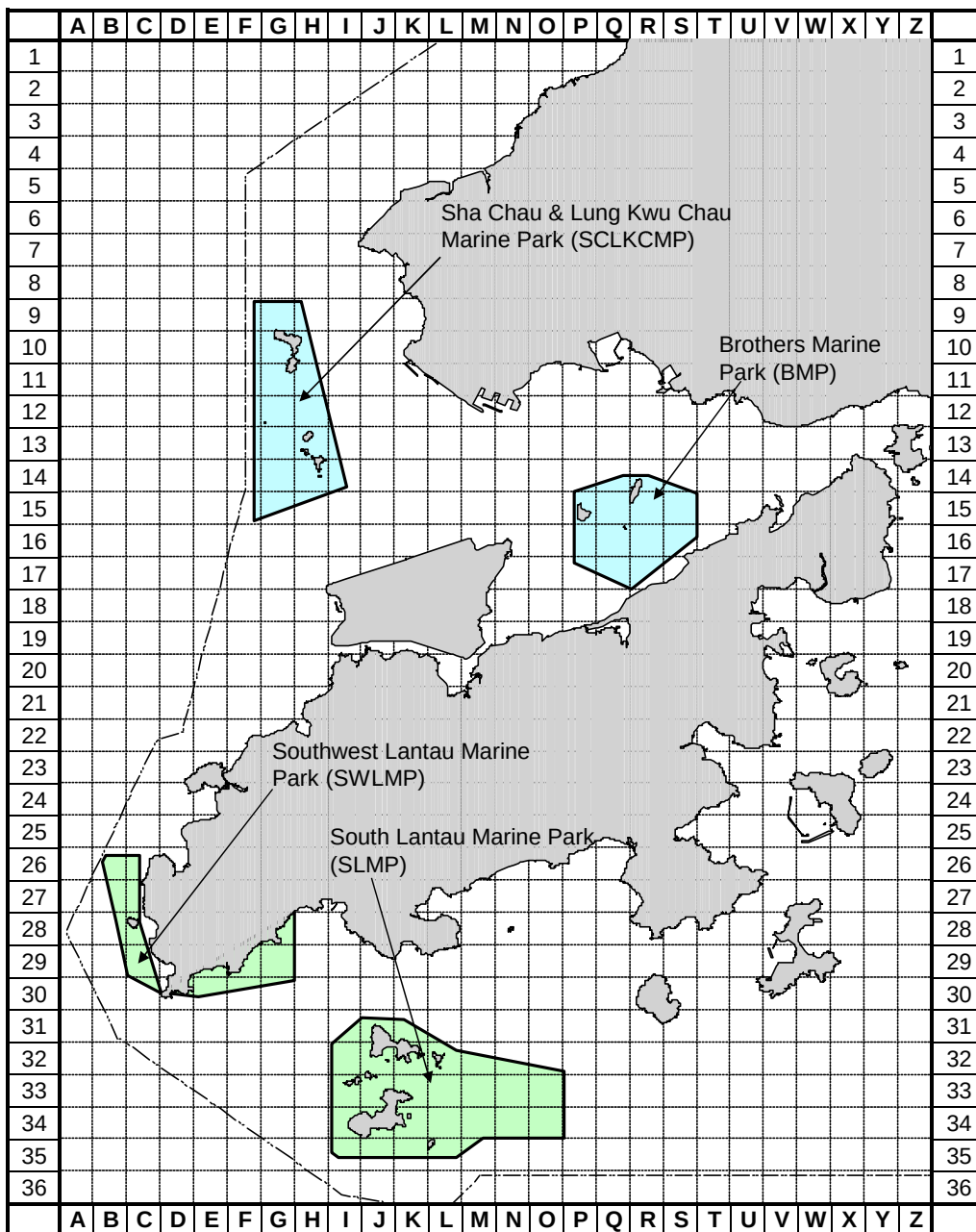
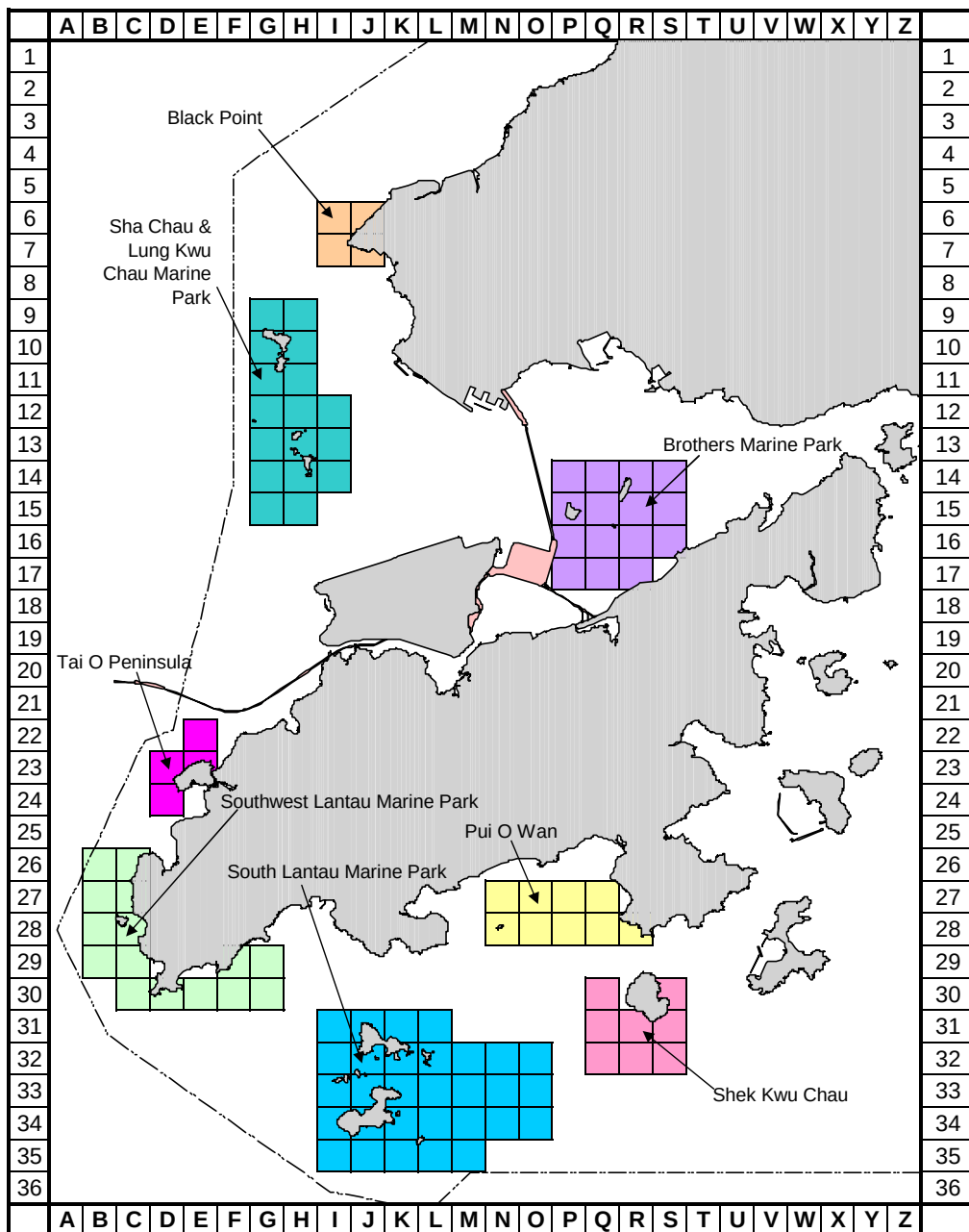


Figure 12. Grids of key marine mammal habitats in western HK waters that were examined for temporal trend in dolphin and porpoise densities

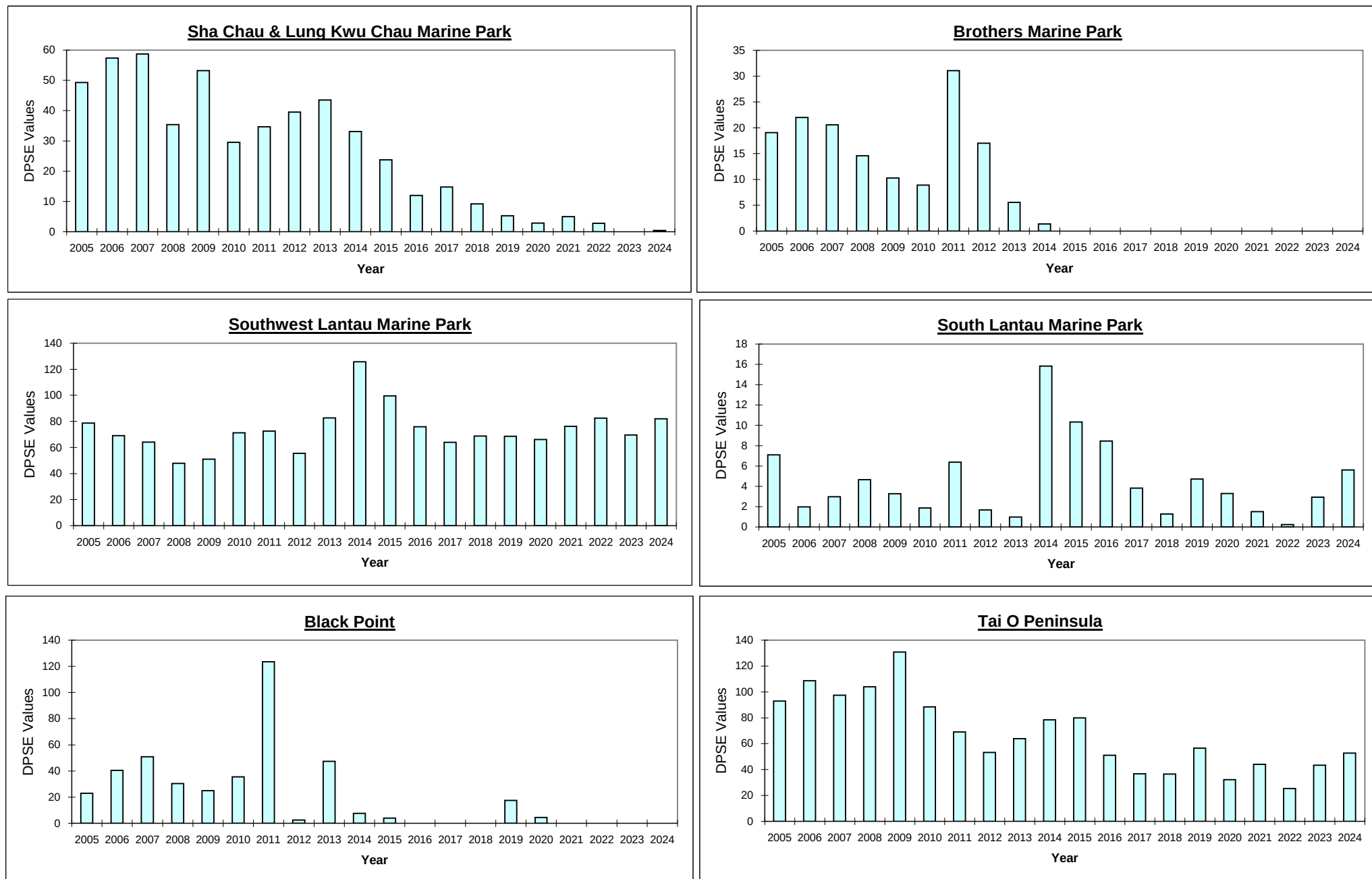


Figure 13. Temporal trend of dolphin densities (DPSE Values) at six key dolphin habitats in Lantau waters in the past two decades (2005-24)

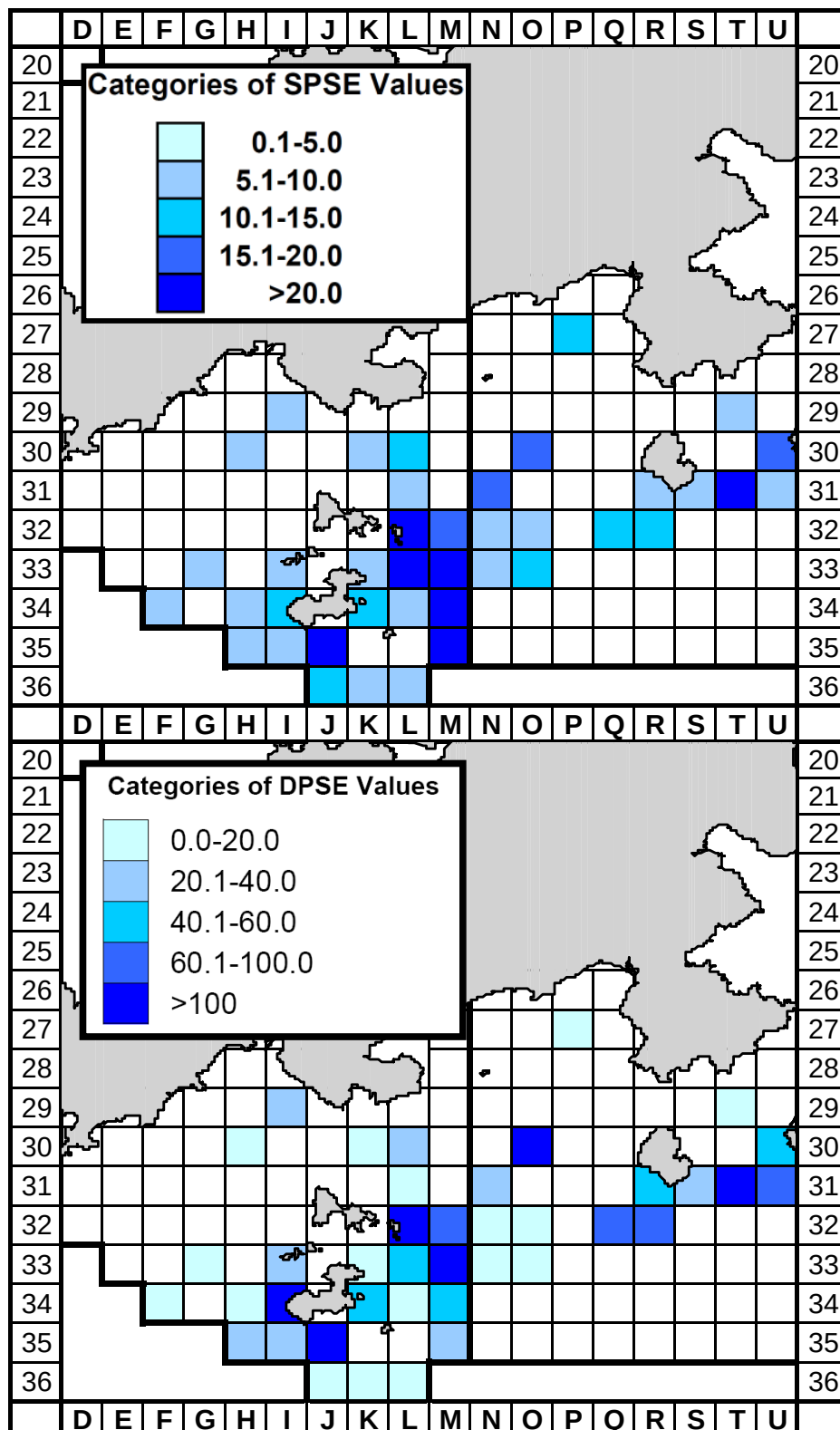


Figure 14. Density of finless porpoises with corrected survey effort per km² in South Lantau waters, using data collected during January - December 2024 (SPSE = no. of on-effort porpoise sightings per 100 units of survey effort; DPSE = no. of porpoises per 100 units of survey effort)

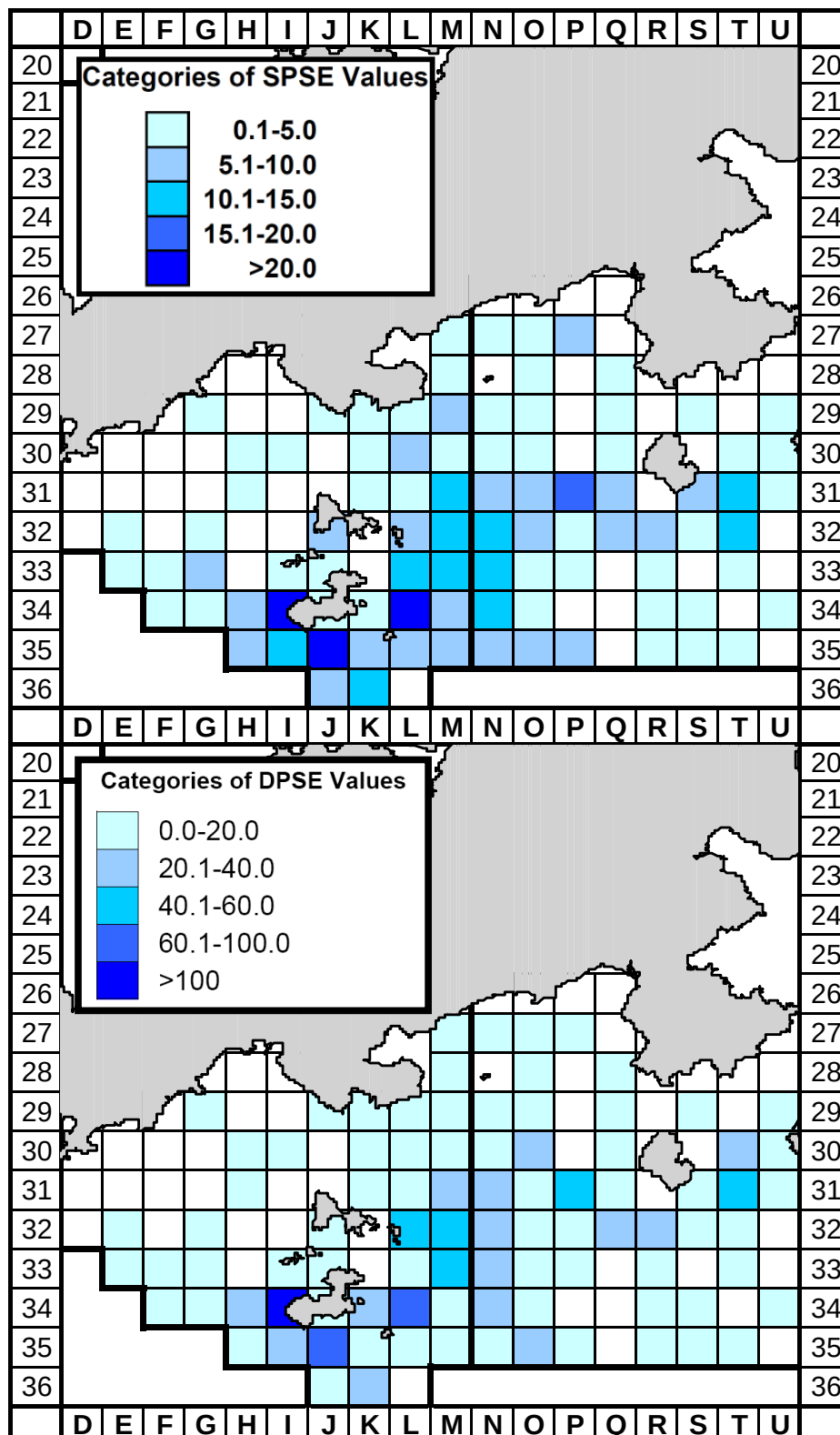


Figure 15. Density of finless porpoises with corrected survey effort per km² in southern waters of Hong Kong during dry season (December to May), using data collected during 2020-24 (SPSE = no. of on-effort porpoise sightings per 100 units of survey effort; DPSE = no. of porpoises per 100 units of survey effort)

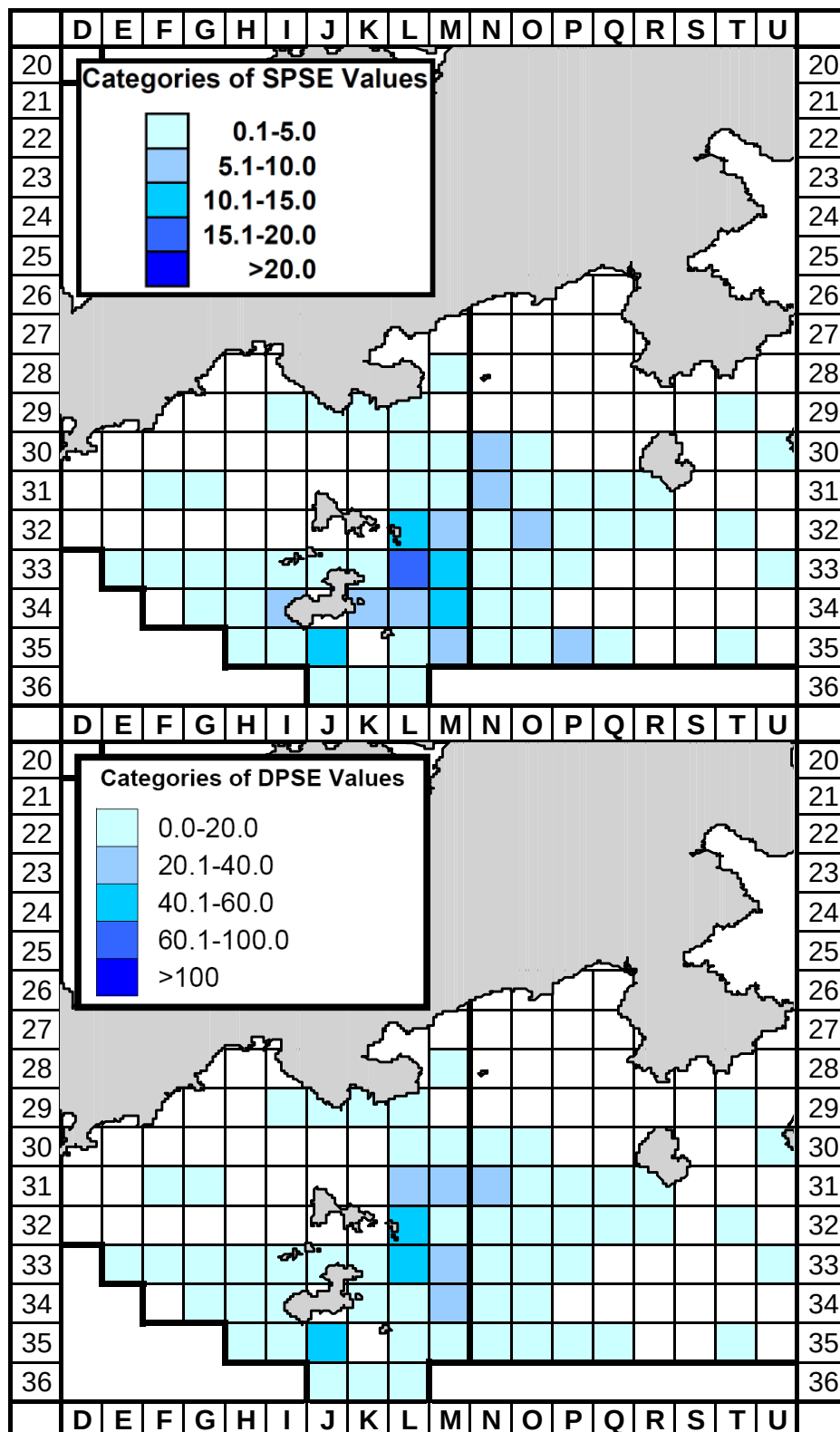


Figure 16. Density of finless porpoises with corrected survey effort per km² in southern waters of Hong Kong during wet season (June to November), using data collected during 2020-24 (SPSE = no. of on-effort porpoise sightings per 100 units of survey effort; DPSE = no. of porpoises per 100 units of survey effort)

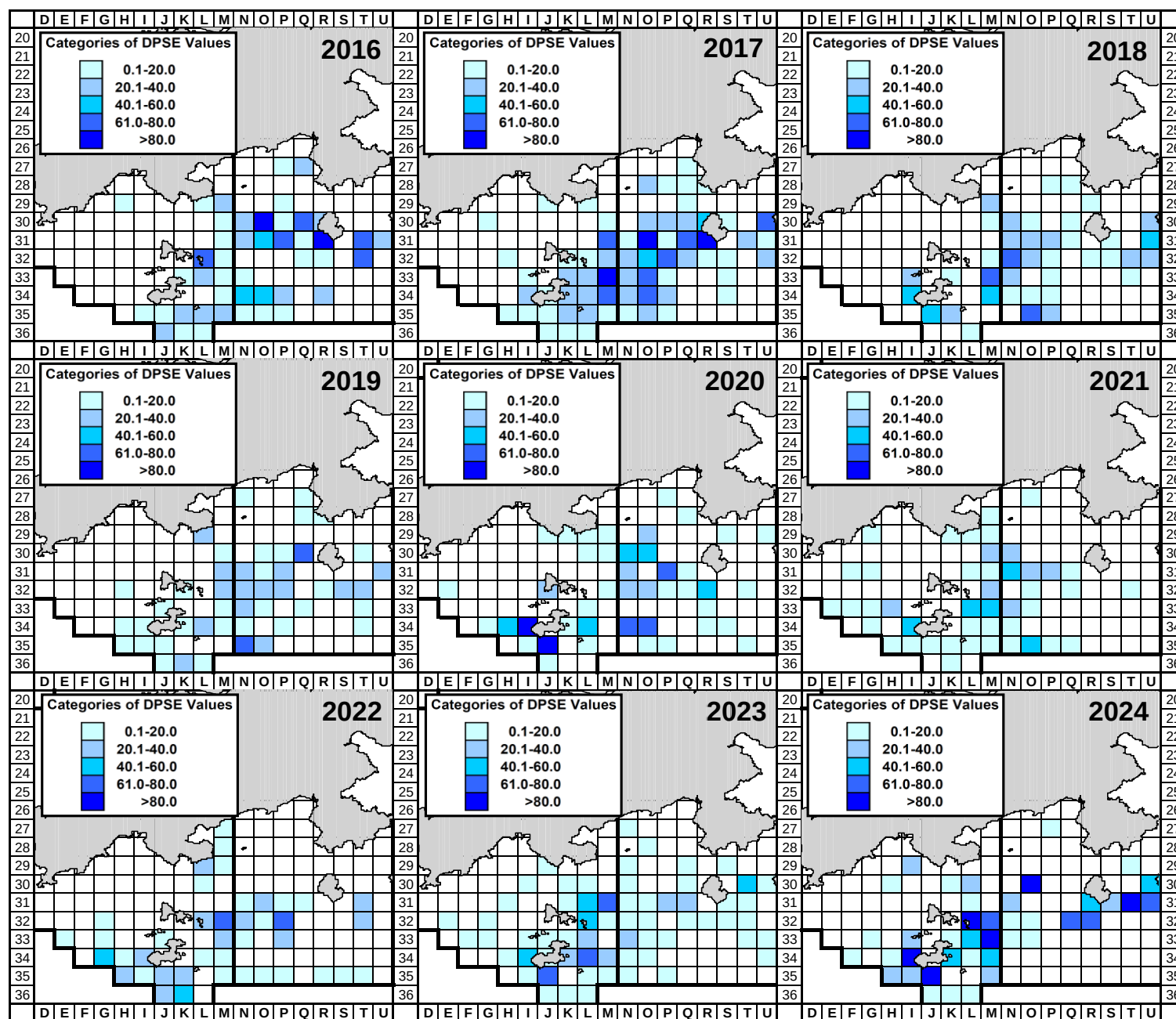


Figure 17. Comparison of porpoise densities with corrected survey effort per km² in South Lantau waters in 2016-24 (number within grids represent "DPSE" = no. of porpoises per 100 units of survey effort)

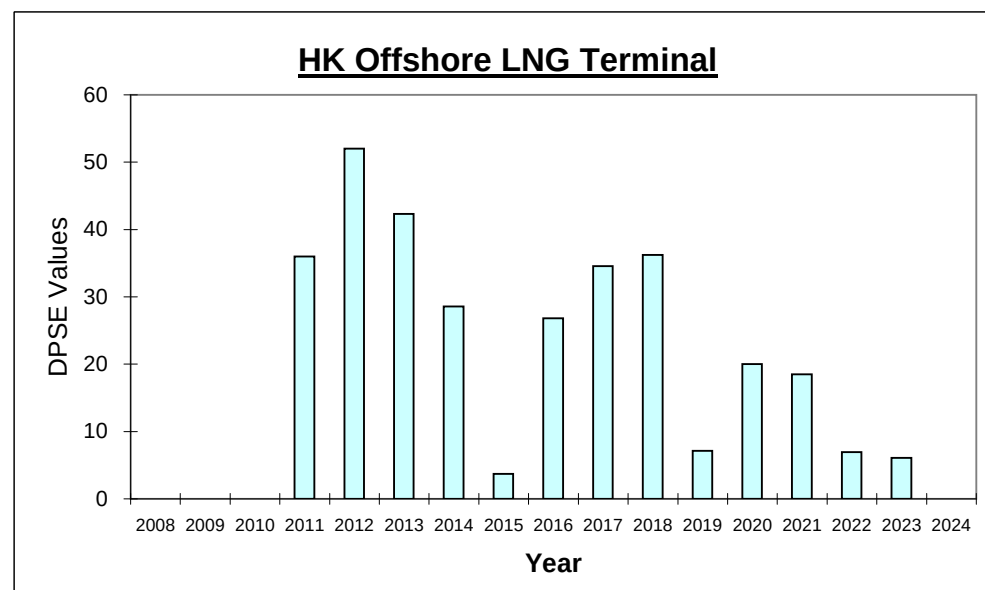
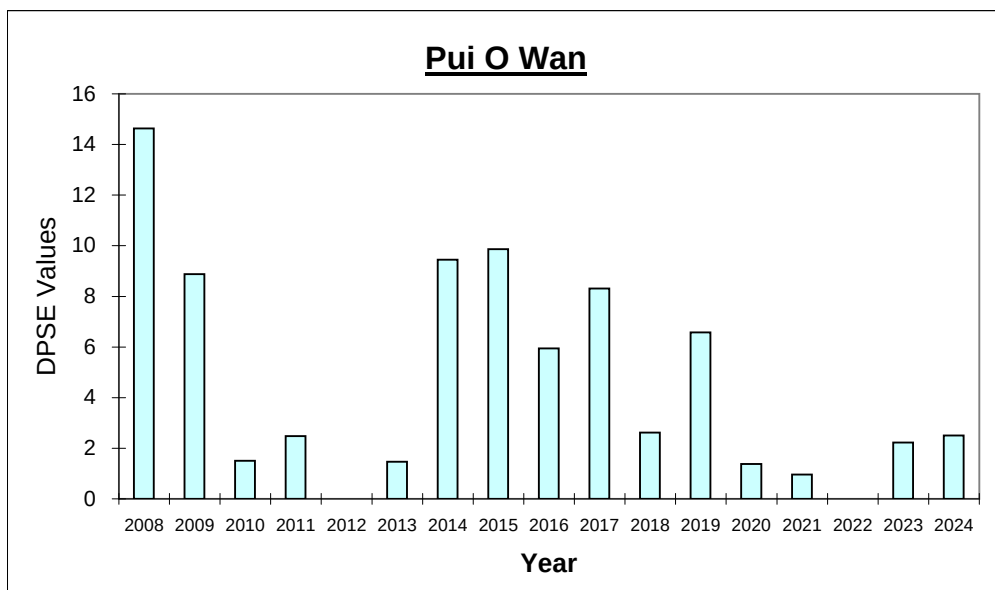
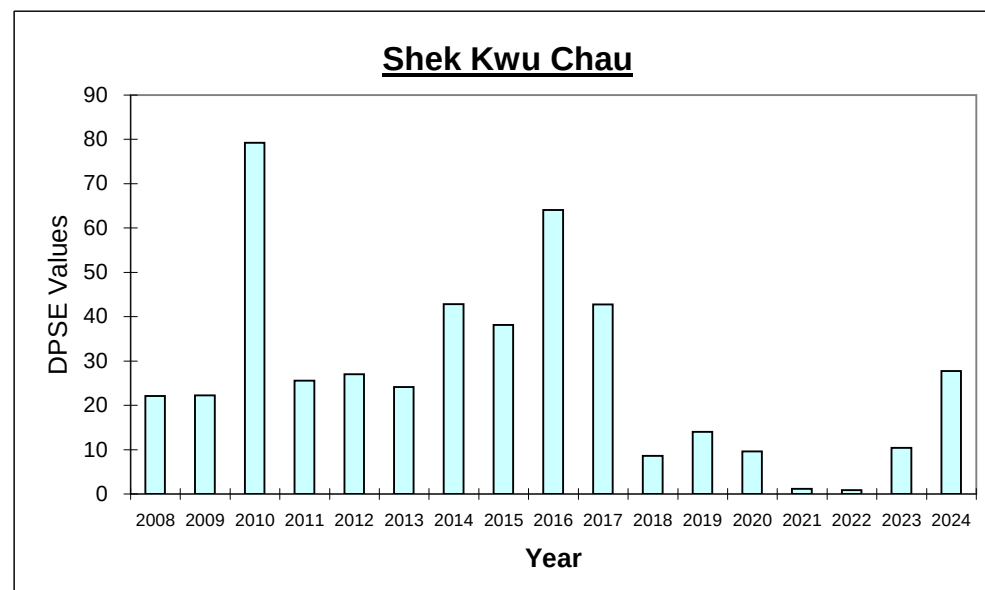
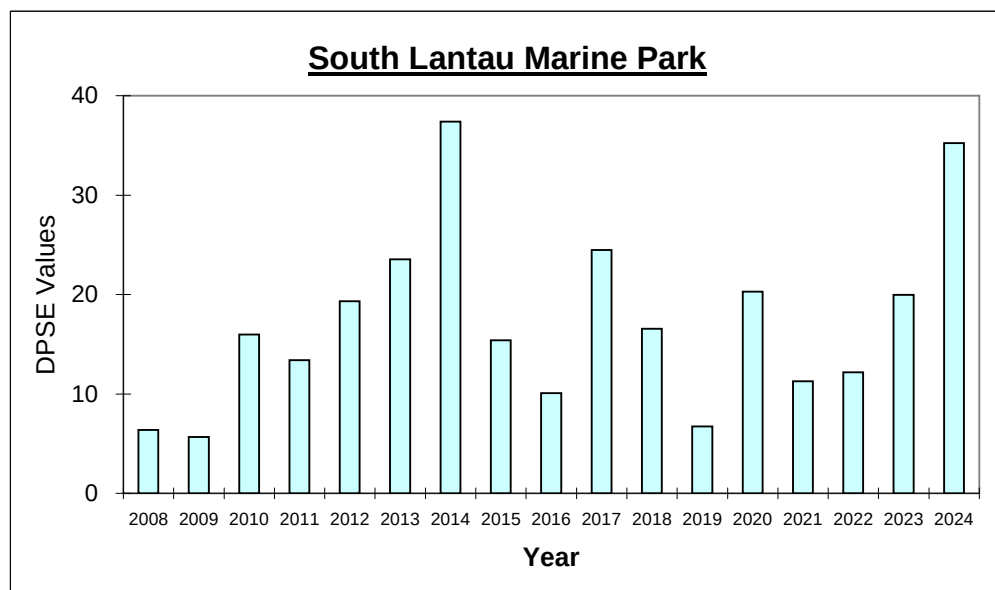


Figure 18. Temporal trend of porpoise densities (DPSE Values) at key porpoise habitats in South Lantau waters

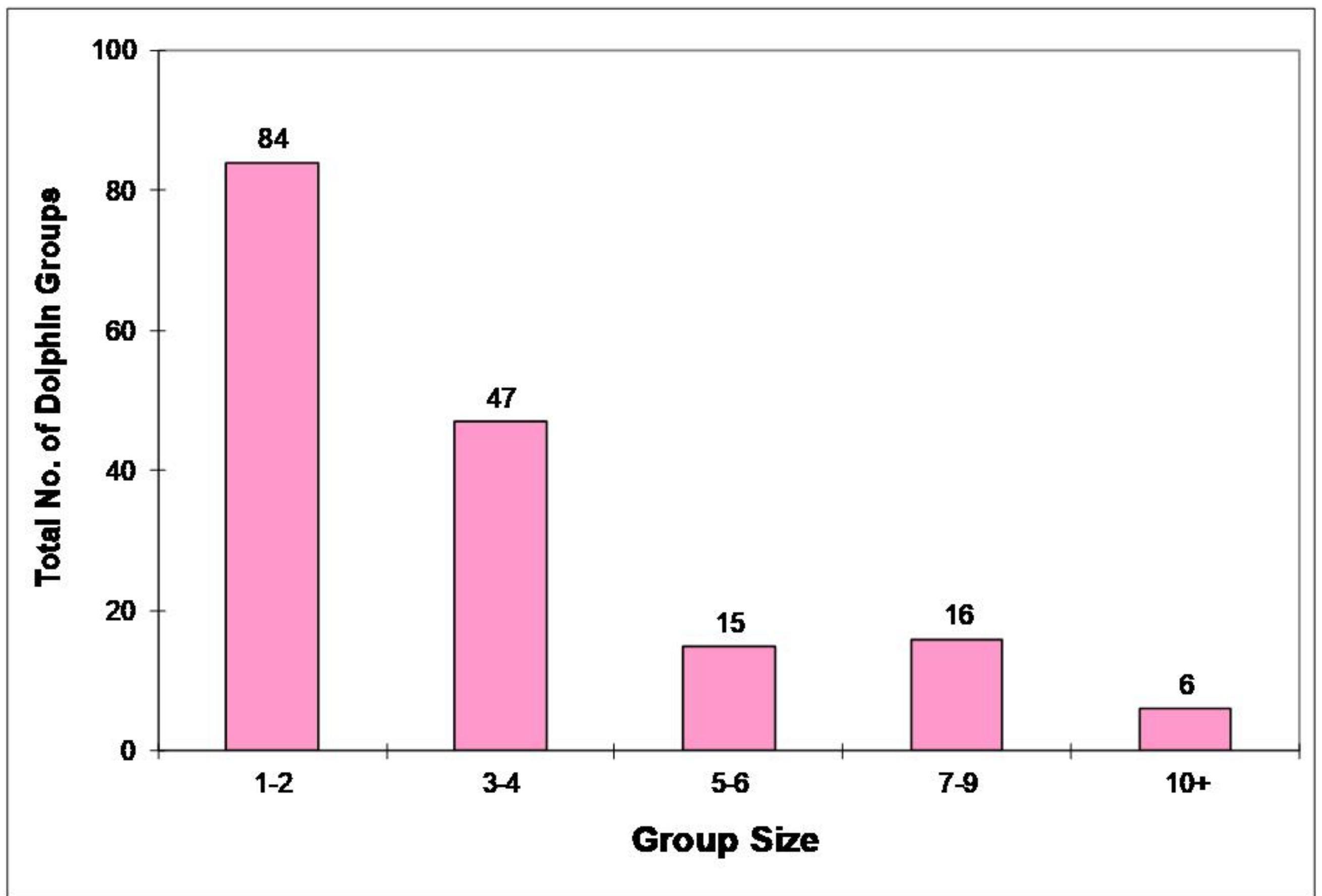


Figure 19. Total number of dolphin groups with different group sizes during April 2024 to March 2025

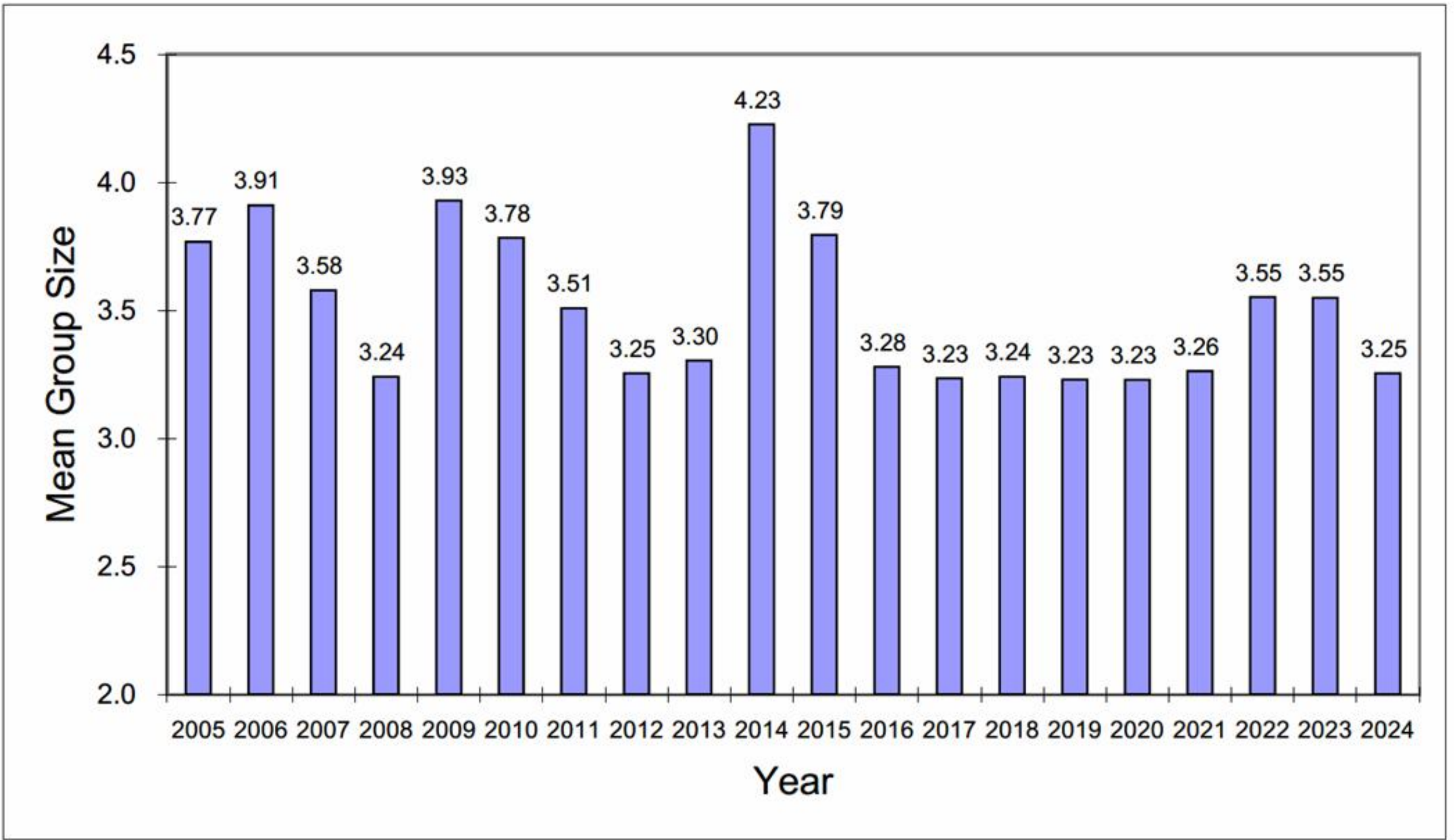


Figure 20. Temporal trend of mean dolphin group size in 2005-24

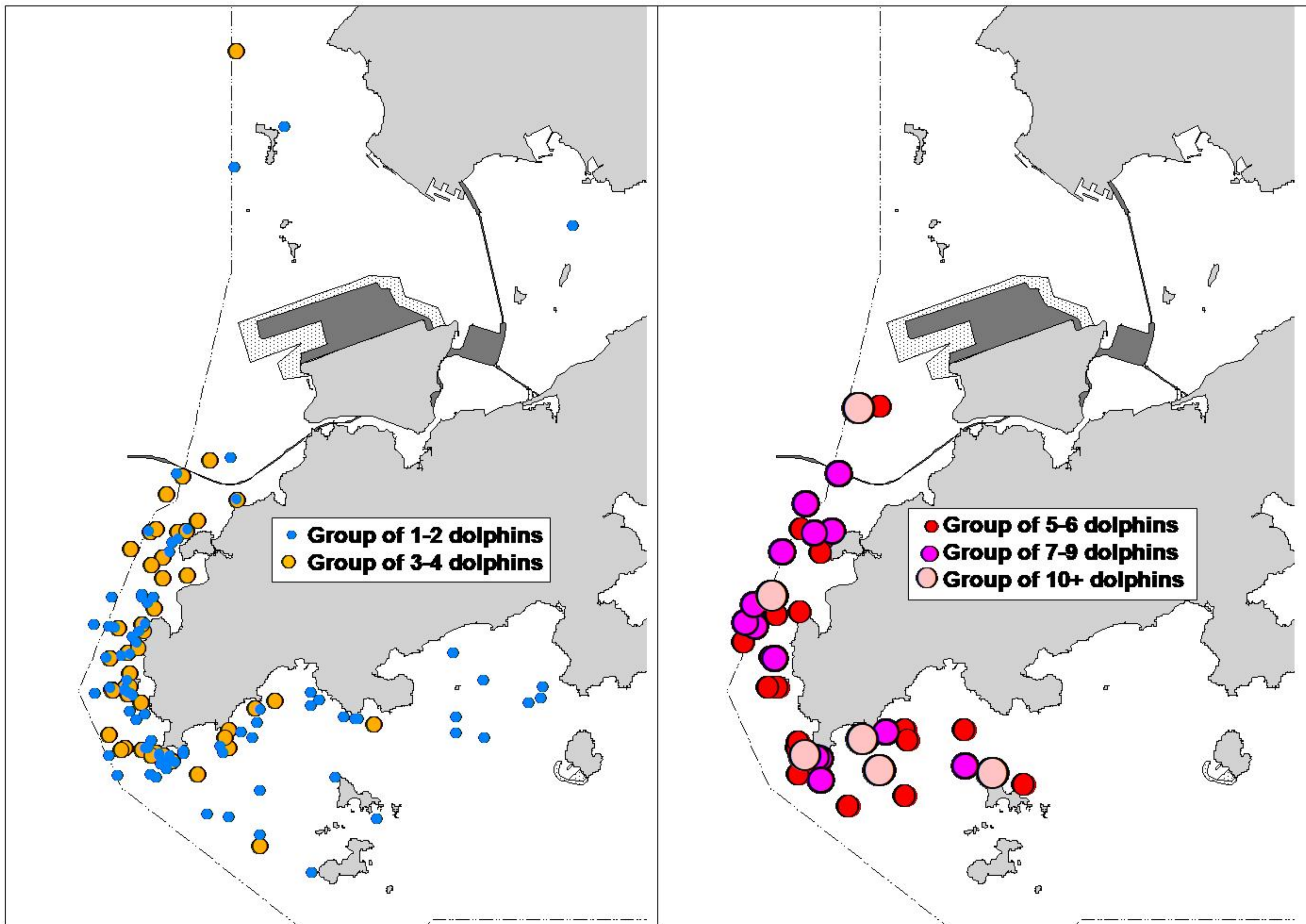


Figure 21. Distribution of Chinese White Dolphins with different group sizes in 2024

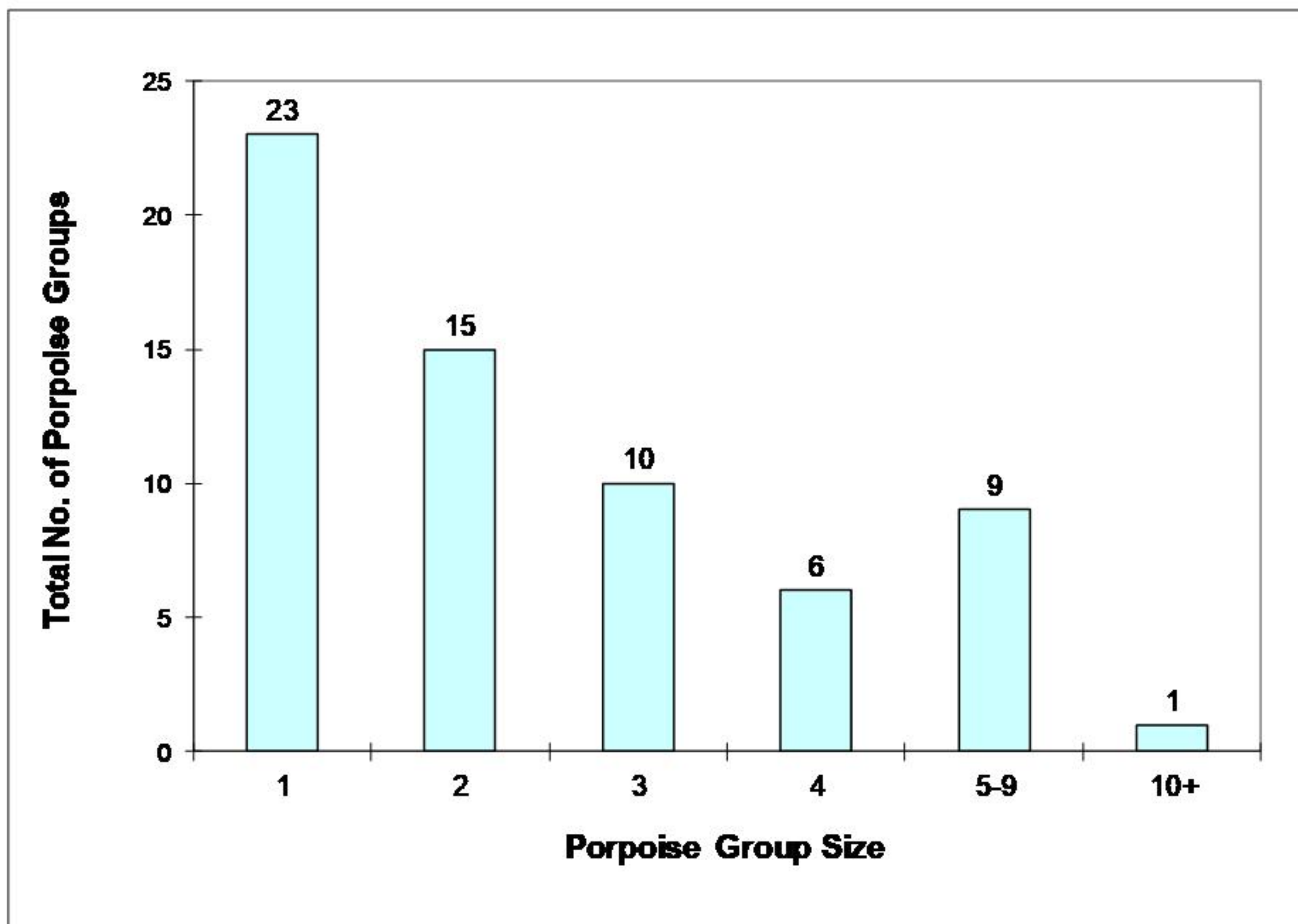


Figure 22. Total number of porpoise groups with different group sizes during April 2024 to March 2025

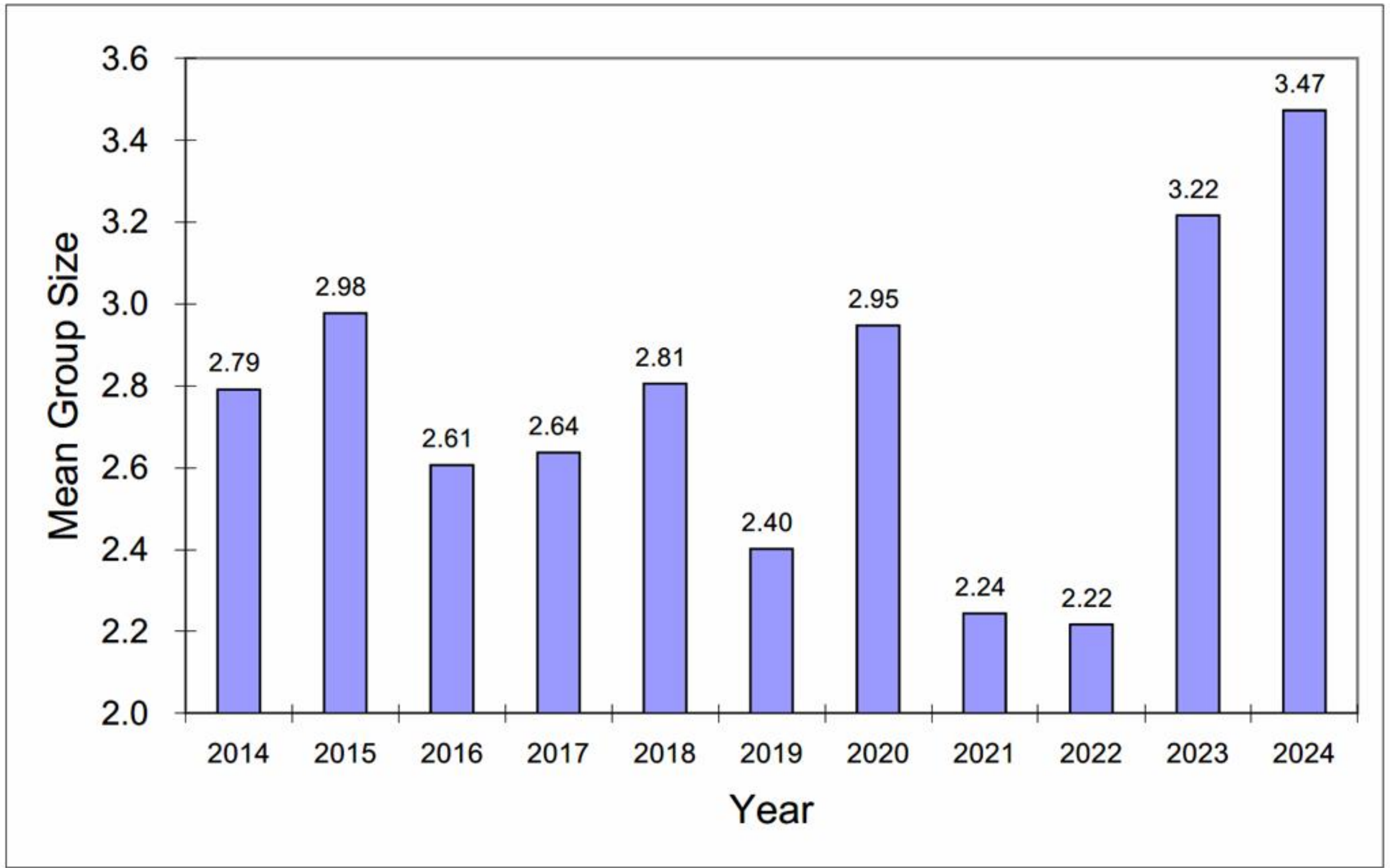


Figure 23. Temporal trend of mean porpoise group size in South Lantau waters in 2014-24

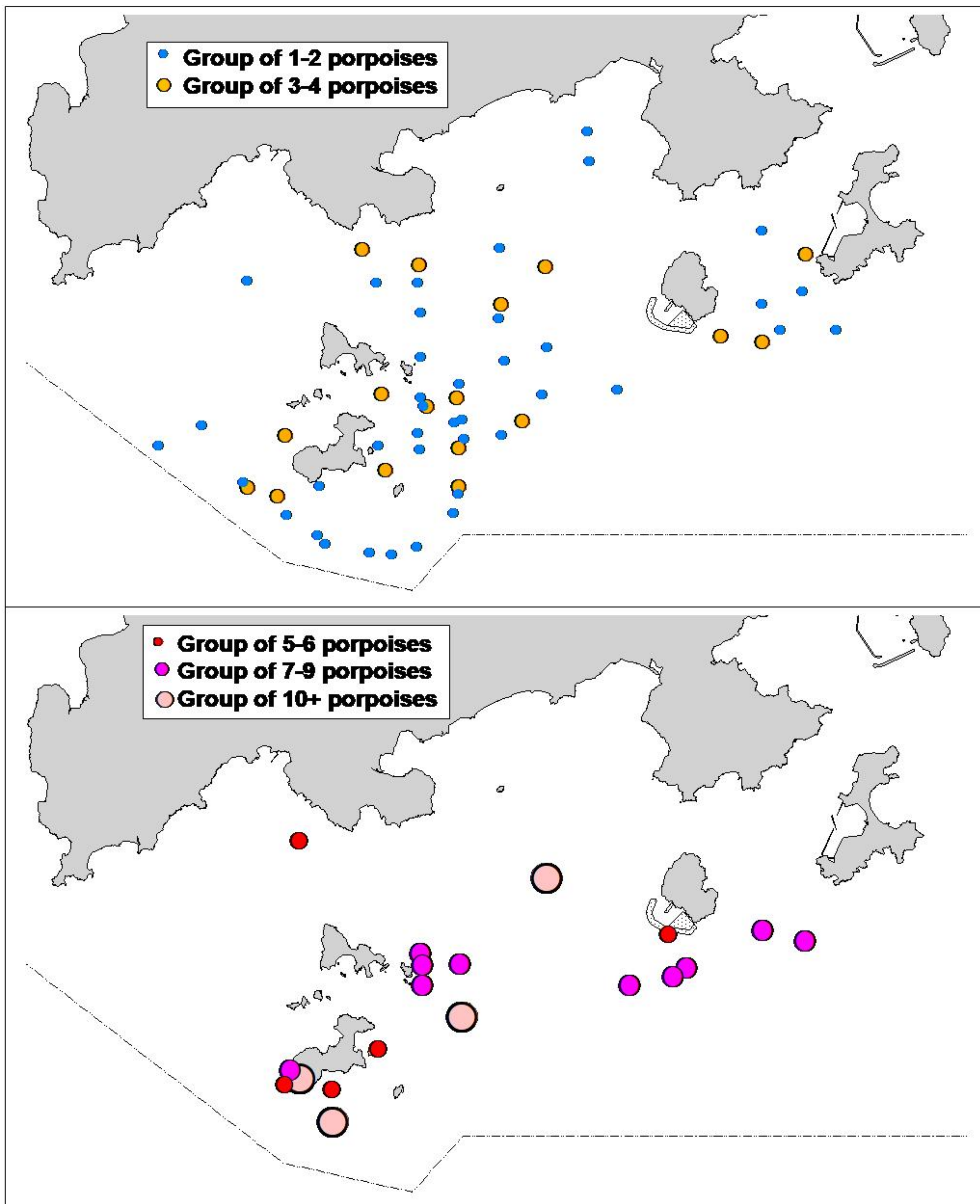


Figure 24. Distribution of finless porpoises with different group sizes in 2024

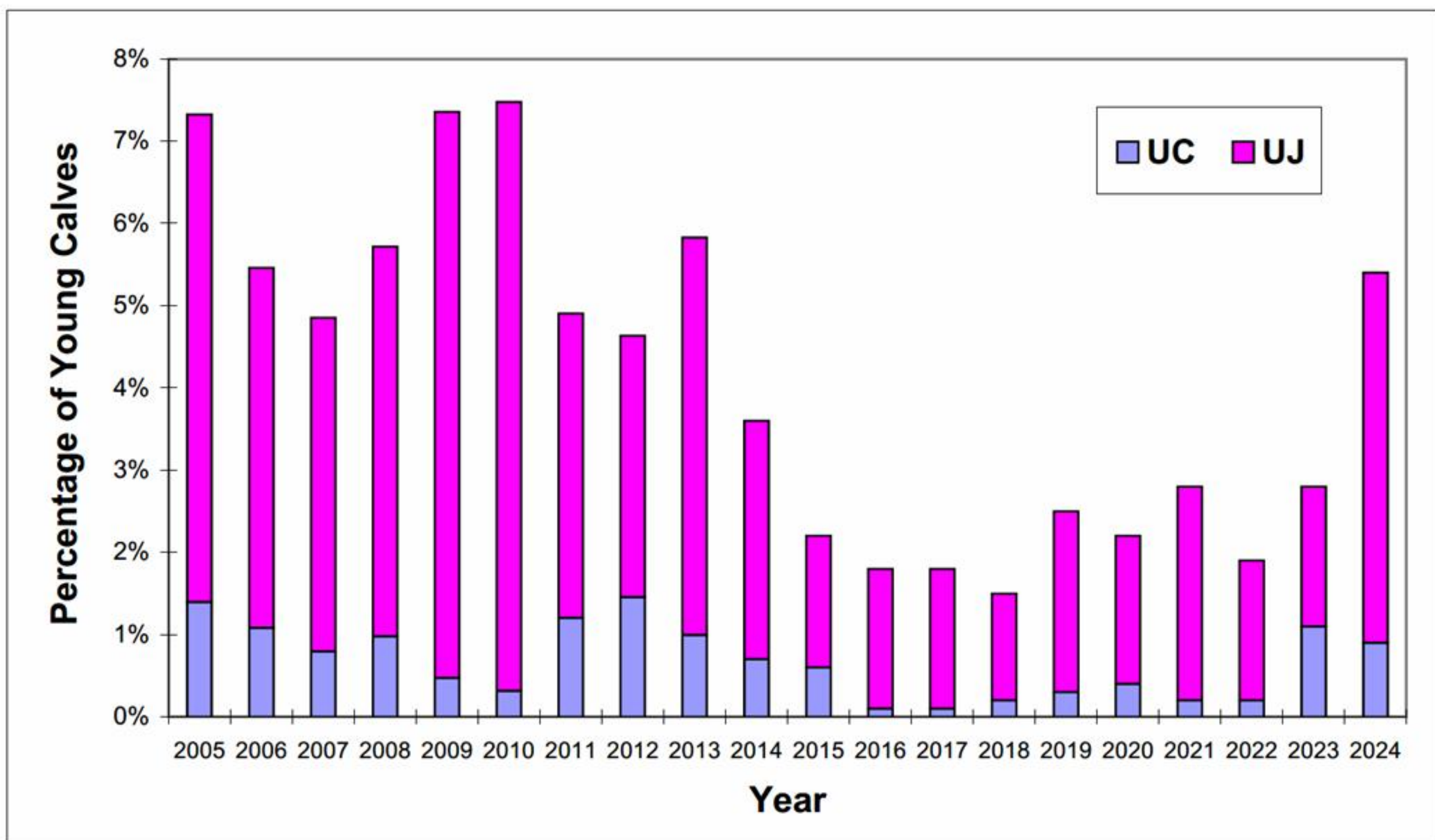


Figure 25. Percentages of young calves (i.e. Unspotted Calves (UC) and Unspotted Juveniles (UJ)) among all dolphin groups during 2005-24

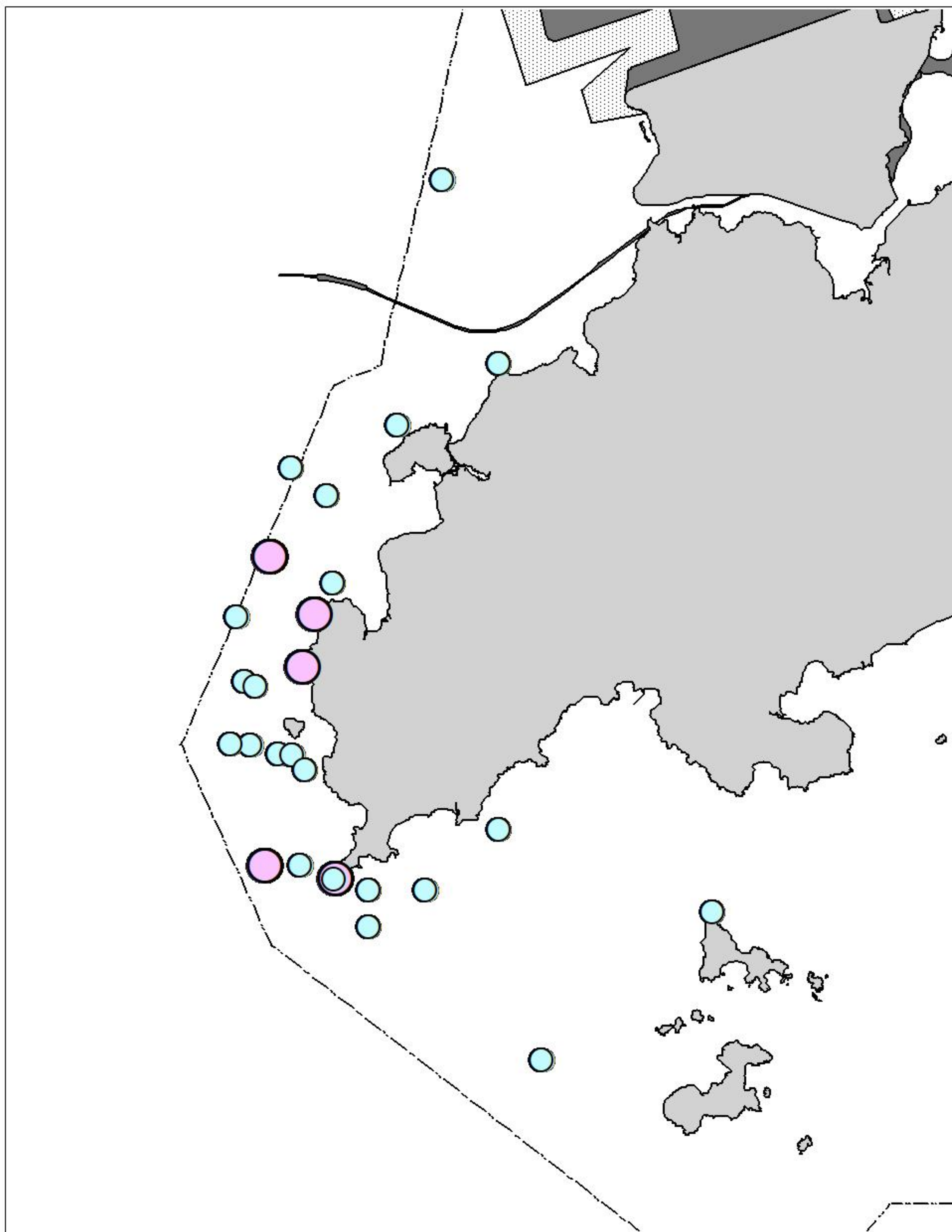


Figure 26. Distribution of Unspotted Calves (purple dots) & Unspotted Juveniles (blue dots) in 2024

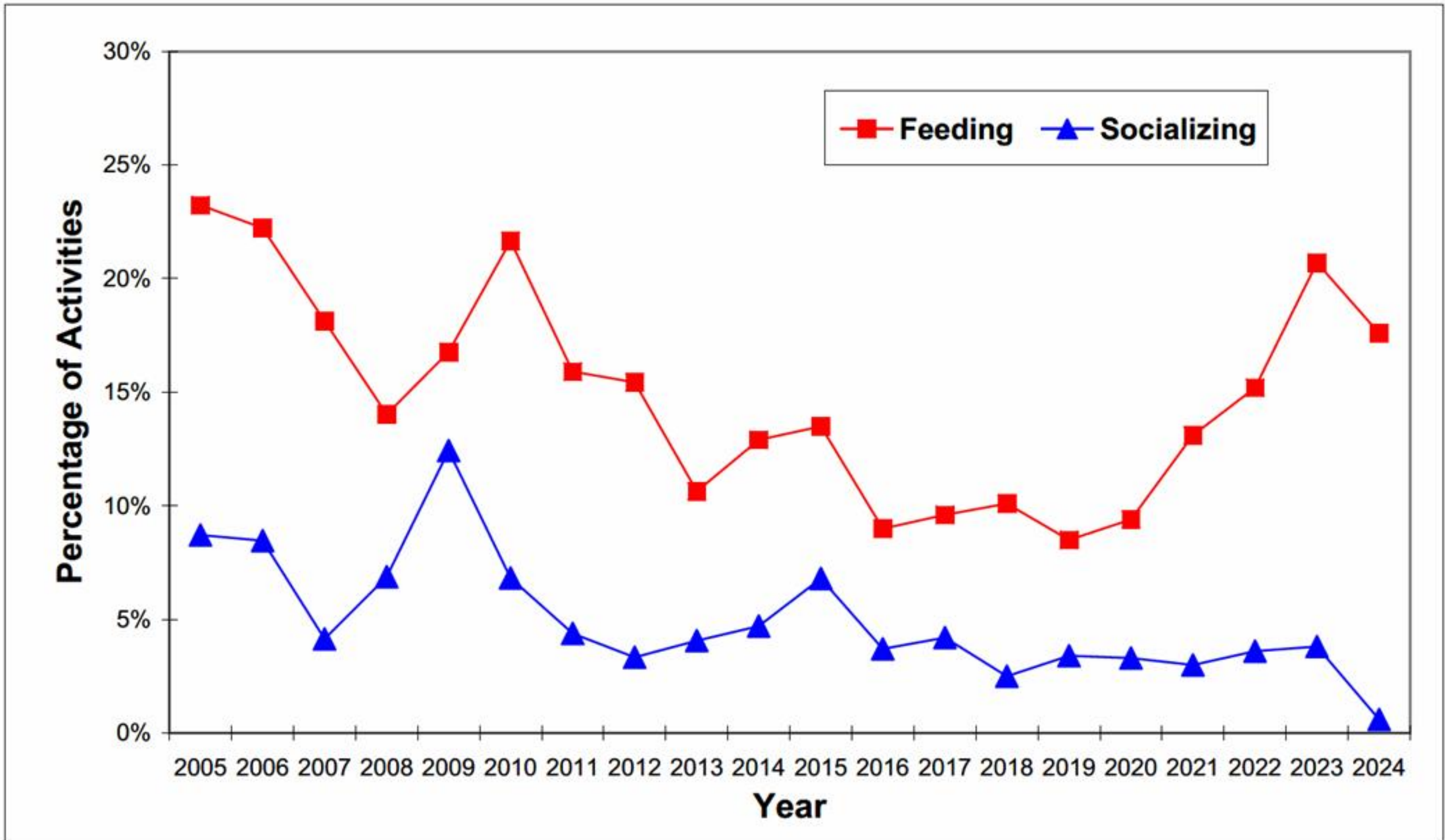


Figure 27. Percentages of feeding and socializing activities among all dolphin groups sighted in Hong Kong during 2005-24

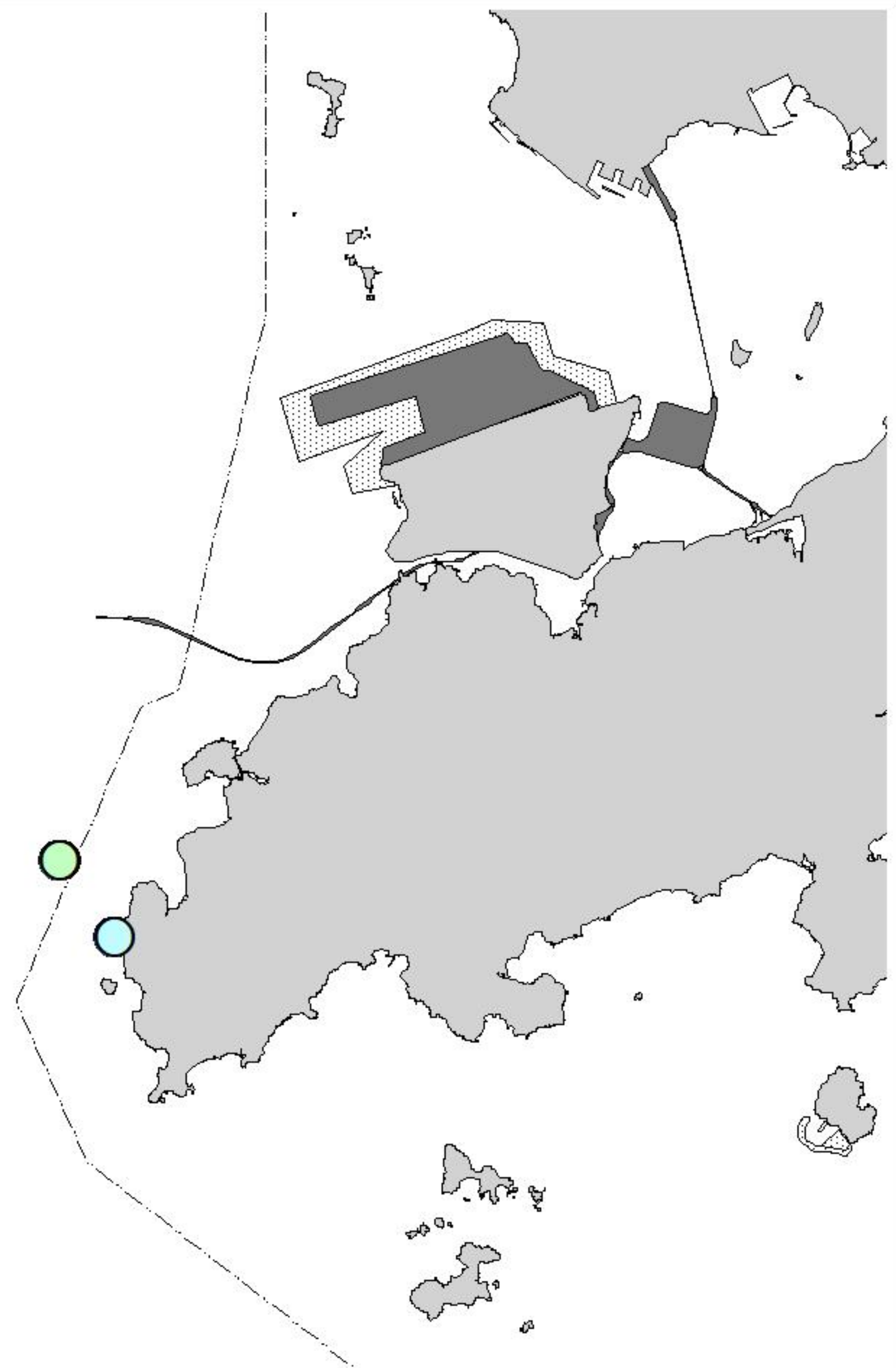
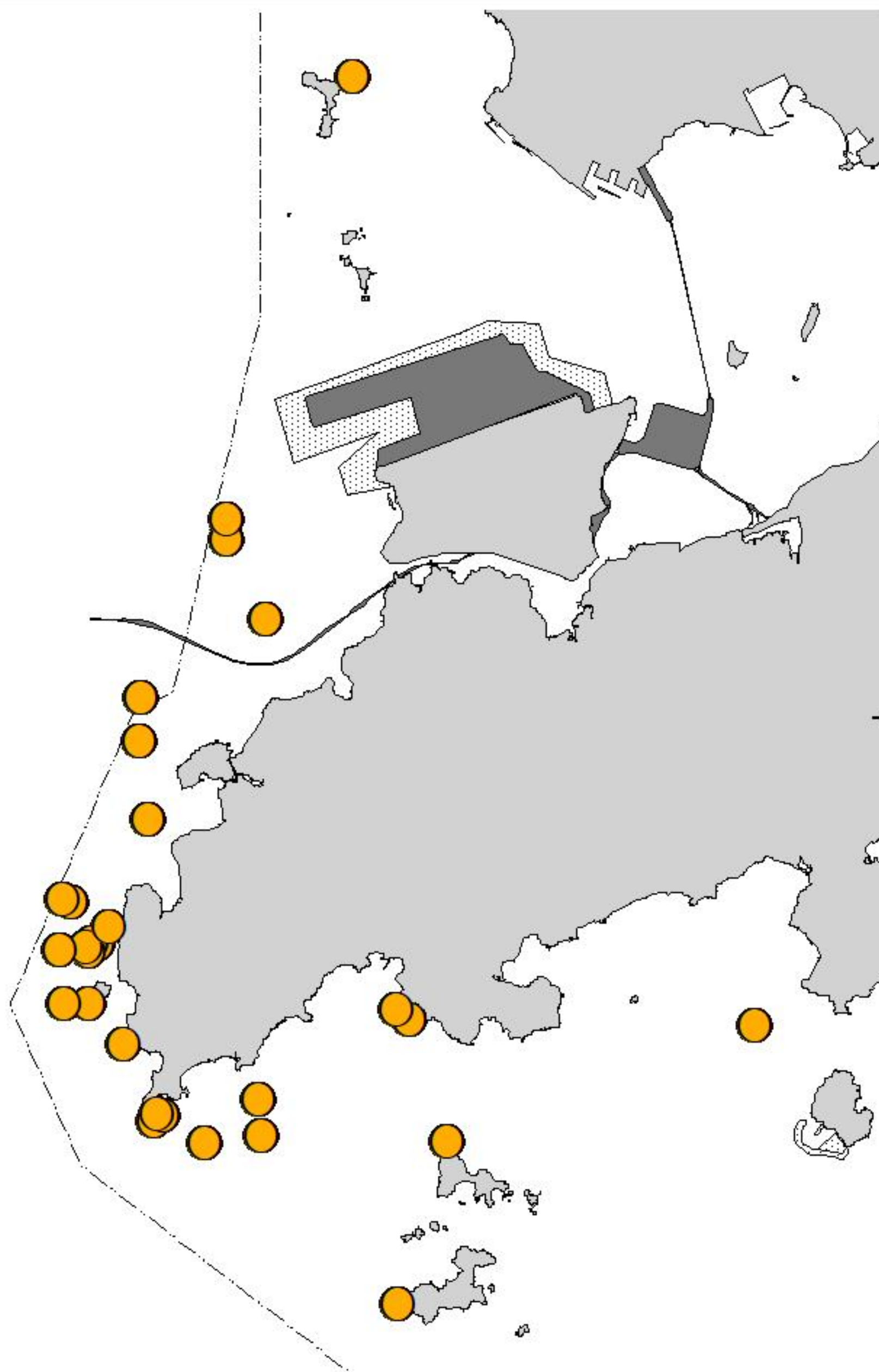


Figure 28. Distribution of Chinese white dolphins engaged in feeding (orange dots), socializing (blue dot) and traveling (green dot) activities in 2024

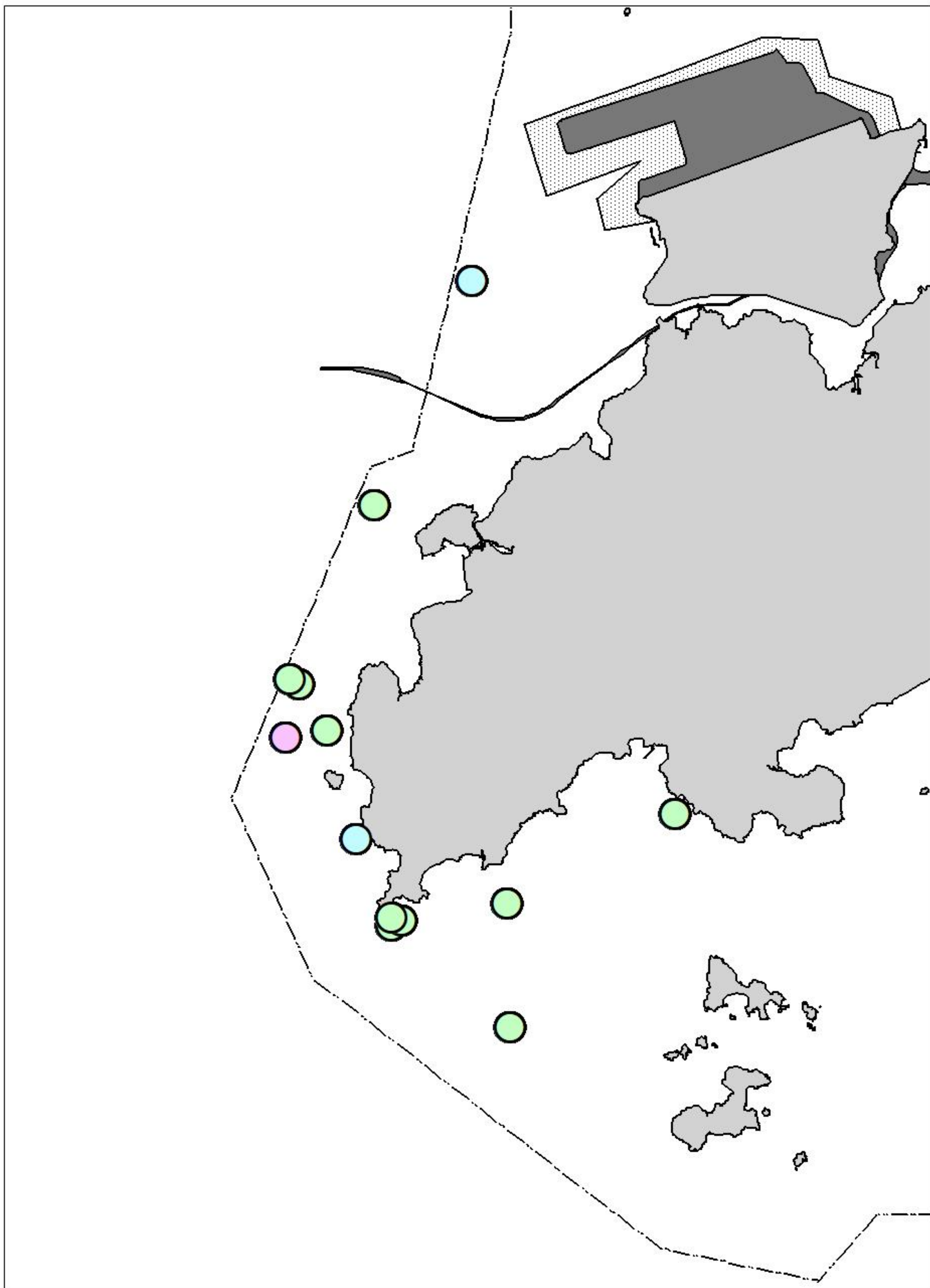


Figure 29. Distribution of dolphin sightings associated with fishing boats (green dots: purse-seiners; blue dots: gill-netters; purple dot: shrimp trawler) in 2024

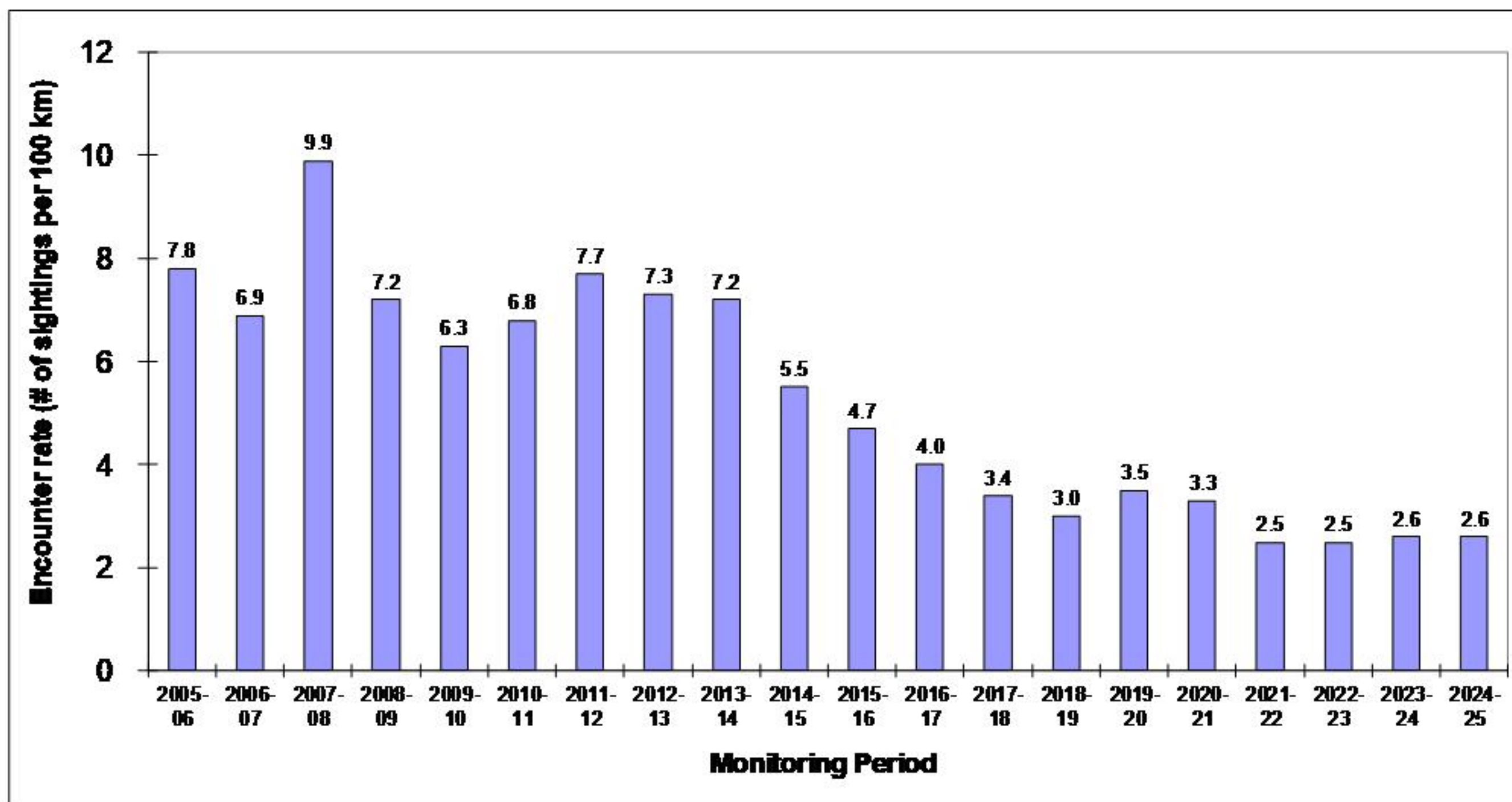


Figure 30a. Temporal trend in encounter rates of Chinese White Dolphins (combined from WL, NWL, NEL and SWL survey areas) in the past 20 monitoring periods from 2005/06 to 2024/25

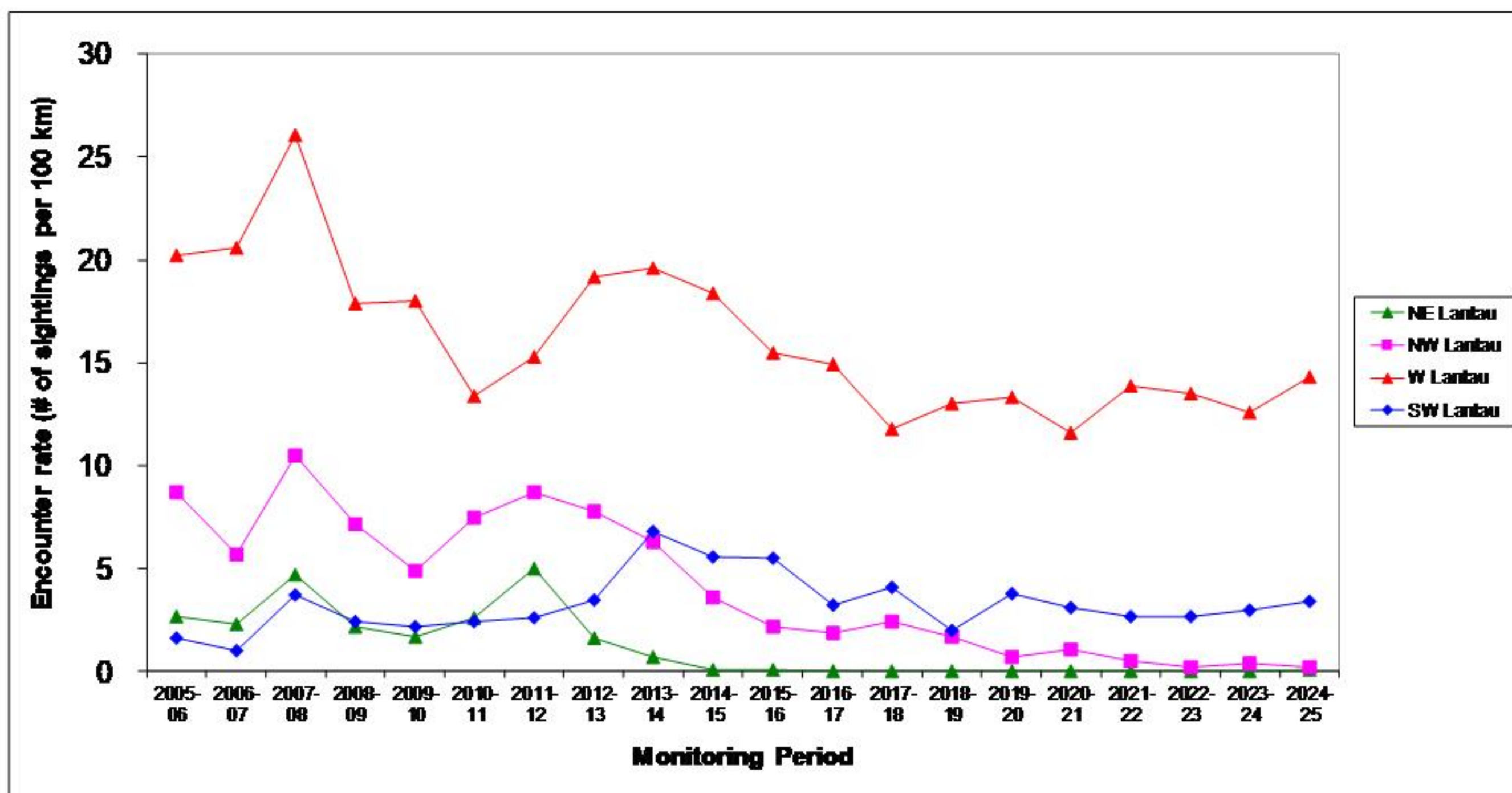


Figure 30b. Temporal trend in encounter rates of Chinese White Dolphins in each of the survey areas in WL, NWL, NEL and SWL waters in the past 20 monitoring periods from 2005/06 to 2024/25

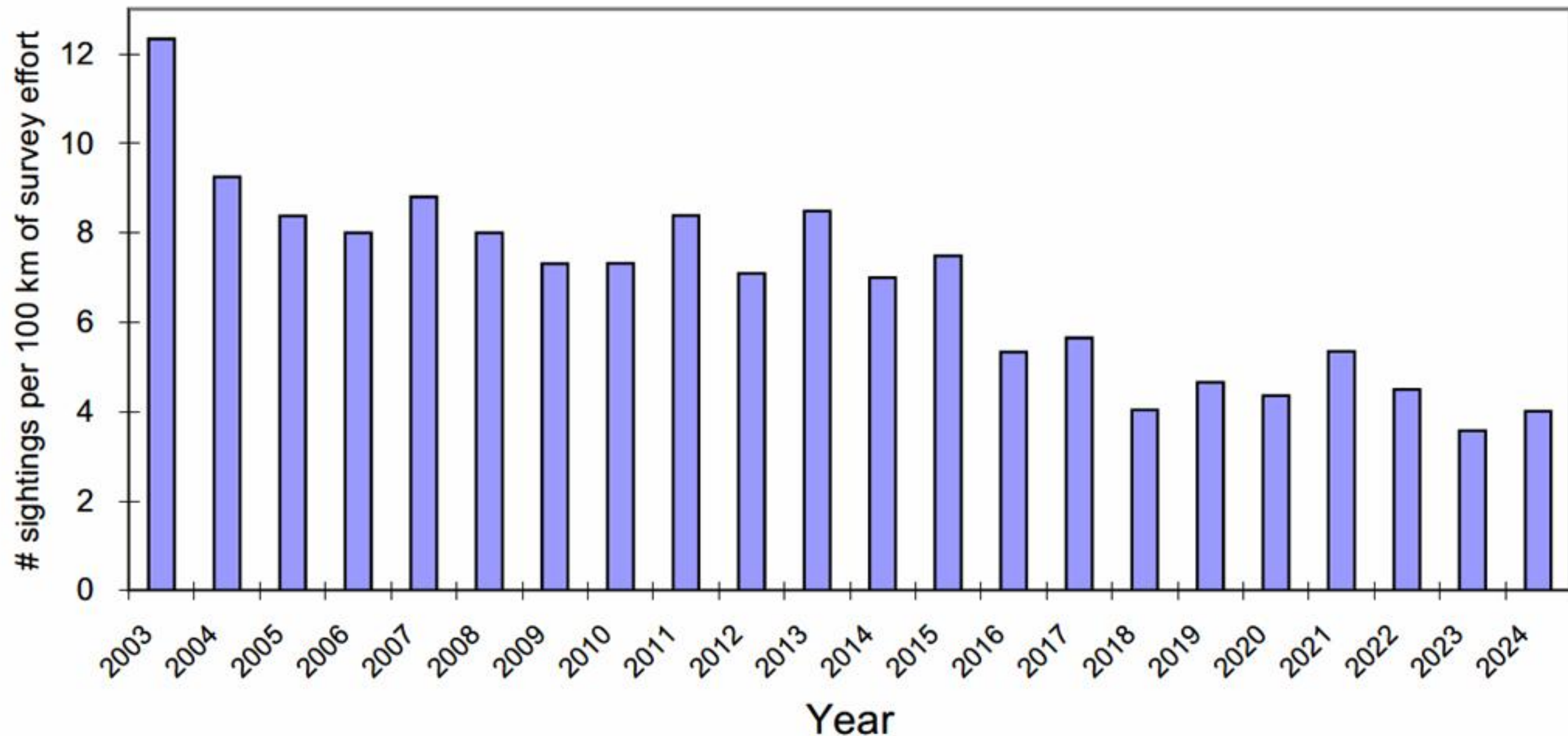


Figure 31a. Temporal trends in combined annual dolphin encounter rates from four survey areas in NEL, NWL, WL and SWL

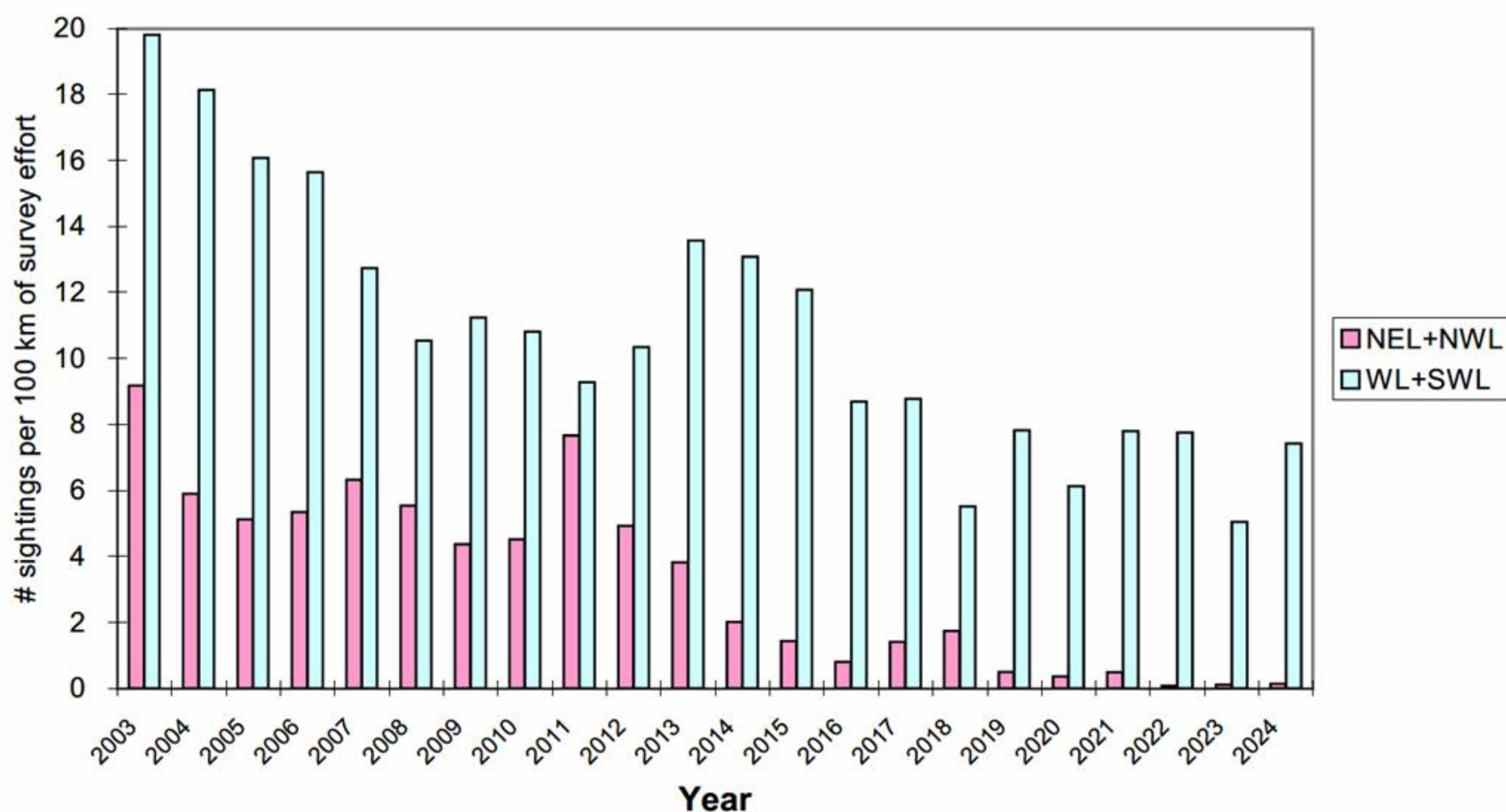


Figure 31b. Temporal trends in annual dolphin encounter rates in North Lantau and West/Southwest Lantau regions

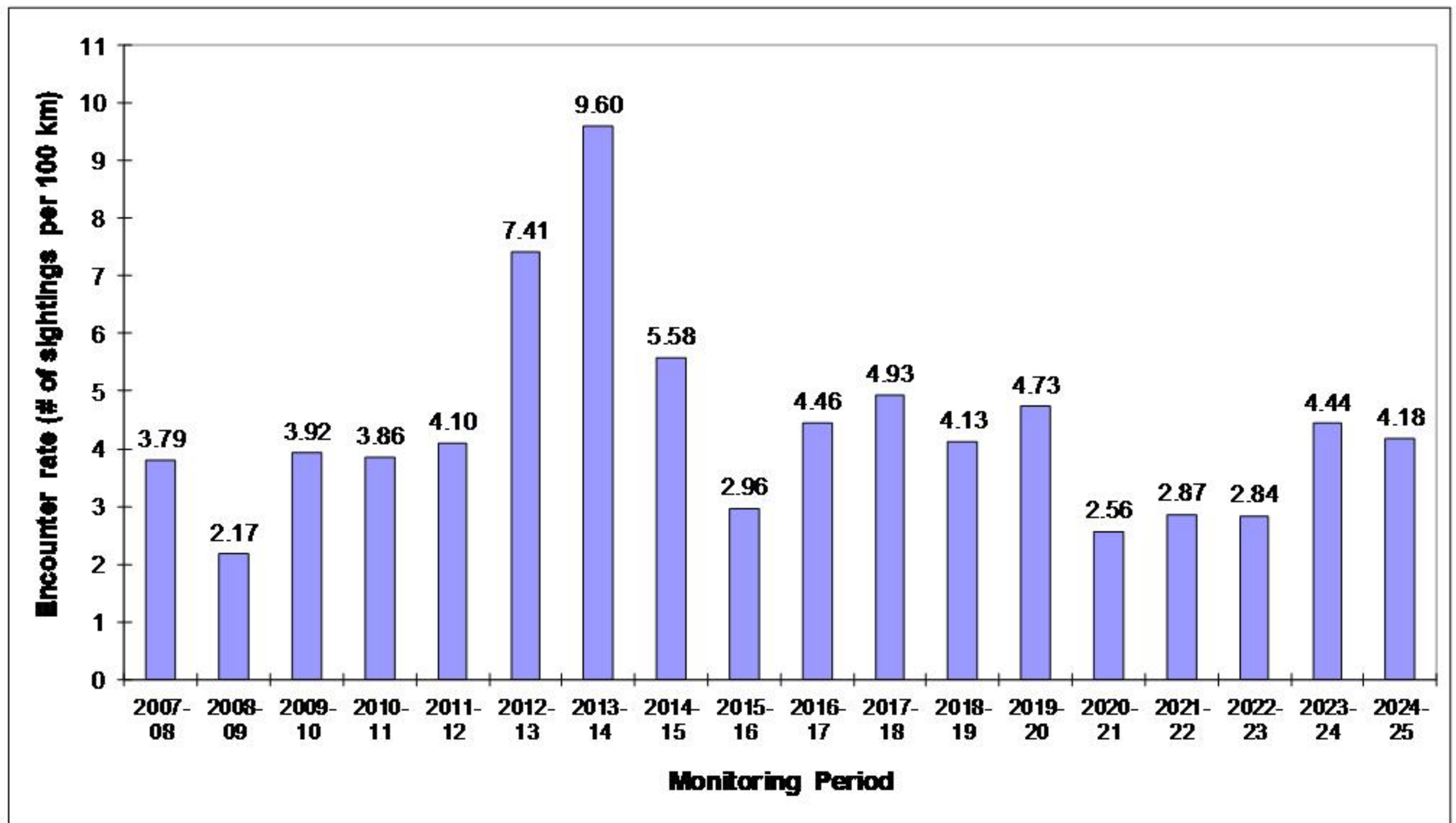


Figure 32. Temporal trend in encounter rates of finless porpoises (combined from SWL and SEL survey areas) in the past 18 monitoring periods from 2007/08 to 2024/25

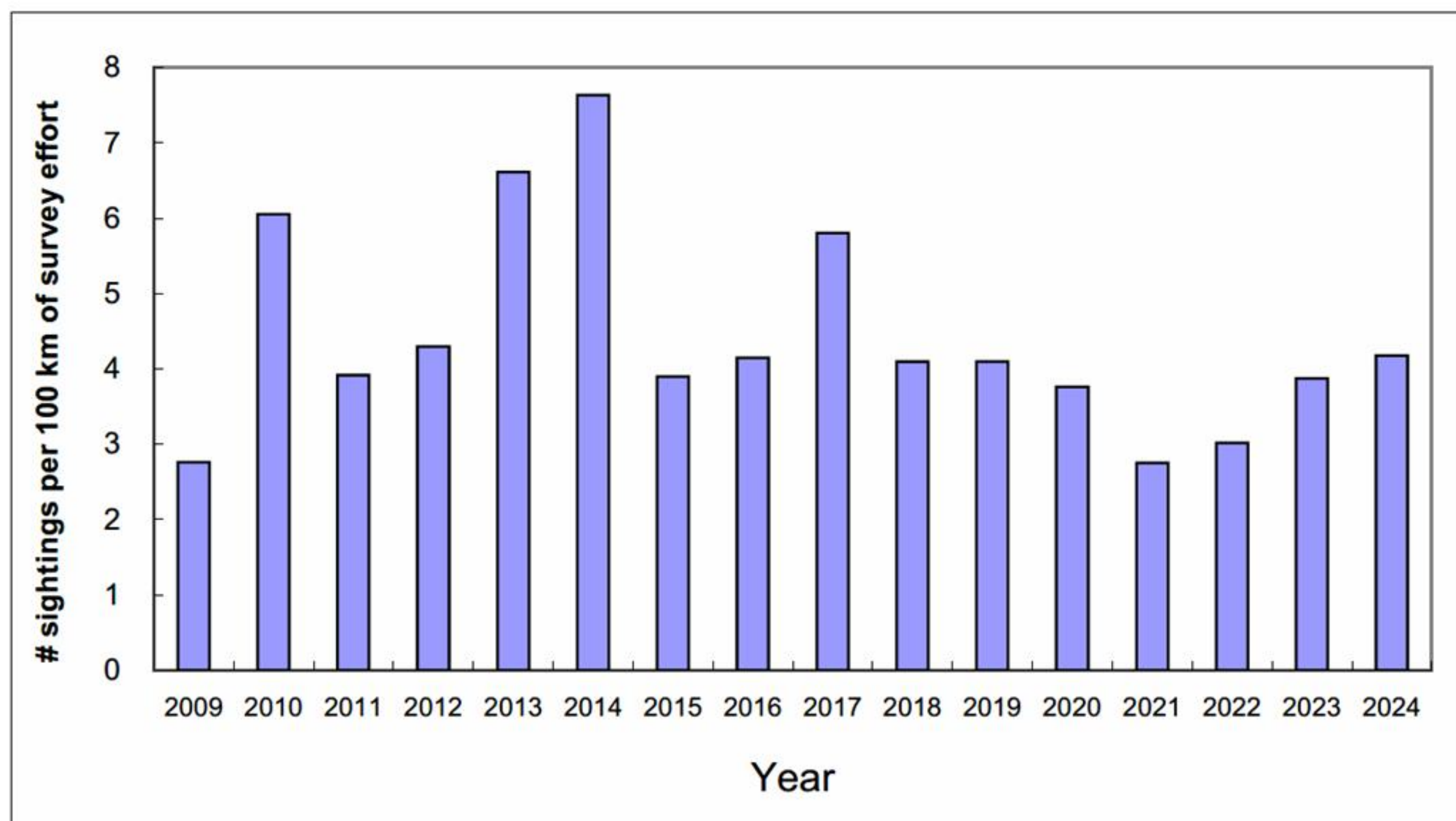


Figure 33a. Temporal trend of annual encounter rates of finless porpoises (combined from SWL and SEL survey areas)

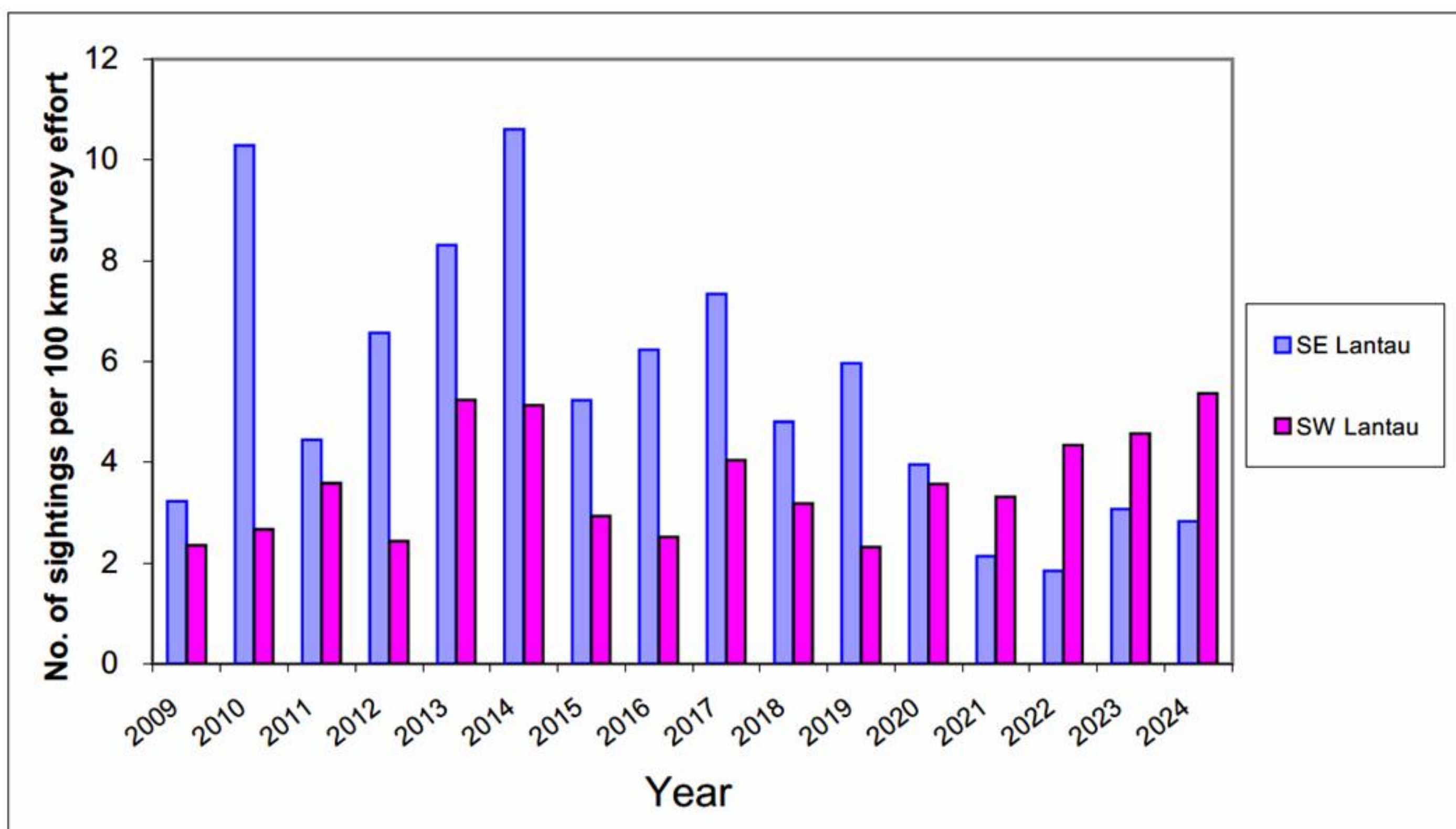


Figure 33b. Temporal trend of annual porpoise encounter rates among the SWL and SEL Lantau survey areas respectively

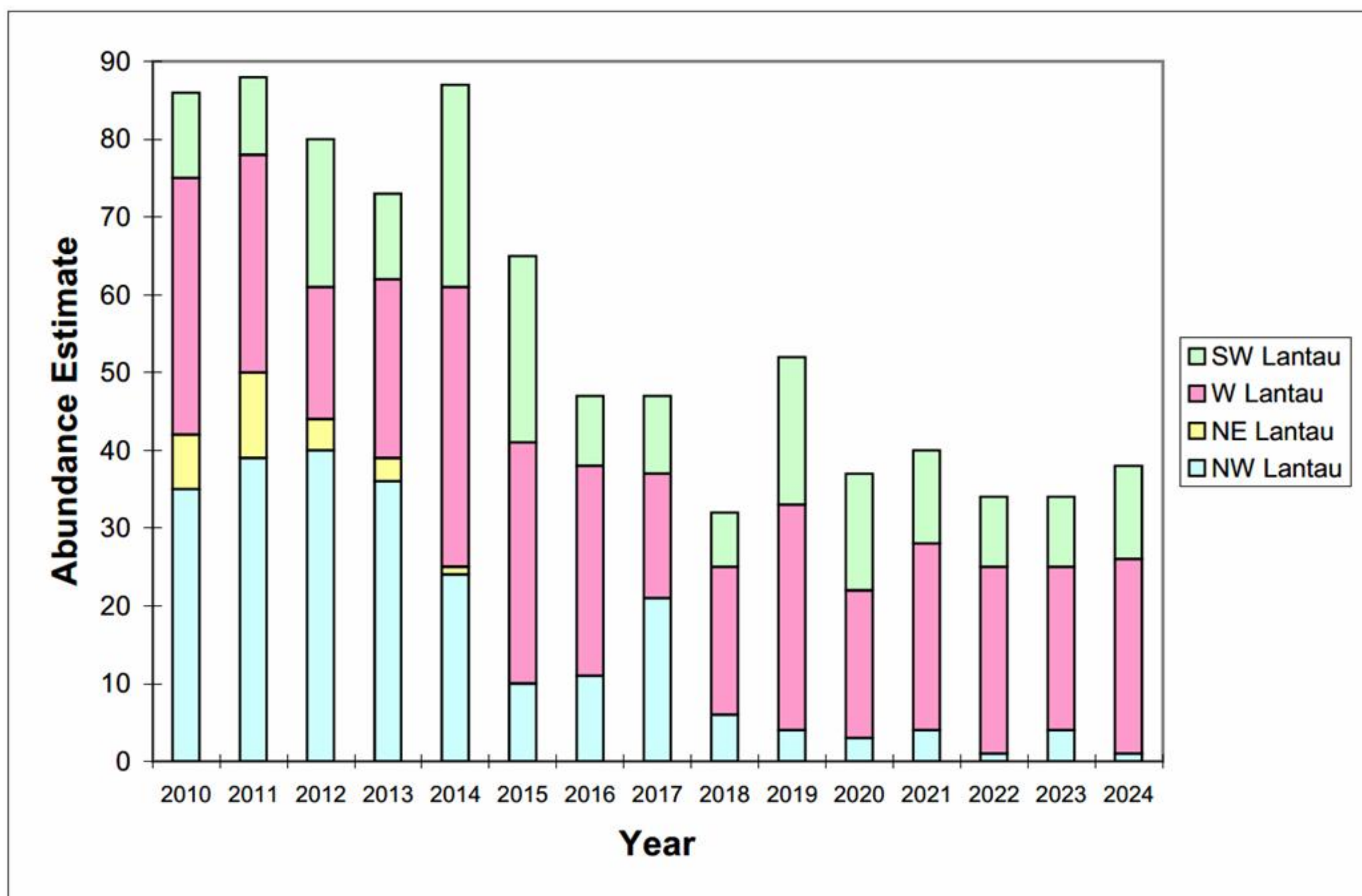


Figure 34. Temporal trends in combined abundance estimates of Chinese White Dolphins in SWL, WL, NWL and NEL from 2010-24

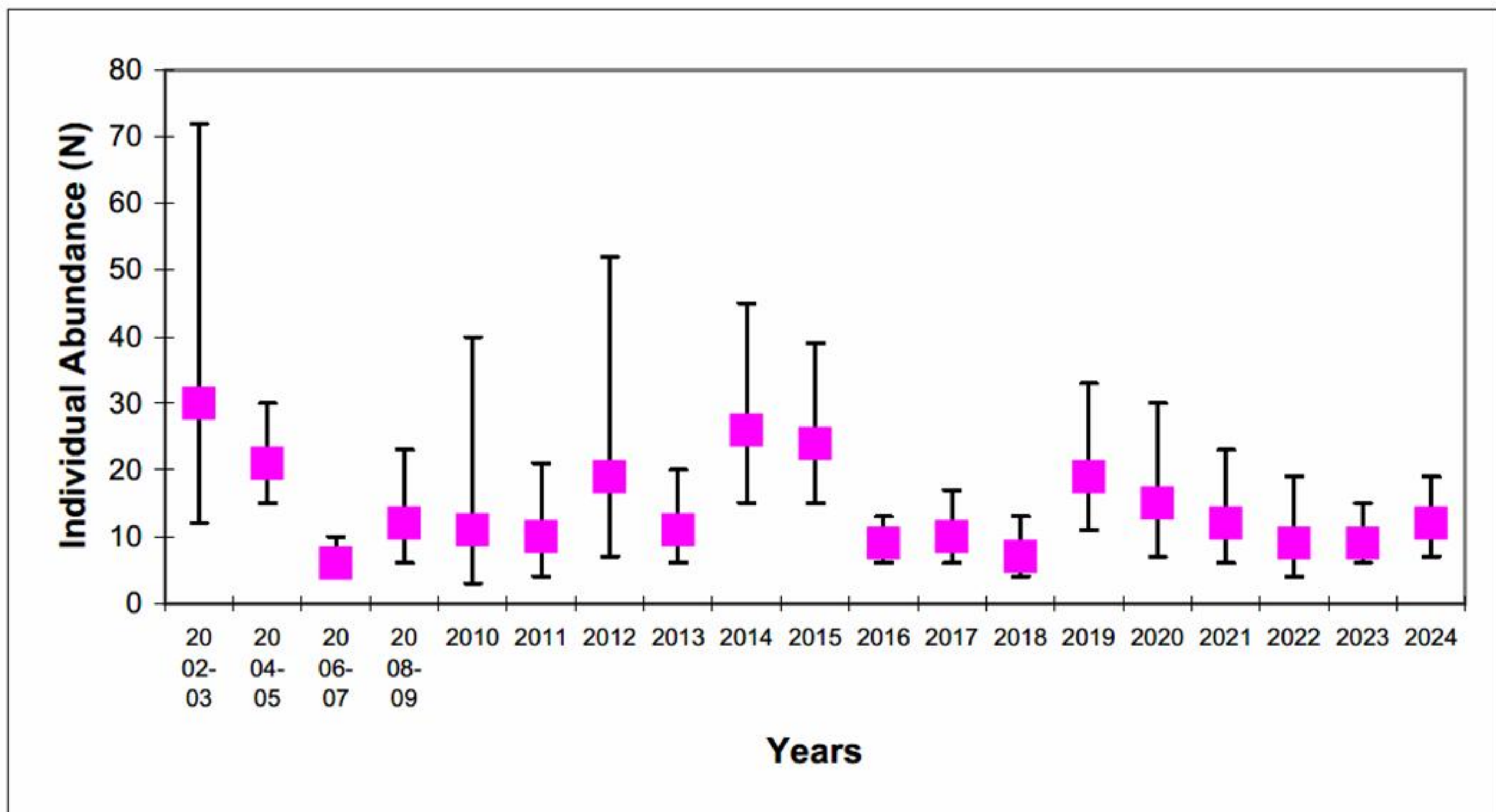


Figure 35. Temporal trend in abundance estimates of Chinese White Dolphins in Southwest Lantau from 2002-24 (error bars: 95% confidence interval of abundance estimates)

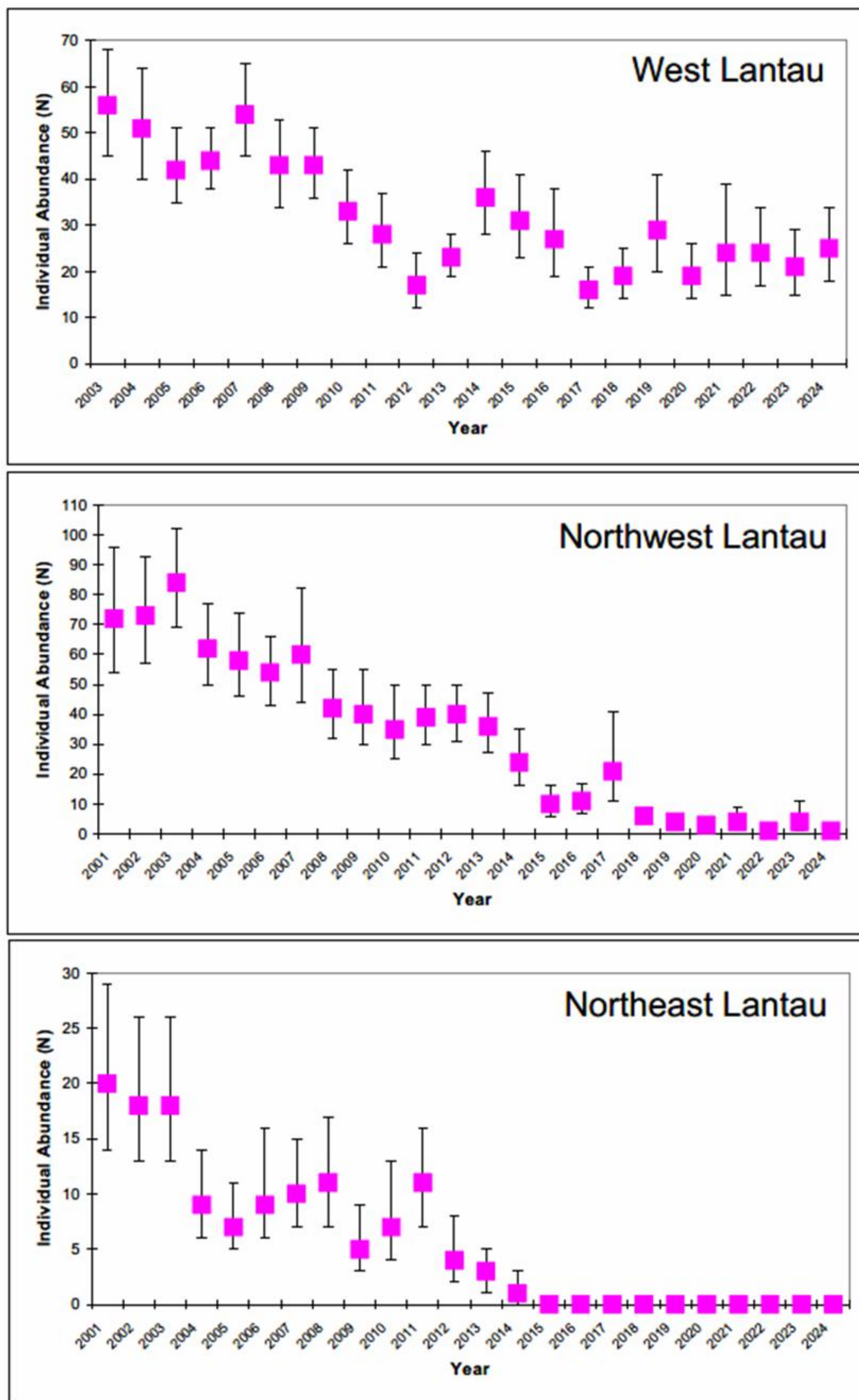


Figure 36. Temporal trends in annual abundance estimates of Chinese White Dolphins in WL (since 2003) & NWL/NEL (since 2001) (error bars: 95% confidence interval of abundance estimates)

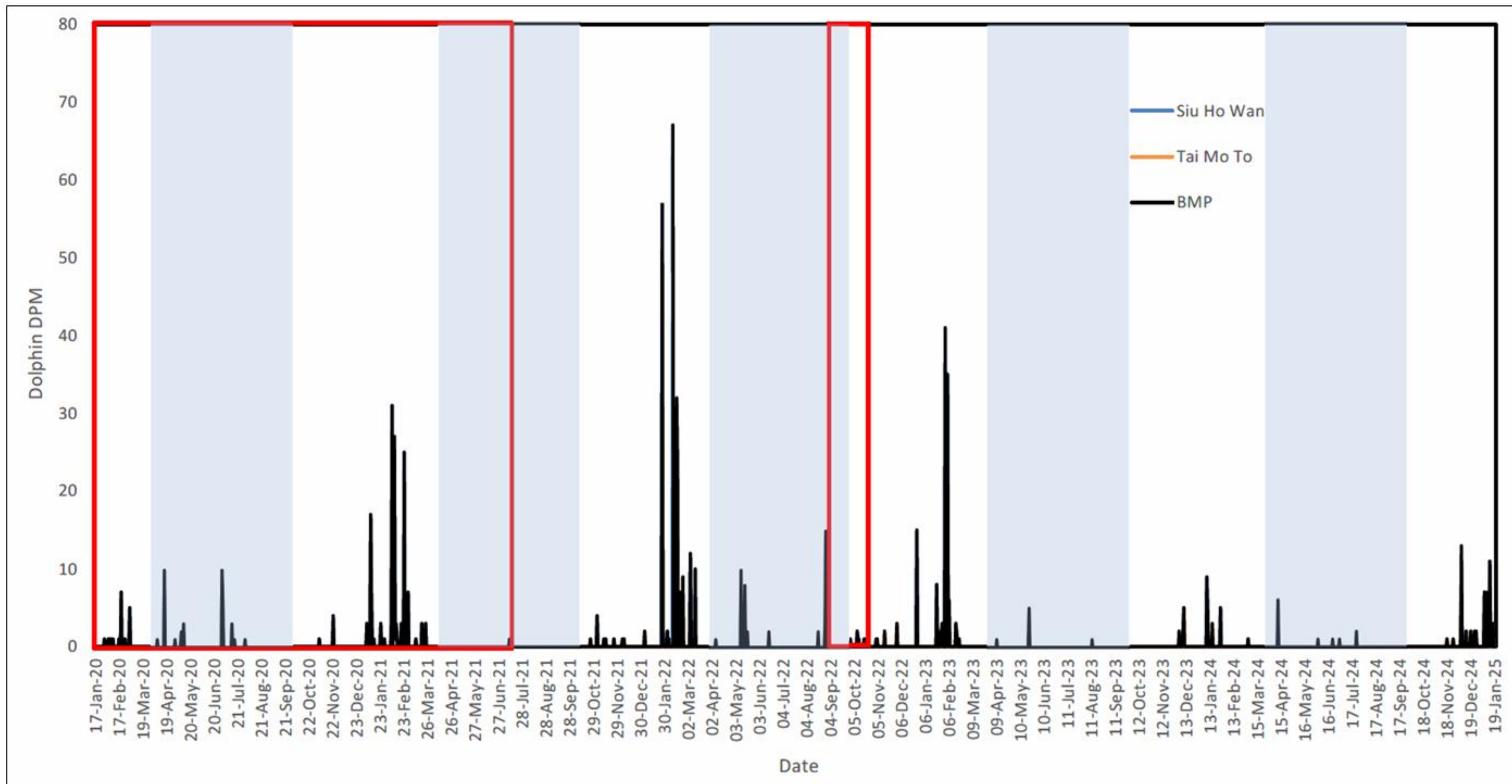


Figure 37. Dolphin DPM per day in the BMP area from January 17th, 2020 to January 20th, 2025 (the shaded and white areas represent the wet and dry seasons, respectively). Note: the red boxes show periods when the Tai Mo To F-POD was not recording data so the observed patterns for BMP should be interpreted accordingly.

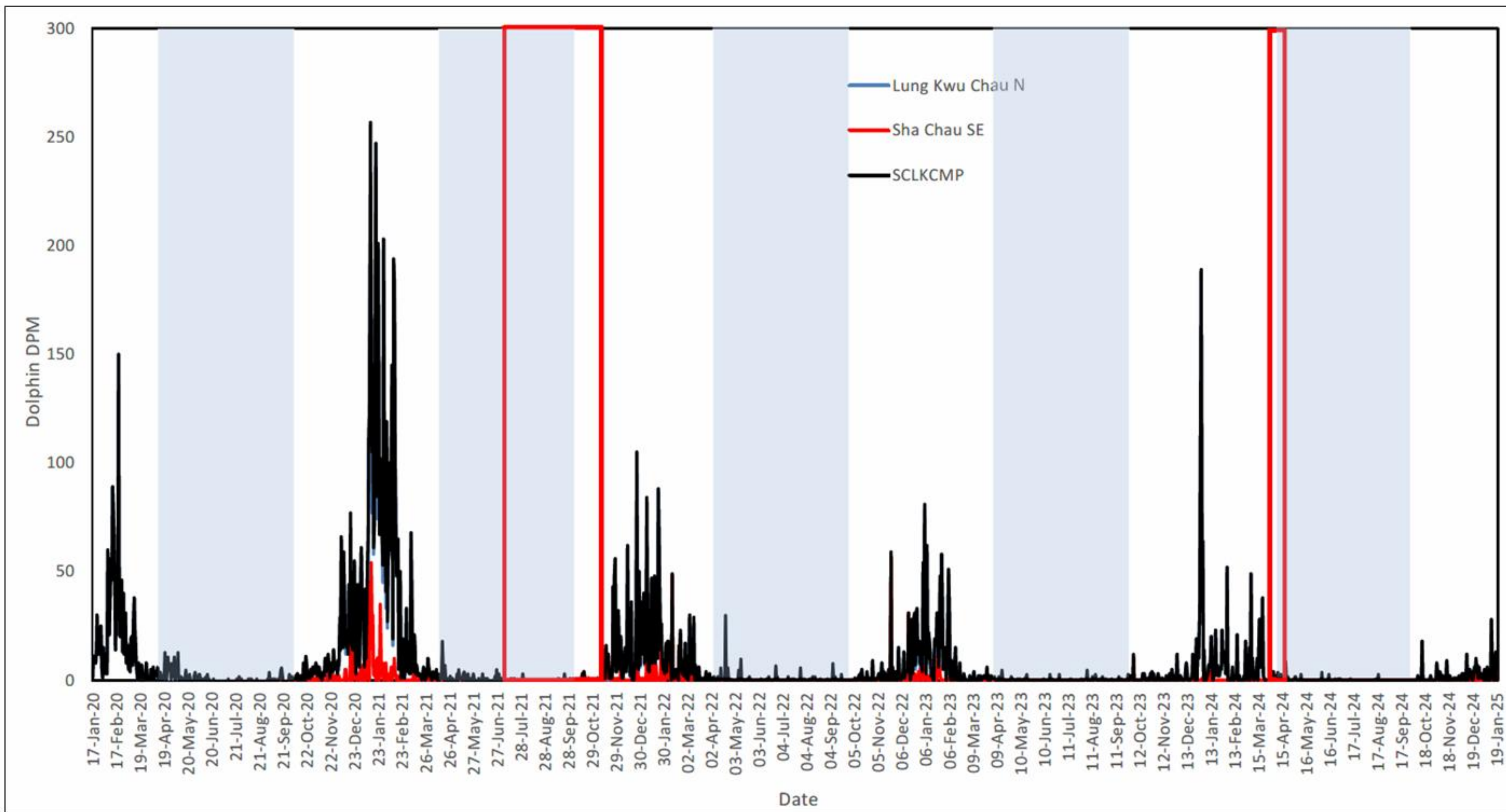
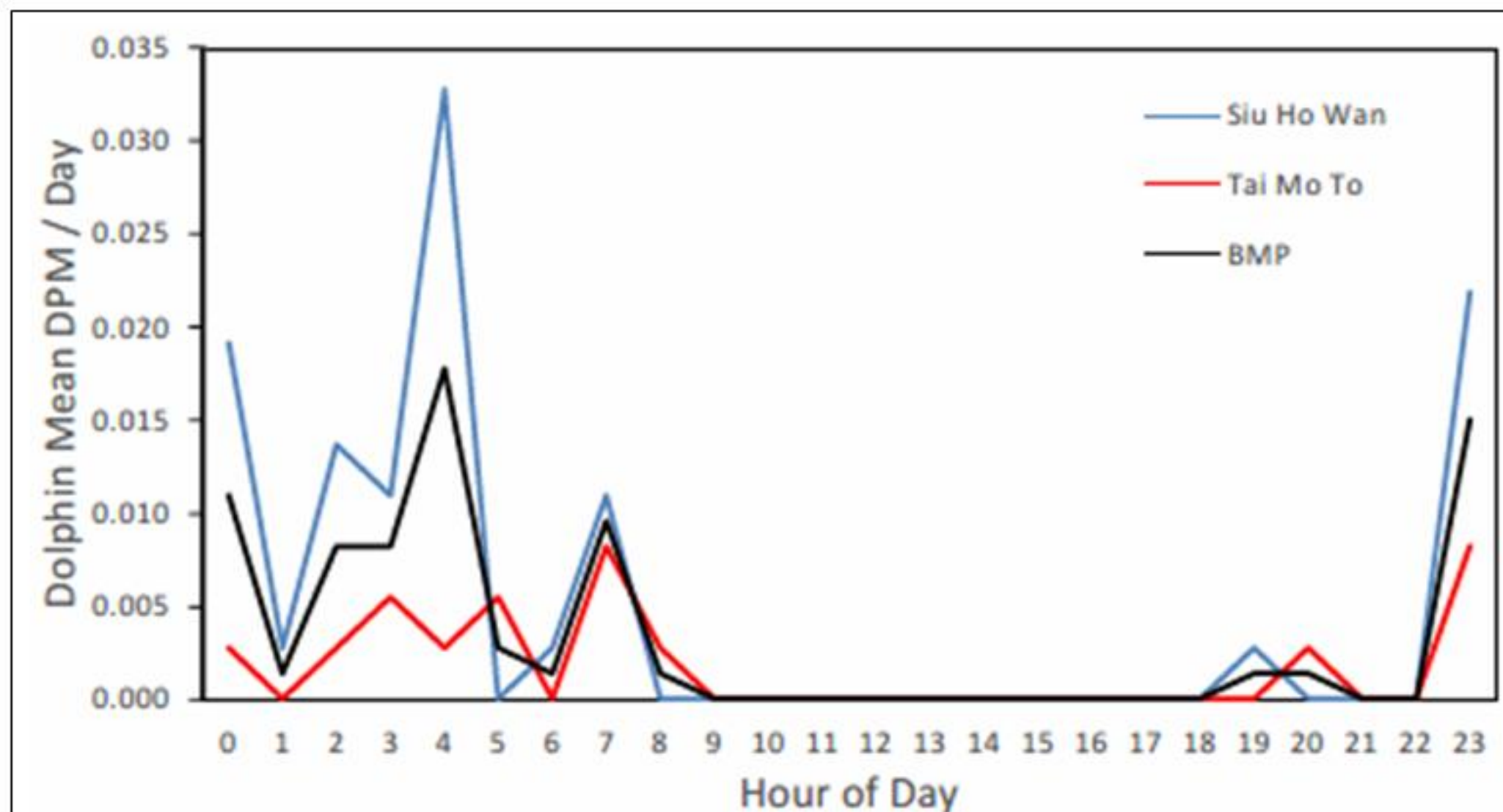
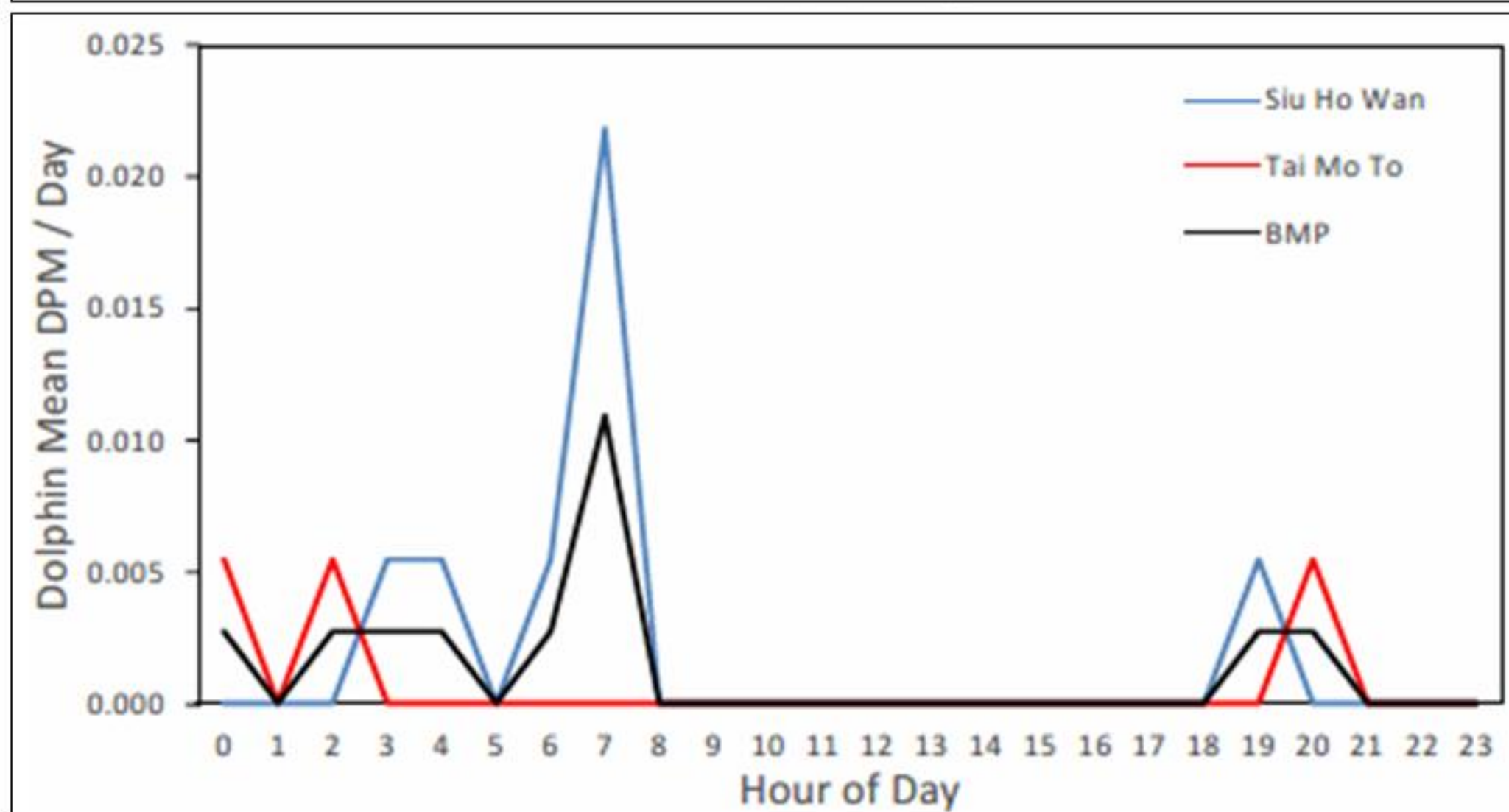


Figure 38. Dolphin DPM per day in the SCLKCMP area from January 17th, 2020 to January 20th, 2025 (the shaded and white areas represent the wet and dry seasons, respectively). Note: the red boxes show periods when at least one F-POD was not recording data so the observed patterns for SCLKCMP should be interpreted accordingly.

(a)



(b)



(c)

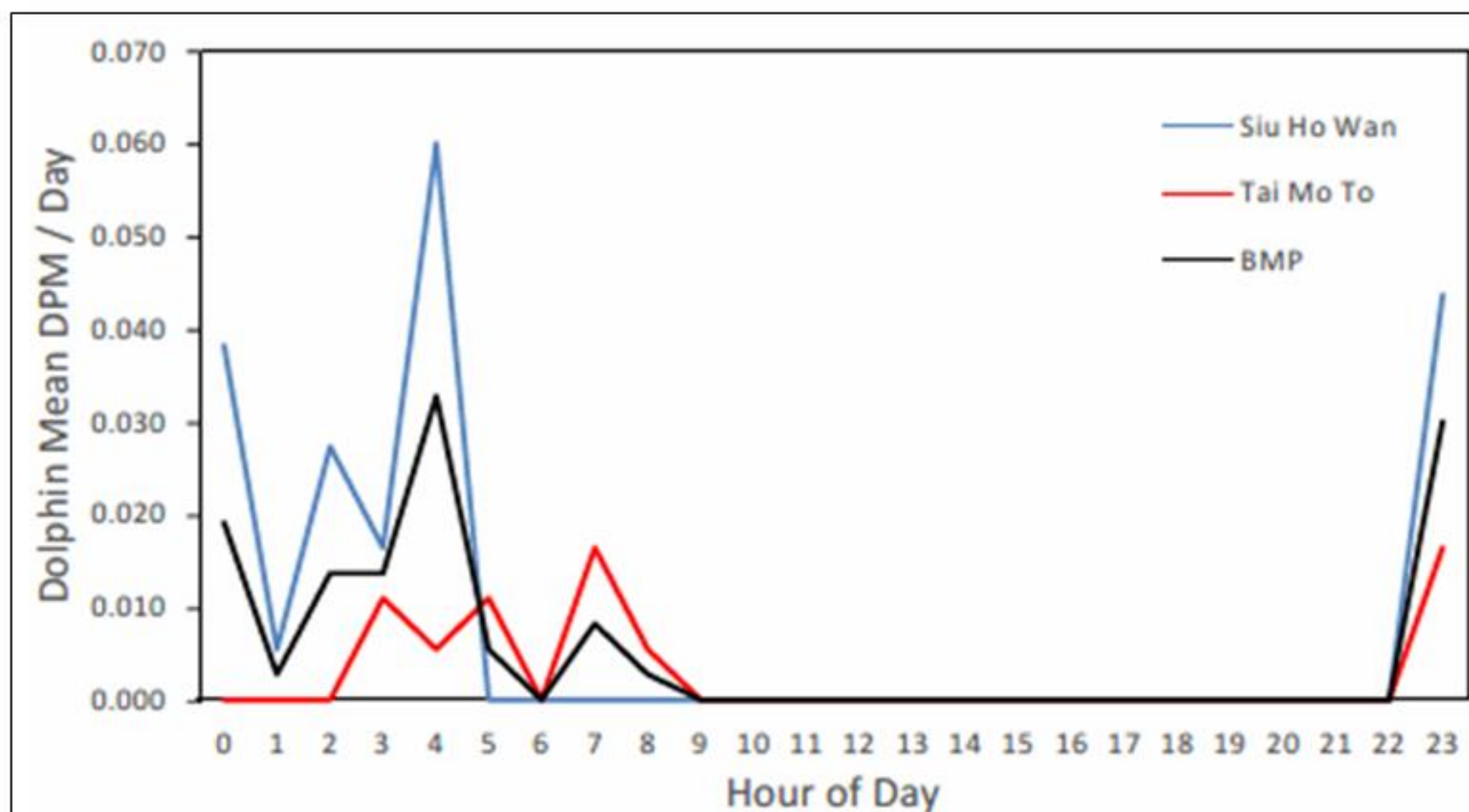
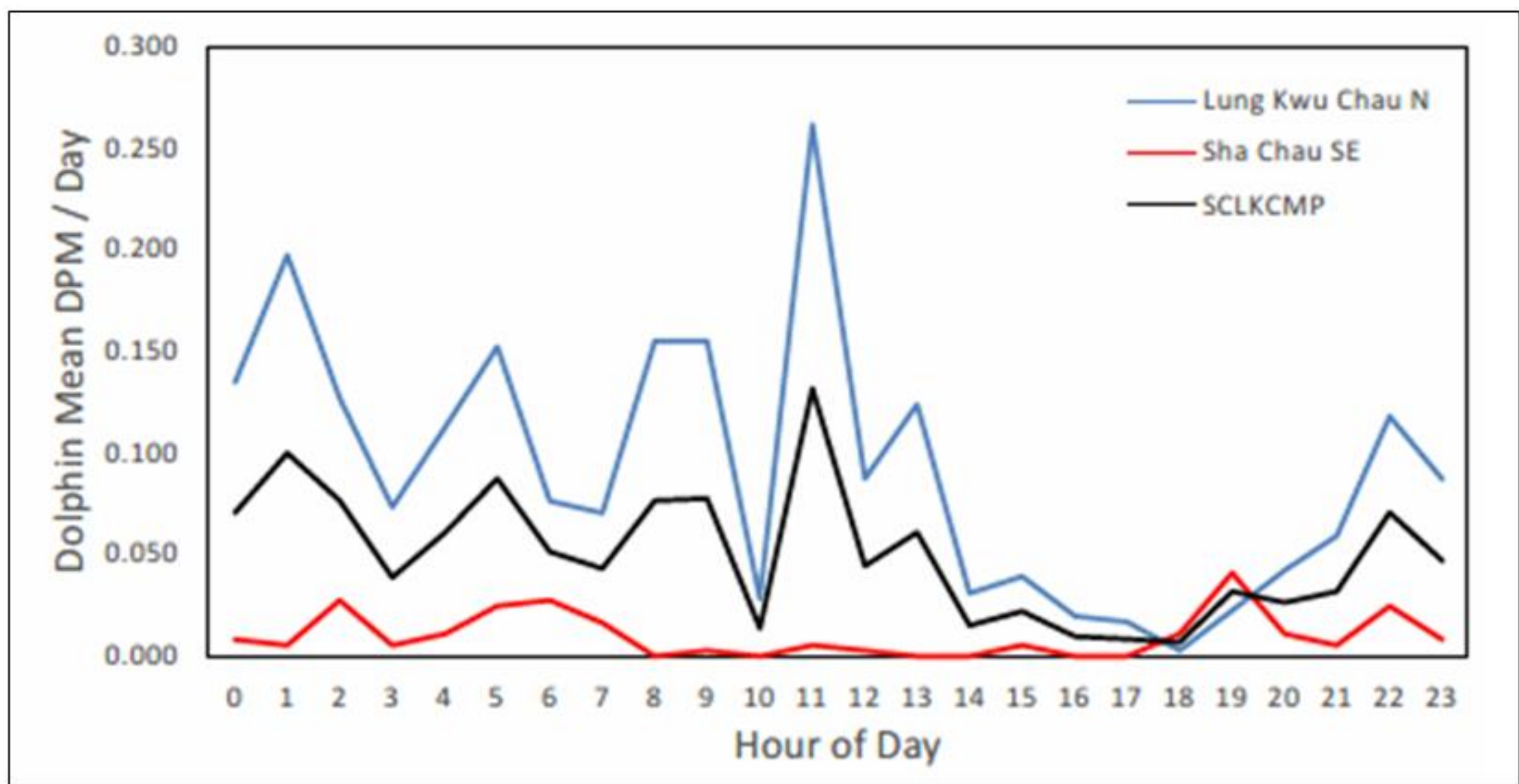
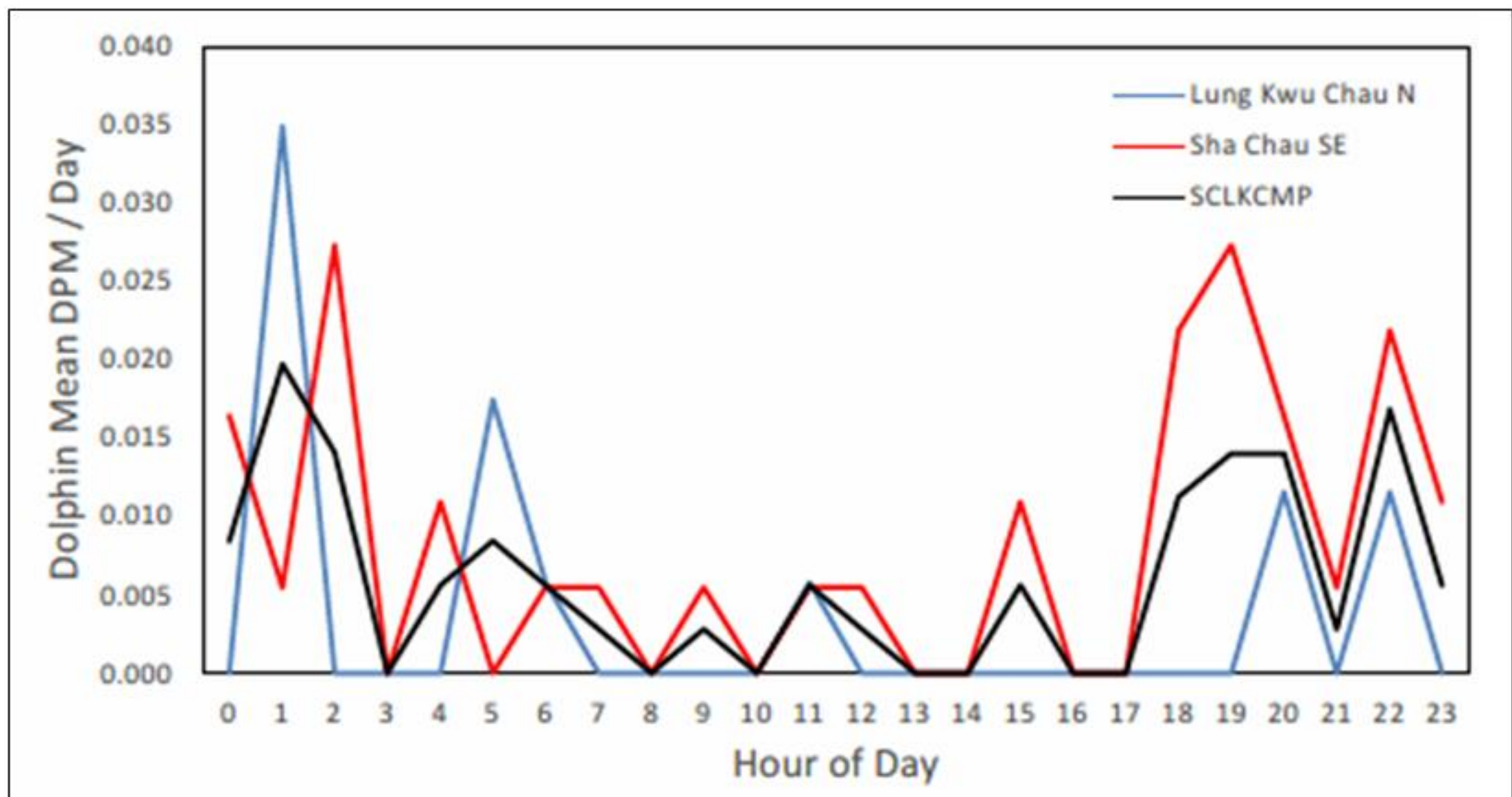


Figure 39. Dolphin DPM diel pattern at BMP (a) from January 1st to December 31st, 2024; (b) in the wet season (April to September, 2024); and c) in the dry season (October to March, 2024)

(a)



(b)



(c)

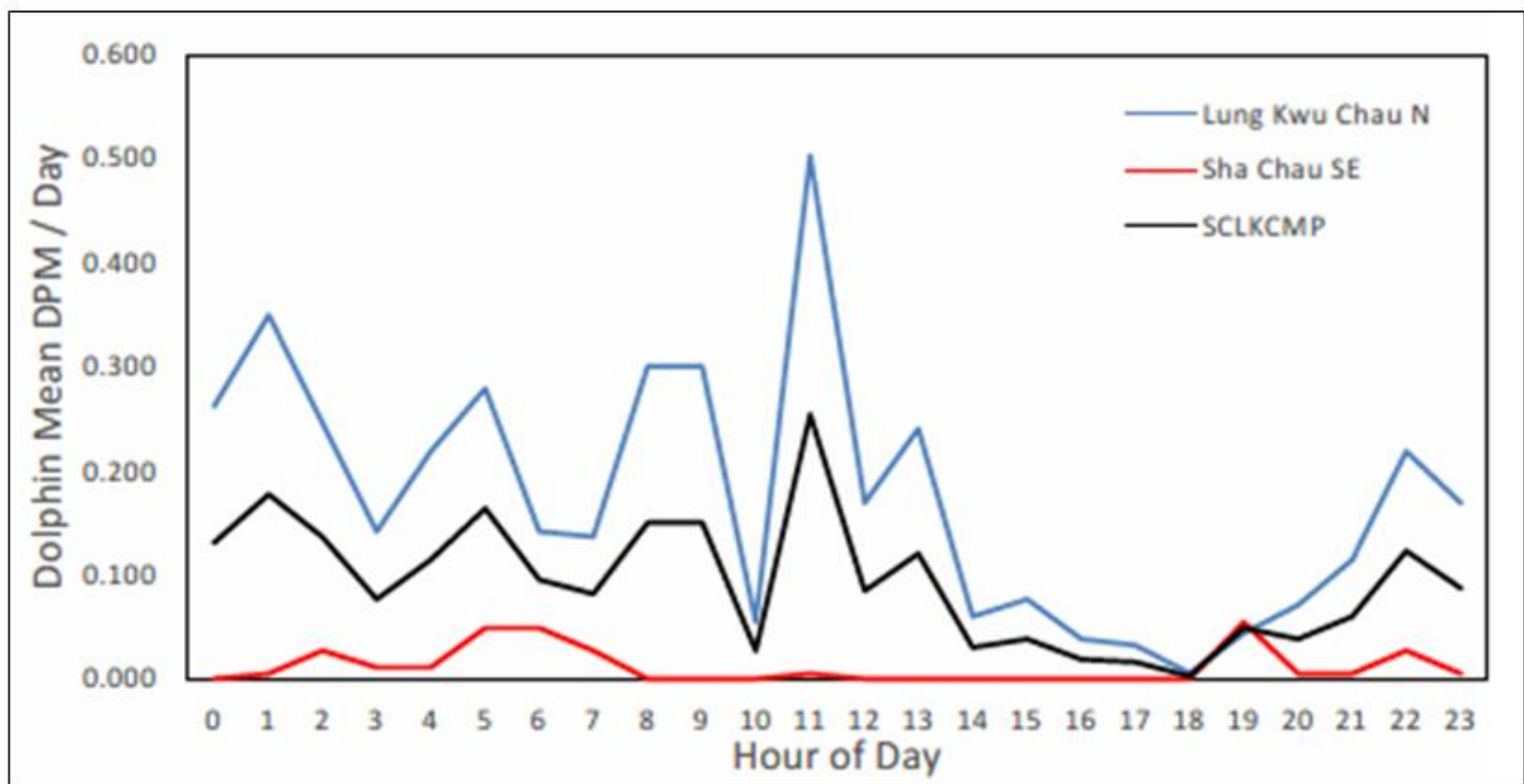


Figure 40. Dolphin DPM diel pattern at SCLKCMP (a) from January 1st to December 31st, 2024; (b) in the wet season (April to September, 2024); and c) in the dry season (October to March, 2024)

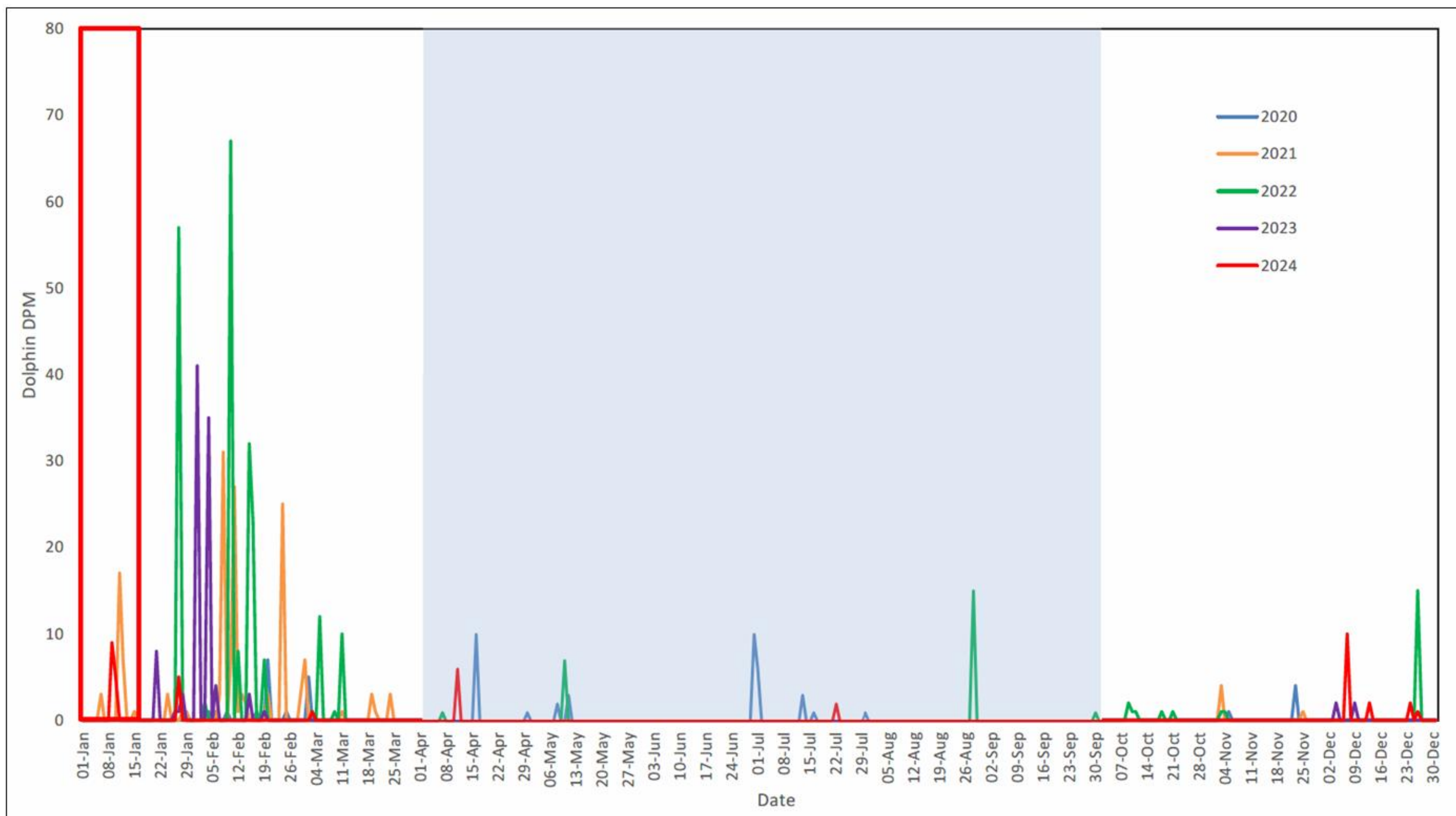


Figure 41. Total dolphin DPM for each day at Siu Ho Wan in 2020, 2021, 2022, 2023 and 2024 (from January 1st to December 31st for each year). The shaded box represents the wet season and the red outlined box represents dates for which data were not available for at least one of the years.

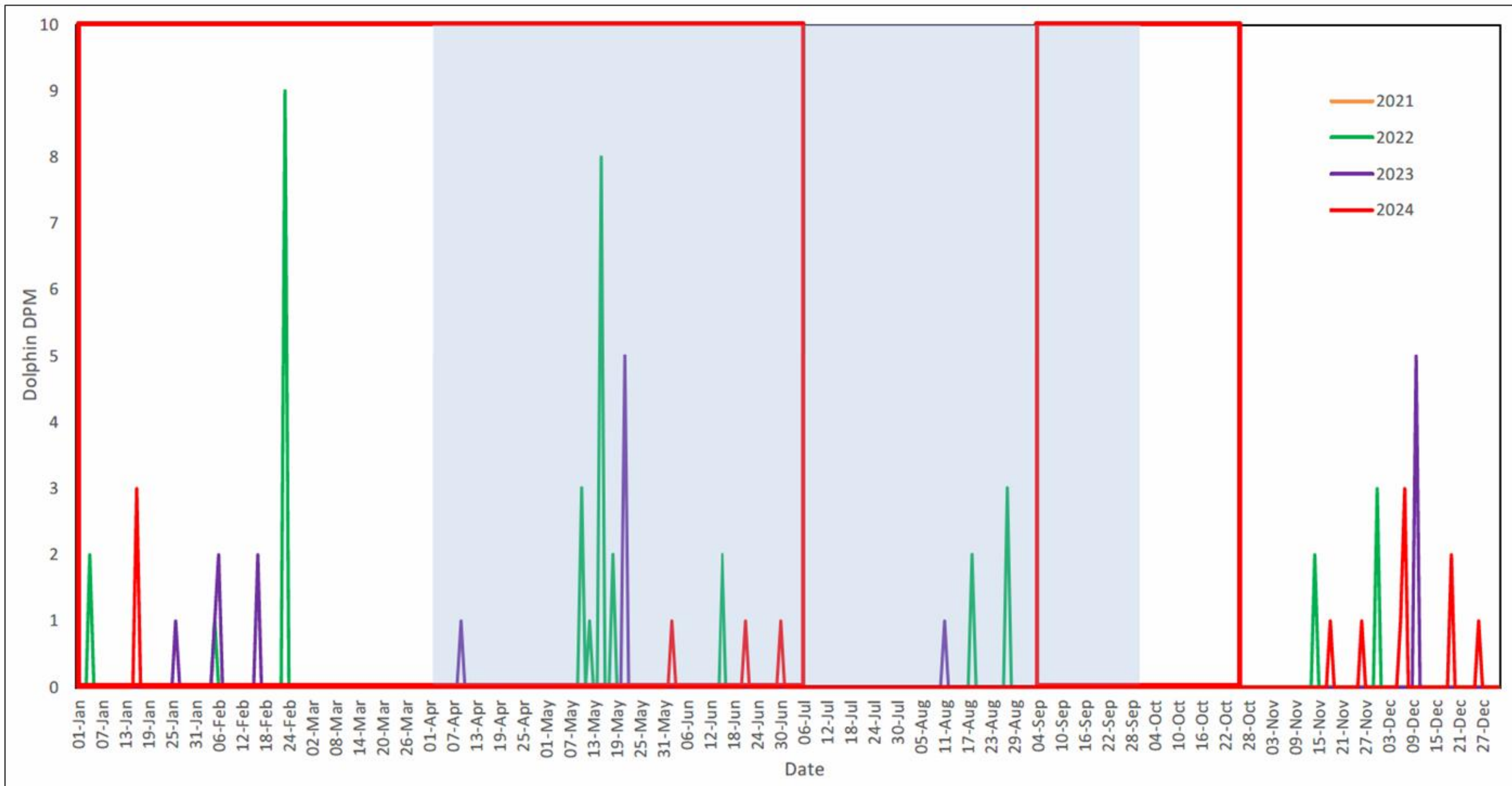


Figure 42. Total dolphin DPM for each day at Tai Mo To in 2021, 2022, 2023 and 2024 (from January 1st to December 31st for each year). The shaded box represents the wet season and the red outlined box represents dates for which data were not available for at least one of the years.

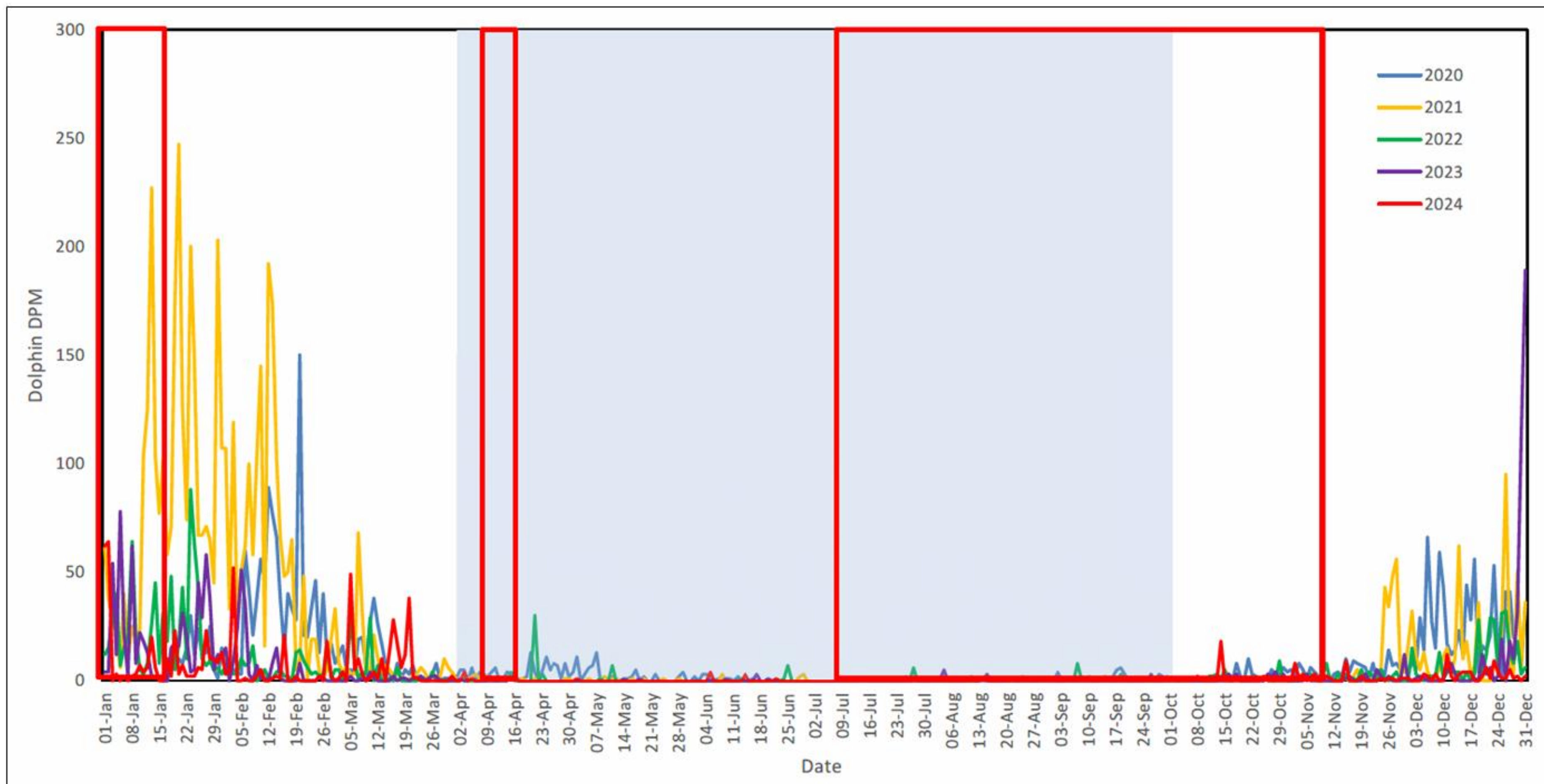


Figure 43. Total dolphin DPM for each day at Lung Kwu Chau N in 2020, 2021, 2022, 2023 and 2024 (from January 1st to December 31st for each year). The shaded box represents the wet season and the red outlined box represents dates for which data were not available for at least one of the years.

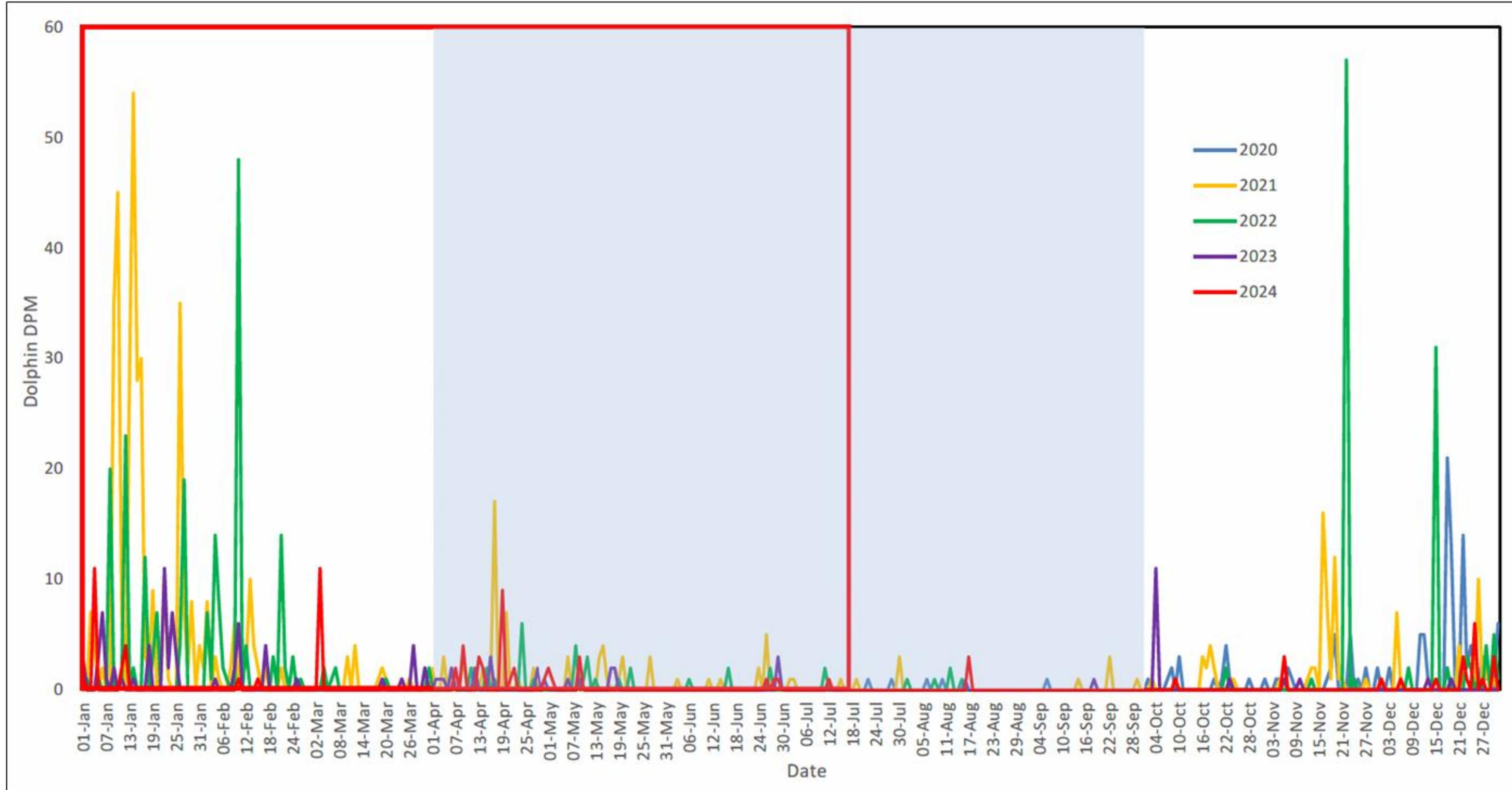


Figure 44. Total dolphin DPM for each day at Sha Chau SE in 2020, 2021, 2022, 2023 and 2024 (from January 1st to December 31st for each year). The shaded box represents the wet season and the red outlined box represents dates for which data were not available for at least one of the years.

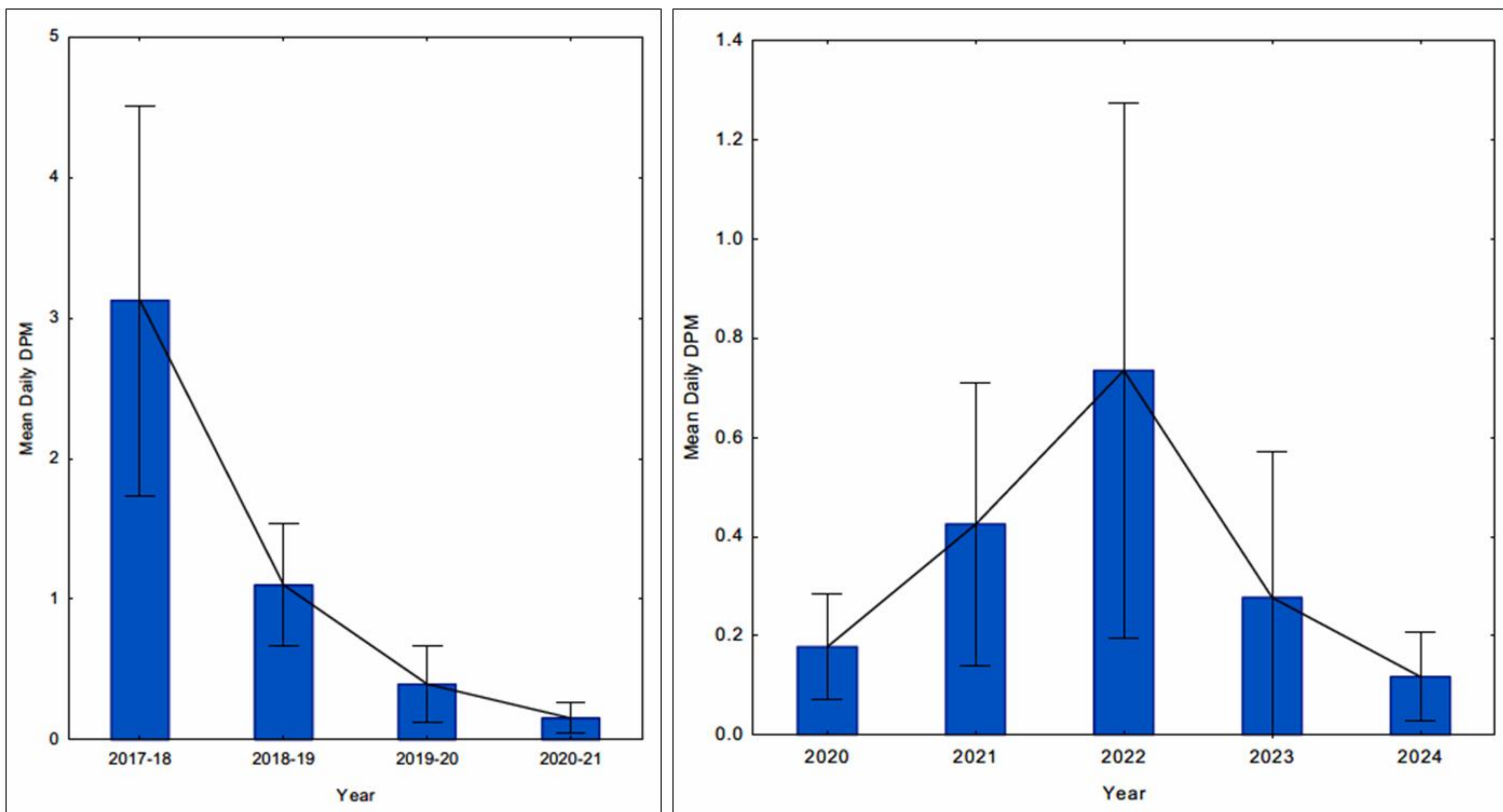


Figure 45. Dolphin mean daily DPMs at Siu Ho Wan as calculated for C-POD data (left) from 2017-18 to 2020-21 and F-POD data from 2020-2024 (right). Error bars represent 95% confidence intervals. Note: the y-axes scales are different.

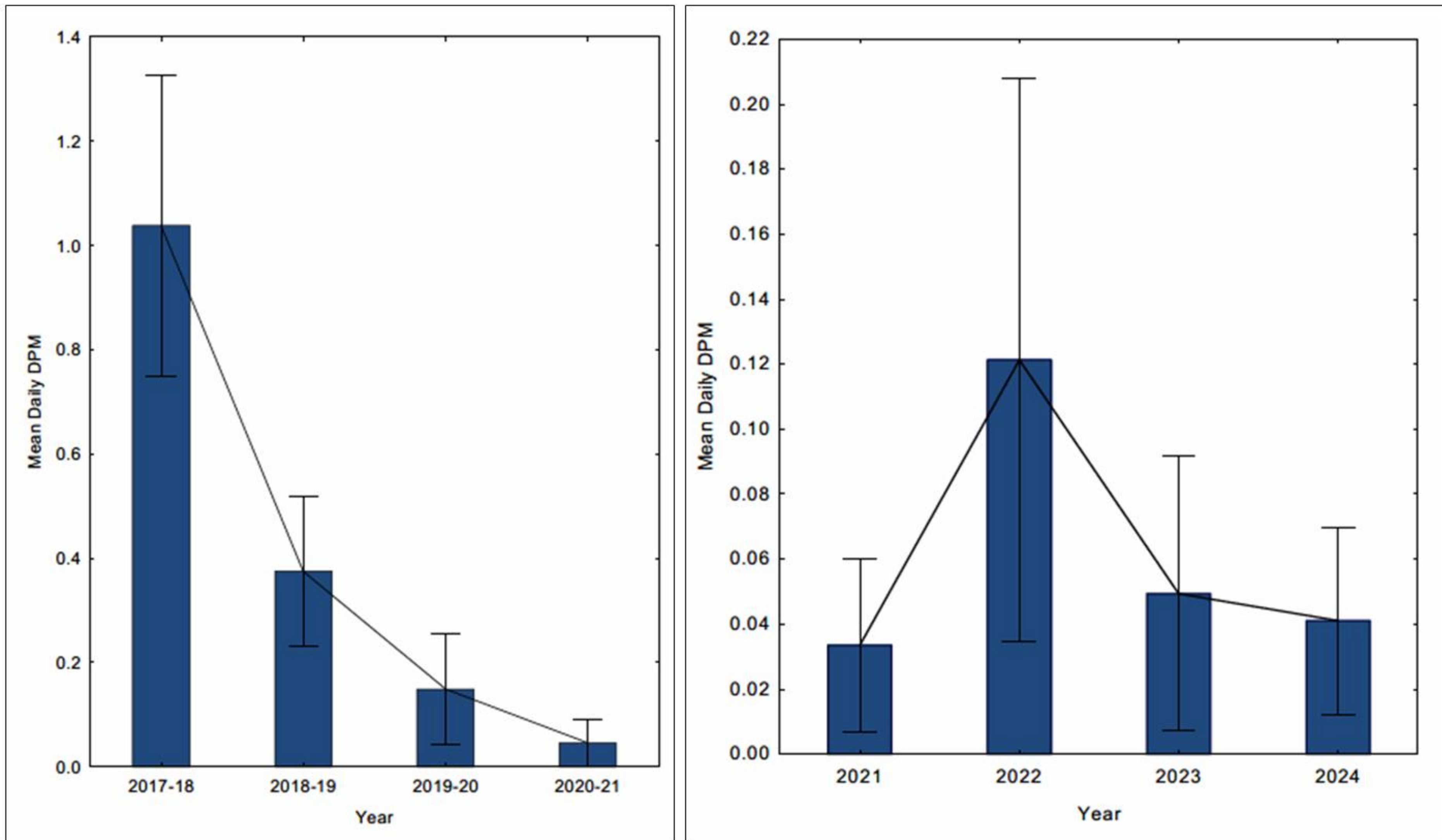


Figure 46. Dolphin mean daily DPMs at Tai Mo To as calculated for C-POD data (left) from 2017-18 to 2020-21 and F-POD data from 2021-2024 (right). Error bars represent 95% confidence intervals. Note: the y-axes scales are different.

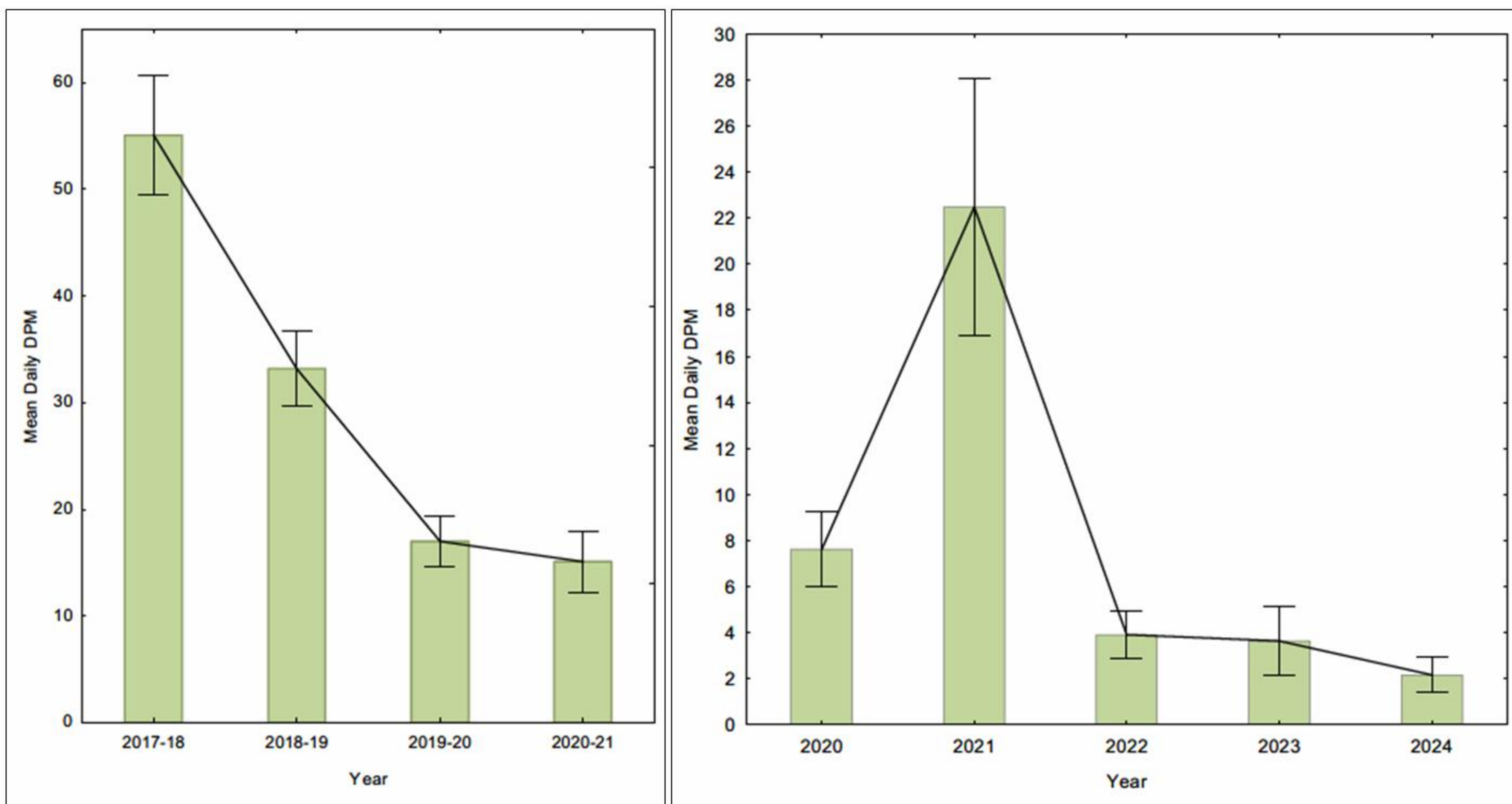


Figure 47. Dolphin mean daily DPMs at Lung Kwu Chau N as calculated for C-POD data (left) from 2017-18 to 2020-21 and F-POD data from 2020-2024 (right). Error bars represent 95% confidence intervals. Note: the y-axes scales are different.

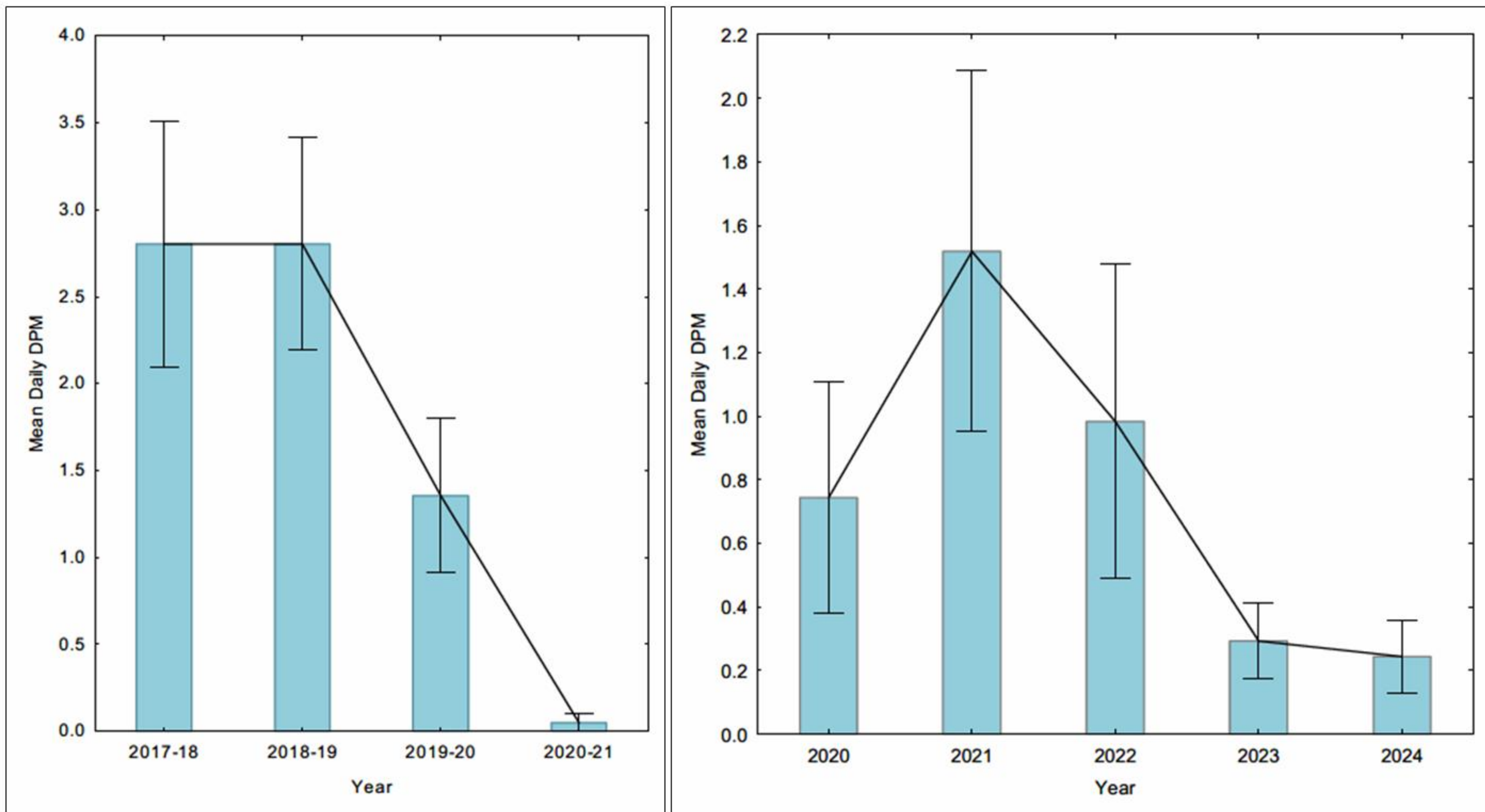


Figure 48. Dolphin mean daily DPMs at Sha Chau SE as calculated for C-POD data (left) from 2017-18 to 2020-21 and F-POD data from 2020-2024 (right). Error bars represent 95% confidence intervals. Note: the y-axes scales are different.

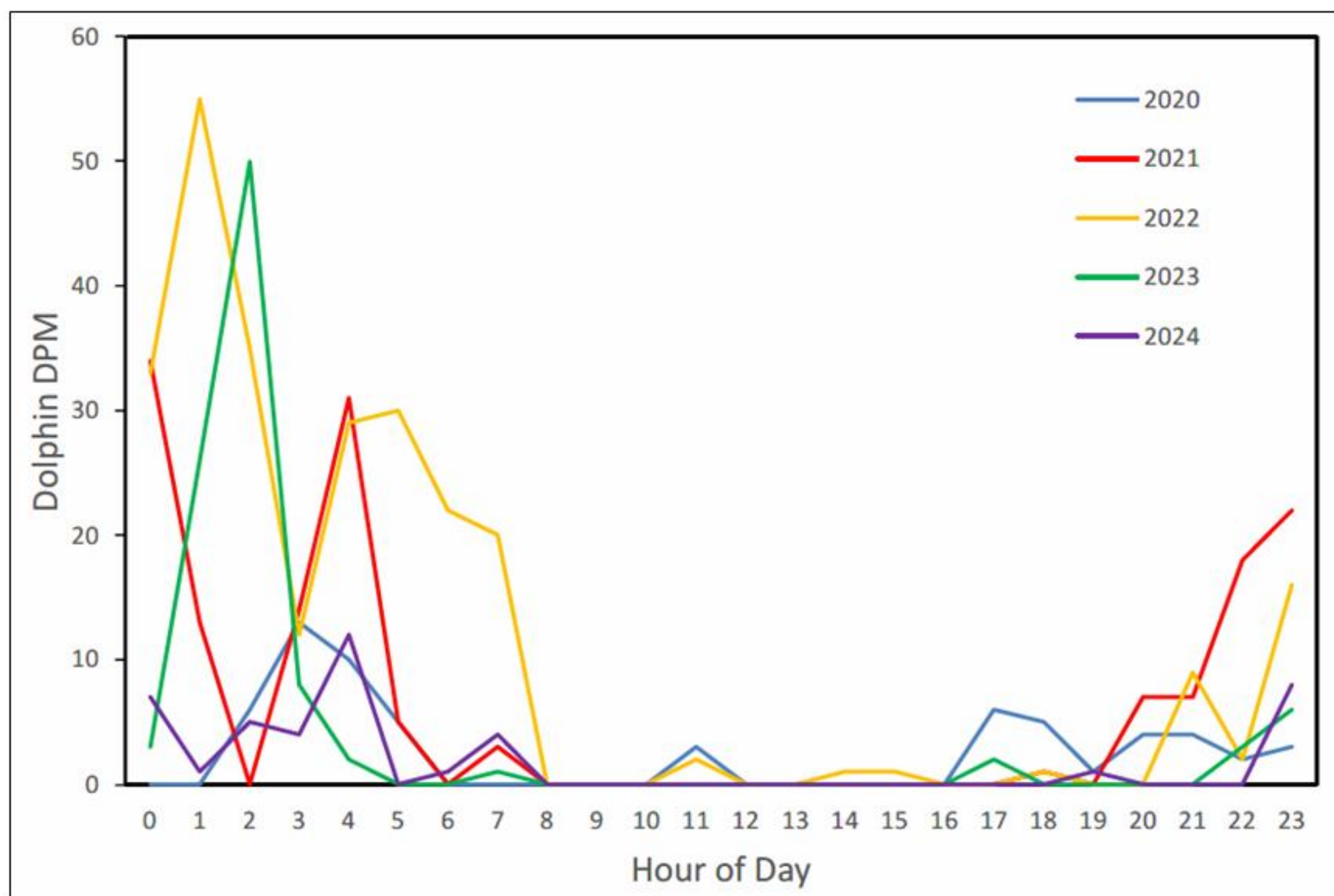


Figure 49a. Dolphin DPM for each hour of day at Siu Ho Wan (2020-24)

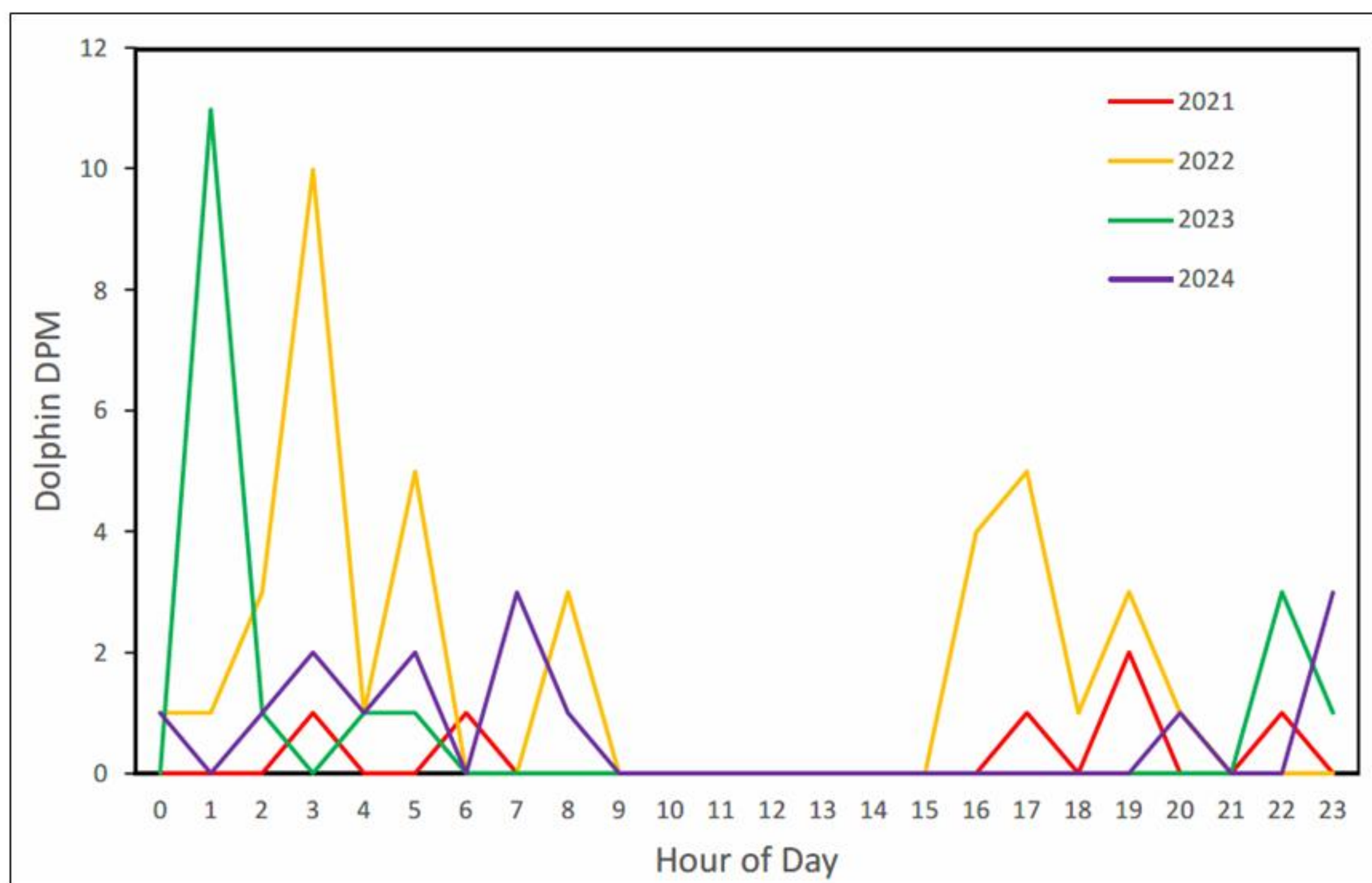


Figure 49b. Dolphin DPM for each hour of day at Tai Mo To (2021-24)

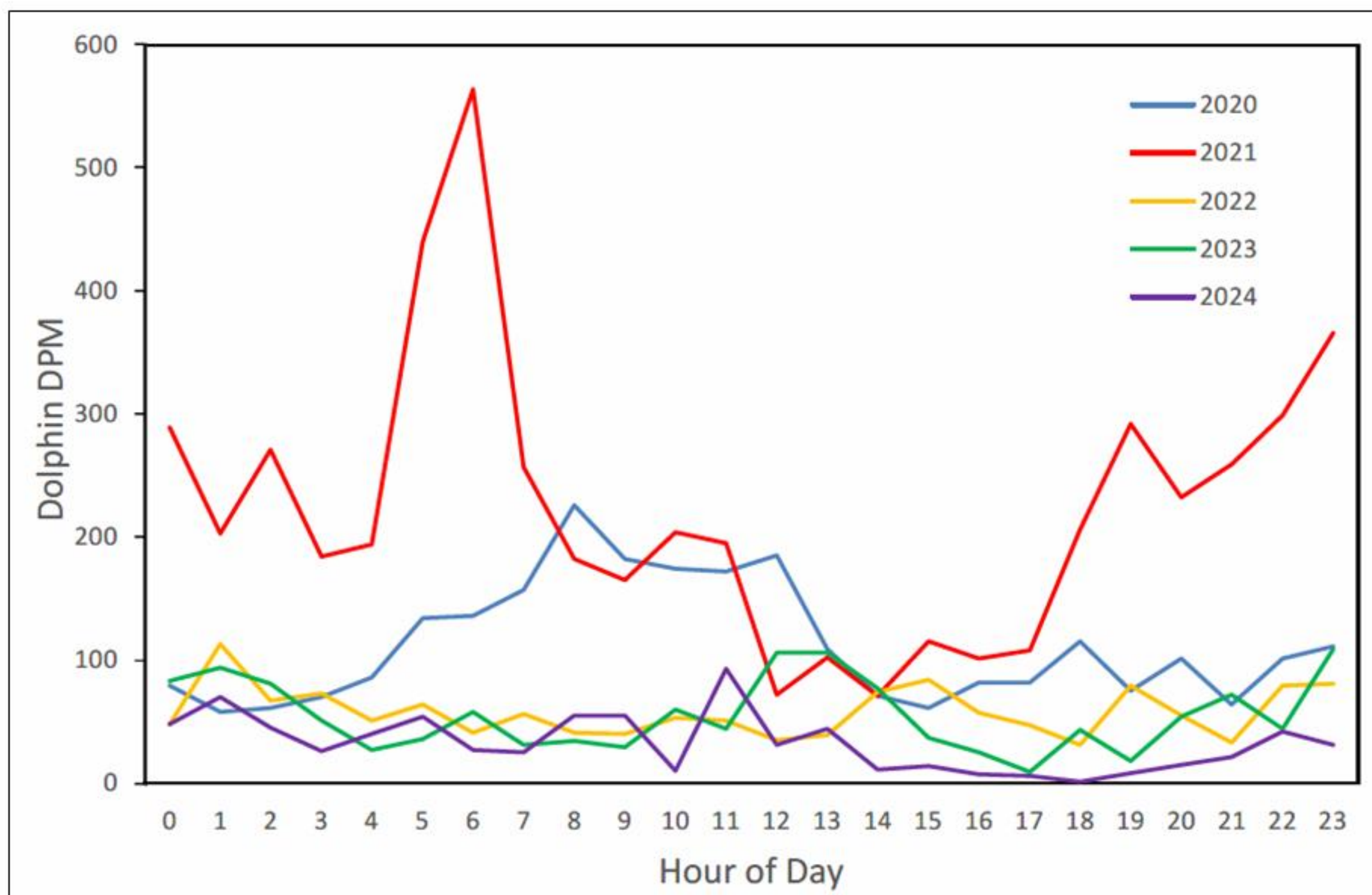


Figure 50a. Dolphin DPM for each hour of day at Lung Kwu Chau N (2020-24)

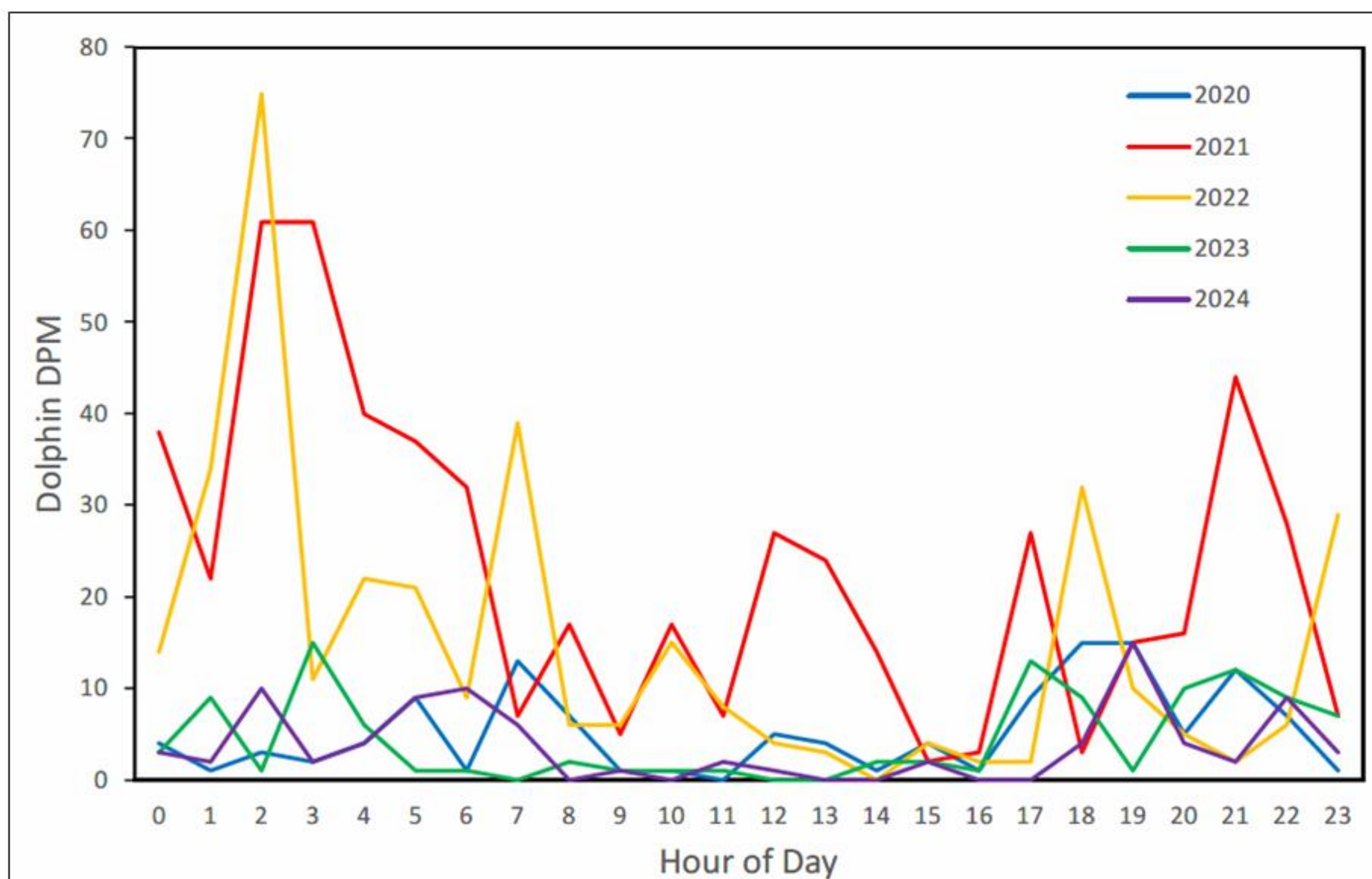


Figure 50b. Dolphin DPM for each hour of day at Sha Chau SE (2020-24)

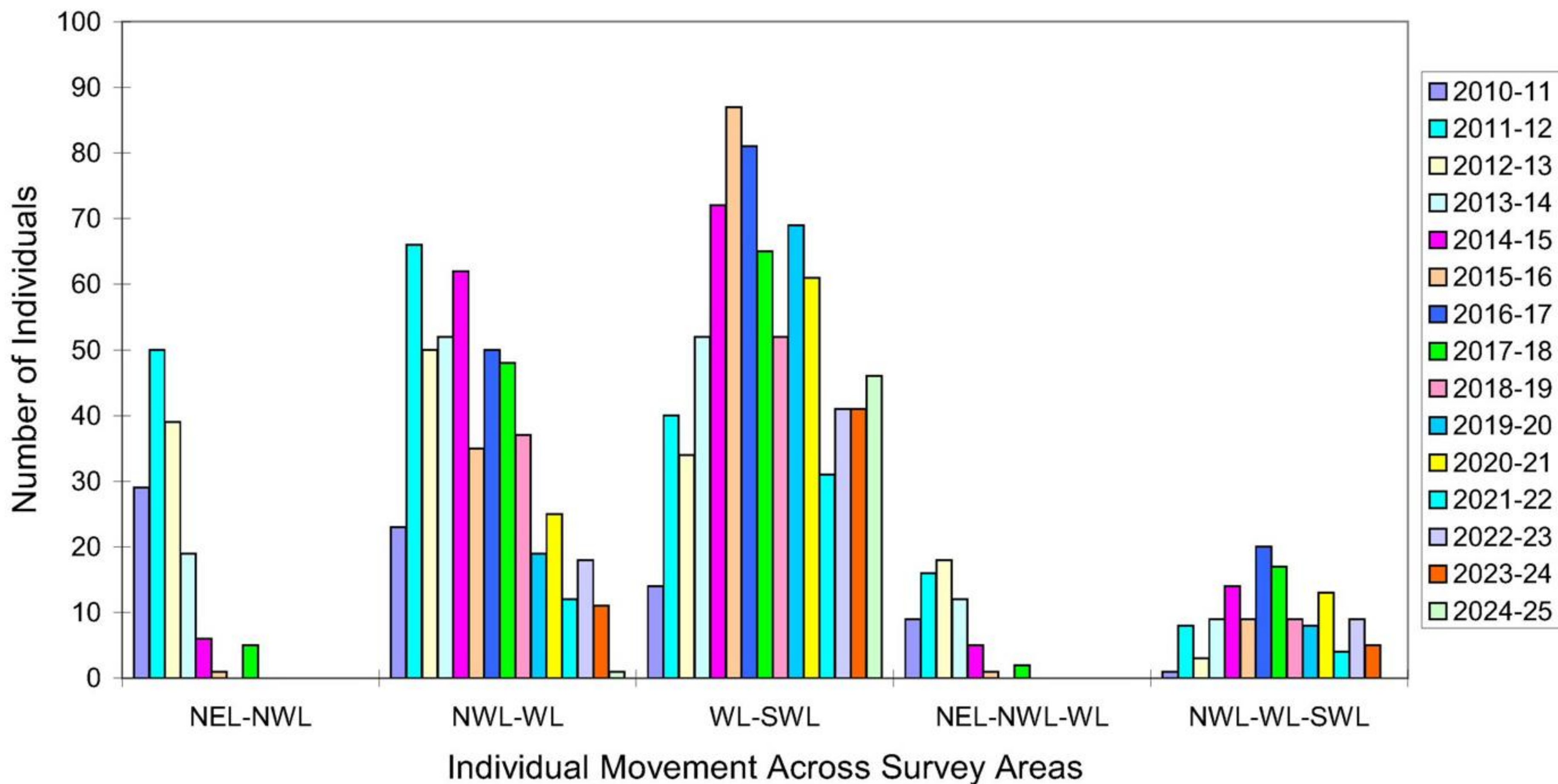


Figure 51. Temporal trends in number of individual dolphins involved in movements across different survey areas around Lantau in the past 15 monitoring periods

Appendix I. HKCRP-AFCD Survey Effort Database (April 2024 - March 2025)

(Note: P = Primary Line Effort where survey effort being conducted along parallel transect lines perpendicular to the coastlines; S = Secondary Line Effort where survey effort being conducted along the connecting lines between parallel lines)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
18-Apr-24	NE LANTAU	2	28.89	SPRING	STANDARD25686	P
18-Apr-24	NE LANTAU	3	2.50	SPRING	STANDARD25686	P
18-Apr-24	NE LANTAU	2	9.21	SPRING	STANDARD25686	S
18-Apr-24	NE LANTAU	3	1.10	SPRING	STANDARD25686	S
24-Apr-24	W LANTAU	1	4.35	SPRING	STANDARD36826	P
24-Apr-24	W LANTAU	2	7.90	SPRING	STANDARD36826	P
24-Apr-24	W LANTAU	1	4.66	SPRING	STANDARD36826	S
24-Apr-24	W LANTAU	2	5.24	SPRING	STANDARD36826	S
24-Apr-24	SW LANTAU	1	5.64	SPRING	STANDARD36826	P
24-Apr-24	SW LANTAU	2	7.78	SPRING	STANDARD36826	P
24-Apr-24	SW LANTAU	1	6.13	SPRING	STANDARD36826	S
24-Apr-24	SW LANTAU	2	4.75	SPRING	STANDARD36826	S
24-Apr-24	SE LANTAU	1	2.50	SPRING	STANDARD36826	P
24-Apr-24	SE LANTAU	2	11.65	SPRING	STANDARD36826	P
24-Apr-24	SE LANTAU	2	5.34	SPRING	STANDARD36826	S
10-May-24	DEEP BAY	2	3.97	SPRING	STANDARD25686	P
10-May-24	DEEP BAY	3	5.66	SPRING	STANDARD25686	P
10-May-24	DEEP BAY	2	5.20	SPRING	STANDARD25686	S
10-May-24	DEEP BAY	3	1.87	SPRING	STANDARD25686	S
10-May-24	NE LANTAU	2	1.40	SPRING	STANDARD25686	P
10-May-24	NE LANTAU	3	28.03	SPRING	STANDARD25686	P
10-May-24	NE LANTAU	3	11.57	SPRING	STANDARD25686	S
13-May-24	SE LANTAU	2	29.89	SPRING	STANDARD25686	P
13-May-24	SE LANTAU	2	7.11	SPRING	STANDARD25686	S
13-May-24	SW LANTAU	1	5.74	SPRING	STANDARD25686	P
13-May-24	SW LANTAU	2	18.62	SPRING	STANDARD25686	P
13-May-24	SW LANTAU	1	2.23	SPRING	STANDARD25686	S
13-May-24	SW LANTAU	2	6.29	SPRING	STANDARD25686	S
16-May-24	W LANTAU	2	2.30	SPRING	STANDARD25686	P
16-May-24	W LANTAU	3	10.74	SPRING	STANDARD25686	P
16-May-24	W LANTAU	4	1.39	SPRING	STANDARD25686	P
16-May-24	W LANTAU	2	3.30	SPRING	STANDARD25686	S
16-May-24	W LANTAU	3	11.43	SPRING	STANDARD25686	S
16-May-24	W LANTAU	4	0.83	SPRING	STANDARD25686	S
16-May-24	NW LANTAU	3	21.30	SPRING	STANDARD25686	P
16-May-24	NW LANTAU	4	11.50	SPRING	STANDARD25686	P
16-May-24	NW LANTAU	3	10.70	SPRING	STANDARD25686	S
16-May-24	NW LANTAU	4	2.70	SPRING	STANDARD25686	S
24-May-24	SE LANTAU	2	16.65	SPRING	STANDARD25686	P
24-May-24	SE LANTAU	3	4.80	SPRING	STANDARD25686	P
24-May-24	SE LANTAU	2	8.48	SPRING	STANDARD25686	S
24-May-24	SE LANTAU	3	2.38	SPRING	STANDARD25686	S
24-May-24	SW LANTAU	2	10.72	SPRING	STANDARD25686	P
24-May-24	SW LANTAU	3	1.31	SPRING	STANDARD25686	P
24-May-24	SW LANTAU	2	6.60	SPRING	STANDARD25686	S
24-May-24	SW LANTAU	3	2.52	SPRING	STANDARD25686	S
28-May-24	W LANTAU	1	1.59	SPRING	STANDARD25686	S
28-May-24	W LANTAU	2	7.10	SPRING	STANDARD25686	S
28-May-24	SW LANTAU	2	13.49	SPRING	STANDARD25686	P
28-May-24	SW LANTAU	3	11.22	SPRING	STANDARD25686	P
28-May-24	SW LANTAU	2	4.60	SPRING	STANDARD25686	S
28-May-24	SW LANTAU	3	3.30	SPRING	STANDARD25686	S
30-May-24	W LANTAU	2	4.80	SPRING	STANDARD25686	P
30-May-24	W LANTAU	3	16.36	SPRING	STANDARD25686	P
30-May-24	W LANTAU	2	2.20	SPRING	STANDARD25686	S

Appendix I. (cont'd.)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
30-May-24	W LANTAU	3	18.20	SPRING	STANDARD25686	S
30-May-24	NW LANTAU	2	16.70	SPRING	STANDARD25686	P
30-May-24	NW LANTAU	3	13.94	SPRING	STANDARD25686	P
30-May-24	NW LANTAU	2	6.50	SPRING	STANDARD25686	S
30-May-24	NW LANTAU	3	5.46	SPRING	STANDARD25686	S
6-Jun-24	SE LANTAU	2	26.69	SUMMER	STANDARD25686	P
6-Jun-24	SE LANTAU	3	3.20	SUMMER	STANDARD25686	P
6-Jun-24	SE LANTAU	2	8.49	SUMMER	STANDARD25686	S
6-Jun-24	SW LANTAU	2	19.27	SUMMER	STANDARD25686	P
6-Jun-24	SW LANTAU	3	8.16	SUMMER	STANDARD25686	P
6-Jun-24	SW LANTAU	2	5.31	SUMMER	STANDARD25686	S
6-Jun-24	SW LANTAU	3	4.62	SUMMER	STANDARD25686	S
7-Jun-24	W LANTAU	2	8.12	SUMMER	STANDARD25686	P
7-Jun-24	W LANTAU	3	9.33	SUMMER	STANDARD25686	P
7-Jun-24	W LANTAU	2	6.02	SUMMER	STANDARD25686	S
7-Jun-24	W LANTAU	3	8.01	SUMMER	STANDARD25686	S
7-Jun-24	NE LANTAU	2	5.79	SUMMER	STANDARD25686	P
7-Jun-24	NE LANTAU	3	2.98	SUMMER	STANDARD25686	P
7-Jun-24	NE LANTAU	3	5.53	SUMMER	STANDARD25686	S
11-Jun-24	SE LANTAU	2	13.24	SUMMER	STANDARD25686	P
11-Jun-24	SE LANTAU	3	14.76	SUMMER	STANDARD25686	P
11-Jun-24	SE LANTAU	2	4.90	SUMMER	STANDARD25686	S
11-Jun-24	SE LANTAU	3	5.40	SUMMER	STANDARD25686	S
11-Jun-24	SW LANTAU	2	11.95	SUMMER	STANDARD25686	P
11-Jun-24	SW LANTAU	3	8.99	SUMMER	STANDARD25686	P
11-Jun-24	SW LANTAU	2	8.06	SUMMER	STANDARD25686	S
11-Jun-24	SW LANTAU	3	5.28	SUMMER	STANDARD25686	S
12-Jun-24	W LANTAU	2	5.69	SUMMER	STANDARD25686	P
12-Jun-24	W LANTAU	3	14.99	SUMMER	STANDARD25686	P
12-Jun-24	W LANTAU	3	11.49	SUMMER	STANDARD25686	S
12-Jun-24	SW LANTAU	2	6.47	SUMMER	STANDARD25686	P
12-Jun-24	SW LANTAU	3	12.39	SUMMER	STANDARD25686	P
12-Jun-24	SW LANTAU	2	2.68	SUMMER	STANDARD25686	S
12-Jun-24	SW LANTAU	3	5.58	SUMMER	STANDARD25686	S
17-Jun-24	W LANTAU	2	0.75	SUMMER	STANDARD25686	P
17-Jun-24	W LANTAU	3	15.56	SUMMER	STANDARD25686	P
17-Jun-24	W LANTAU	2	1.40	SUMMER	STANDARD25686	S
17-Jun-24	W LANTAU	3	8.91	SUMMER	STANDARD25686	S
18-Jun-24	W LANTAU	2	0.71	SUMMER	STANDARD36826	P
18-Jun-24	W LANTAU	3	17.82	SUMMER	STANDARD36826	P
18-Jun-24	W LANTAU	4	1.40	SUMMER	STANDARD36826	P
18-Jun-24	W LANTAU	2	1.12	SUMMER	STANDARD36826	S
18-Jun-24	W LANTAU	3	8.47	SUMMER	STANDARD36826	S
18-Jun-24	W LANTAU	4	1.20	SUMMER	STANDARD36826	S
18-Jun-24	SW LANTAU	2	3.80	SUMMER	STANDARD36826	P
18-Jun-24	SW LANTAU	3	14.14	SUMMER	STANDARD36826	P
18-Jun-24	SW LANTAU	2	4.24	SUMMER	STANDARD36826	S
18-Jun-24	SW LANTAU	3	8.06	SUMMER	STANDARD36826	S
18-Jun-24	SE LANTAU	2	9.40	SUMMER	STANDARD36826	P
18-Jun-24	SE LANTAU	2	3.60	SUMMER	STANDARD36826	S
28-Jun-24	NW LANTAU	2	28.23	SUMMER	STANDARD25686	P
28-Jun-24	NW LANTAU	3	6.20	SUMMER	STANDARD25686	P
28-Jun-24	NW LANTAU	2	3.57	SUMMER	STANDARD25686	S
28-Jun-24	NW LANTAU	3	6.60	SUMMER	STANDARD25686	S
28-Jun-24	DEEP BAY	2	4.00	SUMMER	STANDARD25686	P
28-Jun-24	DEEP BAY	3	2.60	SUMMER	STANDARD25686	P
28-Jun-24	DEEP BAY	2	5.27	SUMMER	STANDARD25686	S
28-Jun-24	DEEP BAY	3	3.13	SUMMER	STANDARD25686	S

Appendix I. (cont'd.)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
28-Jun-24	NE LANTAU	2	11.12	SUMMER	STANDARD25686	P
28-Jun-24	NE LANTAU	3	9.24	SUMMER	STANDARD25686	P
28-Jun-24	NE LANTAU	2	7.00	SUMMER	STANDARD25686	S
28-Jun-24	NE LANTAU	3	3.54	SUMMER	STANDARD25686	S
11-Jul-24	SE LANTAU	2	29.97	SUMMER	STANDARD25686	P
11-Jul-24	SE LANTAU	2	8.93	SUMMER	STANDARD25686	S
11-Jul-24	SW LANTAU	2	20.27	SUMMER	STANDARD25686	P
11-Jul-24	SW LANTAU	3	5.60	SUMMER	STANDARD25686	P
11-Jul-24	SW LANTAU	2	7.30	SUMMER	STANDARD25686	S
11-Jul-24	SW LANTAU	3	2.43	SUMMER	STANDARD25686	S
12-Jul-24	W LANTAU	2	1.67	SUMMER	STANDARD25686	S
12-Jul-24	W LANTAU	3	6.67	SUMMER	STANDARD25686	S
12-Jul-24	SW LANTAU	2	4.28	SUMMER	STANDARD25686	P
12-Jul-24	SW LANTAU	3	2.91	SUMMER	STANDARD25686	P
12-Jul-24	SW LANTAU	2	7.69	SUMMER	STANDARD25686	S
12-Jul-24	SW LANTAU	3	0.11	SUMMER	STANDARD25686	S
23-Jul-24	SE LANTAU	1	1.80	SUMMER	STANDARD36826	P
23-Jul-24	SE LANTAU	2	25.37	SUMMER	STANDARD36826	P
23-Jul-24	SE LANTAU	2	10.83	SUMMER	STANDARD36826	S
23-Jul-24	SW LANTAU	2	13.54	SUMMER	STANDARD36826	P
23-Jul-24	SW LANTAU	2	18.46	SUMMER	STANDARD36826	S
30-Jul-24	W LANTAU	2	5.02	SUMMER	STANDARD36826	P
30-Jul-24	W LANTAU	3	11.94	SUMMER	STANDARD36826	P
30-Jul-24	W LANTAU	2	3.15	SUMMER	STANDARD36826	S
30-Jul-24	W LANTAU	3	8.32	SUMMER	STANDARD36826	S
5-Aug-24	SE LANTAU	2	24.49	SUMMER	STANDARD25686	P
5-Aug-24	SE LANTAU	3	5.20	SUMMER	STANDARD25686	P
5-Aug-24	SE LANTAU	2	9.50	SUMMER	STANDARD25686	S
5-Aug-24	SW LANTAU	2	13.34	SUMMER	STANDARD25686	P
5-Aug-24	SW LANTAU	3	3.10	SUMMER	STANDARD25686	P
5-Aug-24	SW LANTAU	2	5.89	SUMMER	STANDARD25686	S
7-Aug-24	W LANTAU	2	1.71	SUMMER	STANDARD25686	P
7-Aug-24	W LANTAU	3	5.87	SUMMER	STANDARD25686	P
7-Aug-24	W LANTAU	2	3.02	SUMMER	STANDARD25686	S
7-Aug-24	W LANTAU	3	5.22	SUMMER	STANDARD25686	S
7-Aug-24	SW LANTAU	2	5.60	SUMMER	STANDARD25686	P
7-Aug-24	SW LANTAU	2	6.99	SUMMER	STANDARD25686	S
8-Aug-24	W LANTAU	2	0.77	SUMMER	STANDARD25686	P
8-Aug-24	W LANTAU	3	16.21	SUMMER	STANDARD25686	P
8-Aug-24	W LANTAU	2	0.64	SUMMER	STANDARD25686	S
8-Aug-24	W LANTAU	3	9.26	SUMMER	STANDARD25686	S
8-Aug-24	SW LANTAU	2	7.66	SUMMER	STANDARD25686	P
8-Aug-24	SW LANTAU	3	7.35	SUMMER	STANDARD25686	P
8-Aug-24	SW LANTAU	2	1.20	SUMMER	STANDARD25686	S
8-Aug-24	SW LANTAU	3	7.09	SUMMER	STANDARD25686	S
12-Aug-24	W LANTAU	2	8.62	SUMMER	STANDARD25686	P
12-Aug-24	W LANTAU	3	7.28	SUMMER	STANDARD25686	P
12-Aug-24	W LANTAU	2	4.76	SUMMER	STANDARD25686	S
12-Aug-24	W LANTAU	3	4.39	SUMMER	STANDARD25686	S
13-Aug-24	SE LANTAU	1	12.81	SUMMER	STANDARD36826	P
13-Aug-24	SE LANTAU	2	11.53	SUMMER	STANDARD36826	P
13-Aug-24	SE LANTAU	1	3.70	SUMMER	STANDARD36826	S
13-Aug-24	SE LANTAU	2	7.96	SUMMER	STANDARD36826	S
13-Aug-24	SW LANTAU	2	11.34	SUMMER	STANDARD36826	P
13-Aug-24	SW LANTAU	3	10.00	SUMMER	STANDARD36826	P
13-Aug-24	SW LANTAU	2	11.06	SUMMER	STANDARD36826	S

Appendix I. (cont'd.)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
13-Aug-24	SW LANTAU	3	2.50	SUMMER	STANDARD36826	S
22-Aug-24	NW LANTAU	2	22.82	SUMMER	STANDARD25686	P
22-Aug-24	NW LANTAU	2	7.98	SUMMER	STANDARD25686	S
22-Aug-24	NW LANTAU	3	2.90	SUMMER	STANDARD25686	S
22-Aug-24	W LANTAU	2	1.01	SUMMER	STANDARD25686	P
22-Aug-24	W LANTAU	3	17.72	SUMMER	STANDARD25686	P
22-Aug-24	W LANTAU	2	2.44	SUMMER	STANDARD25686	S
22-Aug-24	W LANTAU	3	7.73	SUMMER	STANDARD25686	S
26-Aug-24	W LANTAU	2	13.95	SUMMER	STANDARD25686	P
26-Aug-24	W LANTAU	2	10.01	SUMMER	STANDARD25686	S
26-Aug-24	SW LANTAU	2	25.17	SUMMER	STANDARD25686	P
26-Aug-24	SW LANTAU	3	1.50	SUMMER	STANDARD25686	P
26-Aug-24	SW LANTAU	2	10.37	SUMMER	STANDARD25686	S
26-Aug-24	SW LANTAU	3	0.31	SUMMER	STANDARD25686	S
10-Sep-24	DEEP BAY	2	8.62	AUTUMN	STANDARD36826	P
10-Sep-24	DEEP BAY	2	6.48	AUTUMN	STANDARD36826	S
10-Sep-24	NE LANTAU	2	35.86	AUTUMN	STANDARD36826	P
10-Sep-24	NE LANTAU	2	11.74	AUTUMN	STANDARD36826	S
11-Sep-24	SE LANTAU	1	1.67	AUTUMN	STANDARD25686	P
11-Sep-24	SE LANTAU	2	27.72	AUTUMN	STANDARD25686	P
11-Sep-24	SE LANTAU	2	9.21	AUTUMN	STANDARD25686	S
11-Sep-24	SW LANTAU	2	24.90	AUTUMN	STANDARD25686	P
11-Sep-24	SW LANTAU	2	8.47	AUTUMN	STANDARD25686	S
16-Sep-24	W LANTAU	1	0.23	AUTUMN	STANDARD25686	P
16-Sep-24	W LANTAU	2	2.10	AUTUMN	STANDARD25686	P
16-Sep-24	W LANTAU	2	6.63	AUTUMN	STANDARD25686	S
16-Sep-24	W LANTAU	3	1.71	AUTUMN	STANDARD25686	S
16-Sep-24	SW LANTAU	2	12.05	AUTUMN	STANDARD25686	P
16-Sep-24	SW LANTAU	3	1.80	AUTUMN	STANDARD25686	P
16-Sep-24	SW LANTAU	2	5.61	AUTUMN	STANDARD25686	S
16-Sep-24	SW LANTAU	3	3.60	AUTUMN	STANDARD25686	S
26-Sep-24	SE LANTAU	2	24.76	AUTUMN	STANDARD36826	P
26-Sep-24	SE LANTAU	3	2.17	AUTUMN	STANDARD36826	P
26-Sep-24	SE LANTAU	2	9.39	AUTUMN	STANDARD36826	S
26-Sep-24	SE LANTAU	3	1.27	AUTUMN	STANDARD36826	S
26-Sep-24	SW LANTAU	2	15.68	AUTUMN	STANDARD36826	P
26-Sep-24	SW LANTAU	3	2.81	AUTUMN	STANDARD36826	P
26-Sep-24	SW LANTAU	2	10.69	AUTUMN	STANDARD36826	S
26-Sep-24	SW LANTAU	3	2.68	AUTUMN	STANDARD36826	S
30-Sep-24	NW LANTAU	2	19.90	AUTUMN	STANDARD138716	P
30-Sep-24	NW LANTAU	3	11.50	AUTUMN	STANDARD138716	P
30-Sep-24	NW LANTAU	2	5.43	AUTUMN	STANDARD138716	S
30-Sep-24	NW LANTAU	3	8.87	AUTUMN	STANDARD138716	S
30-Sep-24	W LANTAU	2	8.25	AUTUMN	STANDARD138716	P
30-Sep-24	W LANTAU	3	2.19	AUTUMN	STANDARD138716	P
30-Sep-24	W LANTAU	2	10.60	AUTUMN	STANDARD138716	S
7-Oct-24	SE LANTAU	1	2.20	AUTUMN	STANDARD25686	P
7-Oct-24	SE LANTAU	2	27.30	AUTUMN	STANDARD25686	P
7-Oct-24	SE LANTAU	2	8.08	AUTUMN	STANDARD25686	S
7-Oct-24	SW LANTAU	2	26.58	AUTUMN	STANDARD25686	P
7-Oct-24	SW LANTAU	2	10.00	AUTUMN	STANDARD25686	S
8-Oct-24	W LANTAU	3	16.16	AUTUMN	STANDARD25686	P
8-Oct-24	W LANTAU	2	1.98	AUTUMN	STANDARD25686	S
8-Oct-24	W LANTAU	3	8.66	AUTUMN	STANDARD25686	S
8-Oct-24	SW LANTAU	2	5.58	AUTUMN	STANDARD25686	P
8-Oct-24	SW LANTAU	2	6.17	AUTUMN	STANDARD25686	S
14-Oct-24	W LANTAU	2	3.63	AUTUMN	STANDARD25686	P
14-Oct-24	W LANTAU	3	4.56	AUTUMN	STANDARD25686	P
14-Oct-24	W LANTAU	2	6.24	AUTUMN	STANDARD25686	S
14-Oct-24	W LANTAU	3	2.76	AUTUMN	STANDARD25686	S
14-Oct-24	W LANTAU	4	1.19	AUTUMN	STANDARD25686	S

Appendix I. (cont'd.)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
15-Oct-24	NW LANTAU	2	16.64	AUTUMN	STANDARD25686	P
15-Oct-24	NW LANTAU	3	17.29	AUTUMN	STANDARD25686	P
15-Oct-24	NW LANTAU	2	3.89	AUTUMN	STANDARD25686	S
15-Oct-24	NW LANTAU	3	9.28	AUTUMN	STANDARD25686	S
15-Oct-24	DEEP BAY	2	7.28	AUTUMN	STANDARD25686	P
15-Oct-24	DEEP BAY	2	7.02	AUTUMN	STANDARD25686	S
15-Oct-24	NE LANTAU	2	11.20	AUTUMN	STANDARD25686	P
15-Oct-24	NE LANTAU	3	5.58	AUTUMN	STANDARD25686	P
15-Oct-24	NE LANTAU	2	12.12	AUTUMN	STANDARD25686	S
18-Oct-24	SE LANTAU	2	1.47	AUTUMN	STANDARD25686	P
18-Oct-24	SE LANTAU	3	10.93	AUTUMN	STANDARD25686	P
18-Oct-24	SE LANTAU	4	1.80	AUTUMN	STANDARD25686	P
18-Oct-24	SE LANTAU	2	4.13	AUTUMN	STANDARD25686	S
18-Oct-24	SE LANTAU	3	4.78	AUTUMN	STANDARD25686	S
18-Oct-24	SE LANTAU	4	2.89	AUTUMN	STANDARD25686	S
18-Oct-24	SW LANTAU	3	0.37	AUTUMN	STANDARD25686	P
18-Oct-24	SW LANTAU	4	0.16	AUTUMN	STANDARD25686	P
18-Oct-24	SW LANTAU	3	1.89	AUTUMN	STANDARD25686	S
18-Oct-24	SW LANTAU	4	4.16	AUTUMN	STANDARD25686	S
18-Oct-24	W LANTAU	2	6.88	AUTUMN	STANDARD25686	P
18-Oct-24	W LANTAU	3	2.52	AUTUMN	STANDARD25686	P
18-Oct-24	W LANTAU	4	3.26	AUTUMN	STANDARD25686	P
18-Oct-24	W LANTAU	2	2.05	AUTUMN	STANDARD25686	S
18-Oct-24	W LANTAU	3	8.24	AUTUMN	STANDARD25686	S
22-Oct-24	SE LANTAU	2	25.40	AUTUMN	STANDARD36826	P
22-Oct-24	SE LANTAU	3	0.57	AUTUMN	STANDARD36826	P
22-Oct-24	SE LANTAU	2	10.54	AUTUMN	STANDARD36826	S
22-Oct-24	SW LANTAU	2	4.40	AUTUMN	STANDARD36826	P
22-Oct-24	SW LANTAU	3	14.19	AUTUMN	STANDARD36826	P
22-Oct-24	SW LANTAU	2	2.94	AUTUMN	STANDARD36826	S
22-Oct-24	SW LANTAU	3	7.77	AUTUMN	STANDARD36826	S
22-Oct-24	SW LANTAU	4	2.30	AUTUMN	STANDARD36826	S
29-Oct-24	W LANTAU	2	4.50	AUTUMN	STANDARD25686	P
29-Oct-24	W LANTAU	3	8.92	AUTUMN	STANDARD25686	P
29-Oct-24	W LANTAU	2	0.36	AUTUMN	STANDARD25686	S
29-Oct-24	W LANTAU	3	10.38	AUTUMN	STANDARD25686	S
29-Oct-24	SW LANTAU	2	28.67	AUTUMN	STANDARD25686	P
29-Oct-24	SW LANTAU	3	2.04	AUTUMN	STANDARD25686	P
29-Oct-24	SW LANTAU	2	14.39	AUTUMN	STANDARD25686	S
8-Nov-24	W LANTAU	3	9.33	AUTUMN	STANDARD25686	S
8-Nov-24	SW LANTAU	2	13.64	AUTUMN	STANDARD25686	P
8-Nov-24	SW LANTAU	3	6.02	AUTUMN	STANDARD25686	P
8-Nov-24	SW LANTAU	2	2.95	AUTUMN	STANDARD25686	S
8-Nov-24	SW LANTAU	3	4.70	AUTUMN	STANDARD25686	S
15-Nov-24	W LANTAU	1	0.61	AUTUMN	STANDARD25686	P
15-Nov-24	W LANTAU	2	5.61	AUTUMN	STANDARD25686	P
15-Nov-24	W LANTAU	2	9.26	AUTUMN	STANDARD25686	S
21-Nov-24	NW LANTAU	2	22.55	AUTUMN	STANDARD25686	P
21-Nov-24	NW LANTAU	3	3.61	AUTUMN	STANDARD25686	P
21-Nov-24	NW LANTAU	2	5.48	AUTUMN	STANDARD25686	S
21-Nov-24	NW LANTAU	3	3.26	AUTUMN	STANDARD25686	S
21-Nov-24	DEEP BAY	2	8.19	AUTUMN	STANDARD25686	P
21-Nov-24	DEEP BAY	2	6.81	AUTUMN	STANDARD25686	S
21-Nov-24	NE LANTAU	2	27.12	AUTUMN	STANDARD25686	P
21-Nov-24	NE LANTAU	2	10.18	AUTUMN	STANDARD25686	S
25-Nov-24	W LANTAU	3	11.76	AUTUMN	STANDARD25686	P
25-Nov-24	W LANTAU	3	10.96	AUTUMN	STANDARD25686	S
25-Nov-24	SW LANTAU	1	2.62	AUTUMN	STANDARD25686	P
25-Nov-24	SW LANTAU	2	31.55	AUTUMN	STANDARD25686	P
25-Nov-24	SW LANTAU	2	13.36	AUTUMN	STANDARD25686	S
29-Nov-24	SW LANTAU	2	11.59	AUTUMN	STANDARD25686	P

Appendix I. (cont'd.)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
29-Nov-24	SW LANTAU	3	7.06	AUTUMN	STANDARD25686	P
29-Nov-24	SW LANTAU	2	8.58	AUTUMN	STANDARD25686	S
29-Nov-24	SW LANTAU	3	0.63	AUTUMN	STANDARD25686	S
29-Nov-24	SE LANTAU	2	28.23	AUTUMN	STANDARD25686	P
29-Nov-24	SE LANTAU	2	8.07	AUTUMN	STANDARD25686	S
2-Dec-24	W LANTAU	2	6.42	WINTER	STANDARD25686	P
2-Dec-24	W LANTAU	2	9.03	WINTER	STANDARD25686	S
12-Dec-24	W LANTAU	3	10.00	WINTER	STANDARD36826	S
12-Dec-24	SW LANTAU	2	9.49	WINTER	STANDARD36826	P
12-Dec-24	SW LANTAU	3	10.29	WINTER	STANDARD36826	P
12-Dec-24	SW LANTAU	2	2.10	WINTER	STANDARD36826	S
12-Dec-24	SW LANTAU	3	7.91	WINTER	STANDARD36826	S
3-Jan-25	W LANTAU	3	1.85	WINTER	STANDARD36826	P
3-Jan-25	W LANTAU	3	4.85	WINTER	STANDARD36826	S
3-Jan-25	W LANTAU	4	4.60	WINTER	STANDARD36826	S
3-Jan-25	SW LANTAU	2	6.20	WINTER	STANDARD36826	P
3-Jan-25	SW LANTAU	3	10.03	WINTER	STANDARD36826	P
3-Jan-25	SW LANTAU	2	4.23	WINTER	STANDARD36826	S
3-Jan-25	SW LANTAU	3	2.71	WINTER	STANDARD36826	S
6-Jan-25	SE LANTAU	2	24.31	WINTER	STANDARD138716	P
6-Jan-25	SE LANTAU	3	3.06	WINTER	STANDARD138716	P
6-Jan-25	SE LANTAU	2	9.60	WINTER	STANDARD138716	S
6-Jan-25	SW LANTAU	2	18.28	WINTER	STANDARD138716	P
6-Jan-25	SW LANTAU	2	12.92	WINTER	STANDARD138716	S
8-Jan-25	W LANTAU	2	9.81	WINTER	STANDARD36826	P
8-Jan-25	W LANTAU	2	7.88	WINTER	STANDARD36826	S
8-Jan-25	SW LANTAU	2	13.49	WINTER	STANDARD36826	P
8-Jan-25	SW LANTAU	3	2.90	WINTER	STANDARD36826	P
8-Jan-25	SW LANTAU	2	12.59	WINTER	STANDARD36826	S
8-Jan-25	SE LANTAU	2	1.66	WINTER	STANDARD36826	P
8-Jan-25	SE LANTAU	3	9.31	WINTER	STANDARD36826	P
8-Jan-25	SE LANTAU	3	5.23	WINTER	STANDARD36826	S
13-Jan-25	W LANTAU	2	10.13	WINTER	STANDARD36826	P
13-Jan-25	W LANTAU	2	15.66	WINTER	STANDARD36826	S
13-Jan-25	W LANTAU	3	0.92	WINTER	STANDARD36826	S
13-Jan-25	NW LANTAU	2	26.90	WINTER	STANDARD36826	P
13-Jan-25	NW LANTAU	2	10.80	WINTER	STANDARD36826	S
14-Jan-25	DEEP BAY	2	7.22	WINTER	STANDARD36826	P
14-Jan-25	DEEP BAY	3	1.37	WINTER	STANDARD36826	P
14-Jan-25	DEEP BAY	2	6.51	WINTER	STANDARD36826	S
14-Jan-25	NE LANTAU	1	7.10	WINTER	STANDARD36826	P
14-Jan-25	NE LANTAU	2	21.67	WINTER	STANDARD36826	P
14-Jan-25	NE LANTAU	1	1.90	WINTER	STANDARD36826	S
14-Jan-25	NE LANTAU	2	9.84	WINTER	STANDARD36826	S
16-Jan-25	SE LANTAU	2	20.52	WINTER	STANDARD36826	P
16-Jan-25	SE LANTAU	3	8.42	WINTER	STANDARD36826	P
16-Jan-25	SE LANTAU	2	6.30	WINTER	STANDARD36826	S
16-Jan-25	SE LANTAU	3	1.96	WINTER	STANDARD36826	S
16-Jan-25	SW LANTAU	2	15.34	WINTER	STANDARD36826	P
16-Jan-25	SW LANTAU	3	8.11	WINTER	STANDARD36826	P
16-Jan-25	SW LANTAU	2	5.89	WINTER	STANDARD36826	S
16-Jan-25	SW LANTAU	3	2.20	WINTER	STANDARD36826	S
17-Jan-25	W LANTAU	2	9.96	WINTER	STANDARD36826	P
17-Jan-25	W LANTAU	3	9.20	WINTER	STANDARD36826	P
17-Jan-25	W LANTAU	2	6.98	WINTER	STANDARD36826	S
17-Jan-25	W LANTAU	3	4.72	WINTER	STANDARD36826	S
17-Jan-25	SW LANTAU	2	7.65	WINTER	STANDARD36826	P
17-Jan-25	SW LANTAU	3	4.74	WINTER	STANDARD36826	P
17-Jan-25	SW LANTAU	2	4.92	WINTER	STANDARD36826	S
17-Jan-25	SW LANTAU	3	0.71	WINTER	STANDARD36826	S
17-Jan-25	SW LANTAU	4	0.67	WINTER	STANDARD36826	S

Appendix I. (cont'd.)

DATE	AREA	BEAU	EFFORT	SEASON	VESSEL	P/S
21-Jan-25	NW LANTAU	2	12.76	WINTER	STANDARD36826	P
21-Jan-25	NW LANTAU	3	13.42	WINTER	STANDARD36826	P
21-Jan-25	NW LANTAU	2	0.80	WINTER	STANDARD36826	S
21-Jan-25	NW LANTAU	3	7.62	WINTER	STANDARD36826	S
21-Jan-25	DEEP BAY	2	8.45	WINTER	STANDARD36826	P
21-Jan-25	DEEP BAY	2	6.55	WINTER	STANDARD36826	S
21-Jan-25	NE LANTAU	2	10.80	WINTER	STANDARD36826	P
21-Jan-25	NE LANTAU	3	14.74	WINTER	STANDARD36826	P
21-Jan-25	NE LANTAU	2	3.80	WINTER	STANDARD36826	S
21-Jan-25	NE LANTAU	3	6.36	WINTER	STANDARD36826	S
23-Jan-25	W LANTAU	2	20.57	WINTER	STANDARD36826	P
23-Jan-25	W LANTAU	2	9.91	WINTER	STANDARD36826	S
23-Jan-25	SW LANTAU	1	9.58	WINTER	STANDARD36826	P
23-Jan-25	SW LANTAU	2	14.94	WINTER	STANDARD36826	P
23-Jan-25	SW LANTAU	1	2.33	WINTER	STANDARD36826	S
23-Jan-25	SW LANTAU	2	8.95	WINTER	STANDARD36826	S
11-Feb-25	W LANTAU	2	11.76	WINTER	STANDARD138716	P
11-Feb-25	W LANTAU	3	6.56	WINTER	STANDARD138716	P
11-Feb-25	W LANTAU	2	8.33	WINTER	STANDARD138716	S
11-Feb-25	W LANTAU	3	2.75	WINTER	STANDARD138716	S
12-Feb-25	SE LANTAU	2	26.61	WINTER	STANDARD36826	P
12-Feb-25	SE LANTAU	2	10.17	WINTER	STANDARD36826	S
12-Feb-25	SW LANTAU	1	8.20	WINTER	STANDARD36826	P
12-Feb-25	SW LANTAU	2	10.76	WINTER	STANDARD36826	P
12-Feb-25	SW LANTAU	1	2.50	WINTER	STANDARD36826	S
12-Feb-25	SW LANTAU	2	10.53	WINTER	STANDARD36826	S
19-Feb-25	W LANTAU	2	14.39	WINTER	STANDARD36826	P
19-Feb-25	W LANTAU	3	1.32	WINTER	STANDARD36826	P
19-Feb-25	W LANTAU	2	11.50	WINTER	STANDARD36826	S
28-Feb-25	SE LANTAU	1	5.52	WINTER	STANDARD36826	P
28-Feb-25	SE LANTAU	2	23.59	WINTER	STANDARD36826	P
28-Feb-25	SE LANTAU	1	1.90	WINTER	STANDARD36826	S
28-Feb-25	SE LANTAU	2	6.51	WINTER	STANDARD36826	S
28-Feb-25	SW LANTAU	2	18.66	WINTER	STANDARD36826	P
28-Feb-25	SW LANTAU	2	7.28	WINTER	STANDARD36826	S
12-Mar-25	W LANTAU	1	0.78	SPRING	STANDARD36826	P
12-Mar-25	W LANTAU	2	13.06	SPRING	STANDARD36826	P
12-Mar-25	W LANTAU	3	1.93	SPRING	STANDARD36826	P
12-Mar-25	W LANTAU	2	8.39	SPRING	STANDARD36826	S
12-Mar-25	W LANTAU	3	0.82	SPRING	STANDARD36826	S
18-Mar-25	DEEP BAY	3	8.36	SPRING	STANDARD36826	P
18-Mar-25	DEEP BAY	3	6.24	SPRING	STANDARD36826	S
18-Mar-25	NE LANTAU	2	12.07	SPRING	STANDARD36826	P
18-Mar-25	NE LANTAU	3	13.02	SPRING	STANDARD36826	P
18-Mar-25	NE LANTAU	2	5.81	SPRING	STANDARD36826	S
18-Mar-25	NE LANTAU	3	4.50	SPRING	STANDARD36826	S

Appendix II. HKCRP-AFCD Chinese White Dolphin Sighting Database (April 2024 - March 2025)

(Note: P = sightings made on primary lines; S = sightings made on secondary lines)

DATE	STG #	TIME	HRD SZ	AREA	BEAU	PSD	EFFORT	TYPE	NORTHING	EASTING	SEASON	BOAT ASSOC.	P/S
24-Apr-24	1	1115	12	W LANTAU	1	33	ON	HKCRP	811458	800756	SPRING	NONE	P
24-Apr-24	2	1234	2	W LANTAU	2	186	ON	HKCRP	805576	801753	SPRING	NONE	P
24-Apr-24	3	1240	2	W LANTAU	2	92	ON	HKCRP	805950	802631	SPRING	NONE	P
24-Apr-24	4	1347	1	SW LANTAU	1	47	ON	HKCRP	807400	808893	SPRING	NONE	S
24-Apr-24	5	1504	1	SE LANTAU	2	41	ON	HKCRP	807482	812420	SPRING	NONE	P
16-May-24	1	1133	9	W LANTAU	2	229	ON	HKCRP	809398	800855	SPRING	NONE	P
16-May-24	2	1209	5	W LANTAU	3	435	ON	HKCRP	809354	801050	SPRING	NONE	S
28-May-24	1	1242	1	W LANTAU	2	129	ON	HKCRP	813359	802709	SPRING	NONE	S
28-May-24	2	1253	1	W LANTAU	2	208	ON	HKCRP	811246	801694	SPRING	NONE	S
28-May-24	3	1340	2	SW LANTAU	2	183	ON	HKCRP	804208	804514	SPRING	PURSE-SEINE	P
30-May-24	1	1242	4	W LANTAU	3	571	ON	HKCRP	808424	800832	SPRING	NONE	S
6-Jun-24	3	1359	1	SW LANTAU	2	73	ON	HKCRP	807400	809007	SUMMER	NONE	S
6-Jun-24	4	1453	3	SW LANTAU	3	78	ON	HKCRP	803220	805503	SUMMER	NONE	P
7-Jun-24	1	1129	8	W LANTAU	2	128	ON	HKCRP	810608	799816	SUMMER	NONE	S
7-Jun-24	2	1151	2	W LANTAU	3	522	ON	HKCRP	810385	800506	SUMMER	PURSE-SEINE	P
11-Jun-24	2	1522	3	SW LANTAU	2	183	ON	HKCRP	806456	804457	SUMMER	PURSE-SEINE	P
12-Jun-24	1	1131	2	W LANTAU	3	124	ON	HKCRP	810508	799785	SUMMER	NONE	S
12-Jun-24	2	1142	2	W LANTAU	3	192	ON	HKCRP	810474	800321	SUMMER	PURSE-SEINE	P
12-Jun-24	3	1155	2	W LANTAU	2	448	ON	HKCRP	809542	801030	SUMMER	PURSE-SEINE	P
12-Jun-24	4	1302	2	SW LANTAU	3	125	ON	HKCRP	806029	802260	SUMMER	PURSE-SEINE	S
12-Jun-24	5	1355	1	SW LANTAU	2	265	ON	HKCRP	807295	805480	SUMMER	NONE	P
17-Jun-24	1	1406	4	W LANTAU	3	74	ON	HKCRP	810384	800579	SUMMER	NONE	P
17-Jun-24	2	1421	5	W LANTAU	3	209	ON	HKCRP	809432	800721	SUMMER	NONE	P
17-Jun-24	3	1512	2	W LANTAU	3	190	ON	HKCRP	806451	801590	SUMMER	NONE	P
17-Jun-24	4	1521	1	SW LANTAU	3	ND	OFF	HKCRP	805929	802156	SUMMER	NONE	
18-Jun-24	1	1051	7	W LANTAU	3	185	ON	HKCRP	813580	802864	SUMMER	NONE	P
18-Jun-24	2	1247	7	SW LANTAU	2	334	ON	HKCRP	806139	802446	SUMMER	PURSE-SEINE	S
11-Jul-24	1	1524	3	SW LANTAU	2	144	ON	HKCRP	807021	804427	SUMMER	NONE	S
12-Jul-24	1	1300	2	W LANTAU	3	47	ON	HKCRP	813193	802533	SUMMER	NONE	S
12-Jul-24	2	1324	4	W LANTAU	3	131	ON	HKCRP	809575	800927	SUMMER	NONE	S
12-Jul-24	3	1344	5	W LANTAU	3	129	ON	HKCRP	806915	801880	SUMMER	NONE	S
12-Jul-24	4	1400	7	SW LANTAU	3	192	ON	HKCRP	806184	802270	SUMMER	PURSE-SEINE	S
12-Jul-24	5	1451	3	SW LANTAU	2	194	ON	HKCRP	806788	804324	SUMMER	NONE	P

Appendix II. (cont'd.)

DATE	STG #	TIME	HRD SZ	AREA	BEAU	PSD	EFFORT	TYPE	NORTHING	EASTING	SEASON	BOAT ASSOC.	P/S
12-Jul-24	6	1514	3	SW LANTAU	2	384	ON	HKCRP	807981	806028	SUMMER	NONE	S
23-Jul-24	3	1429	11	SW LANTAU	2	218	ON	HKCRP	805640	808447	SUMMER	NONE	S
23-Jul-24	4	1540	1	SW LANTAU	2	ND	OFF	HKCRP	804143	809692	SUMMER	NONE	
30-Jul-24	1	1312	3	W LANTAU	3	213	ON	HKCRP	813536	802647	SUMMER	NONE	P
30-Jul-24	2	1416	5	W LANTAU	2	624	ON	HKCRP	809398	800865	SUMMER	NONE	P
30-Jul-24	3	1530	1	SW LANTAU	2	ND	OFF	HKCRP	806334	804323	SUMMER	NONE	
30-Jul-24	4	1541	9	SW LANTAU	2	ND	OFF	HKCRP	806987	804747	SUMMER	NONE	
5-Aug-24	1	1238	1	SE LANTAU	3	318	ON	HKCRP	806783	813450	SUMMER	NONE	P
5-Aug-24	3	1434	2	SW LANTAU	2	188	ON	HKCRP	808044	807647	SUMMER	PURSE-SEINE	S
5-Aug-24	4	1510	8	SW LANTAU	2	208	ON	HKCRP	805885	807467	SUMMER	NONE	P
7-Aug-24	1	1248	7	W LANTAU	2	141	ON	HKCRP	814490	801959	SUMMER	NONE	P
7-Aug-24	2	1316	3	W LANTAU	3	14	ON	HKCRP	812740	802161	SUMMER	NONE	S
7-Aug-24	3	1329	2	W LANTAU	2	6	ON	HKCRP	810559	801589	SUMMER	NONE	S
7-Aug-24	4	1358	2	W LANTAU	3	27	ON	HKCRP	808335	800914	SUMMER	NONE	P
7-Aug-24	5	1428	6	SW LANTAU	2	192	ON	HKCRP	805818	802476	SUMMER	NONE	S
7-Aug-24	6	1520	2	SW LANTAU	2	61	ON	HKCRP	803630	805555	SUMMER	NONE	P
7-Aug-24	7	1528	2	SW LANTAU	2	147	ON	HKCRP	805081	805568	SUMMER	NONE	P
8-Aug-24	1	1031	2	W LANTAU	3	95	ON	HKCRP	815452	802683	SUMMER	NONE	S
8-Aug-24	2	1054	6	W LANTAU	2	121	ON	HKCRP	813625	802452	SUMMER	NONE	P
8-Aug-24	3	1155	1	W LANTAU	3	8	ON	HKCRP	811523	801457	SUMMER	NONE	P
8-Aug-24	4	1220	1	W LANTAU	3	21	ON	HKCRP	809509	800772	SUMMER	NONE	P
8-Aug-24	5	1240	6	W LANTAU	3	23	ON	HKCRP	808346	800914	SUMMER	NONE	P
8-Aug-24	6	1325	3	W LANTAU	3	188	ON	HKCRP	806475	800858	SUMMER	NONE	P
8-Aug-24	7	1339	13	W LANTAU	3	123	ON	HKCRP	806196	801878	SUMMER	NONE	S
12-Aug-24	1	1301	5	W LANTAU	2	24	ON	HKCRP	813889	803308	SUMMER	NONE	S
12-Aug-24	2	1418	4	W LANTAU	3	157	ON	HKCRP	808381	800347	SUMMER	NONE	P
12-Aug-24	3	1508	5	W LANTAU	2	68	ON	HKCRP	806407	801291	SUMMER	NONE	P
12-Aug-24	4	1524	2	SW LANTAU	2	ND	OFF	HKCRP	806228	802456	SUMMER	NONE	
13-Aug-24	1	1138	1	SE LANTAU	2	651	ON	HKCRP	808666	813401	SUMMER	NONE	P
22-Aug-24	1	1256	1	W LANTAU	3	152	ON	HKCRP	813624	803070	SUMMER	NONE	S
22-Aug-24	2	1327	1	W LANTAU	3	166	ON	HKCRP	811434	801478	SUMMER	NONE	P
22-Aug-24	3	1352	4	W LANTAU	3	183	ON	HKCRP	810327	801444	SUMMER	NONE	P
22-Aug-24	4	1422	4	W LANTAU	3	91	ON	HKCRP	808358	800419	SUMMER	NONE	P
22-Aug-24	5	1508	2	W LANTAU	3	256	ON	HKCRP	806473	801652	SUMMER	NONE	P

Appendix II. (cont'd.)

DATE	STG #	TIME	HRD SZ	AREA	BEAU	PSD	EFFORT	TYPE	NORTHING	EASTING	SEASON	BOAT ASSOC.	P/S
22-Aug-24	6	1523	1	SW LANTAU	2	ND	OFF	HKCRP	806523	804230	SUMMER	NONE	
22-Aug-24	7	1529	1	SW LANTAU	2	ND	OFF	HKCRP	806986	804943	SUMMER	NONE	
22-Aug-24	8	1607	1	SE LANTAU	2	ND	OFF	HKCRP	807933	814957	SUMMER	NONE	
26-Aug-24	1	1053	3	W LANTAU	2	372	ON	HKCRP	813944	803329	SUMMER	NONE	S
26-Aug-24	2	1124	3	W LANTAU	2	347	ON	HKCRP	812162	802985	SUMMER	NONE	S
26-Aug-24	3	1204	4	W LANTAU	2	240	ON	HKCRP	809752	801268	SUMMER	NONE	S
26-Aug-24	4	1230	5	W LANTAU	2	106	ON	HKCRP	808403	800089	SUMMER	NONE	P
26-Aug-24	5	1243	3	W LANTAU	2	384	ON	HKCRP	806897	800271	SUMMER	NONE	S
26-Aug-24	7	1438	1	SW LANTAU	2	32	ON	HKCRP	808266	807369	SUMMER	NONE	P
16-Sep-24	1	1301	1	W LANTAU	3	50	ON	HKCRP	812905	802419	AUTUMN	NONE	S
16-Sep-24	2	1308	3	W LANTAU	2	12	ON	HKCRP	812053	802118	AUTUMN	NONE	S
16-Sep-24	3	1319	3	W LANTAU	2	94	ON	HKCRP	810493	801404	AUTUMN	NONE	S
16-Sep-24	4	1350	2	W LANTAU	2	338	ON	HKCRP	806221	800311	AUTUMN	NONE	S
16-Sep-24	5	1406	1	W LANTAU	2	140	ON	HKCRP	805509	801939	AUTUMN	NONE	P
16-Sep-24	6	1415	8	SW LANTAU	2	30	ON	HKCRP	805397	802485	AUTUMN	NONE	P
16-Sep-24	7	1439	1	SW LANTAU	2	39	ON	HKCRP	804276	803720	AUTUMN	NONE	P
30-Sep-24	1	1341	9	W LANTAU	2	609	ON	HKCRP	812919	801141	AUTUMN	NONE	S
30-Sep-24	2	1409	3	W LANTAU	2	57	ON	HKCRP	811035	801858	AUTUMN	NONE	S
30-Sep-24	3	1430	2	W LANTAU	2	475	ON	HKCRP	808249	799841	AUTUMN	NONE	P
30-Sep-24	4	1445	4	W LANTAU	2	272	ON	HKCRP	808235	800924	AUTUMN	NONE	P
30-Sep-24	5	1506	2	W LANTAU	2	206	ON	HKCRP	806506	801724	AUTUMN	NONE	S
30-Sep-24	6	1523	1	W LANTAU	2	176	ON	HKCRP	805556	800598	AUTUMN	NONE	S
30-Sep-24	7	1532	2	SW LANTAU	2	ND	OFF	HKCRP	805785	802362	AUTUMN	NONE	
8-Oct-24	1	1445	1	SW LANTAU	2	289	ON	HKCRP	806371	802931	AUTUMN	NONE	S
14-Oct-24	1	1305	4	W LANTAU	2	399	ON	HKCRP	814810	802279	AUTUMN	NONE	S
14-Oct-24	2	1336	3	W LANTAU	2	510	ON	HKCRP	813008	801018	AUTUMN	NONE	S
14-Oct-24	3	1424	1	W LANTAU	3	25	ON	HKCRP	808425	800388	AUTUMN	NONE	P
14-Oct-24	4	1441	3	W LANTAU	2	207	ON	HKCRP	806418	801436	AUTUMN	NONE	P
14-Oct-24	5	1504	4	W LANTAU	3	107	ON	HKCRP	806398	800683	AUTUMN	NONE	P
18-Oct-24	1	1322	2	W LANTAU	3	34	ON	HKCRP	808213	801151	AUTUMN	NONE	S
18-Oct-24	2	1441	2	W LANTAU	3	104	ON	HKCRP	813571	801709	AUTUMN	NONE	P
18-Oct-24	3	1450	8	W LANTAU	2	272	ON	HKCRP	813548	802256	AUTUMN	NONE	P
22-Oct-24	4	1247	1	SE LANTAU	2	205	ON	HKCRP	806951	812409	AUTUMN	NONE	P
29-Oct-24	1	1104	4	W LANTAU	3	117	ON	HKCRP	812464	801748	AUTUMN	NONE	P

Appendix II. (cont'd.)

DATE	STG #	TIME	HRD SZ	AREA	BEAU	PSD	EFFORT	TYPE	NORTHING	EASTING	SEASON	BOAT ASSOC.	P/S
29-Oct-24	2	1126	8	W LANTAU	3	61	ON	HKCRP	811183	800127	AUTUMN	NONE	S
29-Oct-24	3	1213	3	SW LANTAU	2	88	ON	HKCRP	806006	802466	AUTUMN	NONE	P
29-Oct-24	4	1302	1	SW LANTAU	2	10	ON	HKCRP	807716	805563	AUTUMN	NONE	S
8-Nov-24	1	1310	3	W LANTAU	3	15	ON	HKCRP	808490	800987	AUTUMN	NONE	S
8-Nov-24	2	1321	2	W LANTAU	3	9	ON	HKCRP	808235	801089	AUTUMN	NONE	S
8-Nov-24	3	1331	2	W LANTAU	3	56	ON	HKCRP	806705	801797	AUTUMN	NONE	S
8-Nov-24	4	1341	4	SW LANTAU	3	13	ON	HKCRP	806228	802178	AUTUMN	NONE	S
15-Nov-24	1	1341	4	W LANTAU	2	189	ON	HKCRP	813602	802977	AUTUMN	NONE	P
15-Nov-24	2	1432	5	W LANTAU	2	185	ON	HKCRP	809344	800514	AUTUMN	NONE	P
15-Nov-24	3	1506	3	W LANTAU	2	279	ON	HKCRP	807957	801387	AUTUMN	NONE	S
15-Nov-24	4	1535	1	SW LANTAU	2	ND	OFF	HKCRP	806786	805345	AUTUMN	NONE	
25-Nov-24	1	1033	4	W LANTAU	3	11	ON	HKCRP	813615	801916	AUTUMN	PURSE-SEINE	P
25-Nov-24	2	1216	3	SW LANTAU	2	138	ON	HKCRP	805606	803321	AUTUMN	NONE	P
25-Nov-24	3	1325	3	SW LANTAU	2	332	ON	HKCRP	807761	805378	AUTUMN	NONE	P
25-Nov-24	5	1448	3	SW LANTAU	2	52	ON	HKCRP	807233	809501	AUTUMN	NONE	P
29-Nov-24	3	1304	2	SW LANTAU	2	161	ON	HKCRP	805474	808240	AUTUMN	NONE	S
2-Dec-24	1	1412	1	W LANTAU	2	35	ON	HKCRP	809907	801309	WINTER	NONE	S
2-Dec-24	2	1424	2	W LANTAU	2	18	ON	HKCRP	809433	800236	WINTER	SHRIMP	P
2-Dec-24	3	1454	2	W LANTAU	2	91	ON	HKCRP	807659	801026	WINTER	NONE	P
2-Dec-24	4	1512	3	W LANTAU	2	658	ON	HKCRP	806207	801765	WINTER	NONE	S
12-Dec-24	1	1328	10	SW LANTAU	3	428	ON	HKCRP	806734	803901	WINTER	NONE	S
12-Dec-24	2	1511	1	SW LANTAU	2	288	ON	HKCRP	802364	807399	WINTER	NONE	P
3-Jan-25	1	1328	9	W LANTAU	3	19	ON	HKCRP	806329	801889	WINTER	NONE	S
3-Jan-25	2	1347	5	SW LANTAU	3	183	ON	HKCRP	805917	802507	WINTER	NONE	S
3-Jan-25	3	1404	2	SW LANTAU	3	236	ON	HKCRP	806126	803456	WINTER	NONE	P
6-Jan-25	1	1134	1	SE LANTAU	2	103	ON	HKCRP	809350	815001	WINTER	NONE	S
8-Jan-25	1	1035	1	W LANTAU	2	81	ON	HKCRP	813549	801575	WINTER	NONE	S
8-Jan-25	2	1051	3	W LANTAU	2	280	ON	HKCRP	812573	802501	WINTER	GILL-NET	S
8-Jan-25	3	1131	11	W LANTAU	2	196	ON	HKCRP	809477	800515	WINTER	NONE	P
8-Jan-25	4	1233	3	W LANTAU	2	21	ON	HKCRP	806451	801735	WINTER	NONE	P
13-Jan-25	1	1018	2	W LANTAU	2	324	ON	HKCRP	814176	803710	WINTER	NONE	S
13-Jan-25	2	1027	4	W LANTAU	2	21	ON	HKCRP	813016	802461	WINTER	NONE	S
13-Jan-25	3	1041	1	W LANTAU	3	645	ON	HKCRP	811246	801704	WINTER	NONE	S
13-Jan-25	4	1124	1	W LANTAU	2	134	ON	HKCRP	806451	801405	WINTER	NONE	P

Appendix II. (cont'd.)

DATE	STG #	TIME	HRD SZ	AREA	BEAU	PSD	EFFORT	TYPE	NORTHING	EASTING	SEASON	BOAT ASSOC.	P/S
13-Jan-25	5	1131	6	W LANTAU	2	376	ON	HKCRP	806927	801560	WINTER	NONE	S
13-Jan-25	6	1147	5	W LANTAU	2	243	ON	HKCRP	807449	800984	WINTER	NONE	P
13-Jan-25	7	1207	3	W LANTAU	2	466	ON	HKCRP	808560	799543	WINTER	NONE	P
13-Jan-25	8	1226	4	W LANTAU	2	106	ON	HKCRP	810406	800847	WINTER	NONE	P
13-Jan-25	9	1245	4	W LANTAU	2	568	ON	HKCRP	812462	802387	WINTER	NONE	P
17-Jan-25	1	1004	1	NW LANTAU	2	ND	OFF	HKCRP	817639	805006	WINTER	NONE	
17-Jan-25	2	1013	3	W LANTAU	2	275	ON	HKCRP	818417	803823	WINTER	NONE	P
17-Jan-25	3	1056	4	W LANTAU	3	86	ON	HKCRP	814476	803392	WINTER	NONE	P
17-Jan-25	4	1108	2	W LANTAU	2	269	ON	HKCRP	814231	803577	WINTER	GILL-NET	S
17-Jan-25	5	1121	2	W LANTAU	2	176	ON	HKCRP	813569	802864	WINTER	NONE	P
17-Jan-25	6	1146	5	W LANTAU	3	4	ON	HKCRP	812496	802253	WINTER	NONE	P
17-Jan-25	7	1302	1	W LANTAU	3	524	ON	HKCRP	806464	800951	WINTER	NONE	P
17-Jan-25	8	1329	3	W LANTAU	2	ND	OFF	HKCRP	806008	801527	WINTER	NONE	
17-Jan-25	9	1345	1	SW LANTAU	2	103	ON	HKCRP	805563	802516	WINTER	NONE	P
17-Jan-25	12	1440	1	SW LANTAU	2	561	ON	HKCRP	808069	806482	WINTER	NONE	P
23-Jan-25	1	1028	7	W LANTAU	2	246	ON	HKCRP	815164	802538	WINTER	NONE	S
23-Jan-25	2	1124	10	W LANTAU	2	118	ON	HKCRP	811435	801138	WINTER	NONE	P
11-Feb-25	1	1403	1	W LANTAU	2	97	ON	HKCRP	809366	800308	WINTER	NONE	P
19-Feb-25	1	1344	4	W LANTAU	2	988	ON	HKCRP	811138	800230	WINTER	NONE	S
19-Feb-25	2	1352	1	W LANTAU	2	329	ON	HKCRP	810441	800197	WINTER	NONE	P
19-Feb-25	3	1412	1	W LANTAU	2	21	ON	HKCRP	809321	800700	WINTER	NONE	P
28-Feb-25	1	1129	1	SE LANTAU	2	269	ON	HKCRP	809650	814310	WINTER	NONE	S
12-Mar-25	1	1248	1	W LANTAU	2	101	ON	HKCRP	813648	802050	SPRING	PURSE-SEINE	P
12-Mar-25	2	1326	7	W LANTAU	2	428	ON	HKCRP	810674	799940	SPRING	NONE	S
12-Mar-25	3	1401	7	W LANTAU	2	336	ON	HKCRP	809368	799638	SPRING	NONE	P
12-Mar-25	4	1447	2	W LANTAU	2	544	ON	HKCRP	806453	800662	SPRING	NONE	P
12-Mar-25	5	1503	2	W LANTAU	2	174	ON	HKCRP	806440	801590	SPRING	NONE	P

Appendix III. HKCRP-AFCD Finless Porpoise Sighting Database (April 2024 - March 2025)

(Note: P = sightings made on primary lines; S = sightings made on secondary lines)

DATE	STG #	TIME	HRD SZ	AREA	BEAU	PSD	EFFORT	NORTHING	EASTING	SEASON	P/S
13-May-24	1	1321	7	SW LANTAU	1	132	ON	804627	811498	SPRING	P
13-May-24	2	1332	14	SW LANTAU	1	178	ON	803464	811527	SPRING	P
13-May-24	3	1345	3	SW LANTAU	1	57	ON	801914	811483	SPRING	P
13-May-24	4	1410	6	SW LANTAU	2	158	ON	802770	809586	SPRING	P
24-May-24	1	1351	7	SW LANTAU	2	84	ON	804152	810620	SPRING	P
6-Jun-24	1	1015	1	SE LANTAU	2	109	ON	806177	819482	SUMMER	P
6-Jun-24	2	1305	3	SW LANTAU	3	41	ON	803841	811434	SUMMER	P
11-Jun-24	1	1314	1	SW LANTAU	2	124	ON	804750	810590	SUMMER	P
23-Jul-24	1	1321	2	SW LANTAU	2	197	ON	803875	810589	SUMMER	P
23-Jul-24	2	1357	5	SW LANTAU	2	92	ON	801875	808522	SUMMER	P
5-Aug-24	2	1405	3	SW LANTAU	2	170	ON	803943	809681	SUMMER	P
26-Aug-24	6	1402	4	SW LANTAU	2	2	ON	801889	806573	SUMMER	S
11-Sep-24	1	1323	2	SW LANTAU	2	12	ON	803387	811537	AUTUMN	P
11-Sep-24	2	1330	4	SW LANTAU	2	122	ON	802744	811484	AUTUMN	P
11-Sep-24	3	1341	1	SW LANTAU	2	75	ON	801737	811472	AUTUMN	P
11-Sep-24	4	1347	1	SW LANTAU	2	70	ON	801349	811338	AUTUMN	P
26-Sep-24	1	1231	2	SE LANTAU	2	161	ON	804670	812539	AUTUMN	P
26-Sep-24	2	1310	3	SW LANTAU	3	139	ON	806744	810542	AUTUMN	P
26-Sep-24	3	1325	7	SW LANTAU	2	262	ON	804595	810621	AUTUMN	P
26-Sep-24	4	1336	3	SW LANTAU	2	104	ON	803643	810722	AUTUMN	P
26-Sep-24	5	1350	2	SW LANTAU	2	228	ON	800587	810490	AUTUMN	P
26-Sep-24	6	1400	2	SW LANTAU	2	193	ON	800467	809397	AUTUMN	S
26-Sep-24	7	1446	5	SW LANTAU	2	89	ON	807313	807748	AUTUMN	S
7-Oct-24	1	1228	3	SE LANTAU	2	96	ON	806706	813481	AUTUMN	P
7-Oct-24	2	1334	1	SW LANTAU	2	44	ON	800433	809923	AUTUMN	S
7-Oct-24	3	1347	2	SW LANTAU	2	169	ON	802814	809597	AUTUMN	P
8-Oct-24	2	1536	4	SW LANTAU	2	ND	OFF	807089	809223	AUTUMN	
22-Oct-24	1	1012	2	SE LANTAU	2	143	ON	807530	818514	AUTUMN	P
22-Oct-24	2	1105	5	SE LANTAU	2	445	ON	805273	816336	AUTUMN	P
22-Oct-24	3	1239	1	SE LANTAU	2	122	ON	799498	812397	AUTUMN	P
22-Oct-24	5	1332	1	SW LANTAU	3	253	ON	802724	810566	AUTUMN	S
29-Oct-24	5	1457	1	SW LANTAU	2	148	ON	803665	810650	AUTUMN	P
29-Oct-24	6	1527	1	SW LANTAU	2	29	ON	803331	811392	AUTUMN	P
29-Oct-24	7	1534	1	SW LANTAU	2	61	ON	804151	811487	AUTUMN	P
8-Nov-24	5	1522	2	SW LANTAU	2	134	ON	801279	807479	AUTUMN	P
25-Nov-24	4	1412	1	SW LANTAU	2	261	ON	800845	808211	AUTUMN	S
25-Nov-24	6	1528	1	SW LANTAU	2	86	ON	802966	811598	AUTUMN	P
29-Nov-24	1	1228	1	SW LANTAU	2	35	ON	800646	808355	AUTUMN	S
29-Nov-24	2	1238	2	SW LANTAU	2	50	ON	801931	808234	AUTUMN	S
6-Jan-25	2	1207	1	SE LANTAU	2	125	ON	804035	814539	WINTER	P
6-Jan-25	3	1238	1	SE LANTAU	2	155	ON	803219	812403	WINTER	P
6-Jan-25	4	1246	4	SE LANTAU	2	165	ON	804470	812415	WINTER	P
8-Jan-25	5	1309	3	SW LANTAU	2	27	ON	808057	806904	WINTER	S
8-Jan-25	6	1347	3	SW LANTAU	2	136	ON	801622	807418	WINTER	P
16-Jan-25	1	1144	2	SE LANTAU	2	155	ON	805108	815521	WINTER	P
16-Jan-25	2	1215	3	SE LANTAU	2	145	ON	801678	813370	WINTER	P
16-Jan-25	3	1223	5	SE LANTAU	2	71	ON	802354	813351	WINTER	P
16-Jan-25	4	1333	1	SW LANTAU	2	191	ON	803763	811507	WINTER	P
16-Jan-25	5	1337	2	SW LANTAU	2	29	ON	803320	811537	WINTER	P
16-Jan-25	6	1406	1	SW LANTAU	2	182	ON	802150	809575	WINTER	P
17-Jan-25	10	1409	2	SW LANTAU	2	17	ON	803510	804534	WINTER	P
17-Jan-25	11	1417	1	SW LANTAU	2	146	ON	804551	804536	WINTER	P
23-Jan-25	3	1358	2	SW LANTAU	2	87	ON	804018	805432	WINTER	P
23-Jan-25	4	1439	1	SW LANTAU	1	57	ON	804182	806546	WINTER	P
23-Jan-25	5	1449	1	SW LANTAU	1	169	ON	802499	806471	WINTER	P
12-Feb-25	1	1245	4	SE LANTAU	2	69	ON	806320	812542	WINTER	P
12-Feb-25	2	1300	4	SE LANTAU	2	47	ON	808767	812360	WINTER	S
12-Feb-25	3	1342	3	SW LANTAU	2	32	ON	802823	810587	WINTER	P
28-Feb-25	2	1337	1	SW LANTAU	2	288	ON	804051	811538	WINTER	P
28-Feb-25	3	1342	6	SW LANTAU	2	85	ON	803641	811558	WINTER	P
28-Feb-25	4	1404	1	SW LANTAU	2	107	ON	800930	810470	WINTER	S
28-Feb-25	5	1424	2	SW LANTAU	2	276	ON	803312	809639	WINTER	P
28-Feb-25	6	1439	1	SW LANTAU	2	31	ON	805804	809509	WINTER	P
12-Mar-25	6	1539	2	SW LANTAU	2	ND	OFF	806611	810325	SPRING	

Appendix IV. Individual dolphins identified during AFCD surveys (April 2024 to March 2025)
(in bold & italics: new individuals)

DOLPHIN ID	DATE	STG#	AREA
CH12	12/07/24	3	WL
	03/01/25	1	WL
	13/01/25	6	WL
CH38	12/06/24	4	SWL
	23/07/24	3	SWL
	30/07/24	4	SWL
	05/08/24	4	SWL
	08/08/24	7	WL
	26/08/24	4	WL
	14/10/24	5	WL
	29/10/24	2	WL
	12/12/24	1	SWL
	08/01/25	3	WL
	12/03/25	4	WL
CH105	18/06/24	2	SWL
CH108	24/04/24	3	WL
	05/08/24	4	SWL
	08/08/24	7	WL
	12/08/24	3	WL
	26/08/24	4	WL
	16/09/24	3	WL
	30/09/24	4	WL
	29/10/24	2	WL
	12/12/24	1	SWL
	13/01/25	5	WL
CH112	07/08/24	2	WL
CH113	18/06/24	1	WL
	30/07/24	2	WL
	07/08/24	1	WL
	26/08/24	1	WL
CH141	16/09/24	6	SWL
	12/12/24	1	SWL
	17/01/25	2	WL
CH320	07/06/24	1	WL
	18/06/24	2	SWL
	08/01/25	1	WL
NL33	12/07/24	3	WL
NL46	07/08/24	5	SWL
	08/08/24	6	WL
NL104	30/07/24	1	WL
	07/08/24	1	WL
NL123	07/08/24	5	SWL
	07/08/24	7	SWL
NL156	16/05/24	2	WL
	12/07/24	4	SWL
	12/08/24	1	WL
	29/10/24	1	WL
	02/12/24	4	WL
	17/01/25	6	WL
NL202	17/06/24	1	WL
	23/01/25	1	WL
NL224	18/06/24	1	WL
	12/08/24	1	WL
	26/08/24	1	WL
NL256	07/08/24	1	WL
	08/08/24	2	WL
	14/10/24	2	WL
NL261	08/01/25	3	WL
NL269	02/12/24	4	WL
NL293	16/09/24	3	WL
NL296	17/06/24	2	WL

DOLPHIN ID	DATE	STG#	AREA
NL306	24/04/24	5	SEL
	05/08/24	1	SEL
	13/08/24	1	SEL
	22/08/24	8	SEL
	22/10/24	4	SEL
	06/01/25	1	SEL
	28/02/25	1	SEL
NL311	24/04/24	1	WL
	30/07/24	2	WL
NL321	07/08/24	3	WL
NL332	12/08/24	2	WL
SL40	18/06/24	2	SWL
	12/07/24	6	SWL
	05/08/24	3	SWL
	18/10/24	2	WL
	17/01/25	7	WL
	17/01/25	8	WL
	11/02/25	1	WL
SL44	12/06/24	4	SWL
	12/08/24	4	SWL
	14/10/24	3	WL
	17/01/25	3	WL
	23/01/25	1	WL
SL60	24/04/24	4	SWL
	23/07/24	3	SWL
	17/01/25	8	WL
	23/01/25	2	WL
SL66	14/10/24	1	WL
	23/01/25	1	WL
SL68	07/08/24	5	SWL
	26/08/24	5	WL
	12/12/24	1	SWL
WL11	17/06/24	1	WL
WL21	18/06/24	1	WL
	08/08/24	2	WL
WL42	16/05/24	1	WL
	16/05/24	2	WL
	23/07/24	3	SWL
	30/07/24	4	SWL
	08/08/24	5	WL
	16/09/24	6	SWL
	29/10/24	2	WL
	25/11/24	5	SWL
	03/01/25	2	SWL
	13/01/25	7	WL
WL58	12/03/25	3	WL
WL61	16/05/24	1	WL
	16/05/24	2	WL
	28/05/24	2	WL
	14/10/24	5	WL
	18/10/24	1	WL
WL66	08/08/24	2	WL
	14/10/24	2	WL
WL72	16/05/24	1	WL
	16/05/24	2	WL
	22/08/24	4	WL
	18/10/24	3	WL
	29/10/24	2	WL
	15/11/24	1	WL
	25/11/24	2	SWL
	25/11/24	3	SWL
	12/12/24	1	SWL
	03/01/25	1	WL

DOLPHIN ID	DATE	STG#	AREA
WL72	13/01/25	8	WL
	17/01/25	3	WL
	23/01/25	2	WL
	12/03/25	3	WL
WL79	11/06/24	2	SWL
	12/06/24	2	WL
	12/06/24	3	WL
	11/07/24	1	SWL
	12/07/24	4	SWL
	22/08/24	1	WL
	26/08/24	2	WL
	14/10/24	1	WL
	18/10/24	3	WL
	25/11/24	1	WL
	19/02/25	2	WL
	19/02/25	3	WL
	12/03/25	1	WL
WL91	28/05/24	3	SWL
	11/06/24	2	SWL
	12/06/24	2	WL
	11/07/24	1	SWL
	12/07/24	6	SWL
	08/11/24	4	SWL
WL92	02/12/24	2	WL
	12/07/24	2	WL
	30/09/24	1	WL
WL94	12/03/25	2	WL
	12/07/24	3	WL
	07/08/24	5	SWL
	08/11/24	4	SWL
	02/12/24	4	WL
	12/12/24	1	SWL
WL109	03/01/25	1	WL
	13/01/25	6	WL
	16/05/24	1	WL
	30/05/24	1	WL
	23/07/24	3	SWL
	30/07/24	4	SWL
	22/08/24	5	WL
	16/09/24	6	SWL
	30/09/24	2	WL
	18/10/24	3	WL
WL114	29/10/24	2	WL
	08/11/24	2	WL
	15/11/24	1	WL
	12/12/24	1	SWL
	03/01/25	2	SWL
	23/01/25	2	WL
	12/03/25	3	WL
WL118	23/07/24	3	SWL
	30/07/24	4	SWL
WL128	12/07/24	3	WL
	08/11/24	1	WL
	25/11/24	2	SWL
WL129	24/04/24	1	WL
	23/07/24	3	SWL
	08/08/24	5	WL
WL130	03/01/25	2	SWL
	30/07/24	3	SWL
	05/08/24	4	SWL
	30/09/24	7	SWL
	29/11/24	3	SWL
	23/01/25	2	WL

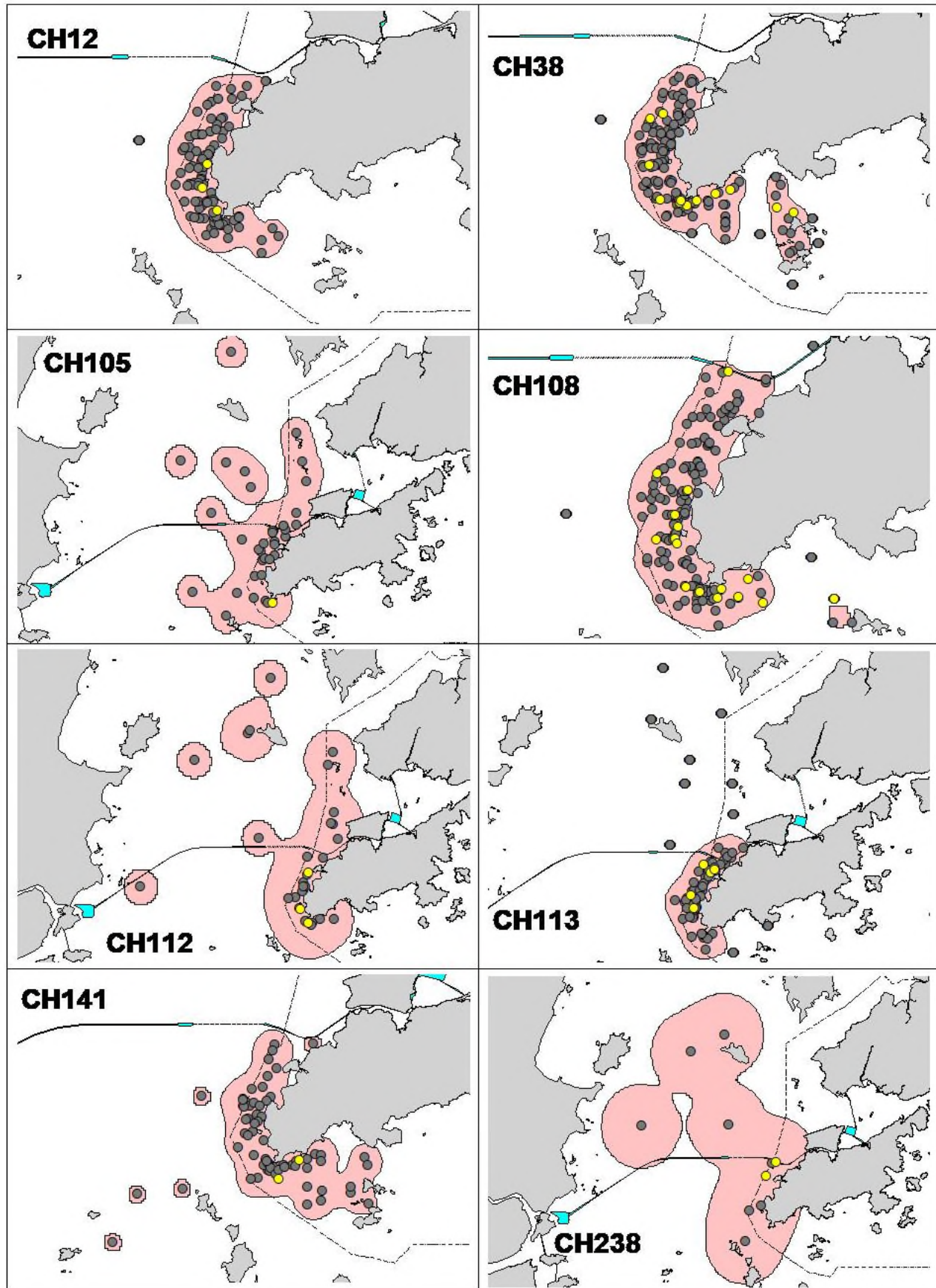
Appendix IV. (cont'd)
(in bold & italics: new individuals)

DOLPHIN ID	DATE	STG#	AREA
WL131	24/04/24	1	WL
	23/07/24	3	SWL
	30/07/24	4	SWL
	30/09/24	1	WL
	15/11/24	2	WL
	15/11/24	3	WL
	12/12/24	1	SWL
	03/01/25	1	WL
	23/01/25	2	WL
WL142	24/04/24	1	WL
	12/07/24	2	WL
	08/08/24	7	WL
	12/03/25	2	WL
WL145	08/08/24	2	WL
	22/08/24	2	WL
	13/01/25	6	WL
WL152	06/06/24	4	SWL
	12/07/24	3	WL
	12/07/24	4	SWL
	12/07/24	5	SWL
	30/07/24	4	SWL
	05/08/24	4	SWL
	22/08/24	5	WL
	16/09/24	6	SWL
	14/10/24	1	WL
	18/10/24	3	WL
	08/11/24	3	WL
	15/11/24	2	WL
	25/11/24	5	SWL
	03/01/25	2	SWL
	13/01/25	4	WL
	12/03/25	5	WL
WL159	08/08/24	5	WL
WL166	24/04/24	1	WL
WL169	06/06/24	4	SWL
WL171	18/06/24	2	SWL
	12/07/24	4	SWL
	30/07/24	4	SWL
	17/01/25	6	WL
WL179	07/06/24	1	WL
	18/06/24	2	SWL
	25/11/24	1	WL
	03/01/25	1	WL
	12/03/25	5	WL
WL200	12/07/24	4	SWL
WL203	30/09/24	5	WL
	30/09/24	7	SWL
WL206	07/06/24	1	WL
WL208	24/04/24	1	WL
	07/08/24	5	SWL
	08/08/24	2	WL
	08/11/24	1	WL
	08/01/25	2	WL
WL213	12/07/24	4	SWL
	08/08/24	7	WL
WL220	28/05/24	1	WL
	11/06/24	2	SWL
	11/07/24	1	SWL
	12/08/24	4	SWL
	26/08/24	4	WL
	16/09/24	7	SWL
	29/10/24	4	SWL
	08/11/24	3	WL

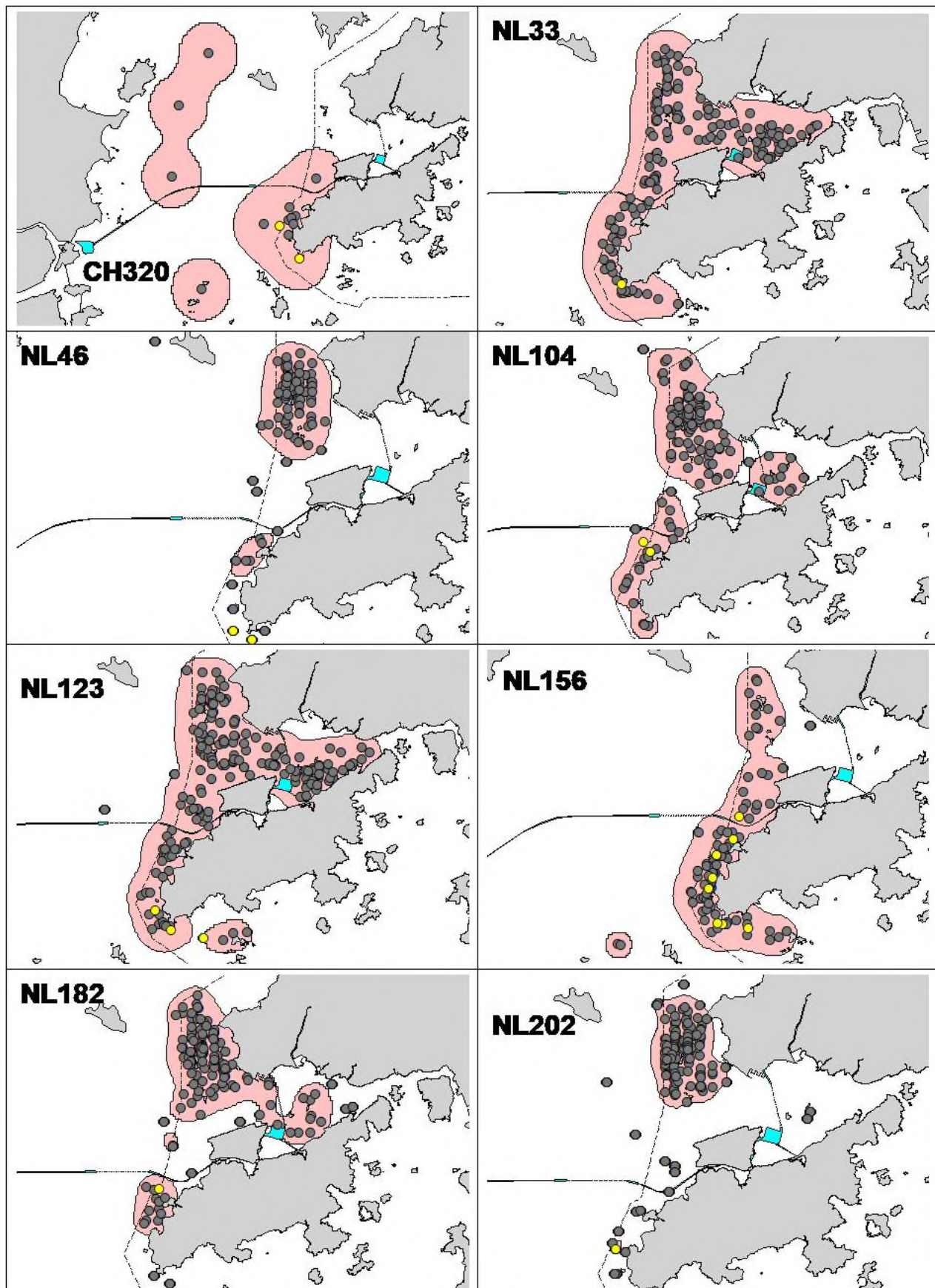
DOLPHIN ID	DATE	STG#	AREA
WL220	12/12/24	2	SWL
	17/01/25	9	SWL
WL221	26/08/24	7	SWL
	08/01/25	3	WL
WL229	26/08/24	5	WL
	16/09/24	3	WL
WL236	08/08/24	7	WL
	30/09/24	1	WL
WL245	30/07/24	2	WL
WL249	24/04/24	1	WL
	12/07/24	2	WL
	13/01/25	8	WL
	23/01/25	2	WL
WL250	18/06/24	2	SWL
	05/08/24	3	SWL
WL254	14/10/24	4	WL
	08/11/24	1	WL
	03/01/25	1	WL
	08/01/25	2	WL
	08/01/25	3	WL
	12/03/25	2	WL
WL256	18/06/24	1	WL
WL259	16/09/24	2	WL
WL272	12/06/24	1	WL
	25/11/24	1	WL
WL273	12/08/24	3	WL
	26/08/24	4	WL
	30/09/24	4	WL
	29/10/24	3	SWL
	03/01/25	3	SWL
	08/01/25	4	WL
WL283	07/08/24	5	SWL
	26/08/24	3	WL
	08/11/24	4	SWL
	25/11/24	1	WL
	17/01/25	5	WL
WL290	17/01/25	5	WL
WL294	24/04/24	1	WL
	17/01/25	2	WL
	12/03/25	3	WL
WL298	12/06/24	1	WL
	08/08/24	4	WL
WL299	24/04/24	1	WL
	30/07/24	2	WL
WL301	23/01/25	2	WL
WL302	07/06/24	1	WL
	30/07/24	1	WL
	07/08/24	4	WL
WL304	08/08/24	5	WL
	22/08/24	3	WL
WL305	02/12/24	1	WL
	08/01/25	3	WL
	23/01/25	2	WL
	07/06/24	1	WL
WL310	07/06/24	1	WL
WL311	08/08/24	5	WL
	23/01/25	2	WL
WL312	13/01/25	9	WL
	17/01/25	4	WL
WL313	17/06/24	2	WL
	13/01/25	7	WL
	23/01/25	2	WL

DOLPHIN ID	DATE	STG#	AREA
WL314	23/07/24	3	SWL
	08/08/24	7	WL
	08/01/25	4	WL
	12/03/25	2	WL
WL317	17/06/24	1	WL
	05/08/24	4	SWL
	14/10/24	4	WL
	29/10/24	1	WL
	08/11/24	4	SWL
	03/01/25	1	WL
	08/01/25	3	WL
	12/03/25	2	WL
WL318	23/01/25	2	WL
WL323	22/08/24	4	WL
	08/01/25	3	WL
WL324	07/08/24	1	WL
	13/01/25	5	WL
WL326	08/08/24	7	WL
	03/01/25	1	WL
	08/01/25	3	WL
	23/01/25	1	WL
WL327	08/08/24	7	WL
WL328	13/01/25	6	WL
WL330	12/07/24	1	WL
	12/08/24	1	WL
	16/09/24	6	SWL
	08/01/25	3	WL
	13/01/25	5	WL
	17/01/25	6	WL
WL331	16/05/24	1	WL
	30/05/24	1	WL
	30/07/24	4	SWL
	16/09/24	6	SWL
	30/09/24	2	WL
	29/10/24	2	WL
	08/11/24	2	WL
	15/11/24	1	WL
WL333	12/12/24	1	SWL
	12/07/24	1	WL
WL335	19/04/24	1	WL
	23/01/25	1	WL
WL336	16/05/24	1	WL
	16/05/24	2	WL
	22/08/24	4	WL
	29/10/24	2	WL
	25/11/24	2	SWL
	12/12/24	1	SWL
	03/01/25	1	WL
	13/01/25	8	WL
	17/01/25	3	WL
	23/01/25	2	WL
	12/03/25	3	WL
WL337	08/08/24	6	WL
	08/08/24	7	WL
WL338	12/07/24	4	SWL
WL339	07/01/25	1	NWL
WL340	12/08/24	2	WL
	26/08/24	1	WL
	08/01/25	3	WL
	17/01/25	6	WL
	23/01/25	2	WL

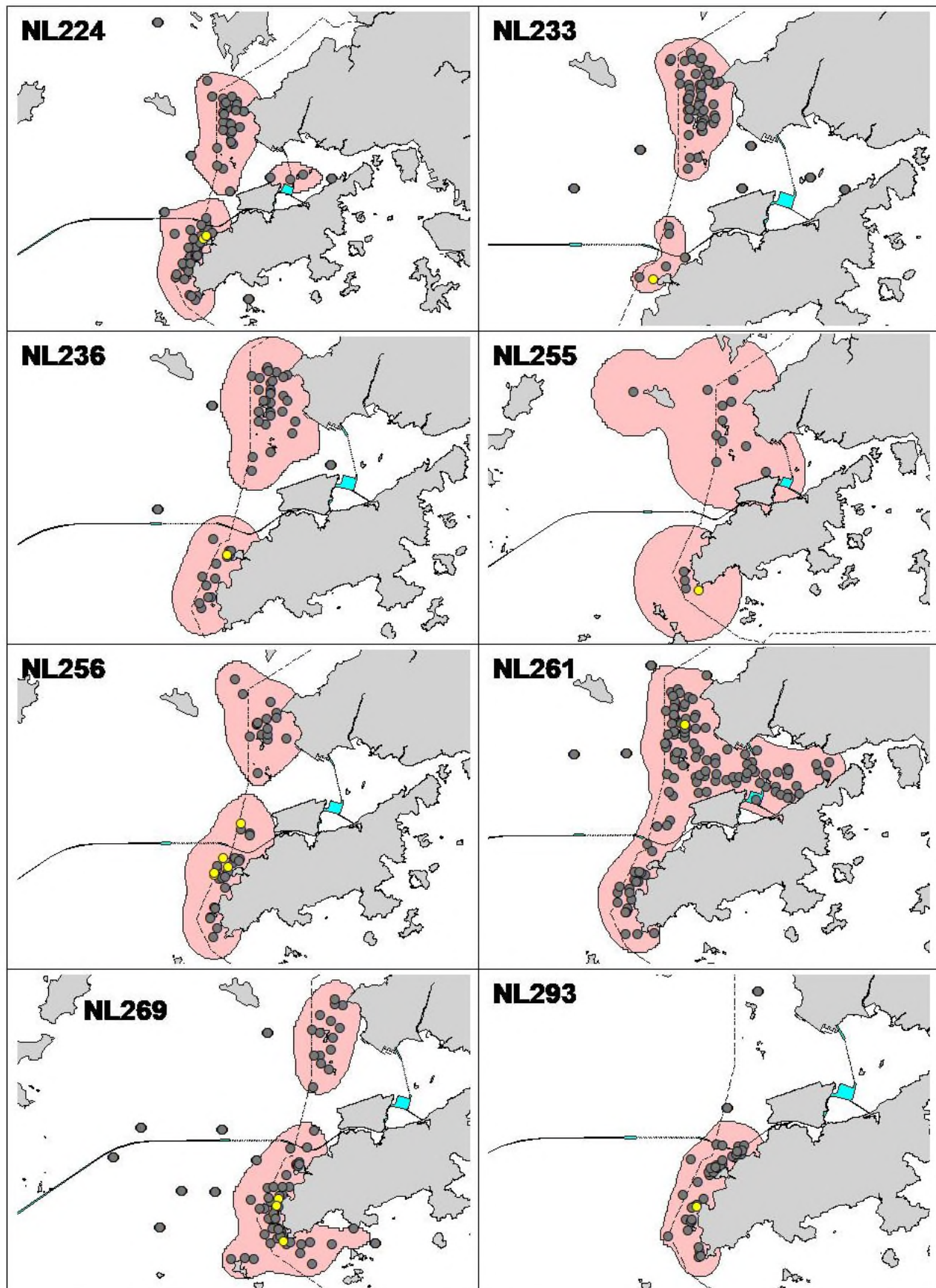
Appendix V. Ranging patterns (95% kernel ranges) of 107 individual dolphins with 10+ re-sightings that were sighted during 2024 (note: yellow dots indicates sightings made in 2024)



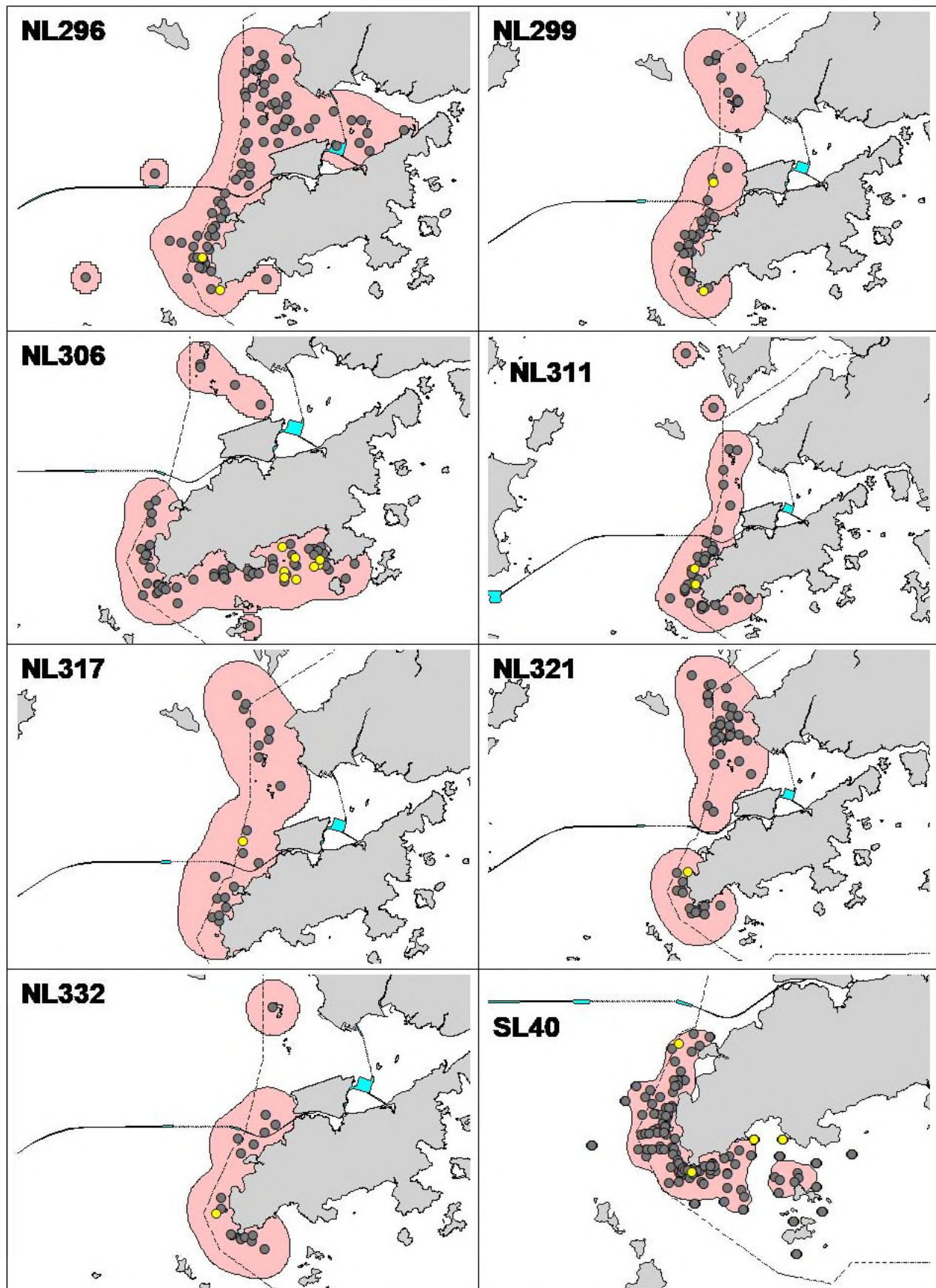
Appendix V. (cont'd).



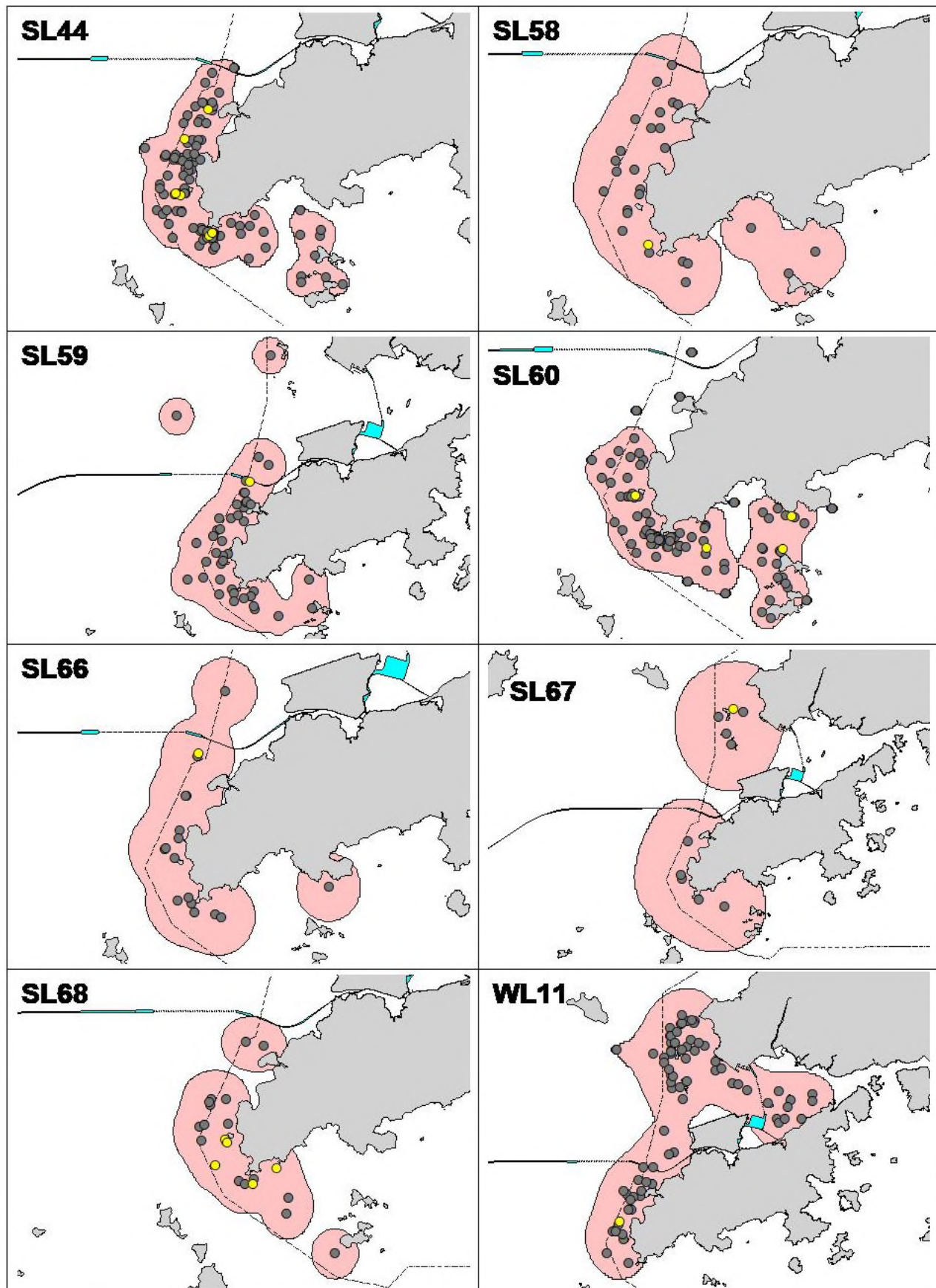
Appendix V. (cont'd).



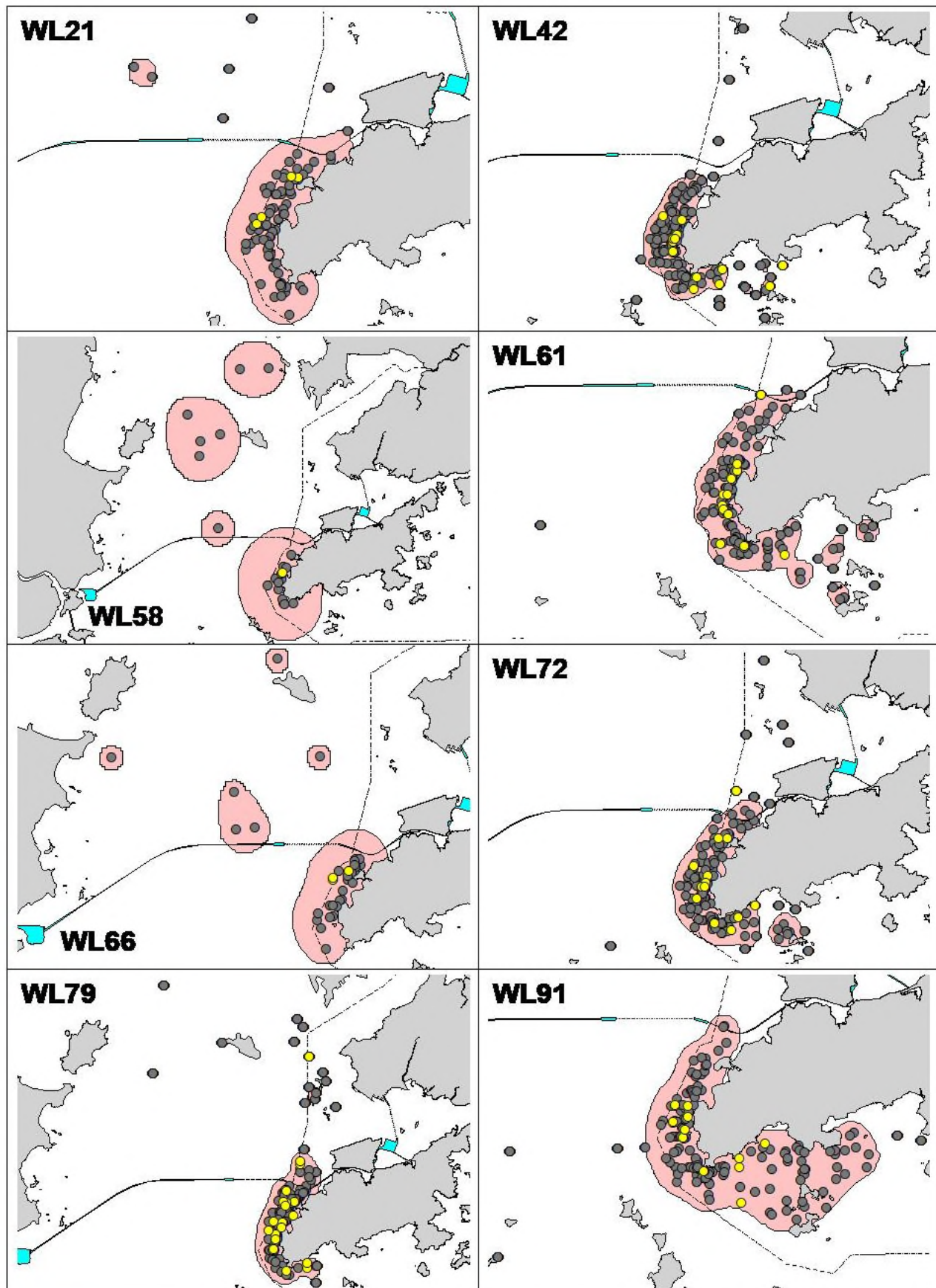
Appendix V. (cont'd).



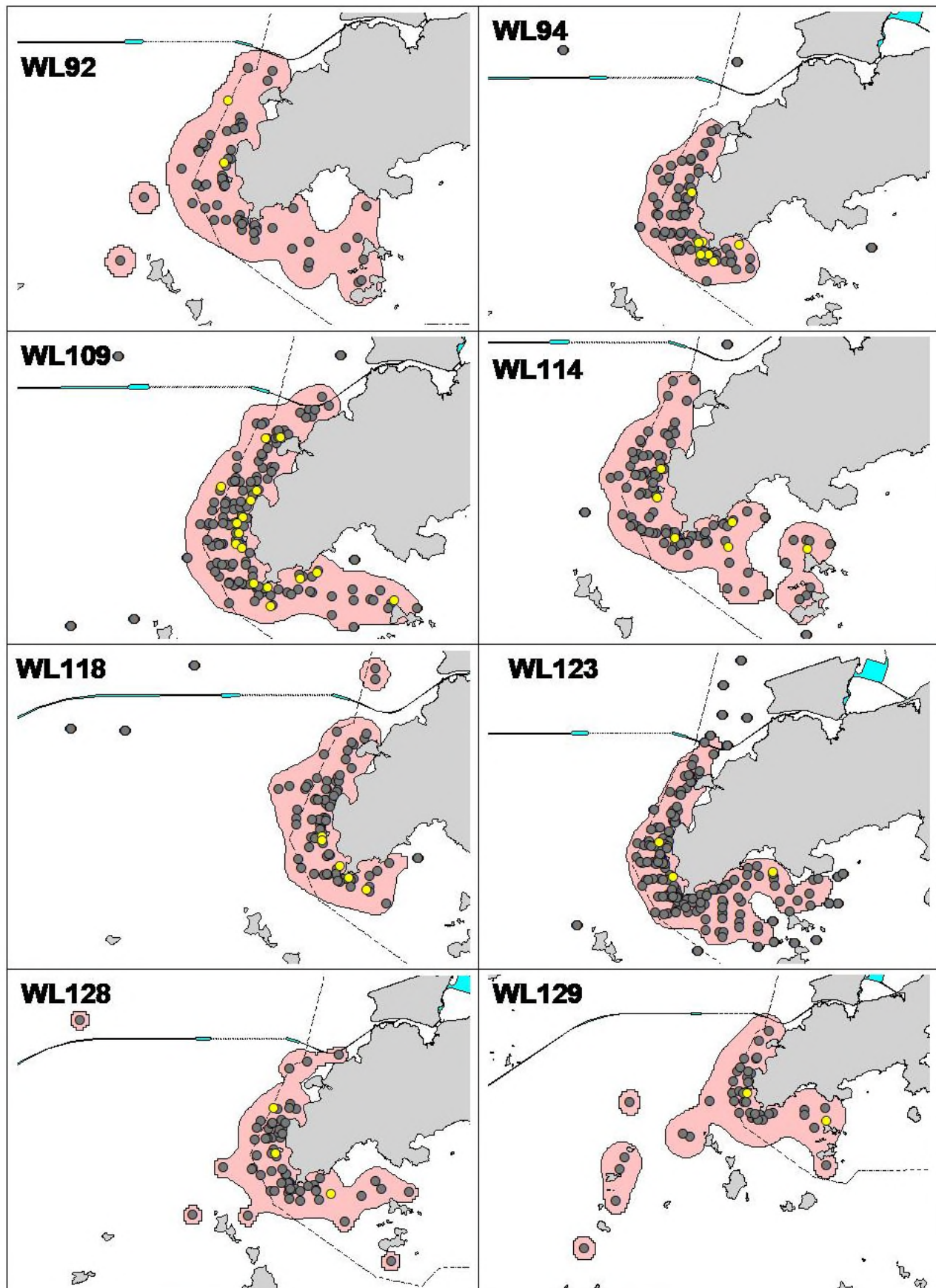
Appendix V. (cont'd).



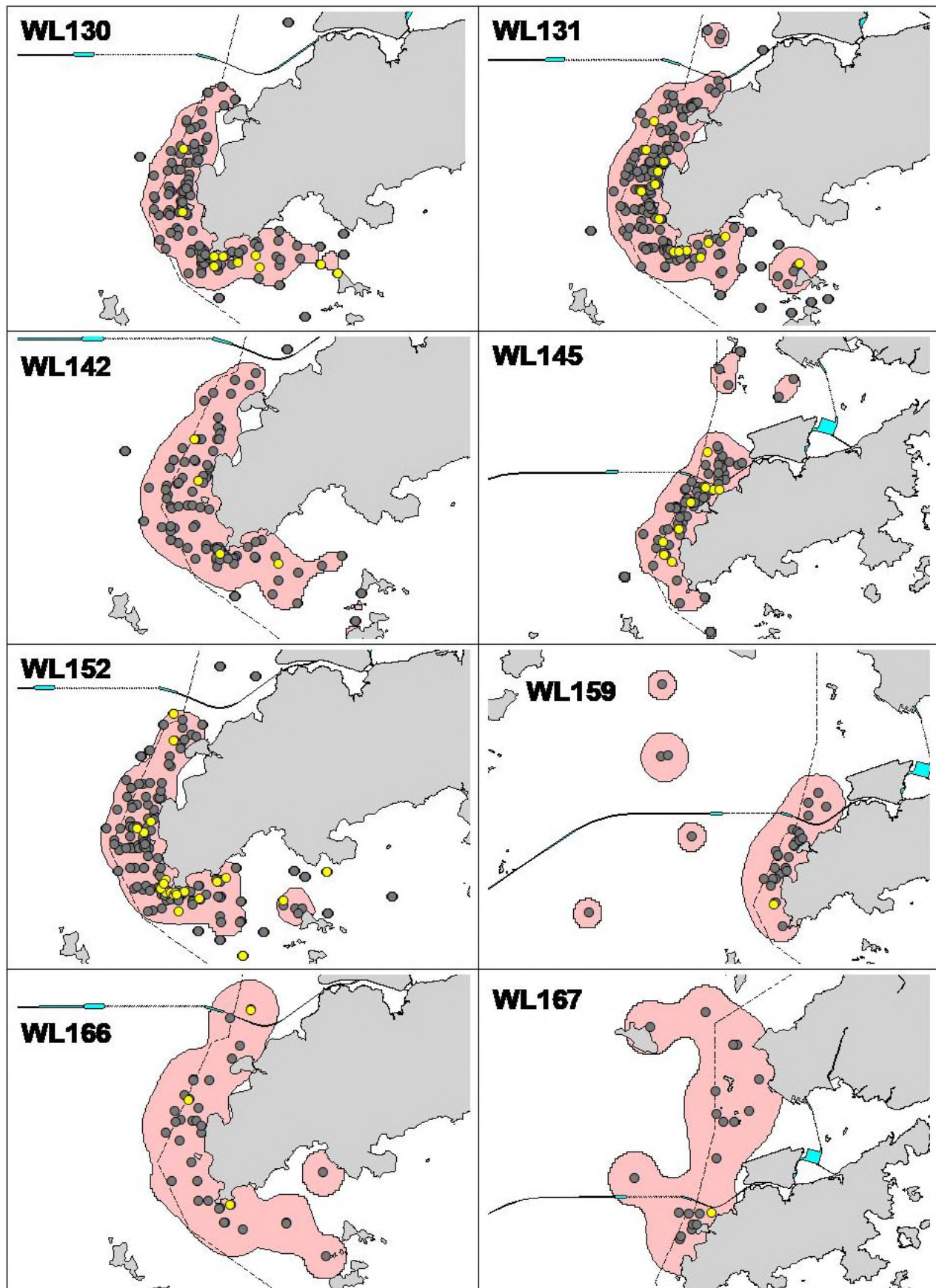
Appendix V. (cont'd).



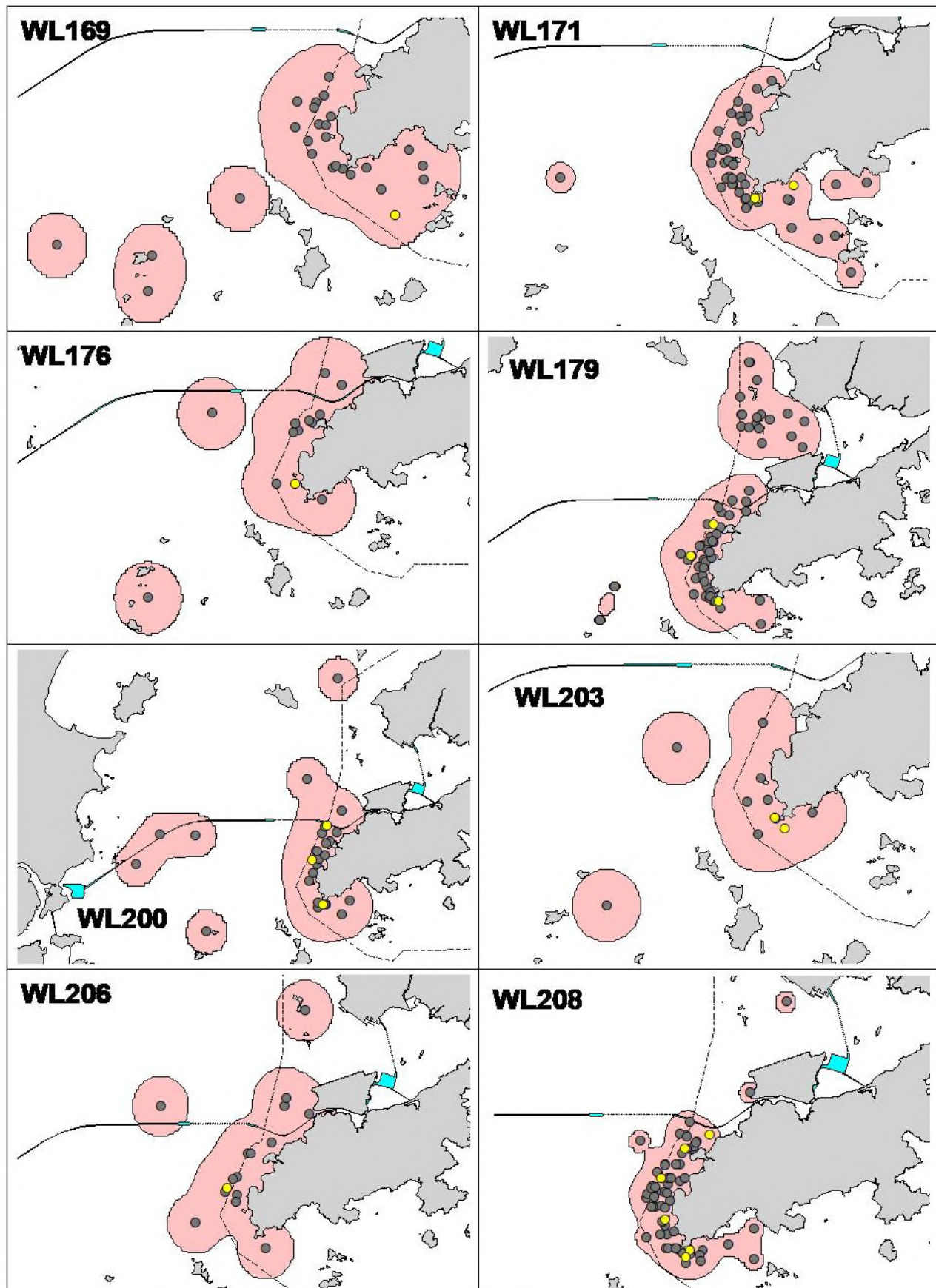
Appendix V. (cont'd).



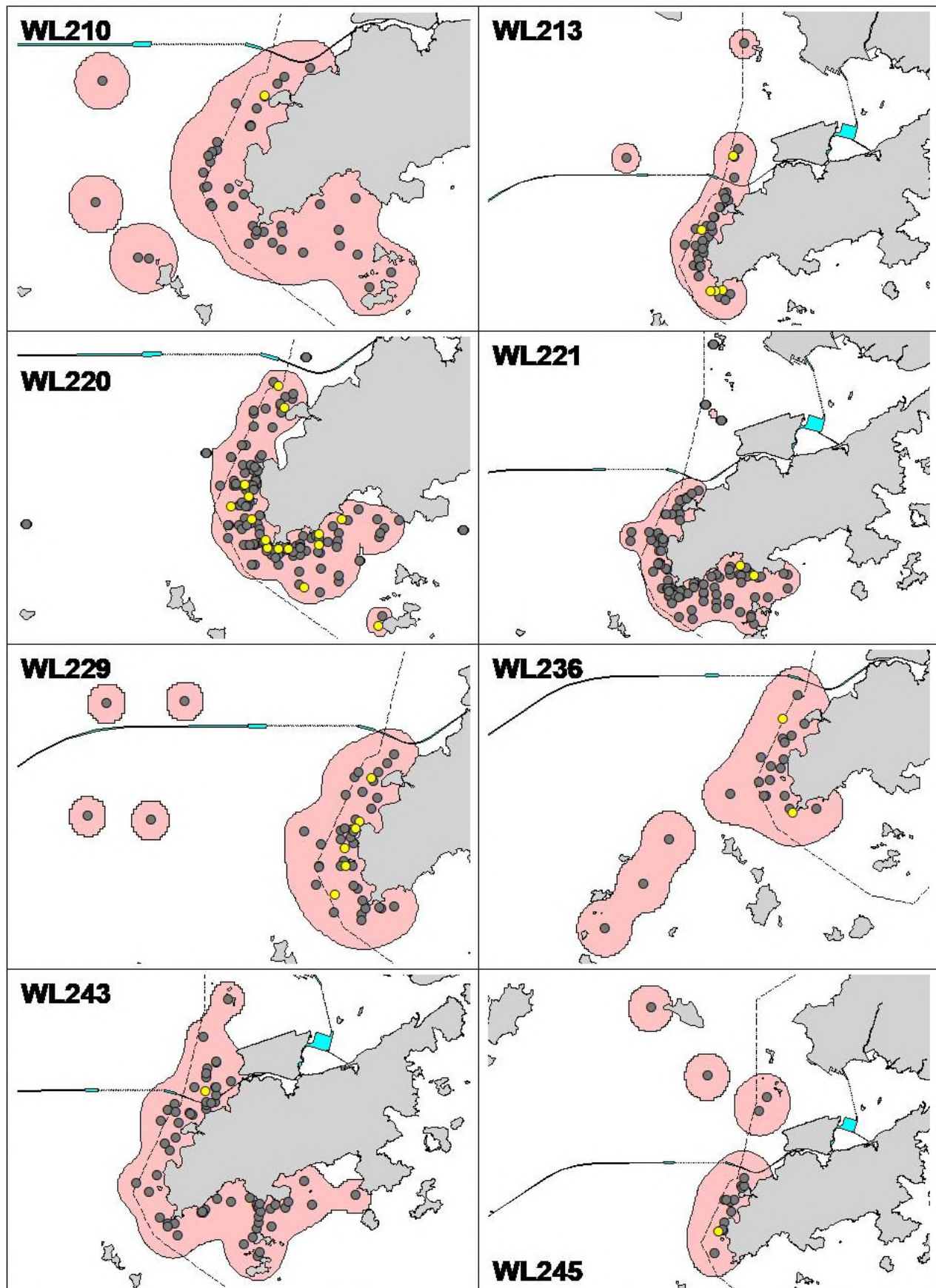
Appendix V. (cont'd).



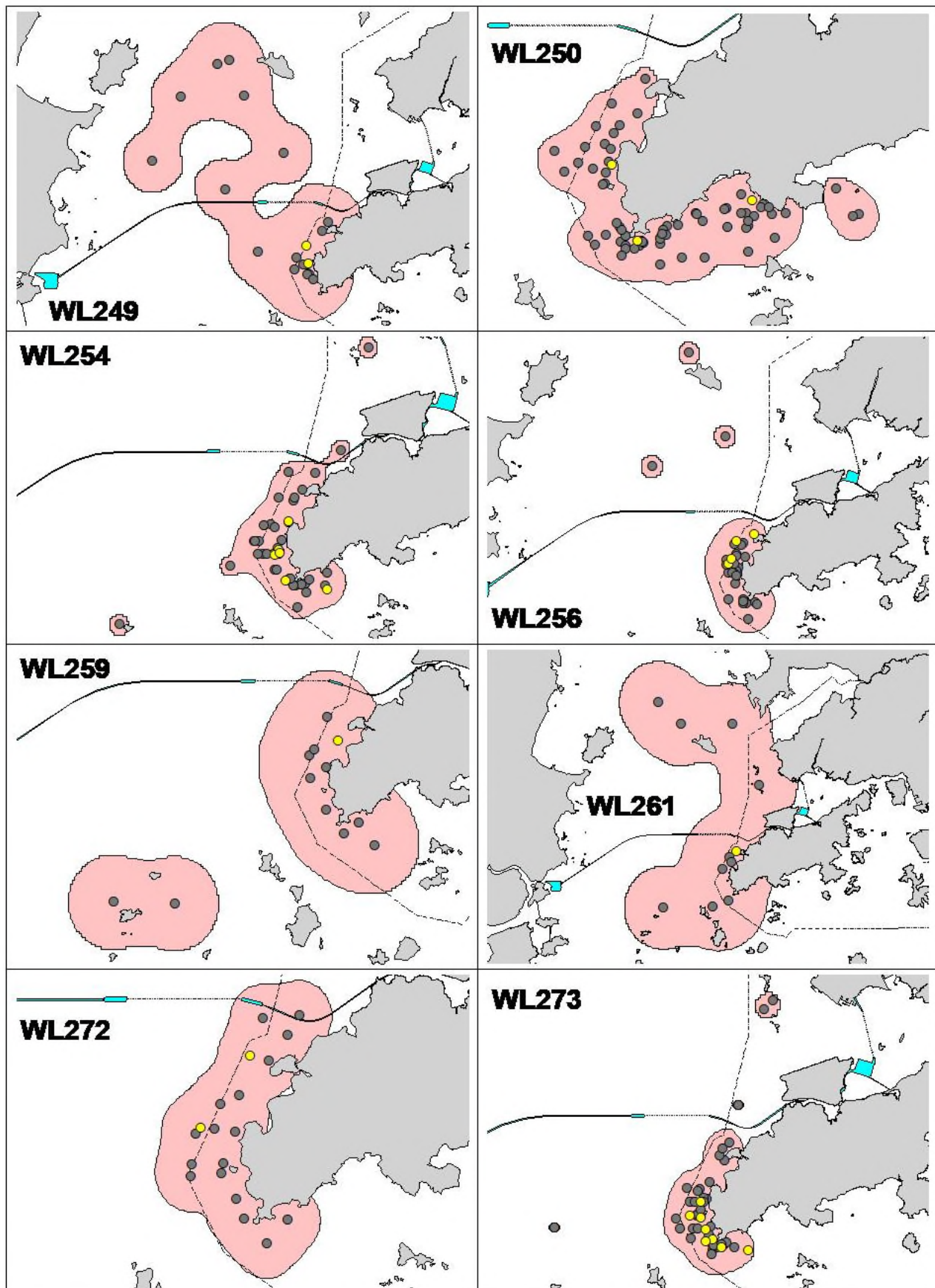
Appendix V. (cont'd).



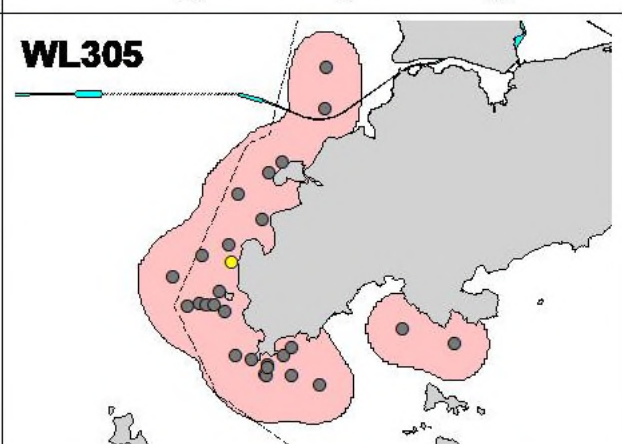
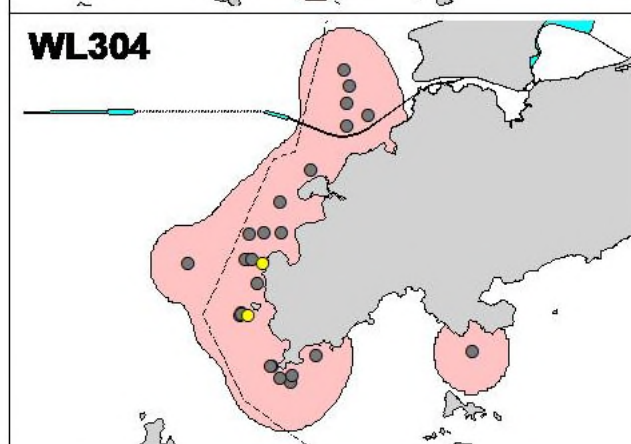
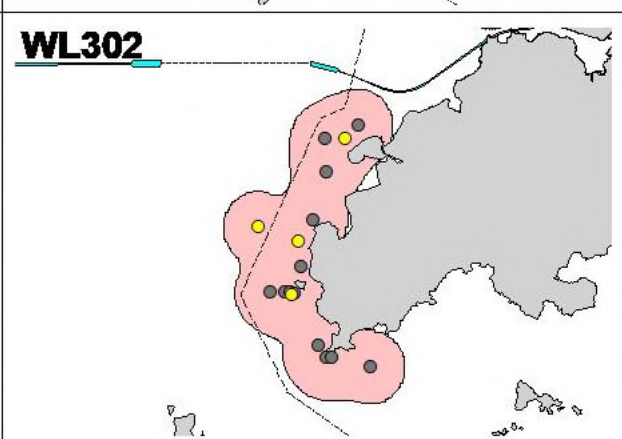
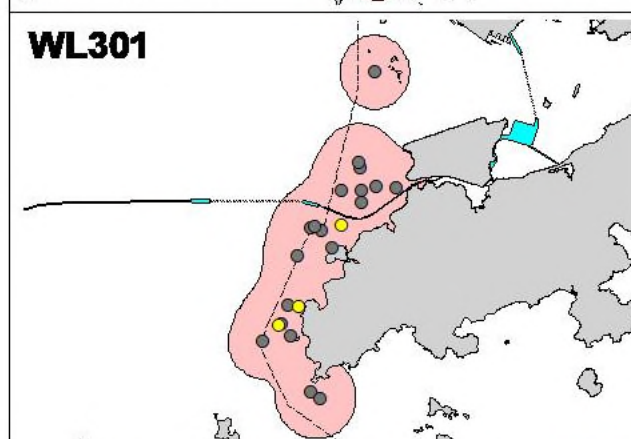
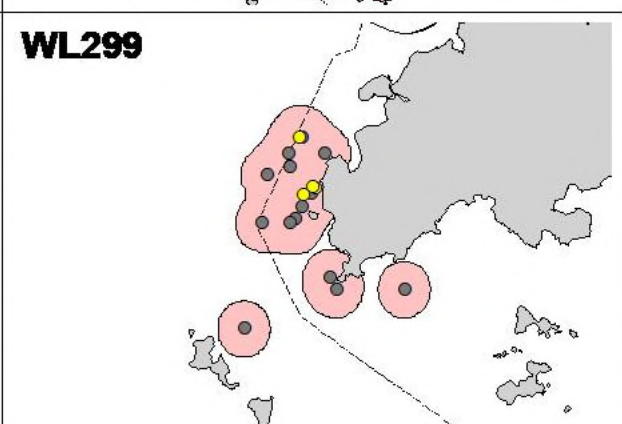
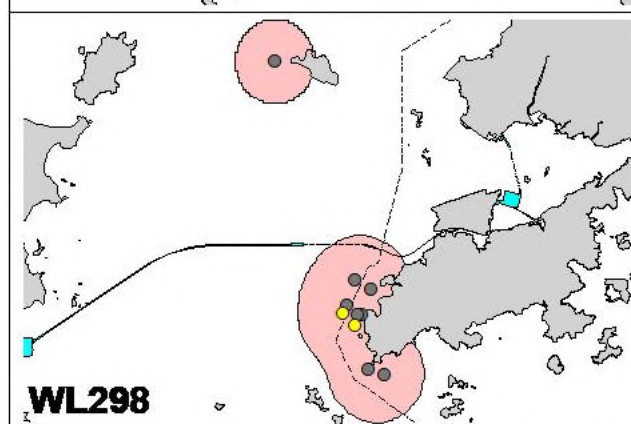
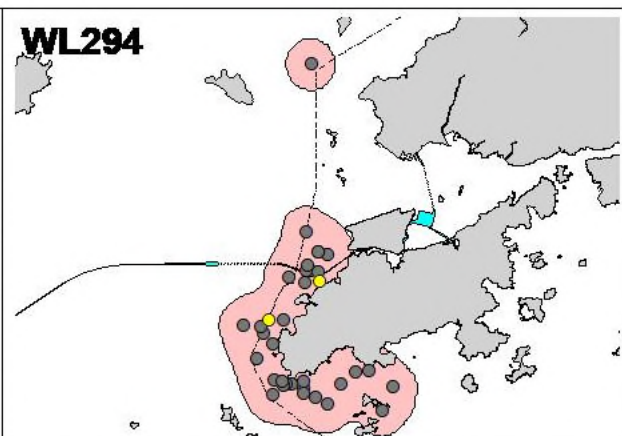
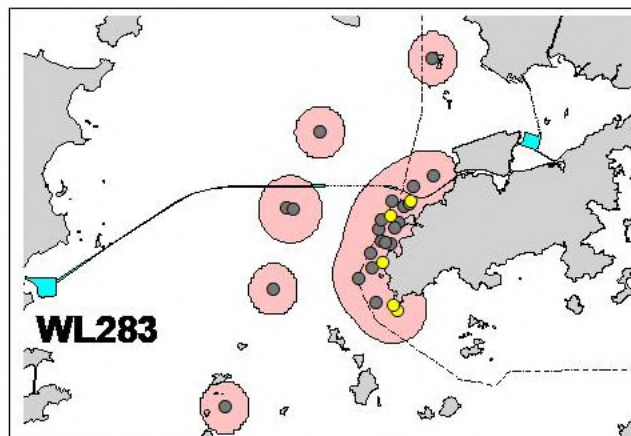
Appendix V. (cont'd).



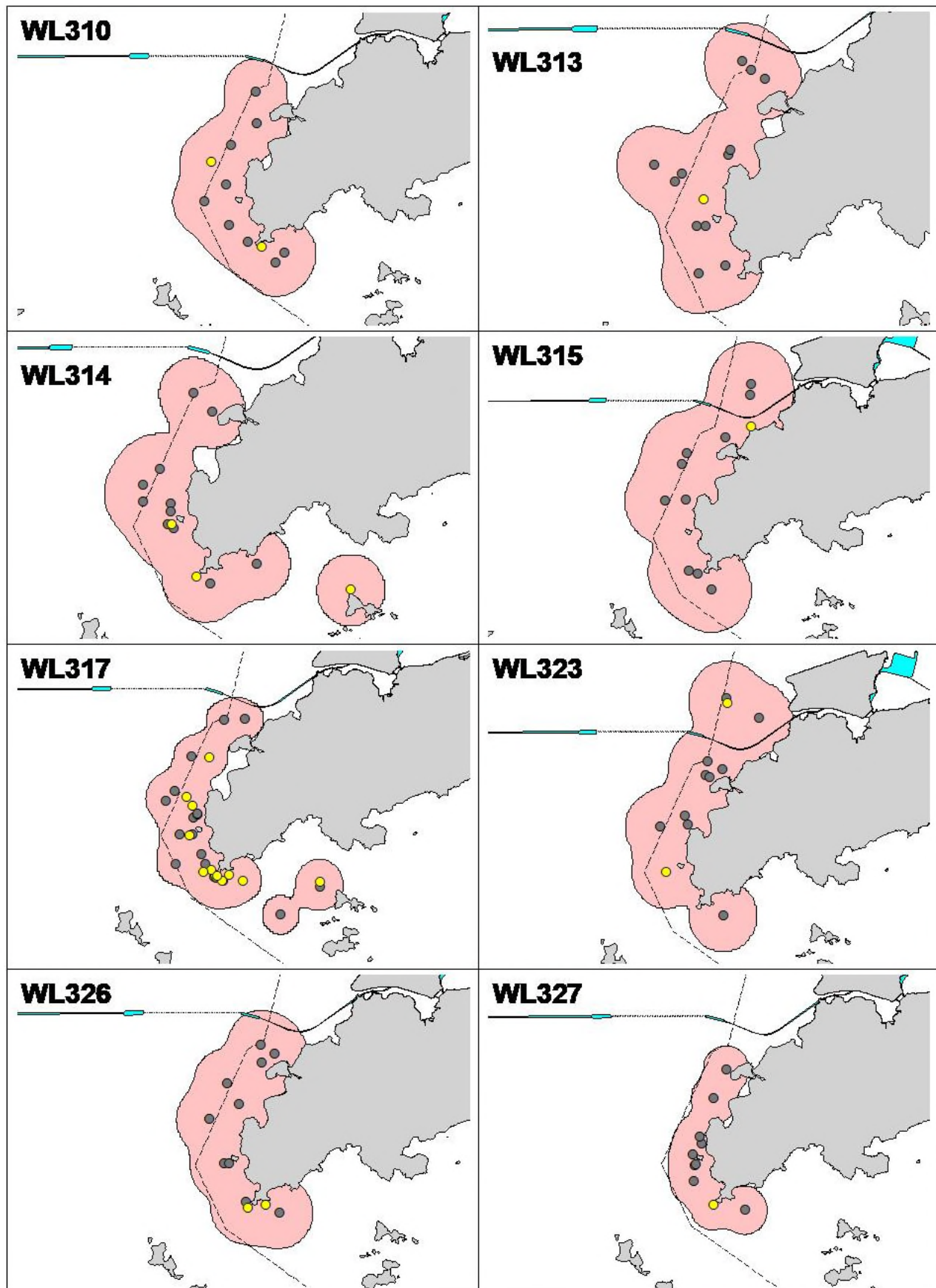
Appendix V. (cont'd).



Appendix V. (cont'd).



Appendix V. (cont'd).



Appendix V. (cont'd).

